wfview

Open Source interface for Icom and Kenwood transceivers

User Manual

This is the official manual for wfview.

If you are just starting, please read the Getting Started chapter first. Most settings should be very intuitive, however, detailed explanations of each button and control are included in the links below.

For support, please visit https://forum.wfview.org/ and sign up.

A downloadable and printable copy of the manual is available here in PDF (version 1.64).

Getting Started:

- First time guide
- Radio Setup:
  - Icom IC-7300
  - Kenwood TS-890S
  - Yaesu FT-710 over USB
- FAQ
- Xiegu X6100 (offsite link)

User Interface:

- Main Window
  - CW Sender
  - Repeater and Split
- Band Switcher
- Frequency Input
- Settings
- Memory Editor
- USB Controllers (Icom RC-28, Stream Deck, sunSDR, Contour Shuttle Express, Contour ShuttlePRO, etc)
- Keyboard Shortcuts
- Appearance Customization with Examples

Interaction with Other Programs:

- Sharing control with other programs (START HERE)
  - Hamlib rigctld emulation (recommended)
  - Virtual Serial Port
- Audio Configuration for Windows/Linux (audio sharing)

Built-in Server:

- Remote Operation (server) with common hardware

Inexpensive Dedicated Server using SBCs like the Raspberry Pi and Inovato Quadra

How-To:

- Send us your logfile
- APRS packet decoding in linux over wfview's network audio
- Serial Port Management (Linux udev rules, Device Manager, etc)
- Inexpensive Dedicated Server using SBCs like the Raspberry Pi and Inovato Quadra

Advanced:

- Command-Line Options
- Logfile
- Preferences File
- Using older (unsupported) radios
- Rig Files and Rig Creator
- Supported Radios and Features
- Headless Server (beta)
- Compiling (see the developer's corner)

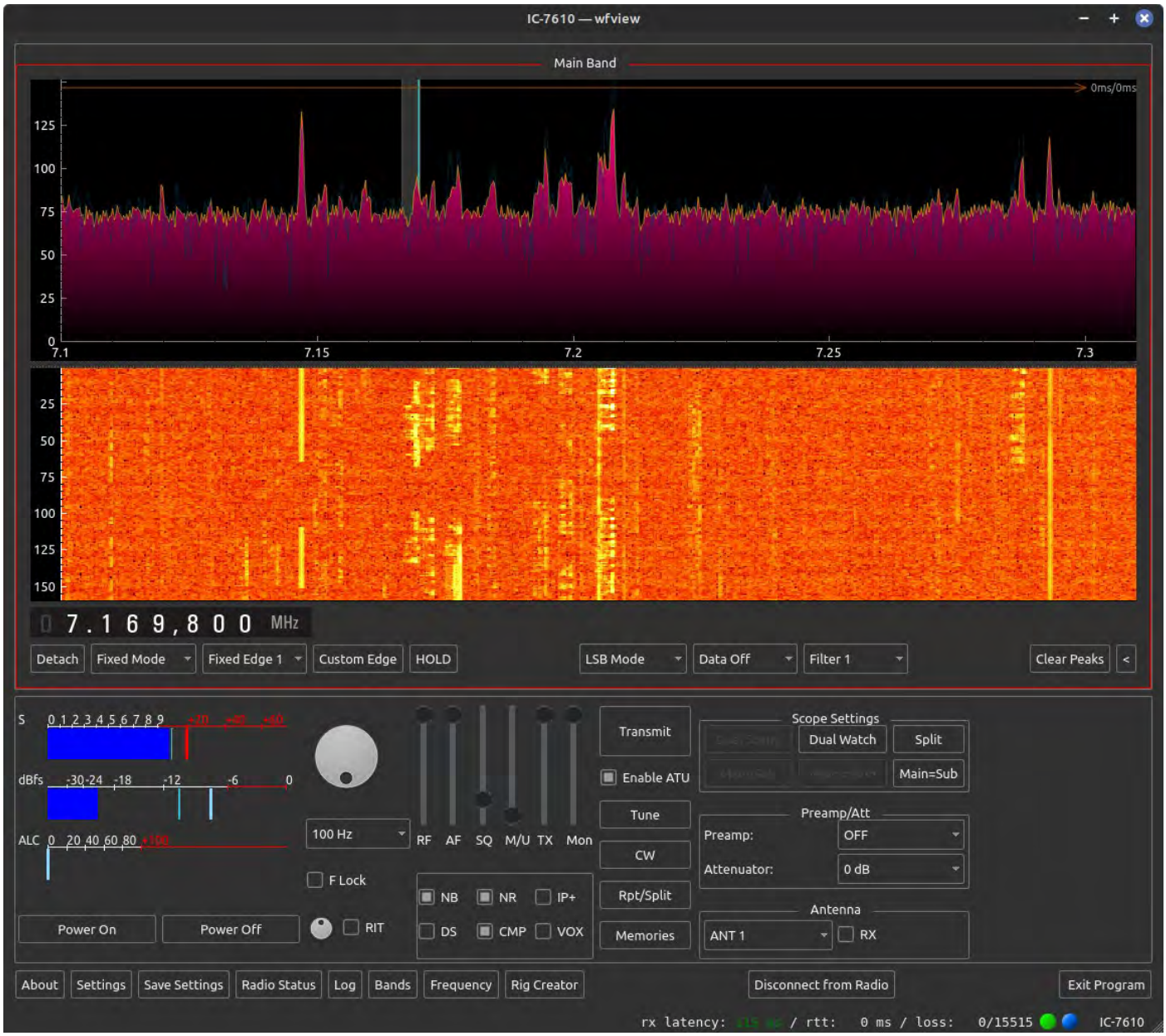
Older versions of the User Manual may be found here.

wfview

Open Source interface for Icom and Kenwood transceivers

Welcome to wfview.org

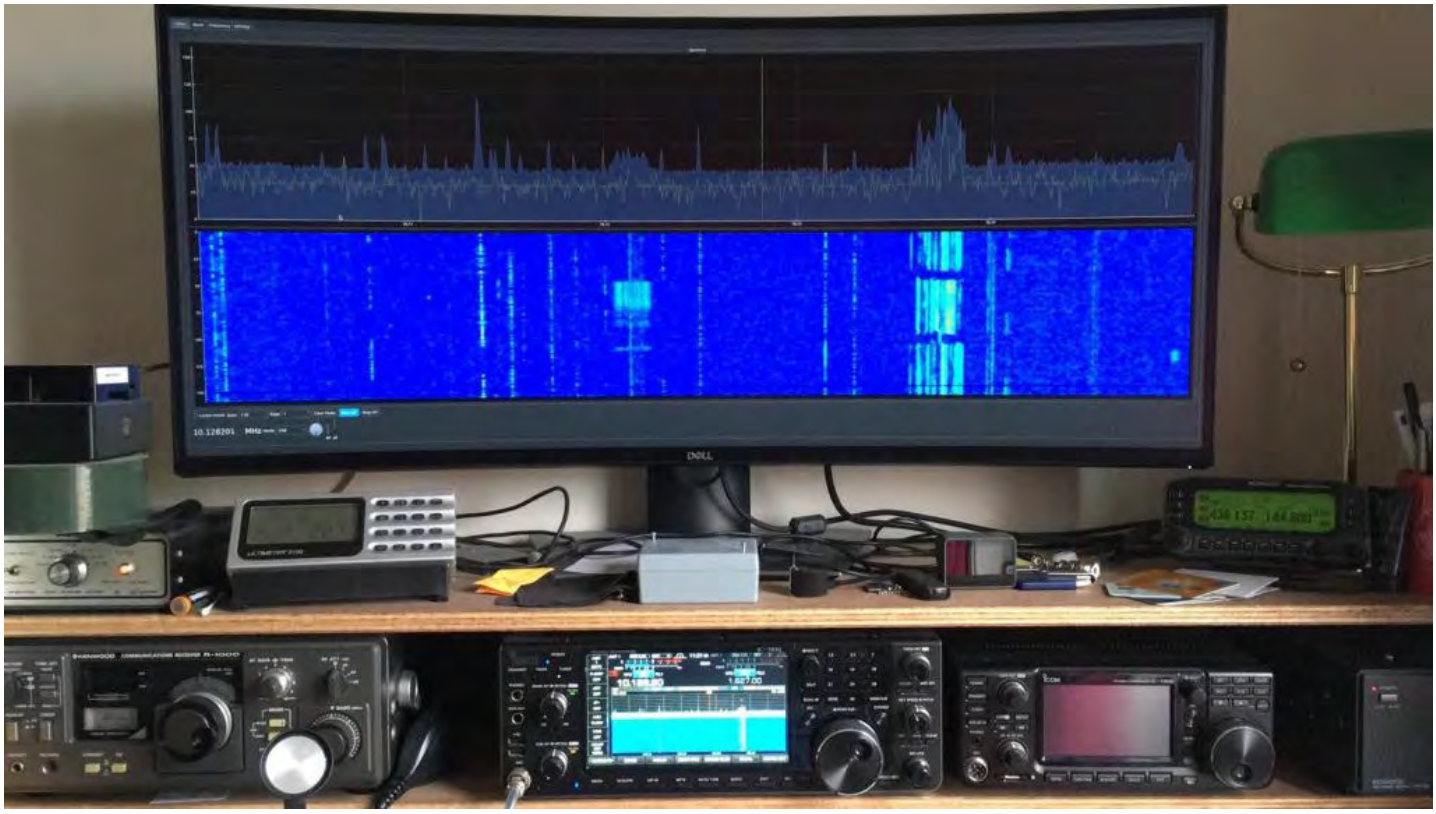
Hello, and thank you for coming here to learn about wfview. wfview is a program developed by amateur radio enthusiasts to control modern Icom and Kenwood ham radios. wfview is free and open source software.



wfview 2.03

wfview controls modern Icom rigs using either a USB serial connection or OEM network (ethernet or wifi) connection. Live, real-time spectrum analyzer data are displayed, and rig controls are presented. Additional programs may tie into the CIV bus using the pseudo-terminal device (linux and macOS), virtual serial port loopback

(windows), or hamlib-compatible rigctld server.



wfview provides the user with controls that may be comfortably operated from a keyboard, mouse, or touch screen interface.

On rigs with built-in network interfaces (such as the TS-890S, IC-705, IC-905, IC-7610, IC-7850, IC-R8600, IC-9700, and Xiegu X6100), live audio streaming is available for both receive and transmit. Special care was taken in the creation of the network code for very low latency audio — ideal for general QSOs as well as time-sensitive modes such as FT8 and JT65.

On the IC-7300 (and other non-network rigs), wfview can serve as a USB to Ethernet “bridge”, connecting the IC-7300 to the network for remote control and live streaming full-duplex audio using another instance of wfview on a remote computer. The server can be a humble Raspberry Pi or a normal desktop computer.

wfview has been developed with an eye towards compatibility. Even though our target platform consists of modern-era transceivers, wfview’s command dictionary is focused on commands with the most compatibility. Many of Icom’s transceivers from the last 20 years will work to some degree with wfview, and we are always adding more.

The user may select between the current desktop theme, an included “dark” theme (qdarkstylesheet), or a user-specified qt stylesheet. All colors can be customized via the built-in User-Defined Color Editor or by specifying a custom stylesheet.

Getting Started

Once wfview has been downloaded, you will need to set up the radio and the program.

Radio Settings (via the Radio's Menus):

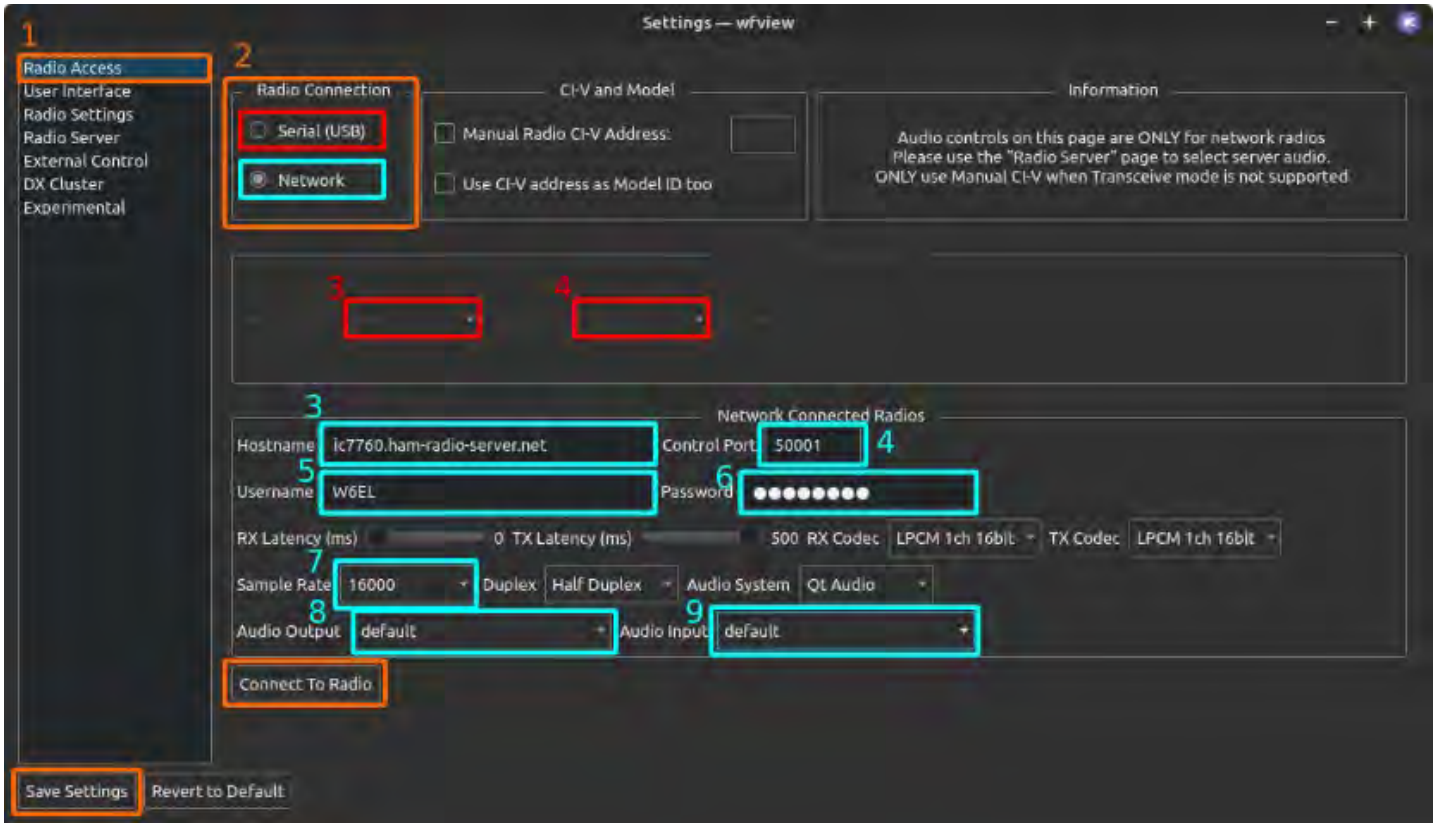
- Icom radios:
  - Set the radio to the default CI-V address.
  - Enable CI-V Transceive. This is required unless the radio does not support this option (older pre-2012 radios).
- For Icom radios with ethernet or wifi (IC-705/905/7610/9700):
  - Set a known username and password
  - **enable** Network Control.
  - Make a note of the radio's ip address, which is visible in the Network Settings screen on the radio.
  - Use the default ports (50001, 50002, and 50003), or use custom ports that you know
- For the Kenwood TS-890:
  - Enable and setup the Network.
  - In the KNS menu, set the administrator username and password
  - Enable KNS and VoIP (it will not be possible to enable these until the username and password are set)
  - A step-by-step guide for the TS-890S is available here.
- For radios with Serial or USB (IC-7100/7200/7300/9100, TS-590S):
  - Set the baud rate, on the radio and in wfview, to the highest supported rate, and make a note of the rate. Here is an in-depth guide to IC-7300 radio settings. If you have difficulty with collisions, try progressively slower baud rates and higher polling rates.

wfview settings:

When you open wfview for the first time, you will be guided to the Radio Access Settings page. However, you can get to the settings page any time by pressing "Settings" at the bottom of the wfview main window and then selecting "Radio Access".

Here is how the settings page looks. Note that when you enable "Network" or "Serial (USB)", different options will be available.

Fill in the following information:



First press [1] “Radio Access” on the left side if it is not already selected.

For Serial (USB Port) radios (IC-7300/7200/7100):

- Select [2]: “Serial (USB)”
- Select the [3] Serial Device.
- Select the [5] Baud Rate.
- **Do not** check “Manual Radio CI-V Address”
- **Do not** check “Use CI-V address as Model ID too”

For Network (ethernet or wifi) radios (IC-705/905/7610/7850/9700):

- Type in the [3] IP address of the radio or radio server. If you know the hostname, you may supply the hostname instead.
- Enter the [4] Control Port. The default is 50001 on most radios.
- Enter the [5] username as set on the radio.
- Enter the [6] password as set on the radio.
- Select [7] the desired sample rate. 16000 Hz is plenty for most applications. Do not leave this blank.
- Select the [8] audio output device you wish to use. Typically this is your headset or speakers. Linux users may simply select “default” for their desktop environment’s default audio choice.
- Select the [9] audio input device you wish to use. Typically this is your microphone. Linux users may simply select “default” for their desktop environment’s default audio choice.
- **Do not** check “Manual Radio CI-V Address”
- **Do not** check “Use CI-V address as Model ID too”
- **Important:** Set the RX Codec and TX Codec to “LPCM 1ch 16bit”. You may experiment with other codecs later.
- **Important:** Set the RX and TX Latency sliders to a middle value of 250ms.

Next:

- Press “Connect to Radio”
- And if successful, press “Save Settings”

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Version	1.0
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Last Updated	7 March 2025

Printable Manual 2023-12-30

OBSOLETE This printable manual is perfect for situations where you might not have internet access, or you wish to search the manual for specific phrases. The manual was made from our usual website manual by wfview user Andy (VE3NVK). NOTE: This manual is for Version 1.64. The current manual may be found HERE.

Kenwood TS-890S Setup

To setup the Kenwood TS-890S for use with wfview, please use the following steps.

KNS is Kenwood's name for remote network access. It will be helpful to write down the Username, Password, and IP Address as you get to those steps. You will need to know them later.

Network Setup

Press MENU. On the right hand side of the screen, the second menu down should read "LAN". If it does not read "LAN", then press the bottom-right side menu button which will read "MORE" and then you should see "LAN" as the second menu item down. Press "LAN".

For beginners, you should set menu item 0, DHCP, to "On". More advanced users may be comfortable with setting this to "Off" and then assigning an appropriate manual IP address. Please do not do use Manual IP addresses unless you really understand what you are doing. Also, please note, the grayed-out IP information shown when DHCP is "On" is not meaningful. The assigned IP address, which you must make a note of, is shown in the bottom-right side of the display.

Press “ESC” to exit the menu.

KNS Setup

Now press “MENU” to enter the menu again.

On the right hand side of the screen, the fourth item down should read “KNS”. If it does not, press the button at the bottom-right side of the screen which is labeled “MORE”. Then press “KNS”.

The screen will show “KNS Status”. Press F4 at the bottom of the screen, which reads “CONFIG”

You will need to setup an “Administrator ID” and “Administrator Password” before proceeding. This is the username and password needed in wfview. For easier typing, plug in a USB keyboard. Set those to values you can remember.

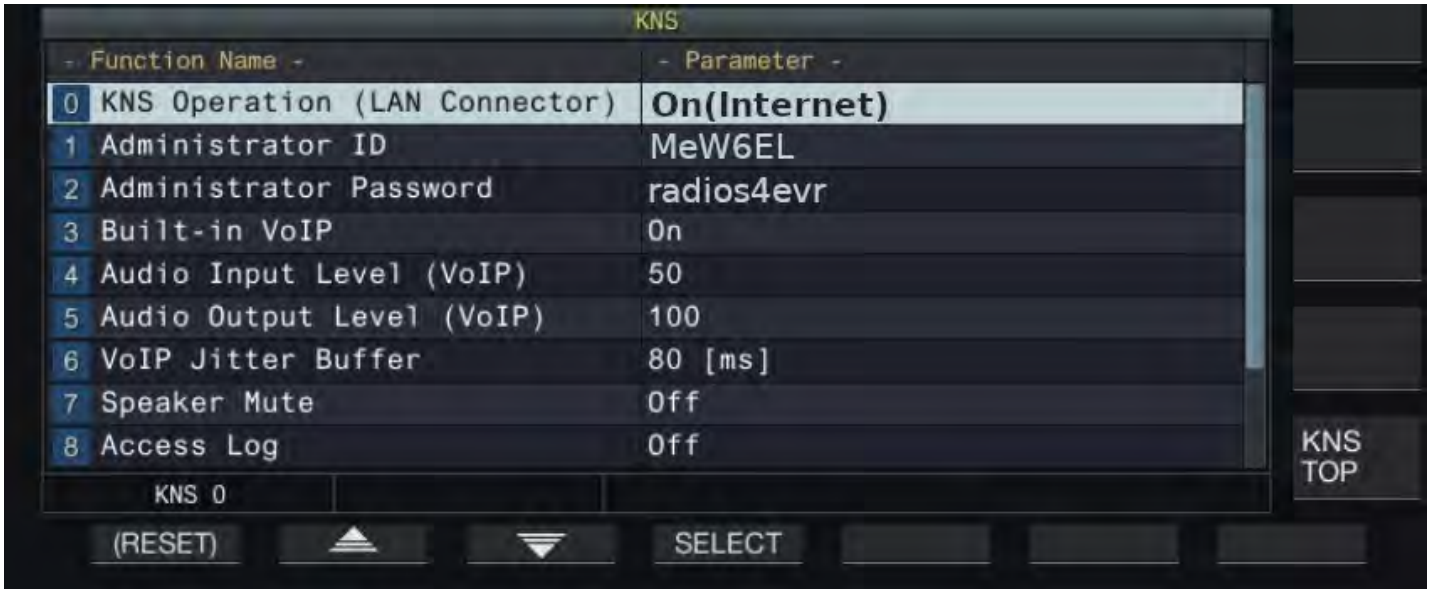
Now, enable “KNS Operation”, which is menu item 0. Set this to “On(Internet)”. You may elect to set it to “On(LAN)”, which makes the waterfall move much faster but causes a lot of traffic on your network. Important: You cannot enable the KNS system until an Administrator username and password are set.

Now enable the “Built-in VoIP” (menu item 3). Important: You cannot enable the VoIP system until an Administrator username and password are set.

On my TS-890, the “Audio Input Level (VoIP)” is set to 13 and the “Audio Output Level (VoIP)” is set to 100. The radio actually runs an ADC and DAC for the VoIP system, thus, it is important to set these so that there isn’t any distortion.

Scroll down to menu item 10, “Session Time” and set it to “Unlimited”. Otherwise you’ll get disconnected by surprise. I suppose you can set that to some other value if you really want to though.

You might also want to set Menu 7, “Speaker Mute” to “On” so as to avoid disturbing people with loud DX at random hours. Or just put a headphone plug in the front of the radio.



Verify

Once these items are complete, you may wish to ping the radio from your computer and verify that it responds. The radio should even respond when it appears to be “off”. This is because the computer inside the radio is always on, even when the radio and display portions are powered down.

Resources

Here is a link to Kenwood’s documentation on using their network software. The document is complex and wonders around between radio and desktop software options. Kenwood shows how to use their software via a USB port to set the radio up, rather than using the radio’s built-in menus. I find this to be needlessly complicated but anyway, here is the documentation: https://www.kenwood.com/i/products/info/amateur/ts_890/pdf/ts890_kns_manual_e.pdf

Chapter 15 of the regular “User Manual” details the KNS Setup much better.
https://manuals.jvckenwood.com/download/TS-890S_~JA_1E8E

Audio

Don’t forget to set the transmit audio source to LAN. You can do this by holding down the “MOD.SEL / DATA” button near the VFO knob on the radio, or use wfview’s “Radio Settings” page to set the modulation source to “LAN”

Remote Recommendations

The software running inside the TS-890S is known to occasionally glitch. This can result in no audio, corrupted waterfall data, and difficulty connecting. The only way to truly reboot the computer inside the TS-890S is to remove the 12V power for a few seconds and then re-apply. If you’re accessing this radio remotely, it would be good to have a way to remotely cycle power. This could be a high-current relay or one of those 120/240V switched outlets (running to a 12V supply).

Yaesu FT-710 Setup

Yaesu FT-710 Field HF/50MHz 100W Portable Transceiver

This page describes how to setup wfview with the Yaesu FT-710 via the USB port interface

On the radio:

Enable the SCU-LAN10. This enables the spectrum output. You don't need to have a SCU-LAN10 to do this, just enable the item in the radio's menu.

On the computer:

You will need to install the driver for the SPI chip inside the FT-710. This driver is from the FTDI company for their FT4222 chipset.

Windows:

Yaesu has a download which has the driver you need.

Go to this page on Yaesu's site.

Click "Files" and then click "FT-710 USB Driver Virtual COM Port Driver (Windows 11/10)"

There is also a PDF you can download which shows the steps, it is listed on the same page under "USB Driver Virtual COM Port Driver (Windows 11/10) Information"

Next, you need to download the "DXX" driver from here

Scroll down the page until you see "Windows (Desktop)" or "Windows (Universal)"

Linux:

The driver can be downloaded from here. Follow the directions from the "00_README.txt" file, which is available at the link.

You will need to set permissions up so that you can access the new SPI device.

As root, create and edit the file `/etc/udev/rules.d/99-ftdi.rules`

Insert the following text:
`SUBSYSTEM=="usb", ATTR{idVendor}=="0403", ATTR{idProduct}=="601c", MODE="0666"`

Note: More advanced users might want a more secure config which owns the device to their username.

Save the file.

Reload the udev subsystem:

```
sudo udevadm control --reload-rules
sudo udevadm trigger
```

Reconnect the radio's USB port

Verify the presence of the serial device for the radio. The device "if00" is the "CAT-1" interface, which we want to use with wfview. This is also the "00" device from the second command.

```
ls -l /dev/serial/by-id/
Silicon_Labs_CP2105_Dual_USB_to_UART_Bridge_Controller_00E1ABCD-if00-port0
Silicon_Labs_CP2105_Dual_USB_to_UART_Bridge_Controller_00E1ABCD-if01-port0
grep . /sys/class/tty/ttyUSB*/device/bInterfaceNumber
/sys/class/tty/ttyUSB0/device/bInterfaceNumber:00
/sys/class/tty/ttyUSB1/device/bInterfaceNumber:01
```

Check for the FT4222 device:

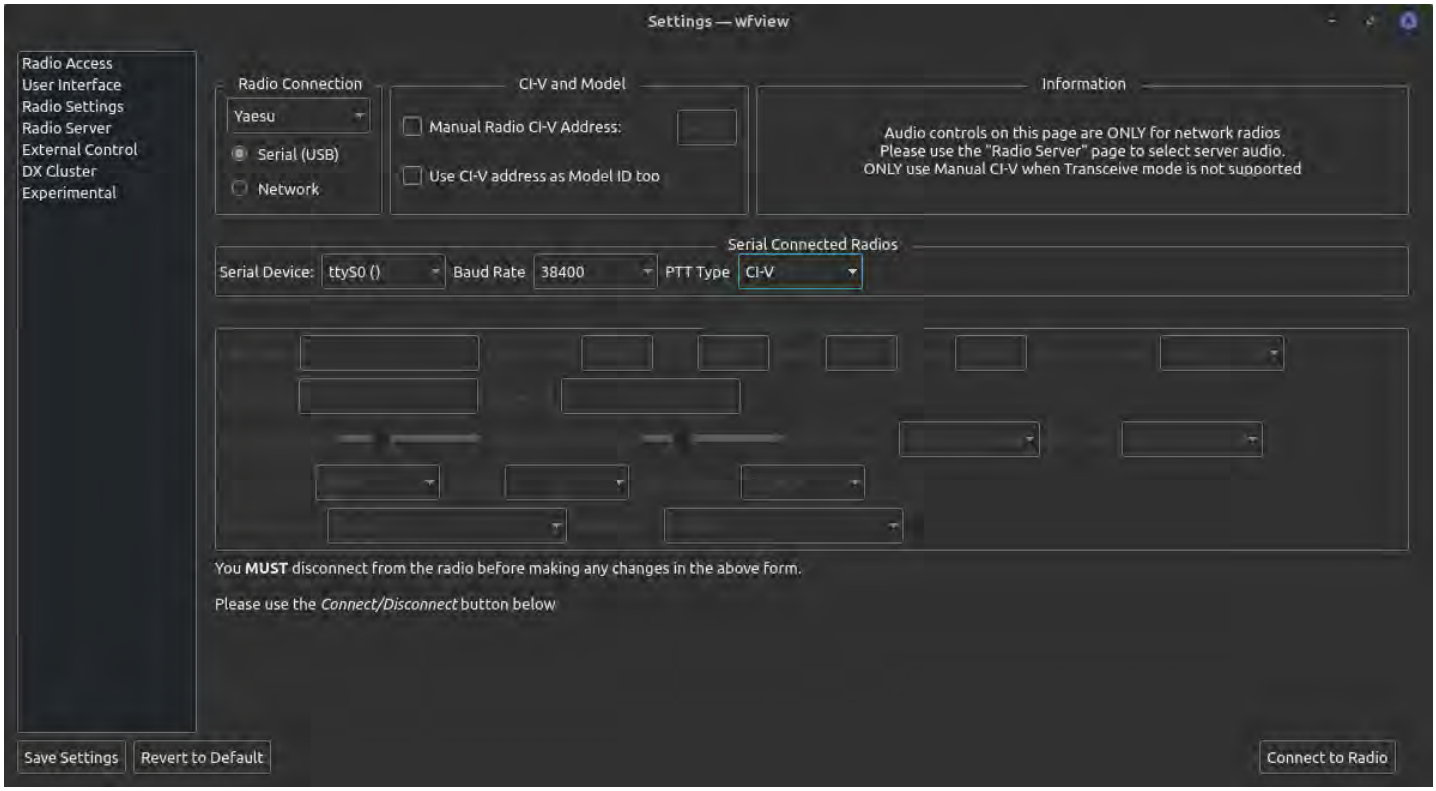
```
lsusb | grep FT4222
Bus 001 Device 005: ID 0403:601c Future Technology Devices International, Ltd FT4222H
```

wfview Setup:

With wfview, you will need only to specify the serial port. wfview can use the FT4222 driver to find the SPI port automatically.

Key points to set in wfview:

- Radio Connection: USB (we support network connections too, if you have the SCU-LAN10)
- Serial Device: Make sure to select the correct port. It will not be the same port as shown in the screenshot!
- Baud Rate: Defaults to 38400, but check on the radio's menu
- PTT Type: CI-V (which means "CAT" for a Yaesu). This just means we send a command for PTT rather than toggling a serial port's analog output signal.



The wfview log will show if the device was found:

```
2026-02-26 19:13:39.362 INF rig: FTDI libraries loaded successfully
2026-02-26 19:13:39.370 INF rig: Init FT4222 as SPI master
2026-02-26 19:13:39.378 INF rig: Init FT4222 as SPI master completed successfully!
```

wfview

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FAQ

This is a collection of questions that we frequently see come up on the support forum. Please, have a look here before posting, it may save you a lot of time.

Jump to: Downloading/Installing | Audio | Waterfall | Sharing Control | Networking | Protocol

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Section 1: Downloading, Installing, and Initial Run:

Q: I'm using Windows and I get an error that "MSVCP140.dll is not found.

A: This happens sometimes. Press "Ignore" if prompted. If there are errors running wfview, simply run the `vcredist_x86.exe` program, which is conveniently located in the wfview application directory (usually `c:\program files (x86)\wfview\`)

Q: I'm using Windows and I get an error that "VCRUNTIME140.DLL" is not found.

A: This happens sometimes. Press "Ignore" if prompted, and then simply run the `vcredist_x86.exe` program, which is conveniently located in the wfview application directory (usually `c:\program files (x86)\wfview\`)

Q: I'm using Windows and I get an error that "vcredist_x86.exe" failed to start or could not be found.

A: This happens sometimes. Press "Ignore" and then, if wfview will not start, simply run the `vcredist_x86.exe` program, which is conveniently located in the wfview application directory (usually `c:\program files (x86)\wfview\`)

Q: I'm using windows and I get an error that "Qt5Xml.dll" is not found.

A: This can happen if wfview isn't in the same folder as that DLL file, and may especially occur since that is a new dependency found only in our daily and weekly builds. The DLL file can be found in our public builds folder. Simply place it in the same folder as wfview.exe and you should be good to go.

Q: I'm trying to use the latest public build and it is not working. It complains that it cannot find portaudio_x86.dll or qcustomplot2.dll (or some other DLL).

A: Any downloaded public build exe must go in the same folder as your existing wfview.exe. Do not rename any files, just move it to the same folder. If you are missing a DLL file, please check the public_builds folder to see if there are any DLL files that you need. Those DLL files need to go in the same folder as wfview.exe. When you have that set up, double-click on the new version of wfview that you downloaded, which will have a filename like "wfview-master-202323332.exe" or similar. Keep the original wfview.exe file around in case you wish to revert to the older version.

Q: I'm trying to use the latest public build and it is not working. It says "The application was unable to start correctly (0xc000007b)".

A: Please make sure that you have all the DLLs from the public_build folder within the same folder as wfview.exe, which is also the same folder that you have placed the downloaded public build exe file. The error means that one (or more) DLL files is missing.

Q: I'm trying to run the linux build script and it says "permission denied" when I try to run the script.

A: Check the directions carefully. Make sure to add execute permission to the script: `chmod 755 fullbuild-wfview.sh`. Do not ever run the build script as root.

Q: I'm trying to run the linux build script and when I run it, it says "./fullbuild-wfview.sh: line 2: syntax error near unexpected token newline" (or similar)

A: Please follow the directions carefully. Most likely, you have inadvertently downloaded some HTML instead of the actual build script. When you click on the link for the script, you must then press the download button on gitlab's web interface. This is also detailed in the video on linux install in our video section.

Q: Linux, I've installed qcustomplot, but when building, the linker says it can't find it

A: This happens because in some versions of linux, the library is named "libQCustomPlot" and in some versions it is named "libqcustomplot". You can fix this with a symlink:

```
cd /usr/lib/x86_64-linux-gnu
sudo ln -s libQCustomPlot.so libqcustomplot.so
```

Q: How can I tell what version of wfview I have?

A: Press the About button.

Q: My wfview doesn't look like the ones I see in screenshots or on youtube. Some features seem missing!

A: Please make sure you're running the latest version. You can check using the About button, and compare with what is offered on our website.

Section 2: Audio

Q: Why isn't there any audio in wfview for my radio connected with USB?

A: wfview does not need to do anything with the audio from a USB radio, unless you are running a wfview server for remote access to your radio. If you need to access the radio's audio, such as for a digital mode like WSJT-X or PSK31, simply select the radio's USB audio directly from the digital mode program. **To use a PC headset or speaker** with your USB-connected radio, you'll need to install some sort of audio loopback utility to direct headset audio into the radio. This is not a wfview setting — it is independent of wfview and will work with or without wfview running. Again, this advice here only applies to the computer physically connected to a USB radio at the radio.

Here are some example google searches which may help you redirect audio from one device to another:

- Windows
- macOS
- Linux

Q: How can I use my computer's speakers, microphone, or headset, with my USB-connected radio?

A: See the above question.

Q: I don't hear any audio on wfview client to my network-connected rig.

A: Check your log (press "Log") and see what audio device wfview is trying to use. Try disconnecting from the

radio, changing the audio system, and then connecting again. You may also want to try other devices.

Q: Why isn't there any transmit audio?

A: Check what audio MOD source is selected on the radio. For convenience, you can also do this within wfview using the Radio Settings page under Settings. For radios with native LAN support, this should be set to "LAN", generally for Voice and Data modes. For radios where you are running your own wfview server, you'll most likely need to set it to "USB" or "ACC" — use whichever is being used by the server computer. Next, verify that the gain for the selected MOD source is at least 50%. The slider will be labeled "LAN" for LAN connected radios, and "USB" for USB connected radios, matching the selected MOD source. Finally, enable the secondary meter under Settings on the User Interface page. Set the secondary meter to "TxAudio". This will cause a second meter to appear on the main wfview page which will always display the transmit audio level as seen by wfview. This is not the same as the radio's audio level, this is the level of the audio being streamed to the radio. This level should be around -12dBfs to -6dBfs. Finally, if that audio is showing, change the secondary meter to ALC and see if the ALC meter is moving during transmit on SSB.

Q: I picked up the mic on the radio and it isn't transmitting any audio! What happened?

A: These days, radios have so many audio inputs for transmit (modulation). You can change the transmit modulation source to "MIC" using the menus on the radio, or conveniently, we have a menu in wfview which also performs this task on the Settings tab under the Radio Settings page. Simply set the "MOD" source to "MIC" for your radio's microphone. You can read more about this here. Set it back to "LAN" or "USB" (IC-7300 users) when you want to switch it back for network use.

Q: The audio is distorted, too quiet, or too noisy

A: See if you can adjust the audio device gain in your operating system's system preferences or control panel. Use wfview's built-in "TxAudio" meter to view the audio level. Speaking should exceed -12dBfs, with some peaks over -6dBfs. Idle sound level should be below -30dBfs. If the audio is redlining in the TxAudio meter, you must make an adjustment outside of wfview, using your operating system's control panels or other adjustments. Once the audio level is under control, use the "LAN" slider (or "USB" for IC-7300 users) to adjust the audio going into the radio's modulation circuits. On sideband, you may want to watch the ALC meter as you adjust. For FM, these radios provide little guidance and you may want an on-air check. Start at 50% and work from there.

Q: I'm using Windows and whenever I adjust the audio levels with Sound Control, wfview drops the audio!

A: Try switching your audio system in wfview between Qt Audio, Port Audio, and RT Audio. One may work better.

Q: The audio (transmit or receive) has clicks, pops, or other artifacts

A: Check the log for messages about packet loss. Real-time UDP streaming audio is hard on network equipment, especially WiFi access points. If there is an excessive amount of warnings about lost packets, you will need to upgrade your network hardware before continuing. If there are no messages in the log about packet loss, continue reading. Make sure your RX and TX latency is set to at least a middle-position (200ms); higher values will result in smoother performance but you may find the latency aggravating. Next, you can try changing the Audio System from the default (QtAudio) to Port Audio or RT Audio. Every computer and audio setup is a little different, so you may need to try each system to find the best performance. Also, try lower the sample rate in wfview to 16khz, as this is plenty of bandwidth for most ham radio uses, and perhaps try the 8-bit uLaw option.

Q: The wfview log says "you may need to install libqt5multimedia5-plugins"

A: This only applies to some linux distributions, and the actual plugin name may vary. The error comes up when no audio devices are found. You may wish to try one of the other audio systems such as Port Audio or RT Audio.

Q: (for network-accessed radios) Which audio device is my microphone? Which audio device is my speaker?

A: This is a tricky one. We present, in wfview, all the audio devices that the computer's operating system and

libraries expose. You may want to experiment with each device one by one, or even use another program to help you test your computer's audio system carefully, so that you know what your devices are named. If you enable the TxAudio meter, it may help you with the microphone search. Remember, you cannot change the device while you are connected — press "Disconnect" first.

Q: (for USB-connected radios, when running wfview as a server) Which audio device is my radio?

A: The exact name will depend a lot on the operating system. Windows often names the device "USB Audio CODEC", and Linux may be similar or may even have "BurrBrown" in the name. If you see several, you may want to disconnect any and all USB audio devices and radios from the computer, close wfview, and then connect only the radio in question. Next, start wfview. If the list is still too cluttered, it can help to reboot with a minimum of audio devices connected.

Q: When I connect to the radio over the network, nobody can hear me and I don't see any transmit power indicated or ALC action.

A: Make sure you switch the radio's Transmit Modulation Input to "LAN" so that the radio transmits network audio. You can do this either using the radio's menus or using wfview's Radio Settings page under the Settings tab. You can read more about this here. Additionally, make sure you have selected your computer's microphone correctly under the "Radio Access" page of the Settings tab.

Section 3: Waterfall Display

Q: Which radios have waterfall display?

A: wfview gets the waterfall data from the radio. Therefore, only radios which provide waterfall data over CI-V will show a waterfall display in wfview. These radios are: IC-705, IC-905, IC-7300, IC-7610, IC-7850/7851, IC-9700, IC-R8600, and the Kenwood TS-890S (beta).

Q: Why is the waterfall display so slow?

A: For a variety of reasons, when you use a serial or USB port for radio connection, the waterfall rate is quite limited. This is a limitation of the radio firmware; wfview displays the data as fast as it is presented over the USB port. Radios which support network connections (Ethernet or WiFi) will have *significantly* faster waterfall when you connect over the network instead of USB.

Q: I have a supported radio for waterfall but I do not see any waterfall.

A: The radio *must* be set at 115200 baud in order for it to generate waterfall data. Additionally, some setup menus must not be open on the radio's screen, and the waterfall must be displayed on the radio display too. These are limitations baked into the firmware of the radio and we cannot change them. Please see our IC-7300 guide for complete setup information if you are having difficulty making this work. Additionally, if you change baud rates to 115200 (as required), you will need to cancel and/or disconnect with wfview and then re-connect when you change baud rate. This is simply a limitation of how serial ports are handled. For Icom radios, don't forget to make sure "Unlink from Remote" is set to "Yes" since "remote" means the 19200 baud CI-V connector on the back of the radio.

Q: Can I lower the CPU usage when drawing the waterfall?

A: Yes. Under Settings, on the User Interface page, disable Anti-Aliasing and disable Interpolation.

Q: Can I turn off the waterfall display:

A: Yes. There is a button for this at the bottom of the waterfall display, "HOLD", which will halt the display.

Q: Why is the waterfall so fuzzy?

A: Make sure to turn off Averaging and also set the Video Bandwidth Filter to "Narrow". This will make a dramatic improvement in image sharpness. These settings are within the radio's "spectrum scope" menus.

Section 4: Sharing Control with Other Programs at the same time

Q: How do I run another program that controls the radio at the same time?

A: See this overview on the topic first. For programs supporting hamlib's rigctld, use wfview's built-in rigctld-compatible server. For programs without this support, use the built-in virtual serial port.

Q: How do I connect two programs at once to the virtual serial port?

A: That isn't supported. The Virtual serial port can only connect to one program at a time. On the other hand, the rigctld server built-in to wfview can handle multiple connections at once.

Q: How do I get wfview's client audio stream into another program?

A: You will need to create a loopback audio device. See this page for audio configuration.

Section 5: Display Issues

Q: The fonts all look too large

A: Check your operating system and see if you have any kind of Interface or Font Scaling enabled.

Q: How do I change the color of something in the plot? How do I make something disappear?

A: Go to Settings, and then click on the "User Interface" page. Use the color editor to select the desired colors. Setting a color to "#00000000" will make a given element invisible. More appearance notes are here.

Q: Where is the SWR/ALC/Audio meter? I see it in videos but I don't see it in my wfview.

A: Double-click the space below the main S-Meter and you can select other types of meters as needed.

Section 6: Networking

Q: What is my ip address?

A: We don't know. Every computer has a different method of showing your computer's ip address. You may want to look in the control panels or system preferences. Remember, your computer's ip address is not the same as your overall public ip address, which is assigned to your home router.

Q: What is my radio's ip address?

A: We don't know it. You can use the menus on the radio to find the ip address.

Q: I can't connect to my radio or radio server

A: The following is a list of things to try and/or check:

- Please see if you can ping from your client computer to the radio or radio server.
- Please check that there isn't a firewall on the client or server side that is preventing the connection (it may help to temporarily turn off any firewalls, just to eliminate this potential issue).
- Some programs, such as AnyDesk, use the same ports as wfview. If you are running a wfview server, you will need to change the wfview ports to something other than their defaults, or, close AnyDesk.
- Make sure the port numbers are correct.
- Make sure the username and password are correct.
- For Icom radios, make sure that "Network Control" is enabled in the radio's menu.
- For Kenwood radios, make sure that "KNS" and "VoIP" are enabled in the radio's menu. More information

here.
- Check the log on both the server and the client side for potential error messages that could be helpful.

Section 7: Protocol

Q: What is the OEM Icom networking protocol?

A: The networking protocol is quite complicated and it is not documented anywhere. wfview is open source software, so you can examine the source code if you want to see how it works. The protocol was first discovered by the kappanhag project. Additionally, this open source project, written in swift, has some good notes on the protocol. The wfview team regularly makes improvements to our implementation of the networking protocol, which is compatible with every available Icom radio with an ethernet or wifi connection to date. The protocol is complex and filled with subtleties which are difficult master, let alone describe. We would welcome help in this area though, since a proper description of the protocol would be helpful to everyone.

Q: What is the OEM Kenwood networking protocol?

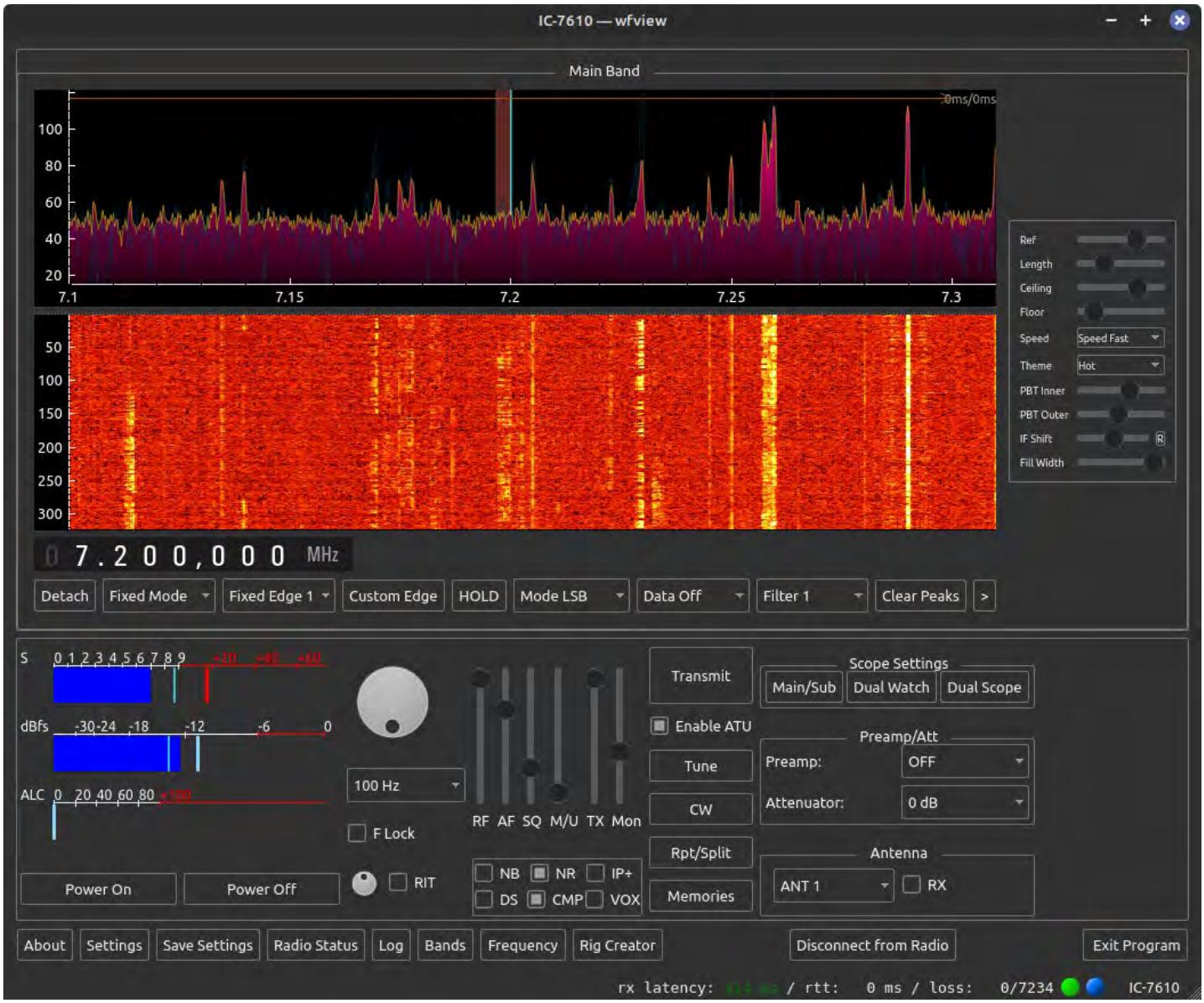
A: The Kenwood networking protocol is well-documented within Kenwood's customer documentation. Tony Collen also documented his work with open890 which you can read here and here.

wfview

Open Source interface for Icom and Kenwood transceivers

Main Window

Here is the main display from wfview. Scroll down for a description of each UI element.



Upper controls:

Spectrum plot and waterfall display:

These two items show the spectrum data passed from the radio to wfview. The vertical axis of the upper plot is likely proportionate to a value in dB, although the exact scale conversion isn't known. Left to right is frequency. The bottom plot's vertical axis is time, with newer values going in at the top. Many adjustments for these plots are available, read on to learn more.

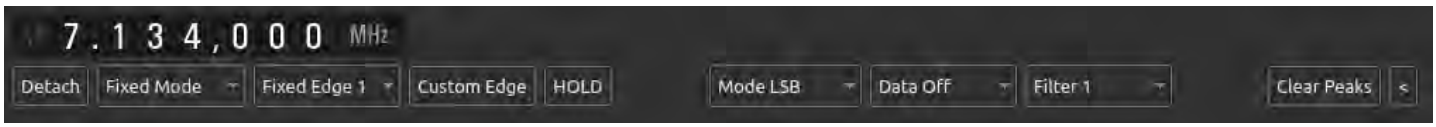
Double-click anywhere in the plot to immediately jump frequency. A single click will simply show the frequency in the lower-left status bar area. Use the scroll wheel to quickly jog around the band at the selected tuning step.

The vertical tuning line in the upper plot indicates the *carrier point* of the radio. This is the frequency the dial indicates as well. The tuning line color can be customized (and even disabled via a transparent color) under the Settings tab on the User Interface page.

There is a divider between the upper plot and the lower waterfall section. You can drag the divider up and down to change the division of area between the plot and the waterfall.

Spectrum Mode and Edge controls:

- Spectrum Mode: This control selects between the radio's built-in four modes: Center, Fixed, Scroll-C, and Scroll-F.
- Span: This control sets the total span for the spectrum, and is only available in Center and Scroll-C modes.
- Edge: In Fixed and Scroll-F modes, this control selects the current edge (1 through 4). Edges are simply the boundaries for the scope data.
- Custom Edge: This button lets you define the scope edges in Fixed mode (in Scroll-F mode, the edge you define is translated over to the current frequency). The Custom Edge dialog box makes it very easy to jump to a desired frequency range.
- ToFixed: This button will take the current center span range and convert it into a fixed range. It is a quick way to keep the same displayed range and allow for tuning without changing the range displayed. Only shown in Center mode.
- Clear Peaks: Clears the spectrum underlay, which can be either Peak Hold, Buffered Peak Hold, or Average (select the desired mode under Settings).
- Press the "<" button at the end of the buttons to show additional controls.



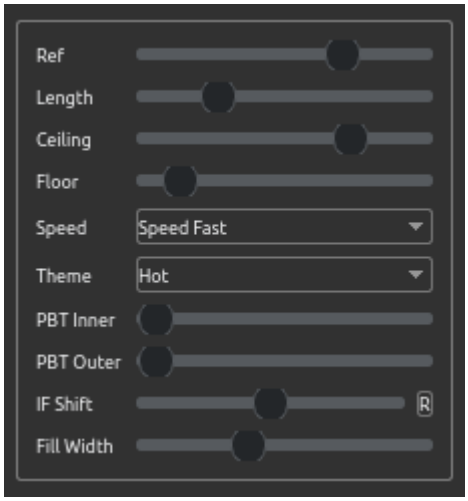
Mode and Filter selection:

- Mode: Select the transmit and receive mode. LSB/USB switching defaults to automatic (see the Settings window).
- Data Off / Data On: When "Data" is turned on, the radio will use, potentially, an alternate modulation input and filter selection. See the radio manual for details.
- Filter 1/2/3: Select any of the three defined receive audio filters that the radio has available.

Detach: Pressing this button will detach the spectrum display from the other controls. This is especially useful for placing the spectrum display on a separate monitor.

Frequency Widget: Press the top or bottom of a given digit to nudge the value. The scroll wheel can scroll any position. Right-click to zero a digit. This widget was borrowed and modified from CuteSDR under the GNU GPL.

Additional Spectrum and filter controls:

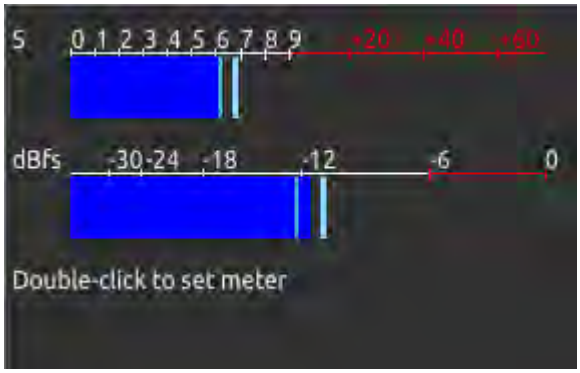


Pressing the “<” or “>” button (to the right of “Clear Peaks”) will show/hide the additional controls.

- Ref: This is the so-called “reference level” for the spectrum display. Increasing this control has the effect of increasing the data value of the spectrum. Note that you can leave this full cranked and then use the Floor and Ceiling controls for a similar effect but with greater display resolution.
- Length: how long the waterfall display is. Shorter values will make the waterfall look a bit faster.
- Ceiling: Top value for the spectrum display and waterfall color mapping
- Floor: Bottom value for the spectrum display and waterfall color mapping. Set this just below the noise floor.
- Speed: How often the radio sends spectral data.
- Theme: Color scheme for the waterfall display. These are defined within qcustomplot and can be previewed here.
- PBT Inner: Pass Band Tunable filter inner position
- PBT Outer: Pass Band Tunable filter outer position
- IF Shift: Performs IF Shift on radios with this function, otherwise, shifts the inner and outer PBT by the same amount
- Fill Width: Filter width for radios with variable filter width

Middle Controls:

Metering:



wfview can display three meters at a time. The first meter is always power output during transmit, and receive strength (“S”) during receive.

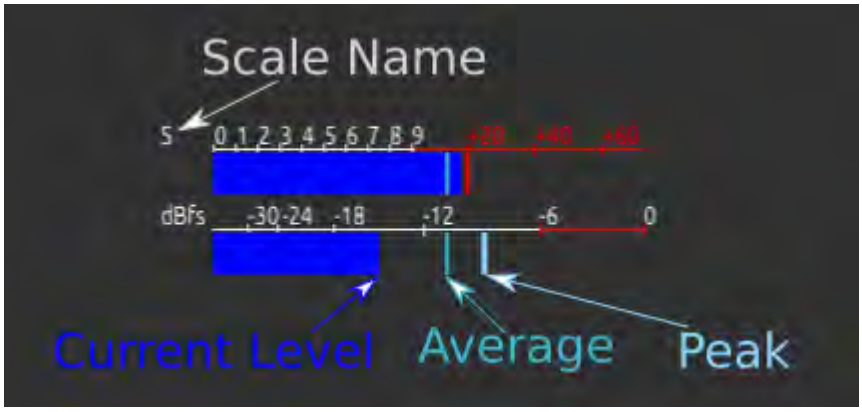
Meters include: SWR, ALC, Compression, Voltage, Current, Center (IC-R8600 only), TxRxAudio, RxAudio, and TxAudio. Please keep in mind that the audio meters will only function for the client side of network radios. The audio meters do not do anything with a locally-connected USB radio.

To change what is metered, simply double-click on the meter (or blank meter space). Click to the right side of the menu to “get out of the menu”.

The meter update speed is proportionate to the radio data polling rate, which can be adjusted in Settings.

The meter displays three numbers at a time.

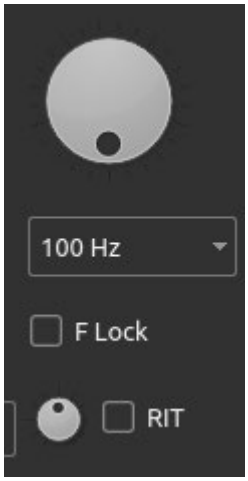
- Instantaneous Level: This is the long solid bar which shows you the same value as the radio's meter.
- Average: A thinner bright bar showing the average of the last ten or so readings.
- Peak: A thicker bright bar showing the maximum of the last ten or so readings.



Radio Power:

Power On / Power Off: For radios with this capability, the radio may be powered on or off remotely. When powered off, the radio enters a low-power standby state. Keep in mind that it can be quite confusing if you connect to a radio which is powered off, you will need to manually power it back on when starting wfview.

Tuning:



wfview has a rather conventional tuning knob (the large circle thing), which can be moved around with the mouse at the selected tuning steps (see the pop-down menu of tuning steps). Shift and control are modifiers for the step as well.

Tuning methods:

- The tuning knob screen widget
- Scrolling the mouse on the spectrum
- Double-clicking within the the spectrum
- +/- keys and other supported keyboard shortcuts
- The "Frequency" window (press * to bring it up quickly).
- Supported hardware interfaces with actual knobs and buttons

F Lock: Prevents the wfview user interface from changing frequencies. Does not lock the radio's physical controls. Does not prevent external control (ie, other programs) from tuning. Just stops accidental mouse clicks within wfview basically.

RIT: Receive Incremental Tuning: applies a delta to the receive frequency

Sliders:

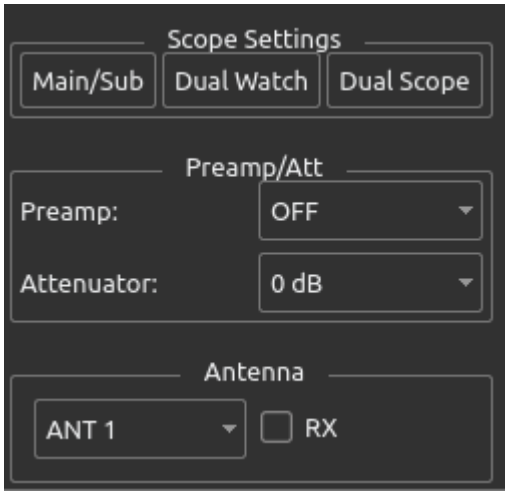
- RF: Receiver RF Gain
- AF: Receive audio gain. For USB and Serial connected radios, this sets the speaker volume on the radio. For network audio radios, this sets the computer's apparent receive audio level.
- SQ: Squelch. If you don't hear any audio, try turning this down.
- MIC / LAN / ACC / USB/ M/U: Slider to adjust the transmit audio gain *for the selected transmit audio source*.
- TX: Transmit RF Power level. This control is in percent.
- Mon: Monitor audio level ("Hear myself" aka "talkback" in some circles). Click the "Mon" text to toggle this feature on and off. Warning: Network latency makes this something you really do not want to leave on.

Square Toggle Controls:

- NB: Noise Blanker
- NR: Noise Reducer (DSP)
- IP+: Basically nothing
- DS: Digi-Sel preselector
- CMP: Transmit audio compressor
- VOX: Voice on Transmit. Use carefully.

Additional:

- Transmit: Toggles PTT. You can also press Control-R and Control-T for transmit and receive. Plus we support several hardware controllers for footswitches and tactile buttons.
- Enable ATU: Enables the radio's ATU function.
- Tune: Cause the ATU to begin a tuning cycle
- CW: Shows the CW window
- Rpt/Split: Shows the repeater and split window (also CTCSS/DPL controls)
- Memories: Shows the memory editor



Scope Settings:

- Main/Sub: Swap the main and sub VFOs
- Dual Watch: Enable receiving two frequencies at once
- Dual Scope: Show a bandscope for each VFO (where supported: 7610 and 7850)

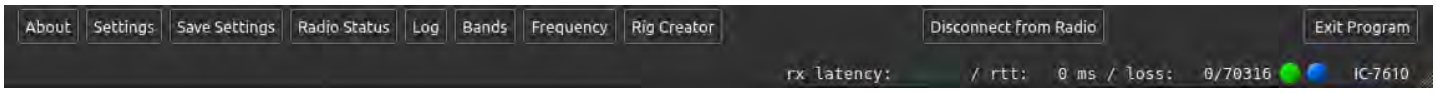
Preamp/Att:

- Preamp: Set the desired preamp
- Attenuator: Set the attenuator

Antenna:

- Select an antenna
- RX: Toggle the “RX” antenna port

Bottom:



Buttons:

- About: Shows the about window with version number, build date, credits, and so on
- Settings: Shows the settings window
- Save Settings: Saves the settings you are currently using for next time.
- Radio Status: Shows detailed radio information about the current connection (network radios only)
- Log: Displays wfview’s internal log data.
- Bands: Shows a band selector with common ham radio bands (and some others)
- Frequency: Frequency entry widget. Press “*” to bring it up for rapid entry
- Rig Creator (if enabled): An advanced option which allows radio definition files to be edited and/or created.
- Connect to Radio / Disconnect from Radio: Causes the serial or network connection to toggle.
- Exit Program: Closes wfview

Status Text:

- rx latency: Actual latency from the radio to wfview.
- rtt: Round trip time in addition to the rx latency.
- loss: Packets lost and total packets sent. For example, 0/7234 means zero packets lost and 7234 packets sent.
- Green circle: Receiving
- Red Circle: Transmitting
- Split Red/Green circle: Split or repeater mode enabled
- “None” or “radio name”: Indicates the detected radio connected. When it says “None”, wfview isn’t connected

or is connected but cannot identify the connected radio.

CW Sender

wfview has a built-in CW Sender which uses the radio's built-in keyer. Text typed in by the user is sent to the radio for keying. A sidetone is not provided at this time, although it is possible to stream the sidetone from USB-connected radios (such as the IC-7300) when used with wfview as a server.

Note: The radio will not actually transmit any CW unless it is in CW or CW-R mode.

CW Sender Controls:

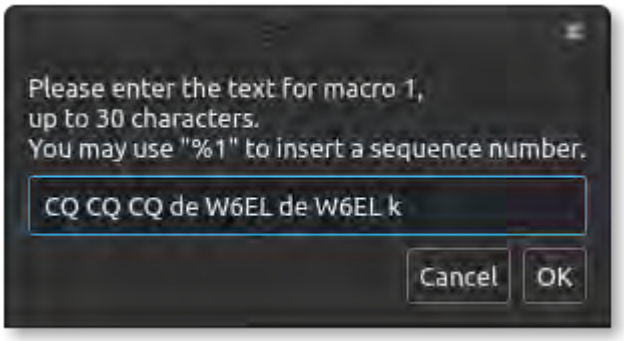
Top level controls as they appear:

- Send: This button sends the user text to the radio. In Full or Semi-Breakin mode, the radio will transmit and begin sending the CW.
- Stop: This button causes the radio to stop sending prior to actually finishing whatever text has been buffered from the user. Otherwise, stopping is automatic.
- The text box allows users to enter text to be sent to the radio. Up to 30 characters may be sent at once. Multiple stanzas of 30 characters may be sent to the radio, which will be sent out in the order they are received. Some characters are not allowed; check the log if you think your characters may be being sent oddly. Characters that aren't allowed get replaced with a "?".
- Break In: This combo box lets you pick the break In mode:
    - Full: The transceiver will switch T/R in between words whenever there is a gap. Some amplifiers may not be able to keep up!
    - Semi: The transceiver will transmit at the beginning of the CW stanza and receive at the end of all the buffered text.

    - Manual: You, the user, must press Transmit at the beginning and press Receive at the end. The radio starts “sending” the CW immediately, so press Transmit quickly!
- WPM: Words per minute. The radio will accept 6 to 48 WPM.
- Pitch: The desired CW pitch, referenced to the tuned carrier frequency.
- Local Sidetone Level: Volume for the locally-generated CW tones.
- Cut Num: Replace numbers with shorter letters, often used in contests for signal reports (in which the constant magical number 599 is sent as 5NN)
- Send Immed: Send the text typed immediately in small chunks rather than waiting for the enter key to be pressed.

Macros

wfview provides ten user-defined macros for common phrases. The macros are saved when the user presses “Save Settings”. Macros may be edited by pressing “Edit Mode” and then clicking on a macro button.



You can type in “%1” (without the quotes) to have wfview automatically replace the %1 with the current sequence number (otherwise referred to as a “serial” number). This is useful in some contests.

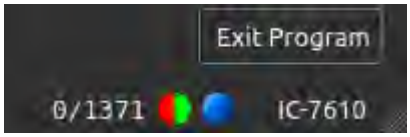
The sequence number may be changed using the up and down spinbox to the right labeled “Seq”.

Uncheck the “Edit Mode” checkbox to use the macros. Simply click a macro button to send it over the air.

Repeater and Split operation

This page describes how to use the SPLIT and Repeater settings of the radio. Press the “Rpt/Split” button in wfview to bring up this window:

Note: When transmitting with split enabled, the transmit indicator “LED” will have a split appearance:



Supported Features

What you see in this window will reflect the features the radio supports. For example, most HF radios do not support the “Repeater Duplex” section but do support using Split Mode. Some radios support Main and Sub VFOs, and some support A and B VFOs. Additionally, exactly how some features on the sub VFO are addressed varies quite a bit. The IC-7610 and IC-7850 have special commands to allow the repeater tone to be set on the sub VFO; the other radios lack these commands.

Repeater Duplex

This section deals with repeater controls, primarily present on radios for accessing VHF and UHF FM repeaters.

- Simplex: transmit and receive frequencies are the same
- Dup+: The transmit frequency is above the receive frequency
- Dup-: The transmit frequency is below the receive frequency
- Auto: The radio should automatically decide which duplex mode to use (may not work correctly depending on your region)
- Set Offset (MHz): Type in the desired offset, in MHz, and press this button to set the default repeater offset for the currently selected band (ie, 2M/70CM/23CM)

Split Mode

This area handles split operation, available on most HF rigs and some VHF/UHF/SHF rigs

- Split On: Turns on the split feature, disabling any other duplex or repeater mode selected
- Split Off: Turns off the split feature. Some radios have both “split off” and “simplex” modes, but I can’t tell you what the difference is.
- Quick Split: Activate the radio’s quick split feature, when available. This feature is supposed to sync the repeater tone, mode, and frequency of the sub VFO. However, it can be “canceled” if the user makes certain adjustments to the sub VFO. The checkbox is available but you may need to experiment with it.
- Offset (KHz): Type in the desired split offset here, for example, 100 for 10M FM repeaters.
- Split+ and Split- buttons: These buttons set the radio’s secondary VFO (“B” or “Sub” depending upon the radio) to the appropriate split frequency and populate the Tx Freq box as a convenience to the operator. If the “Set Rpt Tone” checkbox is checked, and the radio supports such commands (7850 and 7610), then the secondary VFO’s repeater tone and tone mode is also set at this time.
- Tx Freq (MHz): You can manually type in a transmit frequency here and press “Set” any time. But it is not needed when using the Split+ and Split- buttons
- AutoTrack: Checking this box causes the sub VFO to automatically track the Main or “A” VFO with supplied Offset. In FM mode, the repeater tone and repeater tone mode will also track. Use this feature to discover 10M repeaters easily. As this function is similar to Quick Split, do not use both at the same time.

VFO Buttons

These buttons are for selecting, swapping, and equalizing VFOs.

- Sel A, Sel B, Sel Main, and Sel Sub: These buttons “select” a given VFO. On the radios I have, this is the same as touching the given VFO on the screen and does not provide much functionality. But it may be different on other radios.
- A=B: Sets the B VFO to the contents of the A VFO
- M=>S: Sets the Sub VFO to the contents of the Main VFO
- Swap A/B and Swap M/S: Swaps the contents of the two VFOs. Use this feature to rapidly swap between VFOs and to make adjustments to the Sub VFO when needed, for example, with radios that do not support addressing the SUB VFO directly and if you need to set a repeater tone.

I realize some of this seems incomplete or difficult to understand. I recommend playing with these features first with the radio at your side.

Repeater Tone Type

These buttons select the method of repeater tone access.

- None: There is no transmitted tone of any sort and there is not a tone squelch.
- Transmit Tone Only: A subaudable tone is transmitted. No tone squelch.
- Tone Squelch: A tone is transmitted, and the same tone frequency is used as a tone squelch.
- DTCS: Also known as DCS, a digital code is sent with your transmission and is also used as a digital tone squelch.
- Set Sub VFO: Available on the IC-7610 and IC-7850. Use this button to force the sub VFO to a specific tone mode (typically “Transmit Tone only”). This button’s function is automatically run when the “Set Rpt Tone” checkbox is checked.

Tone Selection

Use these controls to select a Tone and/or D(T)CS code. For D(T)CS, the Transmit and Receive code can be inverted.

Press “Set Sub VFO” to force the sub VFO to the specified tone selected. Not available on most radios and not available for D(T)CS. This button’s function is automatically called when setting split and the “Set Rpt Tone”

checkbox is selected.

Radios and Features

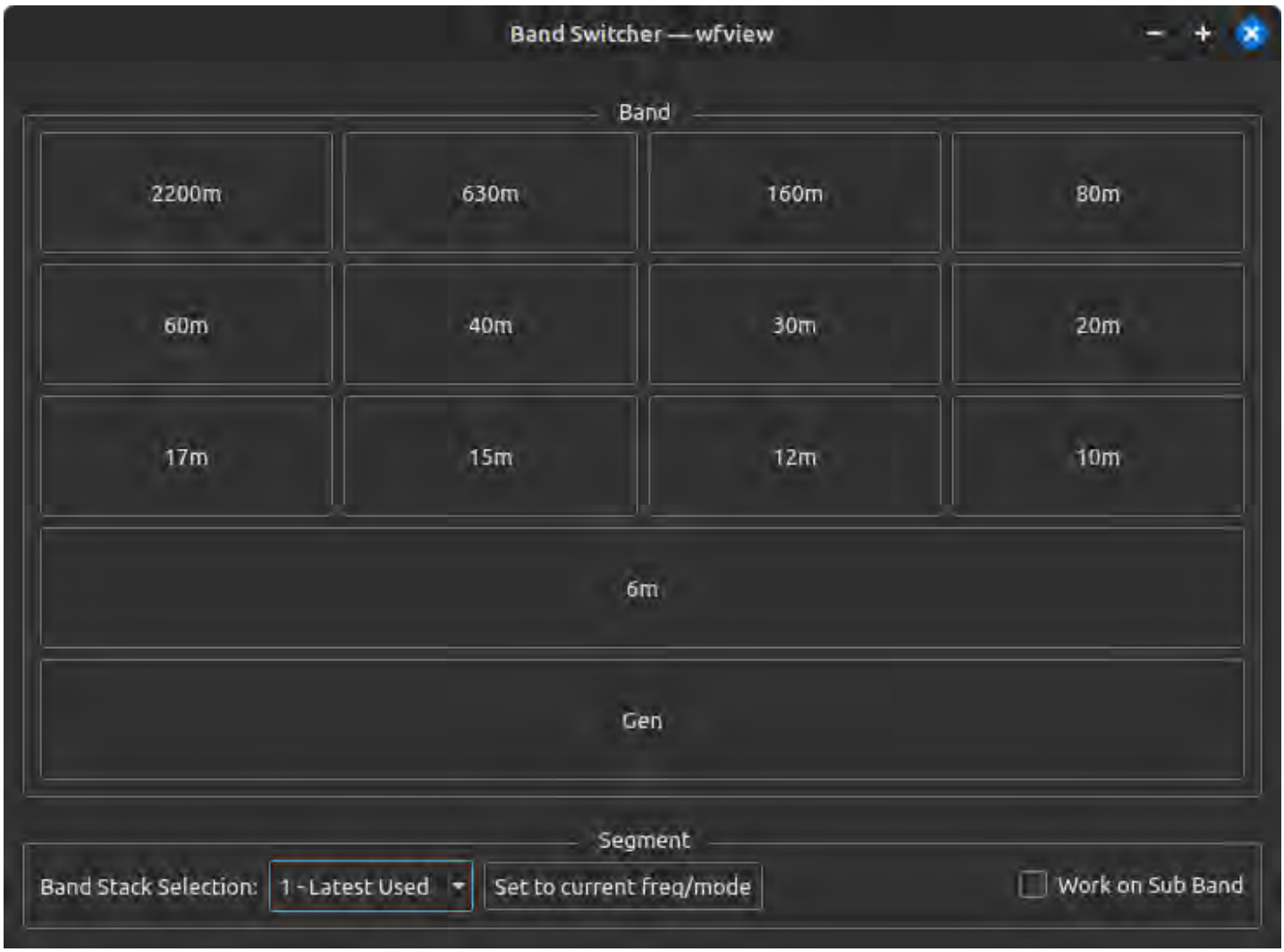
For the curious, here's a short chart on what features are available on what radios:

	VFO A/B/M/S Operations		0x25 0x26	0x29
Model	Has MS 0x07	Has AB 0x07	Spe VFO Bnd/Mode	Specify M/S
IC-705	No	Yes	Yes	No
IC-706	No	Yes	No	No
IC-910H	No	Yes	No	No
IC-7000	No	Yes	No	No
IC-7100	No	Yes	Yes	No
IC-7200	No	Yes	No	No
IC-7300	No	Yes	Yes	No
IC-7410	No	Yes	No	No
IC-7600	Yes	No	No	No
IC-7610	Yes	No	Yes	Yes
IC-7700	No	Yes	Yes	No
IC-7800	Yes	No	No	No
IC-7850	Yes	No	Yes	Yes
IC-9100H	Yes	Yes	No	No
IC-9700	Yes	Yes	Yes	No
IC-R8600	undocumented	undocumented	Yes	No
9700, 9100	For radios with MS and AB: Command 07B0 is Exchange M/S For radios with AB only: Command 07B0 is Exchange A/B For radios with MS only: Command 07B0 is Exchange M/S			

Bands

To select a band of operation, press the Bands button at the bottom of the main window:

The Band Switcher window will show up:



Most of these buttons are probably quite obvious. These buttons will take you to the bands indicated. Most of the buttons have key strokes that you can read about here.

Some bands are not of the common “classic ham radio bands” and may behave a little bit differently. This is because the radio will not have a “last used frequency” remembered.

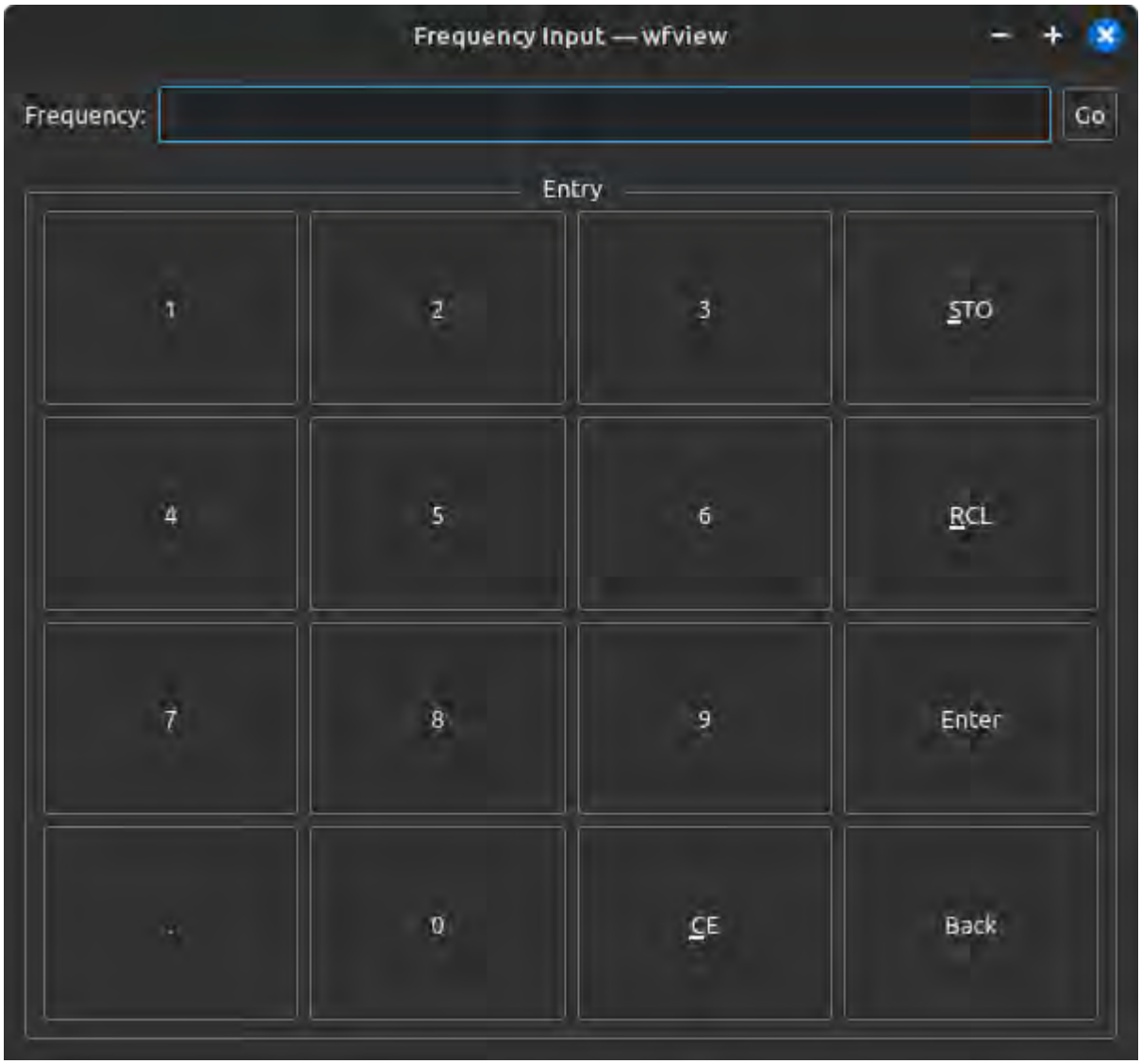
For the most common bands, the radio’s Band Stack Register (or “BSR”) will have stored the last used frequency

(or even the last three depending on the radio), and you can use the Band Stack Selection drop down menu to select which of the three stored frequencies for that band are recalled. You can also press “Set to current freq/mode” to program the BSR manually to the current tuned frequency and mode.

“Work on Sub Band” will cause the selected band to go to the sub receiver (if available).

Frequency Input

Press the “Frequency” button at the bottom of wfview to open the frequency input window (or press the star key on your keypad):



This is the Frequency Input window, and it’s usage should be fairly self-explanatory.

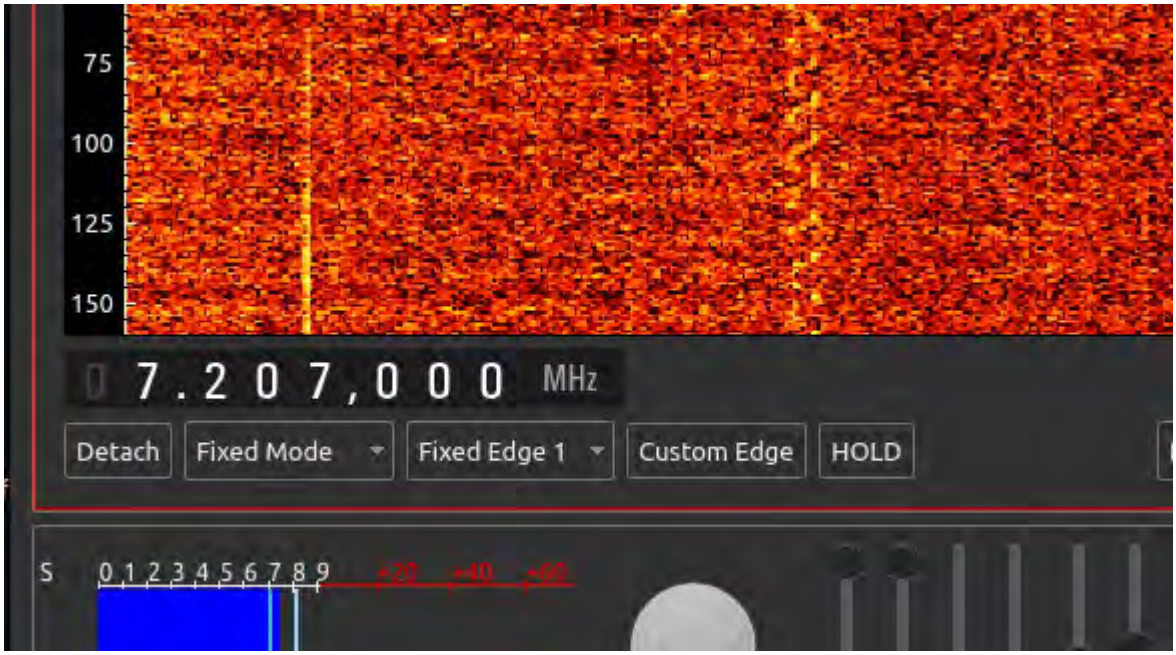
Simply type in any frequency supported by your radio and press enter. If there is no decimal point supplied, wfview will assume the units of entry are KHz, otherwise MHz is assumed.

To quickly store and recall frequencies, use STO and RCL as follows:

To store the current frequency, enter in the memory location (0 through 99) and then press STO.

To recall a memory, enter the memory number (0-99) and then press RCL.

Keep in mind that the Frequency Widget which is part of the main window can also be used:

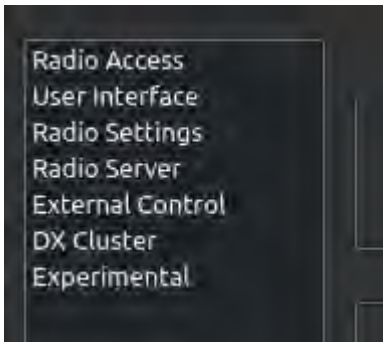


Simply click the top-half of a digit to increase the value, and click the lower half of a digit to decrease the value. The digits can also be scrolled upon individually.

Settings

wfview's settings can be accessed by pressing the "Settings" button in the bottom-left corner of the main window.

On the left side of the Settings window, you will see several categories of settings:



At the bottom of the Settings window, there are the following buttons:

Save Settings: Press this to save your settings

Revert to Default: Press to reset your settings. You can also start wfview with the command-line arguments `--clearconfig CONFIRM`

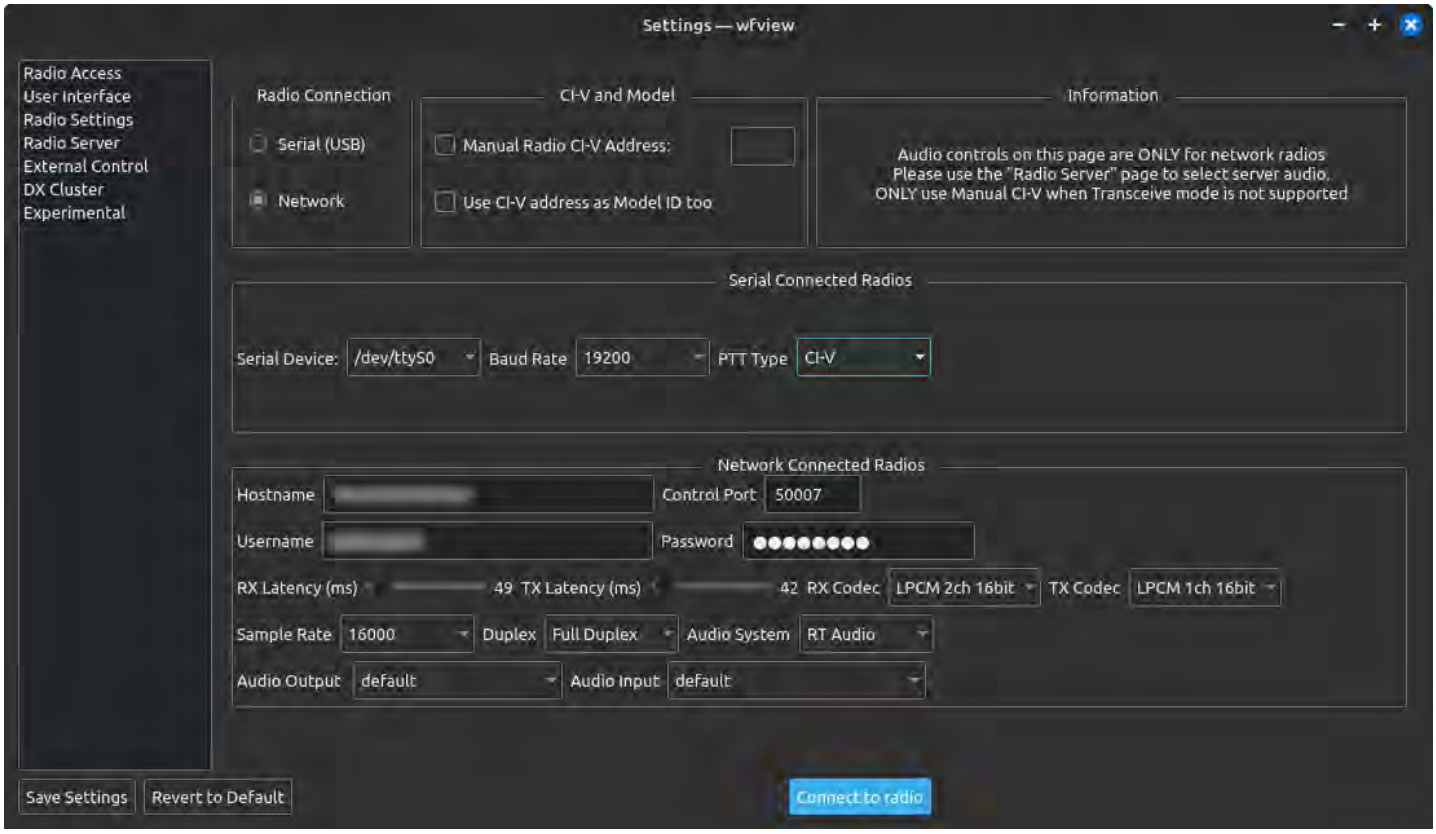
Connect to Radio: Press to connect to your radio (after it has been configured, of course)

Click on one of the following links to read about each of the settings pages:

Radio Access | User Interface | Radio Settings | Radio Server | External Control | DX Cluster | Experimental

Radio Access

These controls tell wfview how to access your radio:



Radio Connection:

- Brand: Select Icom or Kenwood
- Serial (USB): Select this for radios that connect using a USB cable or a form of a serial port cable.
- Network: Select this for radios with built-in WiFi or Ethernet.

Generally, if a radio offers both USB and Networking, the networking connection will provide a better experience.

When Serial or Network is selected, various additional controls will be enabled.

Serial Connected Radios:

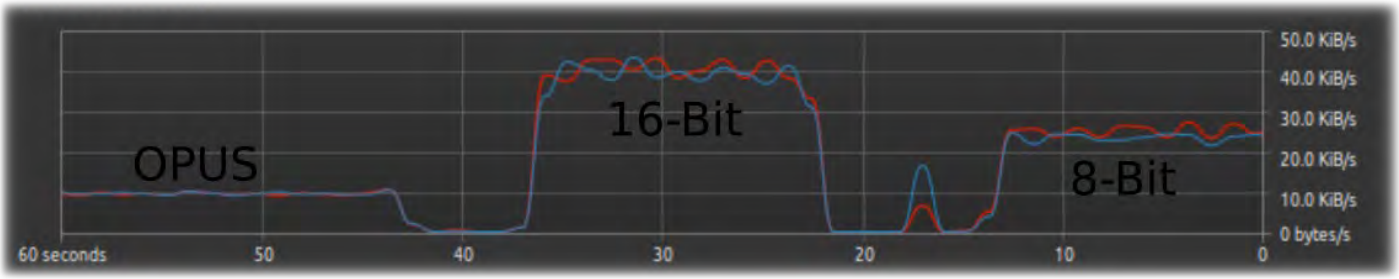
- Serial Device: Select the serial device here. Select “Auto” and wfview will use the first serial port it sees, which may be fine if there is only one such device. Select “Manual” to type in your own serial port. wfview should automatically populate the list of connected devices. Make sure you have permission to access the serial port.
- Baud Rate: The bit rate for the selected serial port. 19200 is common for modern CI-V radios, but older ones may be 9600 or even lower. Always try your radio’s highest supported baud rate first, and make sure to configure your radio prior to connecting. Some radios may need to restart after changing their baud rate.
- PTT Type:
  - CI-V: Modern Icom radios (IC-7100, 7200, 7300) support commands which can be sent to set the PTT on and off.
  - RTS and DTR: Older radios need a hardware PTT switch, and we can use the serial port’s DTR or RTS capability to signal PTT. If you have an adapter that supports hardware PTT, check carefully what signal is being used. NOTE: This is only for older radios (IC-718, for example). Do not use this option if your radio has a CI-V PTT command.
- Tip: Linux users can automatically set useful names for USB serial devices with reasonable default permissions. See this page on how to do it.

Network Connected Radios (WiFi or Ethernet):

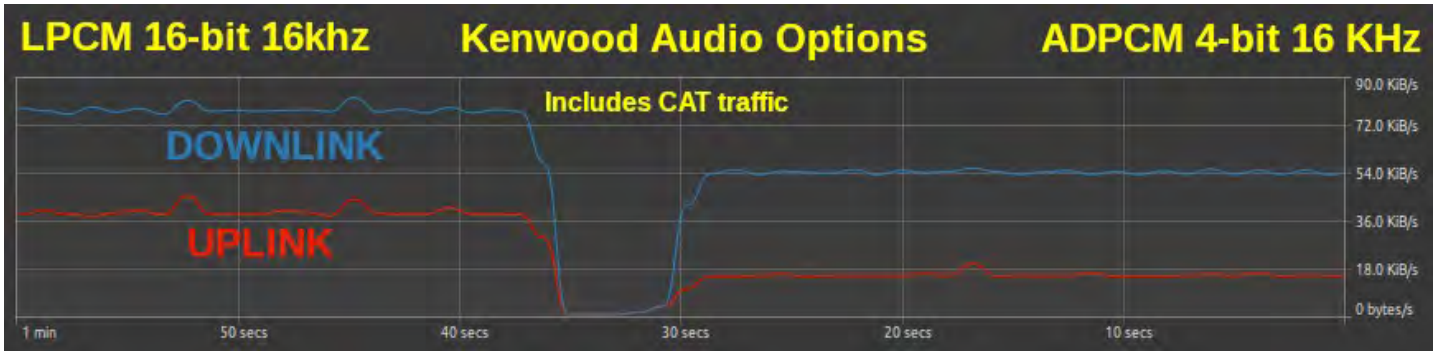
- Hostname: Enter the hostname or IP address of your radio here. You can use the menus on the radio to see what its current IP address is (see the manual for the radio).
- Username: Enter your username as you have set it on the radio. Consult the radio’s manual for how to set

up the username and password.

- Password: Enter your password as you have set it on the radio.
- Control Port: Enter the control port for the radio. The default is 50001. wfview will ask the radio for the other two ports.
- RX Latency: Sets the buffer size for the receive audio stream. Begin with a mid-range setting and increase it if you have audio dropouts. On a very good home network (LAN), you may be able to run this as low as 20ms. On a cellular data hotspot, try more like 300ms.
- TX Latency: Set the buffer size for the transmit audio stream. Generally it is recommended to monitor the RX audio stream and then adjust the TX Latency to the same value as a good RX Latency setting.
- RX Codec and TX Codec: Select the audio encoding scheme to use. The radio must support the selected stream. Use a 2ch ("stereo") stream for radios that have dual VFOs (or if you are only using one VFO, use a 1ch stream. Kenwood users must use "LPCM 1Ch 16bit" or "ADPCM". If you're running your own radio server, you can select any format.
    - LPCM: basic linear PCM encoding. 16 bit sounds a lot better than 8 bit. This is Kenwood's "high quality" audio format.
    - uLaw: an 8 bit format with enhanced dynamic range similar to approximately 14 bit linear PCM. Uses half the bandwidth of a 16 bit stream. Some VPN users may have better success with 8-bit streams due to packet mangling issues.
    - Opus: Opus is a low-loss encoding scheme which delivers a very high quality audio stream with incredibly low bandwidth. This is recommended for users with low bandwidth connections, *however*, it is not supported by any Icom radios. The only situation where an Opus encoding can be used is when you are running your own radio server using another copy of wfview as the server. Opus uses only 10 KiB/sec and has some degree of forward error correction.
    - ADPCM 1Ch: Kenwood's "low quality" audio format.
- Sample Rate: Sets the sample rate for both audio streams. A 16 KHz sample rate will allow for audio up to approximately 8 KHz — plenty for even the widest of ham radio users.



The plot above shows the bandwidth used for a wfview connection to an IC-7300. The bandwidth includes scope data and other CI-V traffic. The plot below shows the bandwidth needed for the Kenwood 890, which also includes CAT (commands) traffic.



- Duplex:
    - Full Duplex: Constantly stream audio for both transmit and receive streams
    - Half Duplex: Mute the transmit audio when not transmitting, and mute the receive audio when not receiving.

Audio System: wfview offers access to three audio systems. If one audio system gives you trouble, consider trying a different one. It is recommended to try the Qt Audio system first.
    - Qt Audio: The default audio system which all Qt-based applications can access.
    - Port Audio: An open-source audio API designed for high compatibility between operating systems
    - RT Audio: An open-source audio API designed with real-time applications in mind.
- Audio Input: Select your computer's microphone here. Or if you are using an external program such as fldigi or wsjt-x, select your audio loopback or virtual audio cables.
- Audio Output: Select your computer's speakers or headphones here. Or if you are using an external program such as fldigi or wsjt-x, select your audio loopback or virtual audio cables. Don't select the same cable as the audio input or you will get nasty bursts of feedback!

Kenwood only notes:

- Connection Type:
    - Ethernet: Maximum performance
    - WiFi and WAN: Lower performance, better for sub-par connections. Waterfall speed will be a lot slower.
- RX and TX Codec notes:
    - Kenwood natively supports "LPCM 1ch 16bit" or "ADPCM 1ch".
- Port: The TS-890S uses TCP/IP port 60000 for control and UDP/IP port 60001 for audio. The audio port cannot be specified on our side.

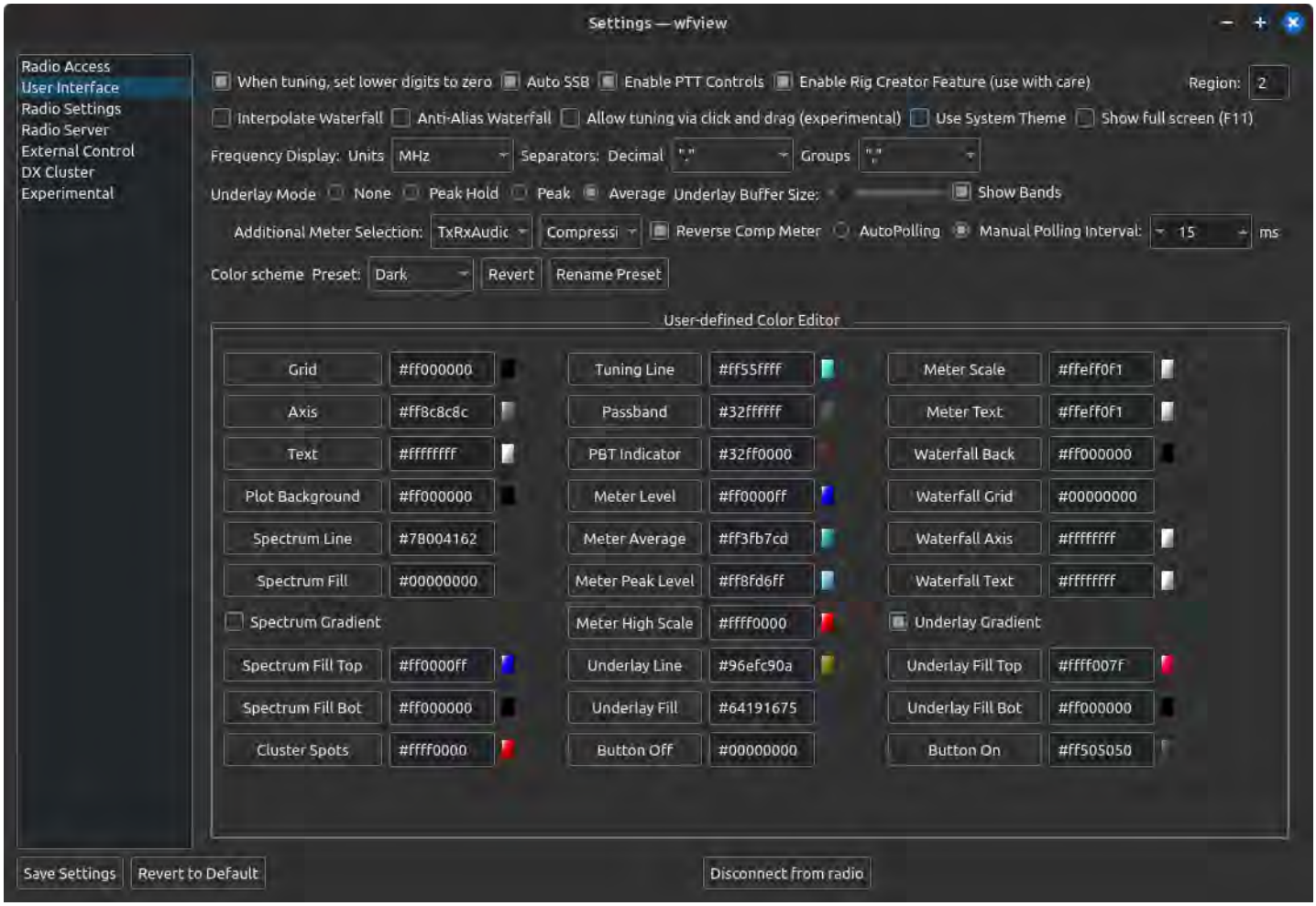
Icom only notes (CI-V and Model):

- Manual Radio CI-V Address: DO NOT set this parameter unless you have an absolutely ancient radio. Modern Icom radios have a feature called CI-V Transceive which, when enabled (please enable it), will cause the radio to self-identify to wfview.
- Use CI-V address as Model ID too: DO NOT set this parameter unless you have an absolutely ancient radio. This will cause wfview to assume that the CI-V address provided corresponds to a radio of the same default CI-V address.
- Please do not turn these parameters on unless you are absolutely sure you need to.

Once you have setup wfview for your radio, you can press "Connect to radio" to see if it works.

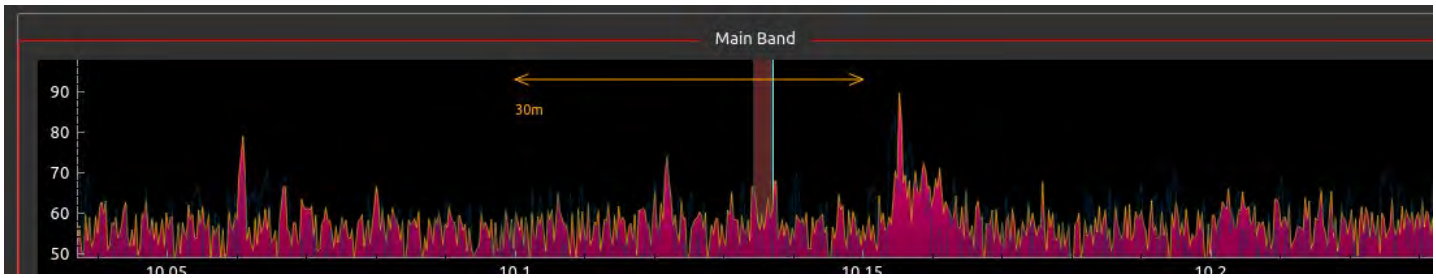
User Interface

These controls adjust how the wfview user interface appears and behaves.



- When tuning, set lower digits to zero: This control causes the tuning dial to automatically set the below-tuning-step digits to zero. This is generally desired. Uncheck this control if you want to dial in and keep an offset applied to all your tuning, for example if you want frequencies always ending in 5 KHz for some reason.
- Auto SSB: This setting will automatically select the “correct” sideband when switching bands. This is only done if you are already in LSB or USB mode.
- Allow tuning via click and drag: This checkbox made it into the master branch accidentally. If enabled, you can try clicking and dragging on the spectrum to tune. It is an experimental option and still under development. Actually, it does not even work in V2.
- Interpolate Waterfall: If you have a big screen, check this box and the pixels of the waterfall will not look quite as blocky and pixelated. Checking this box can add significant CPU use, so do not do this on a humble platform like a Raspberry Pi.
- Anti-Alias Waterfall: Check this box for anti-aliasing on the waterfall display. This seems to have almost no impact on the performance or appearance! But it’s an option if you want it. There is some impact on CPU usage, so uncheck this if your computer is struggling to draw the waterfall.
- Use System Theme: Check this box to switch from wfview’s built-in theme (QDarkStyleSheet) to your desktop environment’s default appearance. Read more about this here.
- Show full screen: Check this box for full-screen mode. You can also use F11 to toggle full screen. Tip: For a dedicated wfview display, use a raspberry pi set to launch wfview on startup under the default user. Check the full screen box and save the settings.
- Underlay Mode: These four options select an “underlay” for the spectrum plot. The underlay is an additional plot shown under the radio’s spectrum data. Two of the underlay options obtain data from a variable-size buffer of received spectra, in which one value is deleted each time a new value is received. The size can be set using the buffer size slider. The available underlay options are:
    - None: no underlay drawn
    - Peak Hold: An underlay is drawn showing the maximum value received. The peaks do not expire except when tuning.

    - Peak: A buffered peak mode. The peak value in the buffer (as set with the Buffer Size slider) is plotted for each frequency.
    - Average: The average value for each frequency is plotted. This provides a “smoothed” spectrum mode which often helps beat down broad-band noise such as RFI.
- Underlay Buffer Size: Sets the size of the underlay buffer. Lower values will be more responsive, higher values will be more “sticky”. Tip: Press “Pop-Out” to see the effects of adjusting this slider in real-time as you make the adjustments.
- Frequency Display:
    - Units: select the units desired for the frequency display. Generally, this is set to MHz.
    - Separators: Select your desired decimal and group values. North American users will likely want to select “.” for the decimal and “,” for the group.
- Show Bands: Check this and wfview will show a line over the top of the spectrum indicating the amateur radio band. You can set the color of this display as well (it’s one of the color buttons below). For example, the 30M in this screenshot:



- Enable PTT Controls: Enables wfview’s transmit button. This checkbox has no effect on the radio’s physical PTT buttons, nor does it block any other radio control programs from sending PTT commands. It is designed to prevent inadvertent transmission, and could be useful during open radio demonstrations. wfview has a built-in 3 minute timeout on transmit.
- Secondary Meter Selection: Select a desired secondary meter here. Some radios do not support some meter types.
    - Available meters: None, SWR, ALC, Compression, Voltage, Current, Center (IC-R8600 only), TxRxAudio (switches Tx/Rx automatically), RxAudio (RX audio all the time), TxAudio (TX audio all the time)
    - See more about the meter under the View Tab manual page.
- Reverse Comp Meter: Check this box to draw the compression meter “broadcast-style” with reduction shown moving to the left as the reduction increases.
- Polling: wfview automatically determines the optimal polling rate for the radio. However, faster polling will make the meters more responsive and is generally fine so long as a secondary program isn’t polling as well. Slower polling rates may increase reliability, especially with older radios using single-wire CI-V. If you are running wfview as a server as well, it can make sense to set the polling period to a high value on the server so that the server isn’t contributing much to the overall polling activity. When set to Auto Polling, wfview may make adjustments to the polling time if the connection exhibits high latency. If you find this annoying, you can set the manual polling interval and it will not be adjusted. Be careful not to be too aggressive with the polling time. Typical polling times for network radios are around 20ms.

Color Scheme:

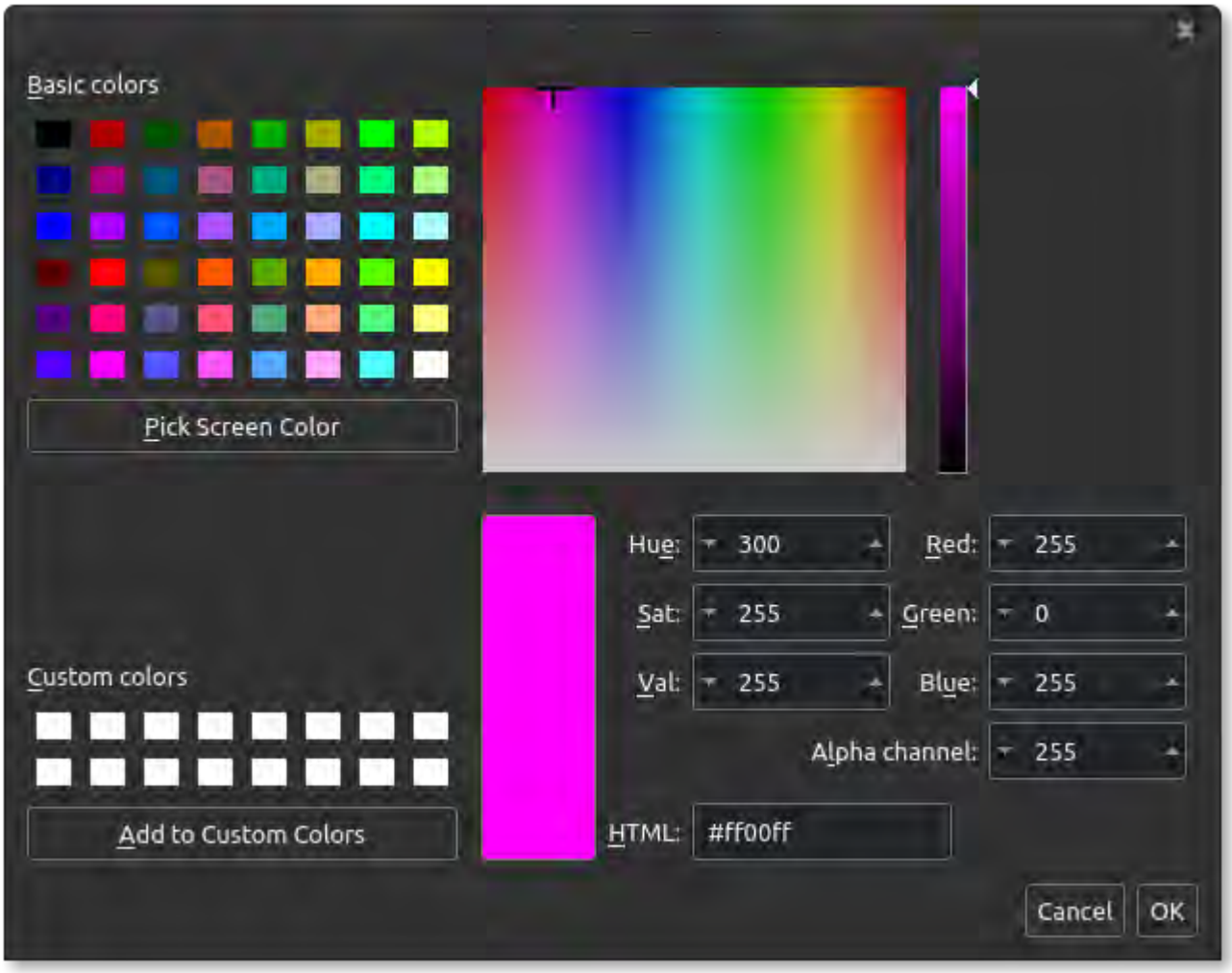
Use these controls to adjust the color scheme. Scoot the settings window over to the side so that you can see the UI change as you make adjustments. More information on customizing the appearance is available here.

- Preset: This control is for recalling the five available color scheme presets. Presets are saved with the settings and may be renamed and reverted.
- Save Preset: This button will save the selected preset’s colors and name to the settings file.
- Revert: Press this button to revert the selected preset to the default values.
- Rename Preset: Press here to rename the selected preset. Names may be up to ten characters long and

may contain spaces and probably even unicode characters.

User-Defined Color Editor Buttons

Use these buttons to change the colors of many of the user-interface elements of wfview. The color editor allows a custom color to be selected. Simply press the button for a given attribute (such as “Grid”, “Axis”, “Text”, etc) and the color picker will be shown:



Alternatively, you may enter in HTML-style color codes. Keep in mind the first two characters represent the “alpha” channel. “ff” means opaque, and “00” means transparent, which is a nice way to turn off any distracting elements that you don’t wish to see (such as the grid lines on the waterfall).

Radio Settings

This settings page is a convenience feature to help you access frequently-used radio settings.

Modulation Source: wfview shows here the radio’s selection of available modulation (transmit) audio sources. For most people, you will want this set to “LAN” while you are using wfview, and set to “MIC” or “M/U” when not using wfview.Icom radios typically have a setting for “normal” data-off modes, and a setting for DATA modes (such as

USB-D). “M/U” means “Mic/USB” and “M/A” means “Mic/Accessory”. Please refer to your Icom manual to see what this means.

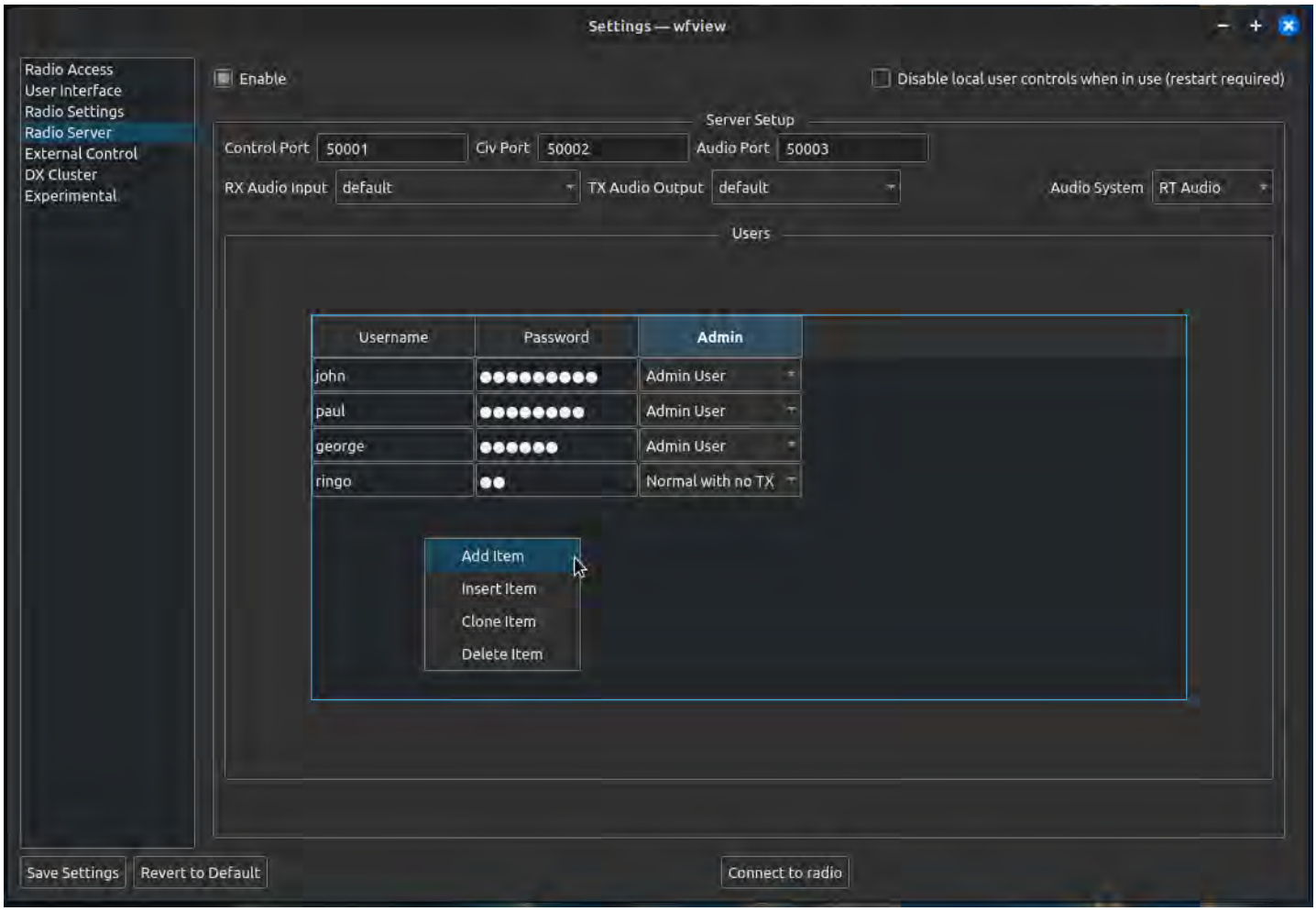
Set Clock: For the IC-7300, this will set the clock to the computer’s clock. The actual clock is set when the seconds go to zero, since there is not a way to send the seconds in the command. So you may need to wait up to a minute for the command to complete.

- Use UTC: Set the radio’s clock to UTC
- Set radio time on connect: Set the radio’s clock every time you connect. This was a requested feature of a 7300 user that was tired of the tremendous clock drift present on the radio.

Manual PTT Toggle: Use these buttons if wfview is somehow confused about the PTT state and PTT Toggle is not working. Honestly, you won’t need it these days.

Radio Server

This settings page is for configuring wfview’s built-in radio server, which allows radios without built-in network connections to become accessible over the internet. For example, an IC-7300 can be connected with a USB port to a dedicated “server” computer and then connected to over the internet for remote use. This is an advanced feature for users already familiar with the concepts of computer networking.



To enable the radio server, check the “Enable” box at the top.

To disable local control within this “server” wfview instance, check the “Disable local controls when in use (restart required)” box, and restart wfview once everything is set up as you want.

- Control Port: The port over which the remote session is managed. The default is 50001 and the protocol is UDP for Icom radios and TCP for Kenwood radios.
- Civ Port: The port used for CI-V commands. The default is 50002 and is UDP for Icom radios.

Audio Port: the port for the audio streaming. The default is 50003 and is always UDP.

Audio settings: Very important. Be careful here and select your radio's audio input and audio output:

- RX Audio Input: Select the radio's receiver audio. This is the radio's audio output and wfview's audio input. Typically this will be "USB CODEC" or some similar name.
- TX Audio Output: Select the radio's transmit audio. This is the radio's audio input and wfview's audio output. Typically this will be "USB CODEC" or similar.

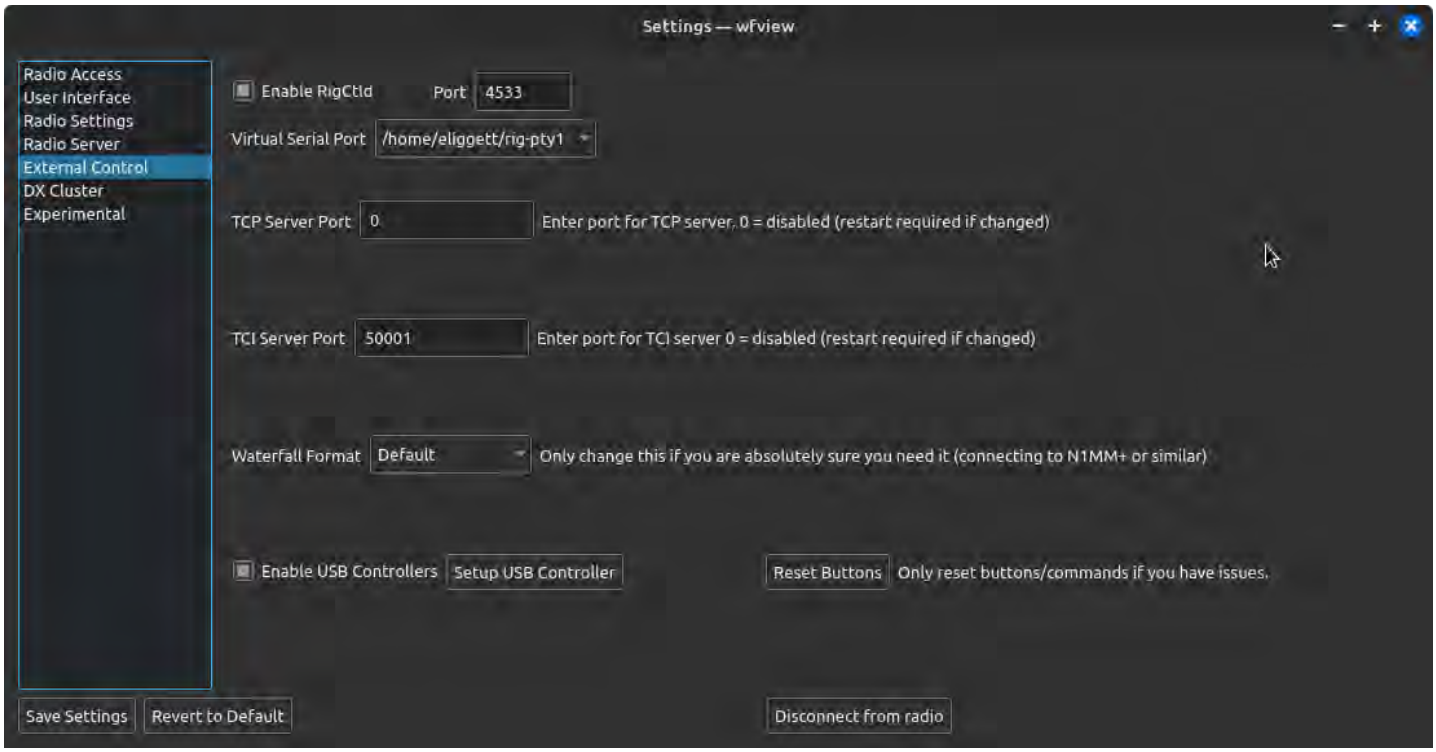
See this page of the manual for more information about remote control.

External Control

These settings configure how external hardware and other programs can control and interact with wfview. Examples of other hardware include USB jog wheels for tuning and PTT buttons. Examples of external software control are programs like WSJT-X and fldigi.

External software control is covered in depth here. Please read it carefully, this topic is important and worth understanding for modern ham radio operators.

External hardware control is covered in depth here.



Enable RigCtld: Check this box to enable wfview's built-in rigctld-compatible server. Don't forget to make a note of the port. The standard port for rigctld is 4532; wfview defaults to 4533 so as to not conflict with any other instance of rigctld you may have running already (but really, you should not unless you have an unusual setup). Read more about this here.

Virtual Serial Port: Select a virtual serial port here which software can connect to which expects to connect directly to the radio. The virtual port will appear just like a serial port to the other application. Please read about this feature carefully here.

TCP Server Port: A raw TCP port which is similar to a virtual serial port. Used by N1MM+ and RUMLogNG

TCI: This is an advanced and rather experimental feature for sharing both audio and control with another application.

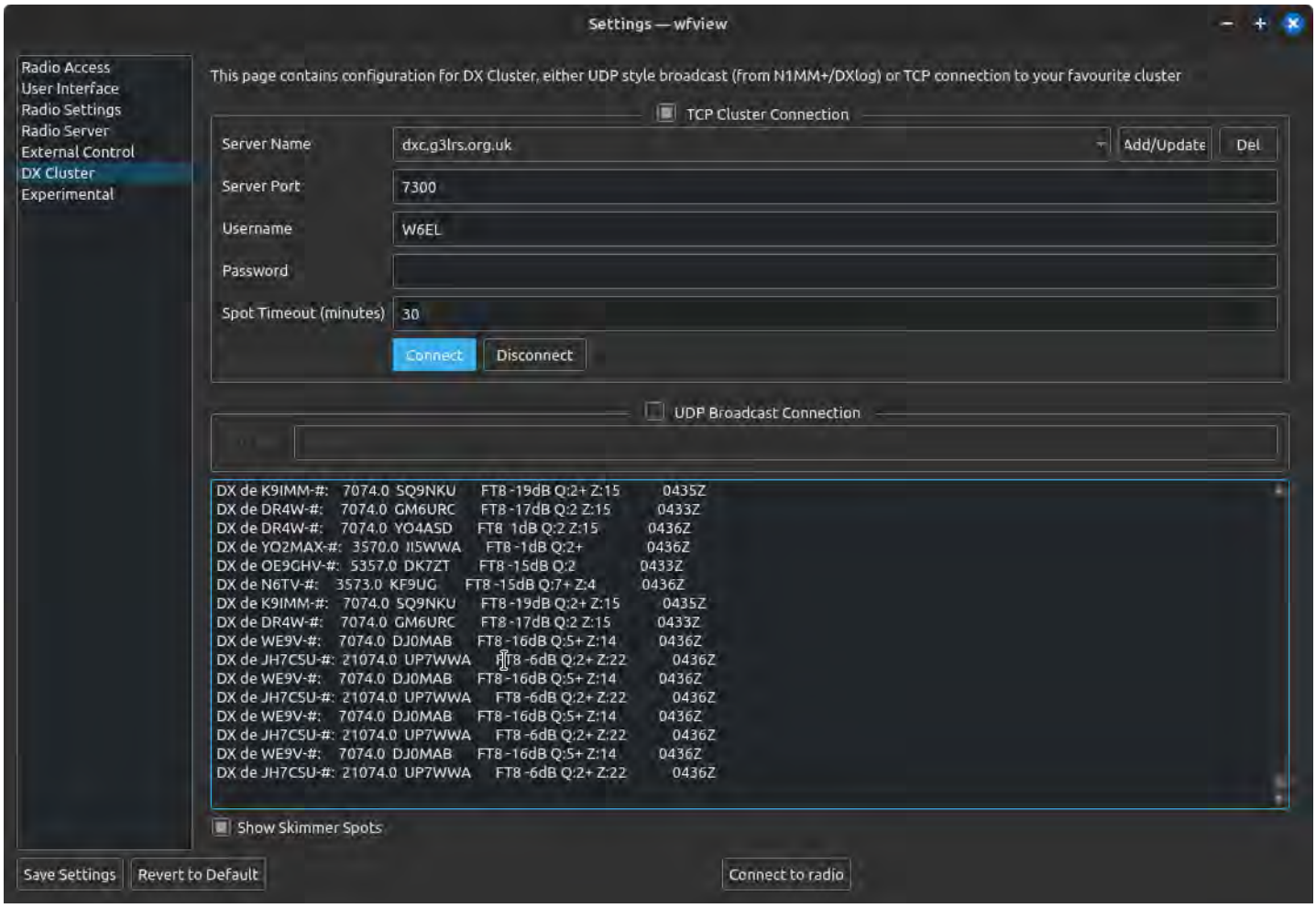
TCI Server Port: Set this to the port you wish to run the TCI server on. Set to zero to disable.

Waterfall Format: Use this feature to emulate LAN or USB-style waterfall data for external applications expecting the format of waterfall data to be different than what you have available. This is a rarely used option, and you should not use it unless you know you need it. Thus far only N1MM+ requires it, and only in some situations.

USB Controllers: There is a page on setting up USB Controllers here. There are many supported controllers, including the Icom RC-28. Press “Reset Buttons” if your button assignments are not working (possible after a major wfview version change).

DX Cluster

These controls are for setting up access to a DX Cluster. A DX Cluster is a server which provides users access to real-time dx reports. The callsigns of the reported stations are overlaid on top of the spectrum view.



To add a DX Cluster server, either select it from the drop-down menu list, or simply type in the server and press “Add/Update”.

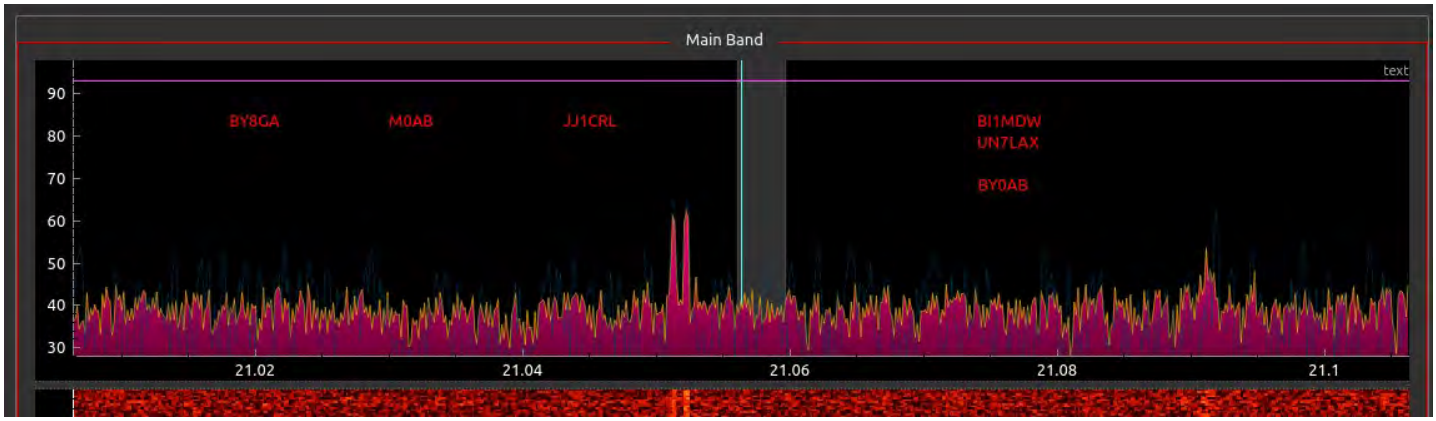
Supply the Server Port, Username, and Password. Many dx clusters do not require a password. Press Connect to use the cluster. Keep in mind it may be a few minutes before you see spots on the band you are currently using.

Set the Spot Timeout (minutes) to the amount of time after a spot is reported when the spot can be dropped from the display.

UDP Broadcast Connection: Enable this option to re-broadcast DX Cluster data to other programs.

Show Skimmer Spots: Show spots which have been “skimmed”

Here is an example of wfview showing DX Cluster results. The station’s reported frequency is on the left side of the drawn text.



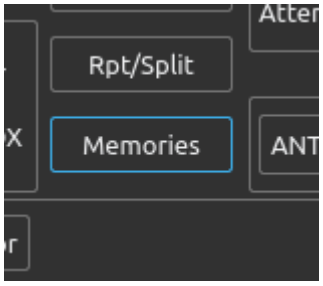
Experimental

These settings are experimental and are not generally enabled. This category is used to briefly test features for developers only.

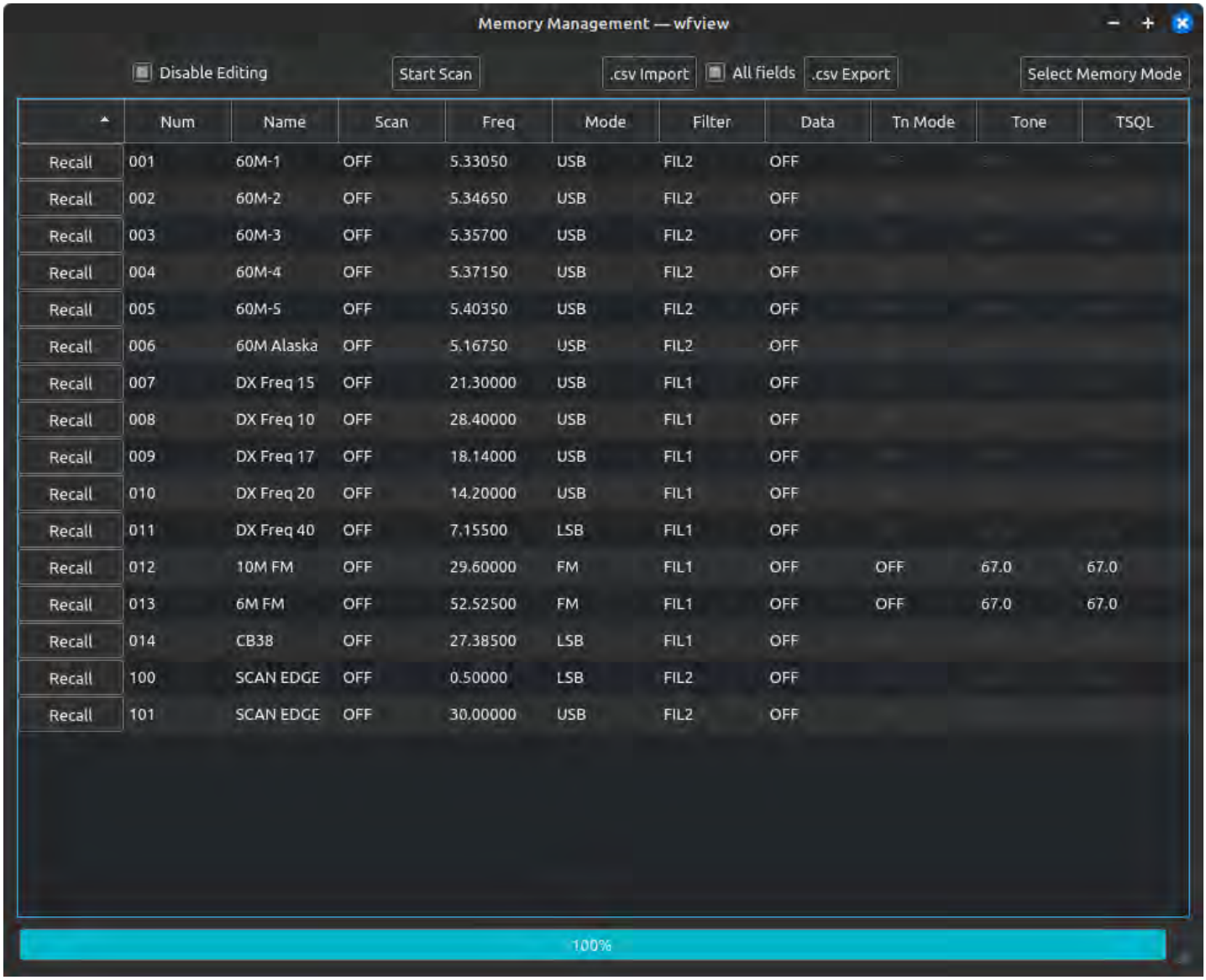
Memory Editor

wfview provides an experimental radio memory editor. This editor lets you select and edit memory channels in a spreadsheet-style. For the IC-9700, Satellite memories can also be edited.

After connecting to the radio, press the “Memories” button near the bottom of wfview:



This will first remind you that the feature is experimental. Once you accept the experimental status, you can see your memories like this:



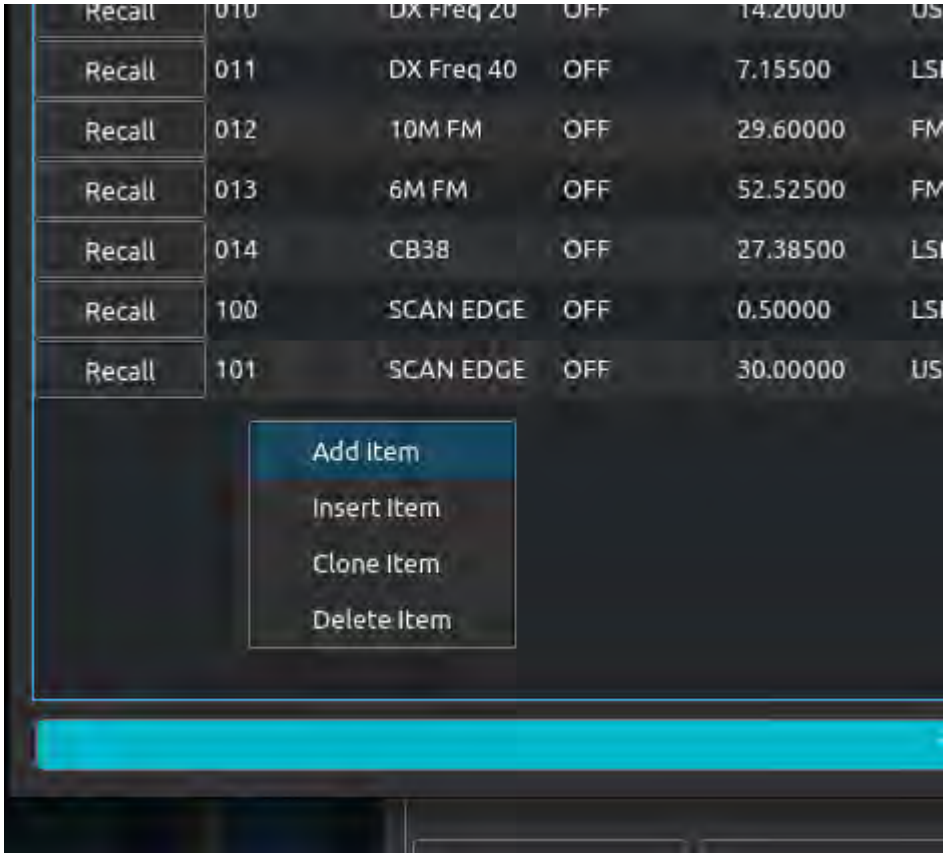
	Num	Name	Scan	Freq	Mode	Filter	Data	Tn Mode	Tone	TSQL
Recall	001	60M-1	OFF	5.33050	USB	FIL2	OFF			
Recall	002	60M-2	OFF	5.34650	USB	FIL2	OFF			
Recall	003	60M-3	OFF	5.35700	USB	FIL2	OFF			
Recall	004	60M-4	OFF	5.37150	USB	FIL2	OFF			
Recall	005	60M-5	OFF	5.40350	USB	FIL2	OFF			
Recall	006	60M Alaska	OFF	5.16750	USB	FIL2	OFF			
Recall	007	DX Freq 15	OFF	21.30000	USB	FIL1	OFF			
Recall	008	DX Freq 10	OFF	28.40000	USB	FIL1	OFF			
Recall	009	DX Freq 17	OFF	18.14000	USB	FIL1	OFF			
Recall	010	DX Freq 20	OFF	14.20000	USB	FIL1	OFF			
Recall	011	DX Freq 40	OFF	7.15500	LSB	FIL1	OFF			
Recall	012	10M FM	OFF	29.60000	FM	FIL1	OFF	OFF	67.0	67.0
Recall	013	6M FM	OFF	52.52500	FM	FIL1	OFF	OFF	67.0	67.0
Recall	014	CB38	OFF	27.38500	LSB	FIL1	OFF			
Recall	100	SCAN EDGE	OFF	0.50000	LSB	FIL2	OFF			
Recall	101	SCAN EDGE	OFF	30.00000	USB	FIL2	OFF			

To recall a memory channel to the VFO, just press “Recall”.

To enter “Memory mode” press “Select Memory Mode”. To get back to normal VFO mode, press the button again (it will say “Select VFO mode”).

If you wish to edit an existing channel or add one, you first must uncheck “Disable Editing” at the top of the window. Once unchecked, you can click around in the memory channels and edit the contents quite easily.

To add a new channel, first enable editing as above. Then right-click in the blank area below the last channel and choose “Add Item”



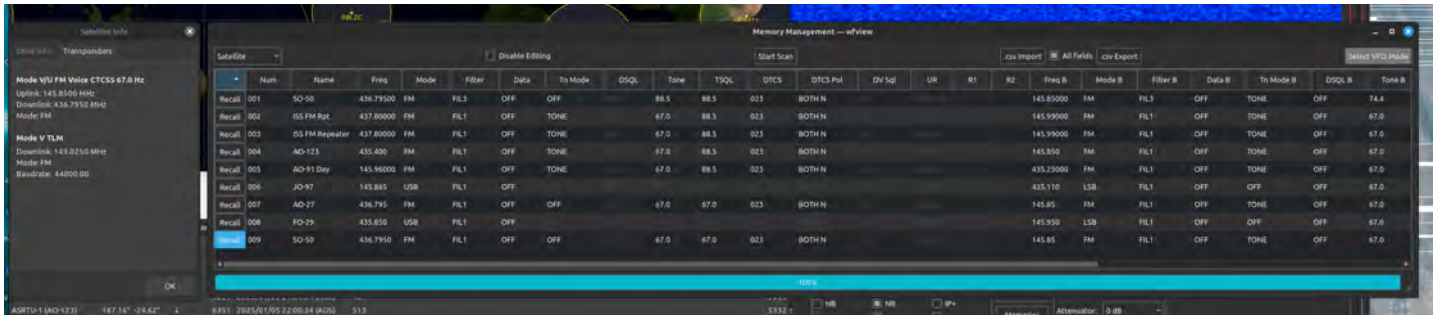
You may need to double-click in the blank mode column area for a new entry to get the mode selection box to pop up.

If you'd like to edit the file in a spreadsheet program such as Excel or Google Sheets, first Export to CSV using the ".csv Export" button. Then make edits as needed and import back to the radio using the ".csv Import" button. Keep in mind that each radio has slightly different features, so you should always begin with an exported template before importing.

For the IC-9700 (and possibly other radios), memory groups are available. These appear, when available, in the upper-left of the memory editor as shown here:

For satellite memories on the IC-9700, the Downlink (your receive) frequency goes under the "Freq" column, and

the Uplink (your transmit) frequency goes under the “Freq B” column:



This is a reliable method of programming the uplink frequency, which is usually only accessible from the computer by swapping the Main and Sub bands, reading and/or editing the new Main frequency, and then swapping back. This is exactly what happens when using Hamlib with gpredict, for example.

wfview

Open Source interface for Icom and Kenwood transceivers

Using USB Controllers (RC-28, Shuttle, Stream Deck, sunSDR, etc)

This page describes how to use USB-connected controllers with wfview to control your radio.

Hardware controllers offer an immensely-improved user experience for tuning, changing modes, PTT, and more.

Supported Controllers

wfview supports the following USB controllers natively:

- Icom RC-28
- Contour Shuttle Xpress aka Shuttle Express
- Contour ShuttlePRO V2
- XBox USB controller TODO: Link for supported device
- P.I. Engineering USB 3 Switch Interface SK3SWI
- sunSDR Pro Ecoder Plus

- Xencelabs Quick Keys wireless
- Stream Deck:
    - Mini, MiniV2, Original, Original V2, Original MK2, XL, XLV2, Pedal, Plus
- Stream Deck clones by Mirabox on AliExpress (inexpensive), sometimes branded “FHOOU:
    - 293, 293S, N3

You can have a look at the top portion of usbcontrollers.cpp for the complete list, if you are curious about it.

Preliminary Setup

You will need to connect the USB device to your computer. It is important that there are **no other programs using the device**. Generally **drivers are not necessary** since these devices are “HID Compliant”. Therefore, **do not** install any drivers or other software. wfview needs access to the USB device without any other programs or software using it at the same time.

Windows and macOS users: You should not need any preliminary setup. Please continue to the section wfview Controller Setup.

Linux Setup for ShuttlePro

Linux users will need to stop their desktop from using the ShuttlePRO wheel as a scroll wheel. This can be done per the following bash shell script taken from this reddit post:

```
#!/bin/bash
for DEV in `xinput --list | grep -i ShuttlePro | cut -d"=" -f2 | awk {'print $1'}`
do
     xinput disable ${DEV}
done
```

Linux users will also need the following udev rule, placed in /etc/udev/rules.d, which gives the user permission to access the shuttle device:

Contents of `/etc/udev/rules.d/99-shuttle.rules`:

```
# for newer PRO model
ATTRS{name}=="Contour Design ShuttlePro" MODE="0644"
# for older PRO model
ATTRS{name}=="Contour Design ShuttlePRO v2" MODE="0644"
# for the Xpress model
ATTRS{name}=="Contour Design ShuttleXpress" MODE="0644"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="0b33", ATTRS{idProduct}=="0020", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="0b33", ATTRS{idProduct}=="0030", MODE="0666"
```

To activate the new rules, just pull the USB plug out and push it back in, that simple.

wfview Controller Setup

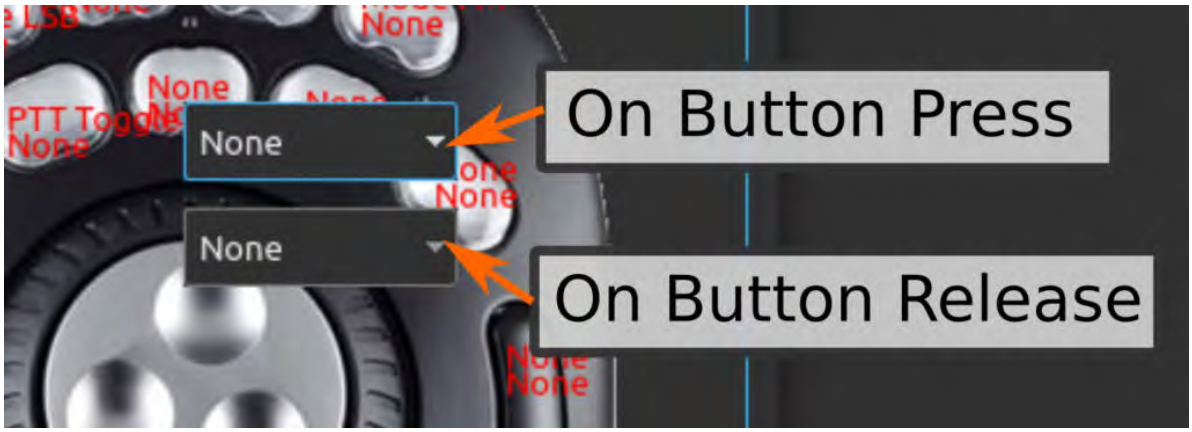
You will need wfview version 1.57 or greater. The latest release is recommended.

Connect the USB controller, and then, launch wfview. Click “Settings” followed by “External Control”. Check the box “Enable USB Controllers”. Then press “Setup USB Controller”.

The controller setup window will appear. This window is different for each type of supported USB Controller. Here is how it looks for the Contour ShuttlePRO V2:

To assign a function to a button, right-click on the desired controller button. Two pop-up menus will appear. The top-most pop-up menu lets you select a function for when the selected button is pressed down. This is what most

users will want for most functions. The bottom menu lets you select what will happen when the button is released. Generally, the bottom menu should be set to “None”. Sometimes you will need to right-click in an empty space in order to remove the menu and reveal other buttons (we’re working on a better way to do this).



To program a traditional PTT button, like a handheld microphone, where the button is held down while you are transmitting, select “PTT On” for the press (top) function and “PTT Off” for the release (bottom).

Here’s a screenshot of a traditional PTT button setup:

For a PTT “sticky toggle” function like the radio’s chassis transmit button on an Icom, select “PTT Toggle” for the top menu and select “None” for the release (bottom) menu. This will make it transmit when you tap it and receive when you tap it again.

Press “Ok”

Important: Do not forget to press “Save Settings”!!

Keystrokes

wfview has keystrokes for almost all commonly-accessed controls one would need to have a QSO or browse the bands.

Not only is this convenient for rapidly manipulating the controls, it also may enable blind operators to use modern touch-screen radios if a computer (or even a Pi) is set up to run wfview in full-screen mode on boot.

Don't forget to check out our supported USB Controllers, which offer even more hands-on controls.

Many users find that a stand-alone numeric keypad provides sufficient controls (frequency, mode, and tuning are available).

For users of the vi editor, we have added HJKL support, with modifiers for J and K being shift and control. H and L follow the selected tuning step.

And for those on the opposite end of the spectrum, we've also made sure that wfview works well on touchscreen devices. wfview looks great running full-screen on a 10″ touch screen connected to a Raspberry Pi, for example.

NOTE: Mac users will need to substitute the Command (⌘) key for the Control key mentioned here.

User Interface Navigation:

Key	Function
F1	Main Window
F2	Band Window
F3	Frequency Window
F4	Settings Window

Numeric Keypad Operation:

Key	Function
/	Cycle through common modes
*	Enter frequency
+	Increase frequency by 100 Hz
-	Decrease frequency by 100 Hz
number and decimal	Used to enter frequency

Band Window:

Key	Function
6	Jump to 6 meters
1	Jump to 10 meters
t	Jump to 12 meters
5	Jump to 15 meters
7	Jump to 17 meters

2	Jump to 20 meters
3	Jump to 30 meters
4	Jump to 40 meters
s	Jump to 60 meters
8	Jump to 80 meters
L	Jump to 160 meters
g	Jump to last general coverage frequency
A	Jump to VHF Air band
V	Jump to 2 meters
U	Jump to 70cm
W	Jump to Wideband FM band

Settings Window:

Use these keyboard shortcuts to change the current page of the Settings window

Key	Function
Shift+F1	Radio Access Page
Shift+F2	User Interface Page
Shift+F3	Radio Settings Page
Shift+F4	Radio Server Page
Shift+F5	External Control Page
Shift+F6	DX Cluster Page
Shift+F7	Experimental Page

Other:

Key	Function
F5	Activate mode LSB
F6	Activate mode USB
F7	Activate mode AM
F8	Activate mode CW
F9	Activate mode USB-D
F10	Activate mode FM
F11	Toggle full screen
F12	Send speech command to radio
F	Tell radio to speak current frequency
M	Tell radio to speak current mode
Control-T	Transmit
Control-R	Receive
Control-P	Toggle PTT
Control-I	Enable/Disable the ATU
Control-U	Run ATU tuning cycle
H	Decrease frequency by selected tuning step
J	Same as -, with modifiers of shift and control too
K	Same as +, with modifiers of shift and control too
L	Increase frequency by selected tuning step
+	Increase frequency by 100 Hz
-	Decrease frequency by 100 Hz
Shift +	Increase frequency by 1KHz
Shift -	Decrease frequency by 1 KHz
Control +	Increase frequency by 1 KHz
Control -	Decrease frequency by 1 KHz

Page Up	Increase frequency by 1 MHz
Page Down	Decrease frequency by 1 MHz

Appearance Customization

The appearance of wfview can be customized fully using several methods.

1. Use the built-in color pickers in Settings, under User Interface in the “User-defined Color Editor”. In addition to assigning colors, you can also make a color fully transparent to hide the element.
2. Select an alternate waterfall theme using the “Theme” combo box at the waterfall bottom-right. Many of these themes will look dramatically different as you set the Top and Bot sliders to change the mapping of data range to color range. The color scales come from QCustomPlot and can be seen here.
3. Toggle the “Use System Theme” checkbox to switch between your default desktop theme and the built-in stylesheet.
4. (Advanced users only) Specify an alternate stylesheet by manually editing the Settings file (or registry as it may be).

User-defined Colors

The built-in color picker looks like this:

Pressing any of the color buttons will let you edit the color for the respective attribute. Colors can also be manually input using the #AARRGGBB format, where “AA” is the alpha channel (ff = fully opaque, 00 = fully transparent). To help visualize the color changes, the “Pop-Out” button can be pressed to temporarly pop-out the settings tab. This lets you see the changes on the View tab right away. Press pop-out again (which will say “Re-attach”) to move the settings tab back into place. The settings can be saved using the “Preset” combo box. Press “Rename” to rename the setting, and press “Revert” for the default name and default colors for the selected preset. Don’t forget to save the preset, otherwise the preset will revert the next time you start wfview.

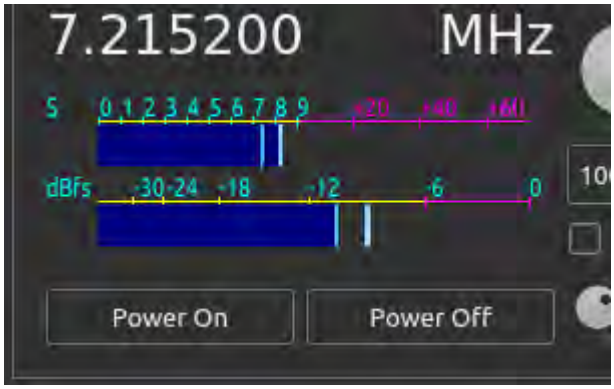
Stylesheet

wfview comes with a stylesheet built-in called QDarkStyle. This stylesheet can either be edited (linux only) or, you may download your own copy of the stylesheet, edit the stylesheet, and specify the stylesheet using the Settings file.

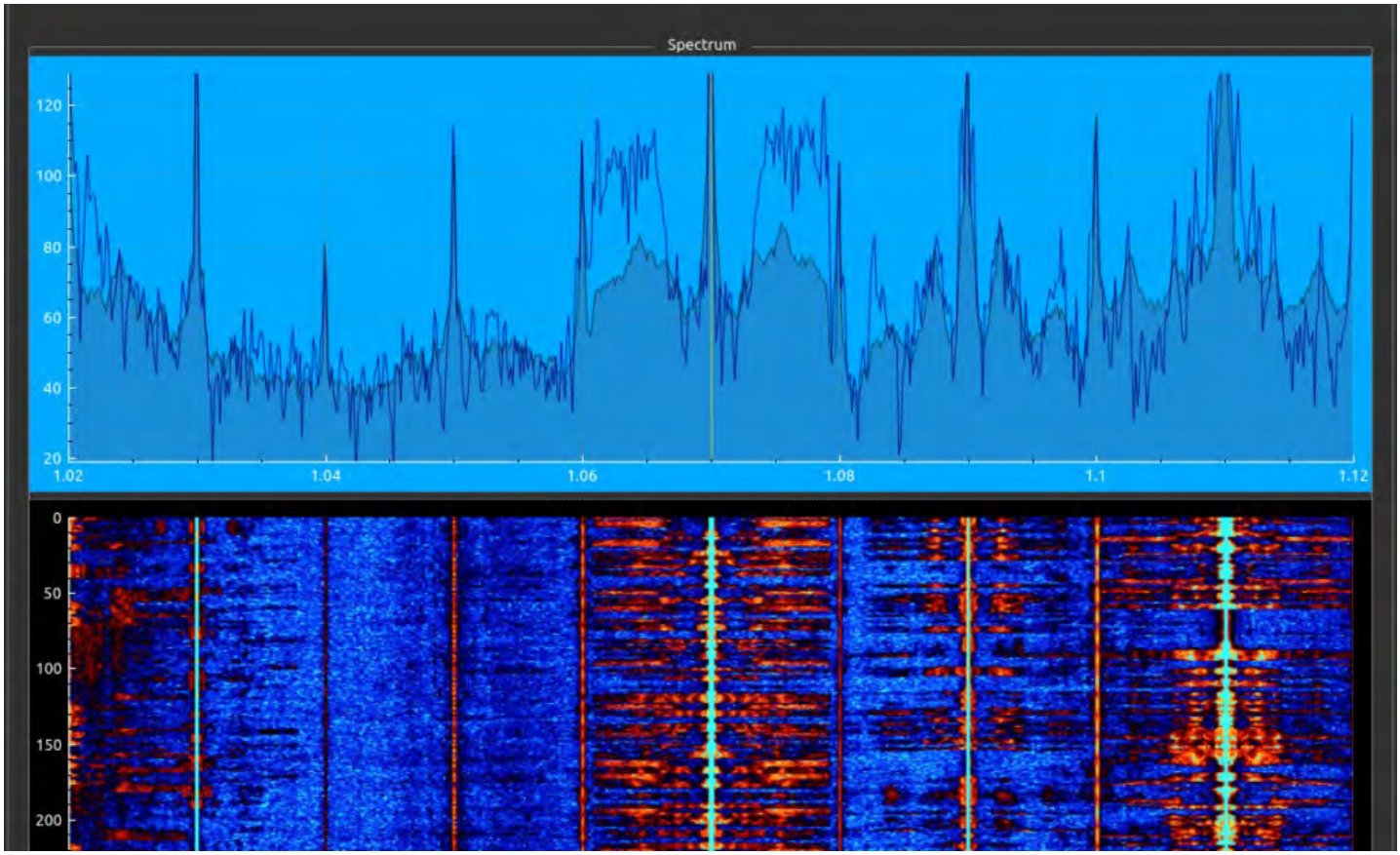
Gallery

Here are some interesting color examples.

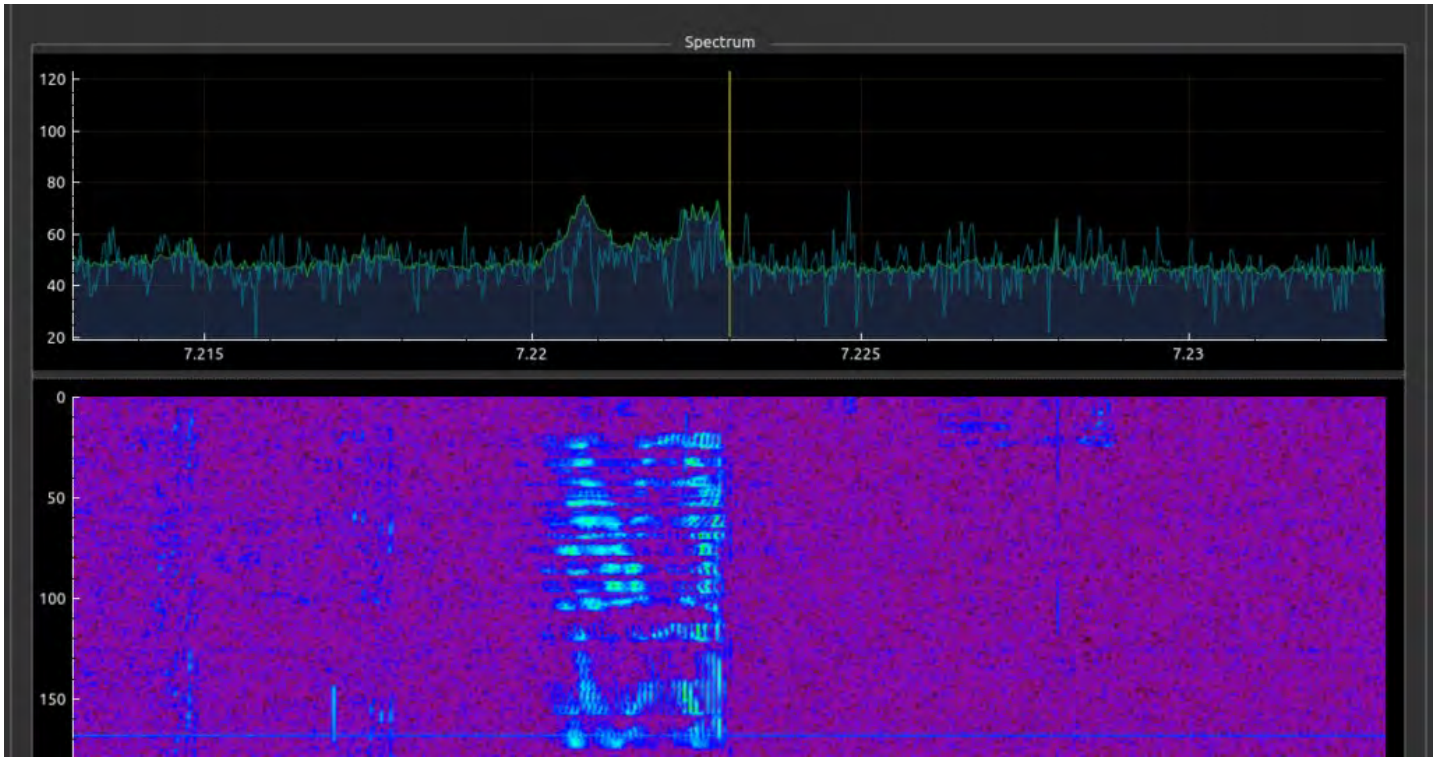
The meters can be customized fully:



You can invert the colors on the spectrum (bright background), for example:



You can also change the tuning line color and the line and fill options:



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Log in

Sharing Control Overview

wfview provides several methods of sharing the control of your radio with other programs. For example, you can connect to your radio with wfview, and then connect fldigi and MacLoggerDX to wfview. In all cases, connect wfview to the radio first. This is because wfview runs some commands that may not be compatible with other software programs, and also needs high-speed access to the waterfall data (which some programs would filter out). rigctld, which is built-in to wfview, is generally the best control option as it supports multiple programs at once and is generally simpler to configure. The virtual serial port option may be better for users running unusual commands that aren't in our implementation of rigctld yet. Don't forget to check our guide on how to share the audio with other programs, needed for network-access audio only of course.

The three principle methods of control are (click the links to learn more):

- built-in hamlib-compatible rigctld server: recommended, supports multiple programs at once.
- Virtual Serial Port interface: Emulates a genuine serial port in software. Only one program may use it at a time.
- Raw TCP/IP socket: Works for a few programs such as N1MM+ and possibly RUMLogNG.

These options can be enabled on the Settings tab under the External Control page.

For 3rd party programs that need radio audio, such as fldigi and wsjt-x:

- If the computer is connected to the radio using a USB cord, simply use the audio device within the 3rd party program, as you would without wfview.
- If you are accessing the radio remotely using wfview over a network, please see our page on Audio Configuration.

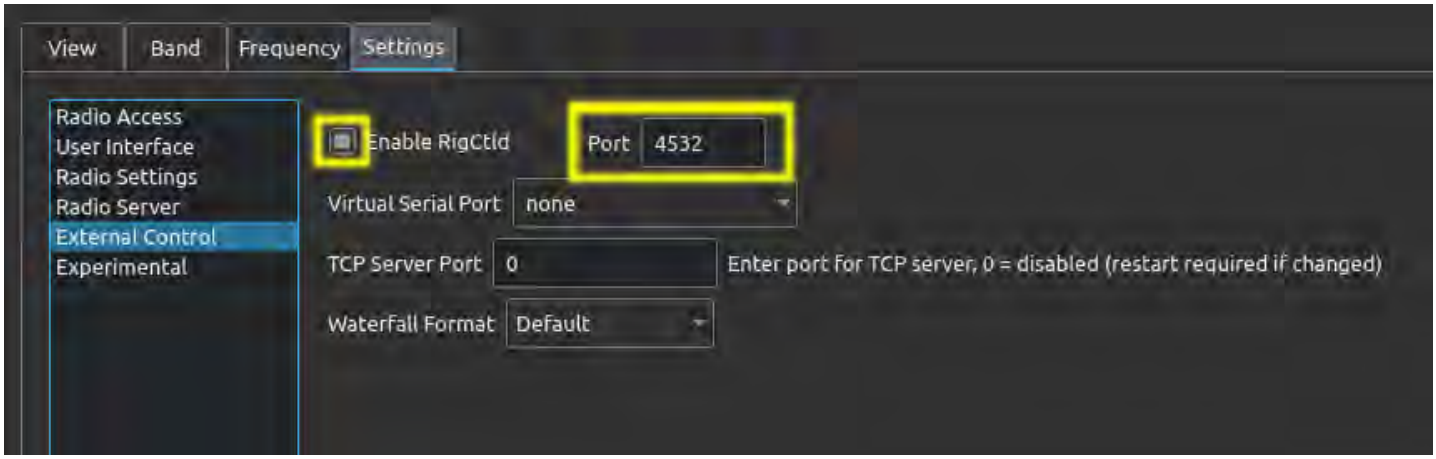
The following is a table of programs we have tested. For programs listing "rigctld", you will need to select "Hamlib NET rigctl" (or similar) as the radio model. See the rigctld page for more information.

Software	Version	Platforms Tested	Working?	Connector	Comments

WSJT-X	2.3.1	Windows, Mac, Linux	YES	rigctld	
Log4OM	2.11	Windows	YES	Omni-Rig	
Minos	2.4.1	MacOS, Windows	PARTIAL	rigctld	Not Linux/MacOS
fldigi and flrig	Current	Linux	YES	rigctld (preferred) and Pseudo-Term	
MacLoggerDX	6.35b5	MacOS	YES	rigctld (preferred) and Pseudo-Term	Requires com.dogparksoftware.MacLoggerDX dp_virtual_serial_port configuration option
QLog	Current	Linux	YES	rigctld	Use model "Hamlib NET rigctl". Don't forget Equipment -> Connect Rig

Hamlib rigctld emulation

wfview has the ability to emulate a subset of the Hamlib rigctld protocol which allows a number of applications to connect for rig control purposes. When supported, this is now the recommended/preferred method of connecting software to wfview (rather than virtual serial port/pseudo tty) as this utilizes a caching mechanism which avoids 'spamming' the rig with large numbers of duplicate CI-V requests.



To enable rigctld emulation, simply put a check next to "Enable RigCtld" within the wfview Settings tab, on the "External Control" page. The default port is set to 4533 so as not to conflict with any existing rigctld that is running. If you are confident that no other rigctld is running, you can set this to 4532 which will be the default port on most software. The port number must be greater than 1024. The port can be changed without restarting wfview, simply uncheck/check Enable RigCtld to restart the rigctld emulation.

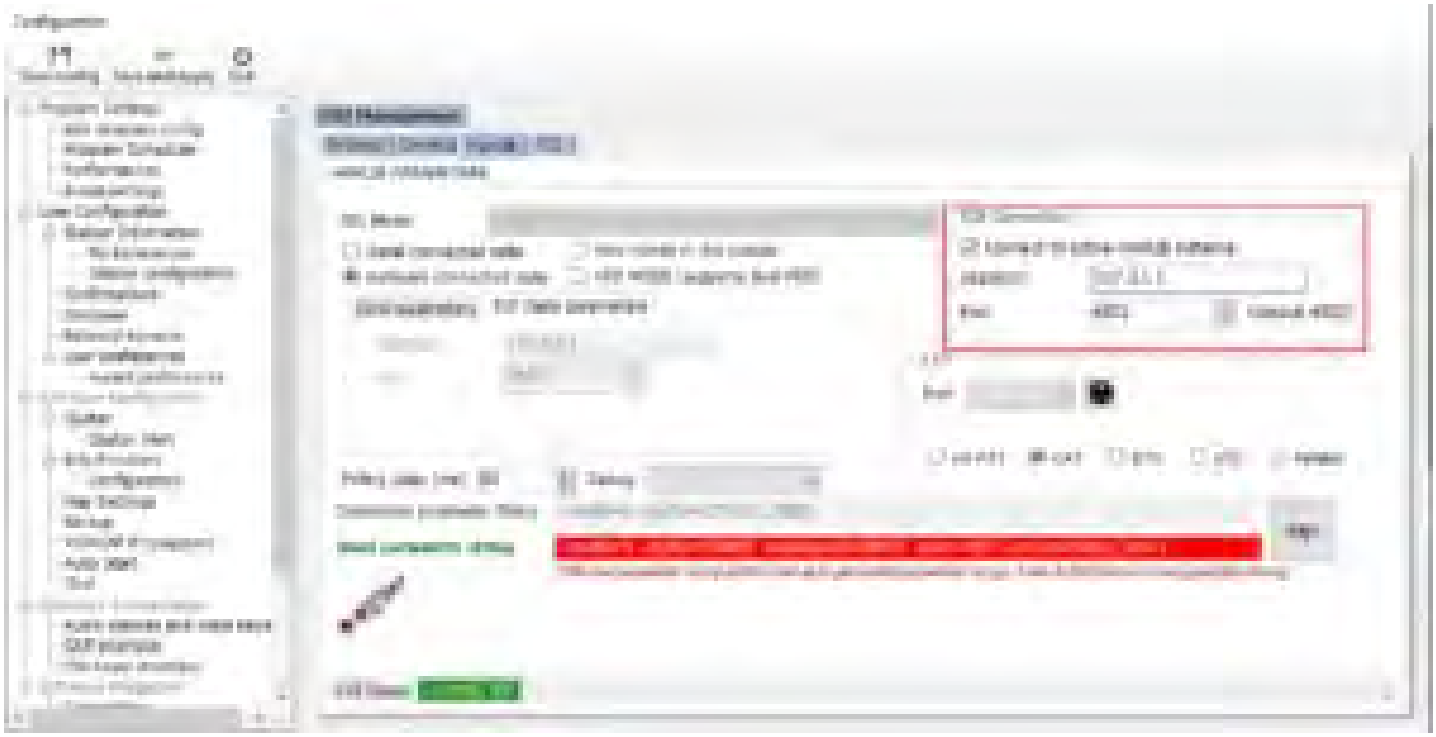
No other configuration of wfview is required, below are the required settings for various software that has been tested so far:

Jump down: Log4OM 2, WSJT-X, MacLoggerDX, fldigi

Log4OM 2 (NextGen) configuration

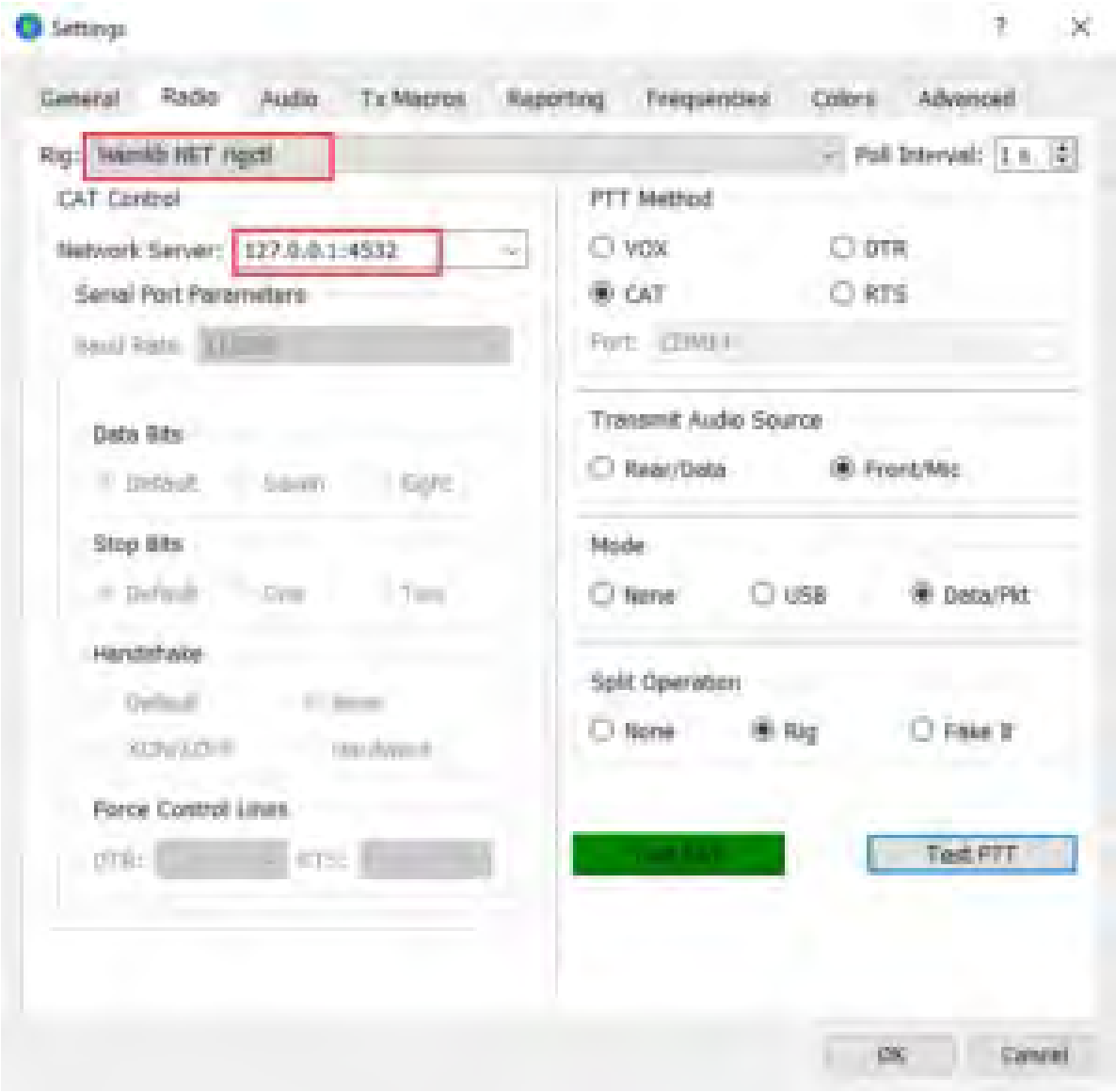
Firstly you must be running the latest version of Log4OM 2 (currently 2.17.0.0). Within "Settings/Program Configuration/CAT Interface" select Hamlib as the CAT engine.

Once done, select the Hamlib tab and enter the address and port of your wfview computer (if the same machine then 127.0.0.1). If you haven't changed the port in wfview to 4532, you will need to enter 4533 in the Port box below.



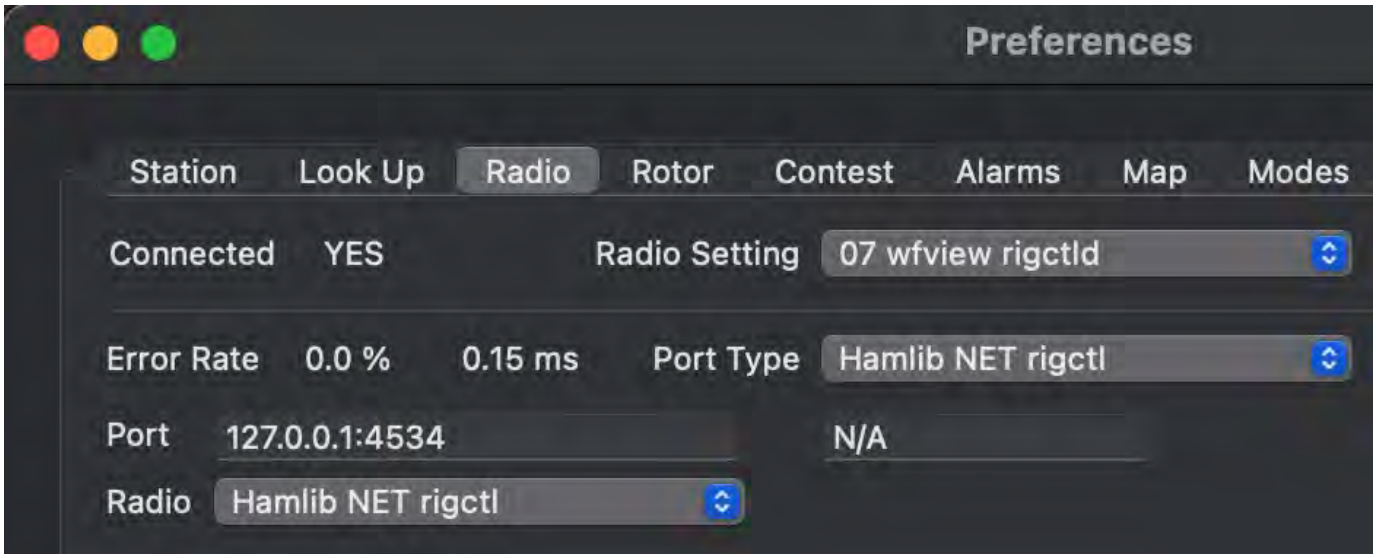
Once the above is saved, I found that I needed to restart Log4OM before it was able to connect to the wfview rigctld.

WSJT-X



Setup of WSJT-X is very straight-forward (this should be the same for the majority of software that is based on WSJT-X like JTDX and JS8-Call). Simply select the Rig: of “Hamlib NET rigctl” and enter 127.0.0.1:<port> into Network Server. Pressing Test CAT should result in the button turning green, which means it is talking to rigctld. Various other settings are optional but I would recommend setting PTT method to CAT, Mode to Data/Pkt and Split to either Fake It or Rig (rig based split support is still experimental in wfview though).

MacLoggerDX

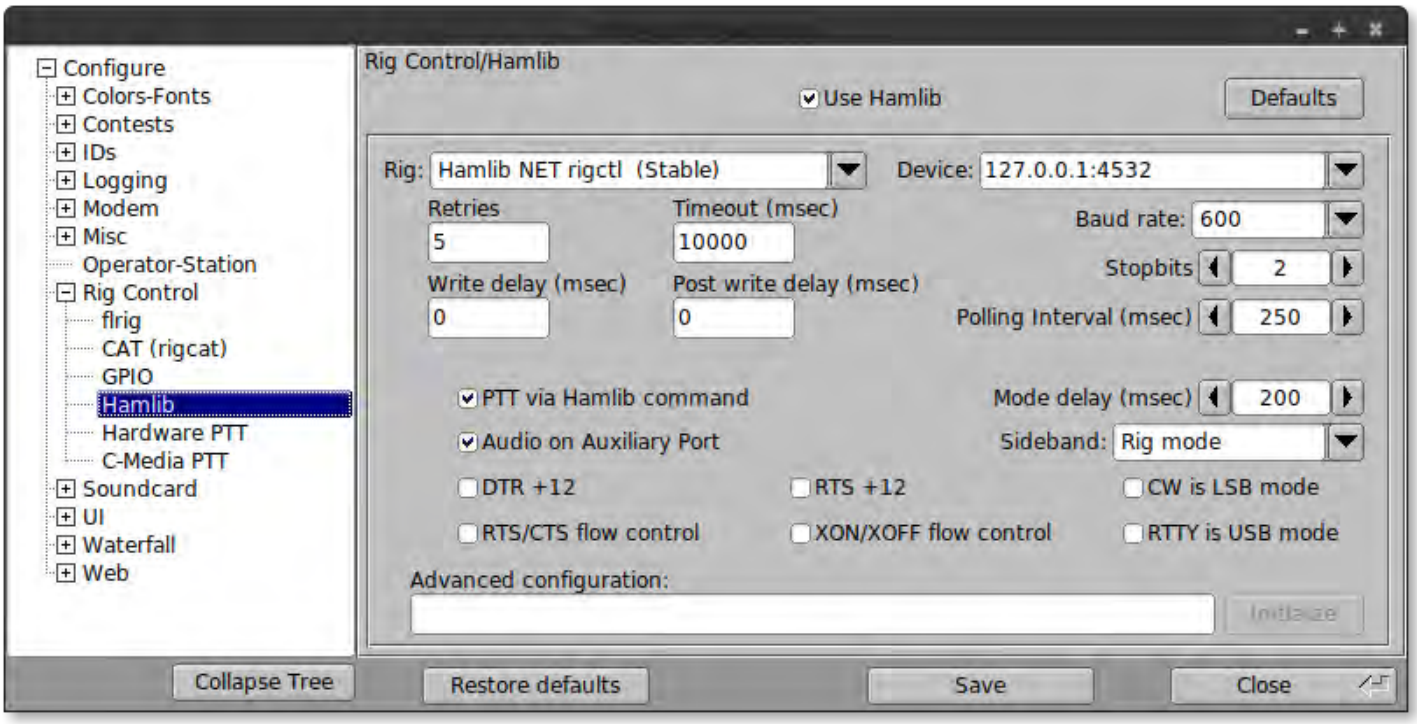


The Radio Setting name is user-defined, set it as you wish. For port type, select “Hamlib NET rigctl”. For the port, use “127.0.0.1:4534”, replacing “4534” with your port as specified in wfview.

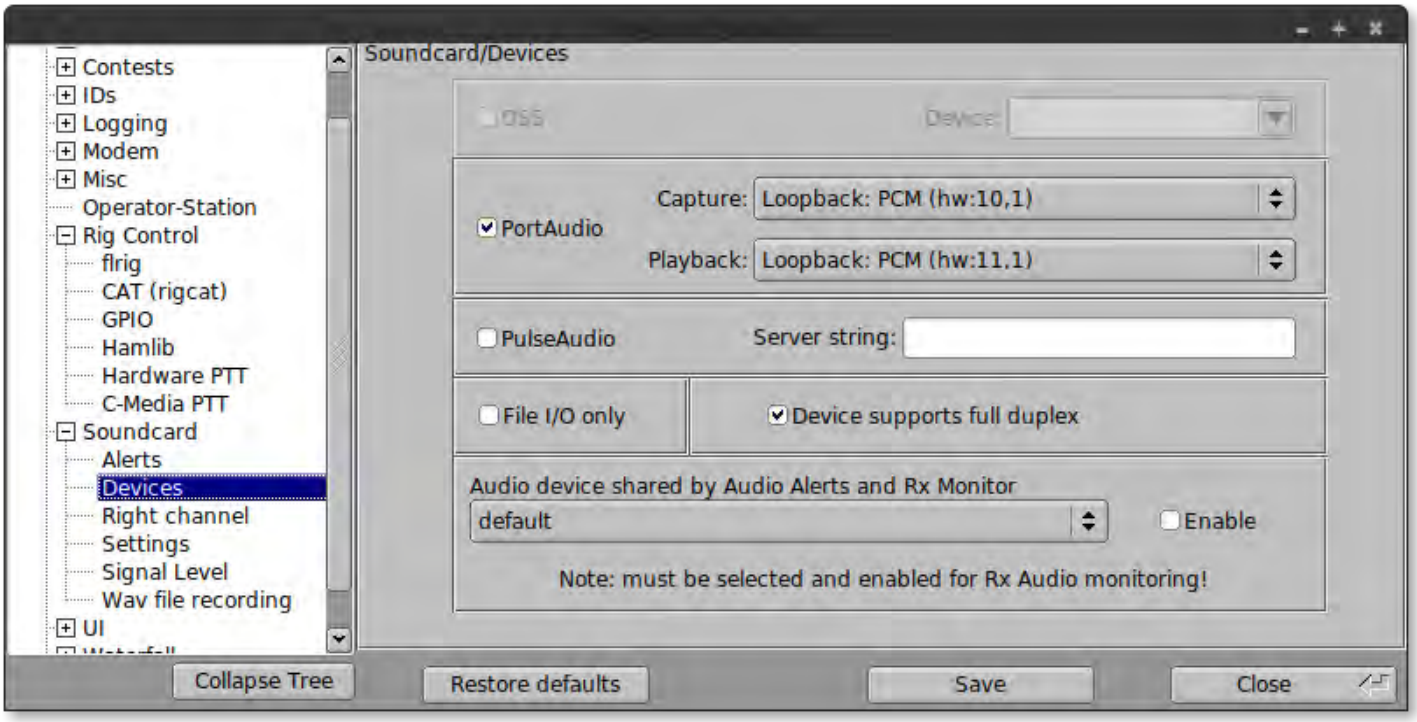
fldigi

Fldigi can easily connect to wfview using wfview’s built-in rigctld emulation. Note that fldigi needs to have been compiled with Hamlib support, which is commonly done.

To configure the radio control, turn off or disable any other rig control configured (flrig, RigCAT, etc), and then configure the Hamlib interface as shown. Be sure to check the “Use Hamlib” box:



For audio, enable a loopback audio device per our Audio Configuration guide, and then set as shown (this will vary some with each platform, the screenshot is from Linux):



Virtual Serial Ports

The built-in virtual serial port option of wfview lets a second program control the radio over a pseudo-terminal device. Using this method, many programs will work “out of the box” without knowledge that they are connected to a virtual serial device. Some programs will need additional configuration. Under Windows, a virtual serial port program is required to make the “loopback” connection between the wfview virtual serial port and the second program. Linux and Mac create genuine pseudo-terminal devices.

When possible (supported), it is generally better to use the built-in rigctld server.

Jump to: Windows Linux Mac

Windows

Virtual Serial Ports

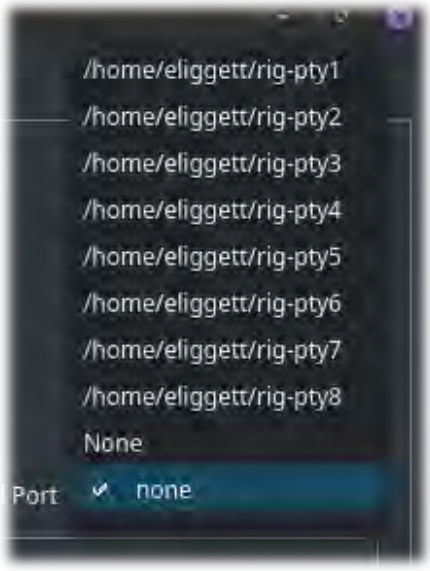
If you wish to connect other software such as digital mode interfaces or logging programs to wfview, this can be accomplished by creating a serial port pair. There are a number of Windows programs that can achieve this, here are a small selection:

- Eterlogic VSPE http://www.eterlogic.com/Products.VSPE.html
- Eltima Virtual Serial Port Driver https://www.eltima.com/products/vspdxp/
- com0com This is a FOSS driver but a Windows signed binary can be downloaded from https://pete.akeo.ie/2011/07/com0com-signed-drivers.html

Each solution operates in (broadly) the same way, you create a “pair” of virtual serial ports which are interconnected in software so any data sent to one port, automatically appears at the other. wfview is therefore configured with one “end” of the virtual serial port pair in the “Virtual Serial Port” combobox within the settings page. Your digital mode software/logging software is configured with the other “end” of the pair. Your software can then communicate with the rig as if it was directly connected.

Linux and Mac

On Linux and macOS, wfview provides a pseudo-terminal device out of the box, which may be used as a normal serial port by other applications. Previous versions created the device at `/tmp/rig` but newer releases of wfview allow the user to specify the device under the Settings tab, using the Virtual Serial Port combo menu:



wfview's pseudo-terminal selection

Users can either select one of the pre-defined names, or type within the box to define a custom name (for example, /home/user/ic-7300). Don't forget to press Save Settings after changing this box. To use the pseudo-terminal in a "client" program (such as fldigi, Ham Radio Deluxe, etc), you will need to either enter the device into the settings for the program, or, for programs that don't allow free-form entry, close the program and edit the program's settings files, manually specifying the same pseudo-terminal port name.

Audio Configuration

NOTE: This page only applies to radios connected via a network connection. USB radios do not need any additional work — just use the USB audio within the program you wish to use it in.

To be able to use wfview with digital modes like FT8, PSK etc. It is necessary to create “virtual audio” connections. These are effectively loopback sound cards that the transmit and receive audio are sent via wfview and allow your digital mode software (WSJT-X, Fldigi etc) to connect to it.

We have tested configurations in both Windows and Linux but similar functionality can be achieved in MacOS using a tool called SoundFlower https://github.com/mattingalls/Soundflower

The following are only required for users who are connected to their rig via LAN/WiFi. For USB connected rigs, you can simply use the USB audio devices provided by your rig.

Windows Virtual Audio Cable

For Microsoft Windows users, we have tested Virtual Audio Cable https://vac.muzychenko.net/en/index.htm which must be registered and costs $30 for non-commercial use.

A similar product called “VB-Cable” is a free alternative that can be downloaded from https://vb-audio.com/Cable/index.htm. Please note that it is not possible to transmit audio using the free VB-Cable. It may seem like it is, but it will not work because it uses the same virtual audio line for receive audio. You have been warned. The developer sells licenses for only $5, probably worth it.

In VAC You will need to create two “Cables” as per the image below:

The key settings are the Sample Rate (SR) which should be 48000 and BPS Range which should be 16bit. I only enabled Line in the “Connected Source Lines”.

Once you have this running, it is simply a case of selecting the Audio Output and Input devices in wfview and your other software. In wfview, Audio Output should be “Line 1 (Virtual Audio Cable” and Audio Input “Line 2 (Virtual Audio Cable)”. In your software, for example WSJT-X, under the Audio tab, Input wants to be “Line 1 (Virtual Audio Cable)” and Output “Line 2 (Virtual Audio Cable)”. Basically, the output of wfview connects to the input of WSJT-X and the output of WSJT-X connects to the input of wfview.

Loopback Audio Devices in Linux

With Linux, no additional software is required however it will need some configuration completing. The following was completed on Ubuntu 18.04 but should be similar for any Debian derived version of Linux.

The basic concept is similar to Windows, you create two loopback devices (virtual audio cables).

If it doesn’t exist already, create a file called /etc/rc.local which will contain various commands that must be run at startup.

```
sudo nano /etc/rc.local
```

Add the following into the file (if the file already exists then just add the sudo line.)

```
#!/bin/sh
#
sudo modprobe snd-aloop

exit 0
```

Newer versions of Ubuntu do not run rc.local by default so this must be enabled.

```
sudo systemctl enable rc-local.service
```

If you receive an error about the unit files having no [Install] section, you will need to do the following. If you don’t receive an error, skip to “**Configuration for snd-aloop device**”

```
sudo nano /etc/systemd/system/rc-local.service
```

Then add the following to this file:

```
[Unit]
 Description=/etc/rc.local Compatibility
 ConditionPathExists=/etc/rc.local

[Service]
 Type=forking
 ExecStart=/etc/rc.local start
 TimeoutSec=0
 StandardOutput=tty
 RemainAfterExit=yes
 SysVStartPriority=99

[Install]
 WantedBy=multi-user.target
```

You will also need to set the executable permission on the rc.local file and enable the service

```
sudo chmod +x /etc/rc.local
sudo systemctl enable rc-local.service
```

If all worked, you will not receive an error on either of these commands!

Configuration for snd-aloop device

The snd-aloop device also needs configuration, this is done in /etc/modprobe.d/virtual-sound.conf (the file can actually have any name as long as it has the .conf extension)

```
sudo nano /etc/modprobe.d/virtual-sound.conf
```

Add the following into this file. This creates 2 loopback devices (10 and 11) You can add more devices here if needed:

```
alias snd-card-0 snd-aloop
options snd-aloop enable=1,1 index=10,11
```

Now Reboot your computer!

Once rebooted, you should be able to run wfview and see a number of new device.

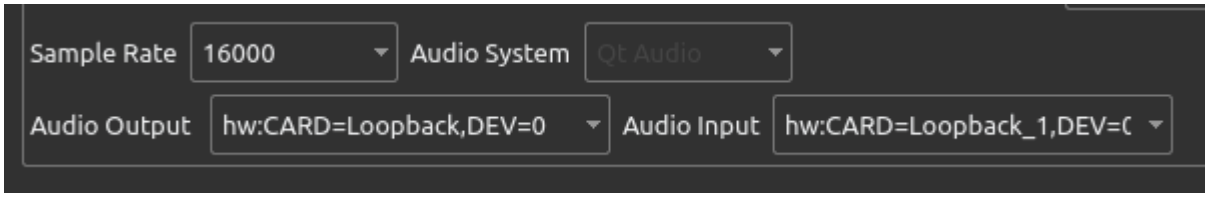
In the Audio Output combobox, select: Be careful, the wfview side of both loopback devices is “DEV=0”

```
hw:CARD=Loopback,DEV=0
```

Then in the Audio Input combobox, select:

```
hw:CARD=Loopback_1,DEV=0
```

Here is a screenshot of wfview properly configured for loopback audio:



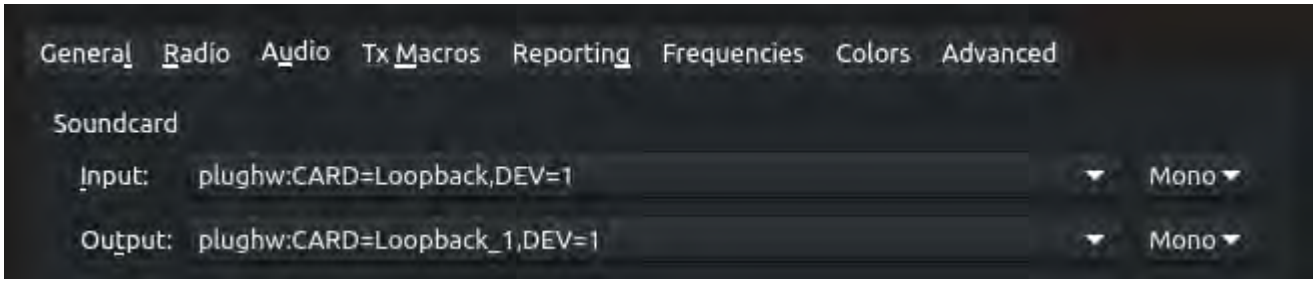
For most applications, select the following for Input: Be careful, the application side has “DEV=1”.

```
hw:CARD=Loopback,DEV=1
```

For Output:

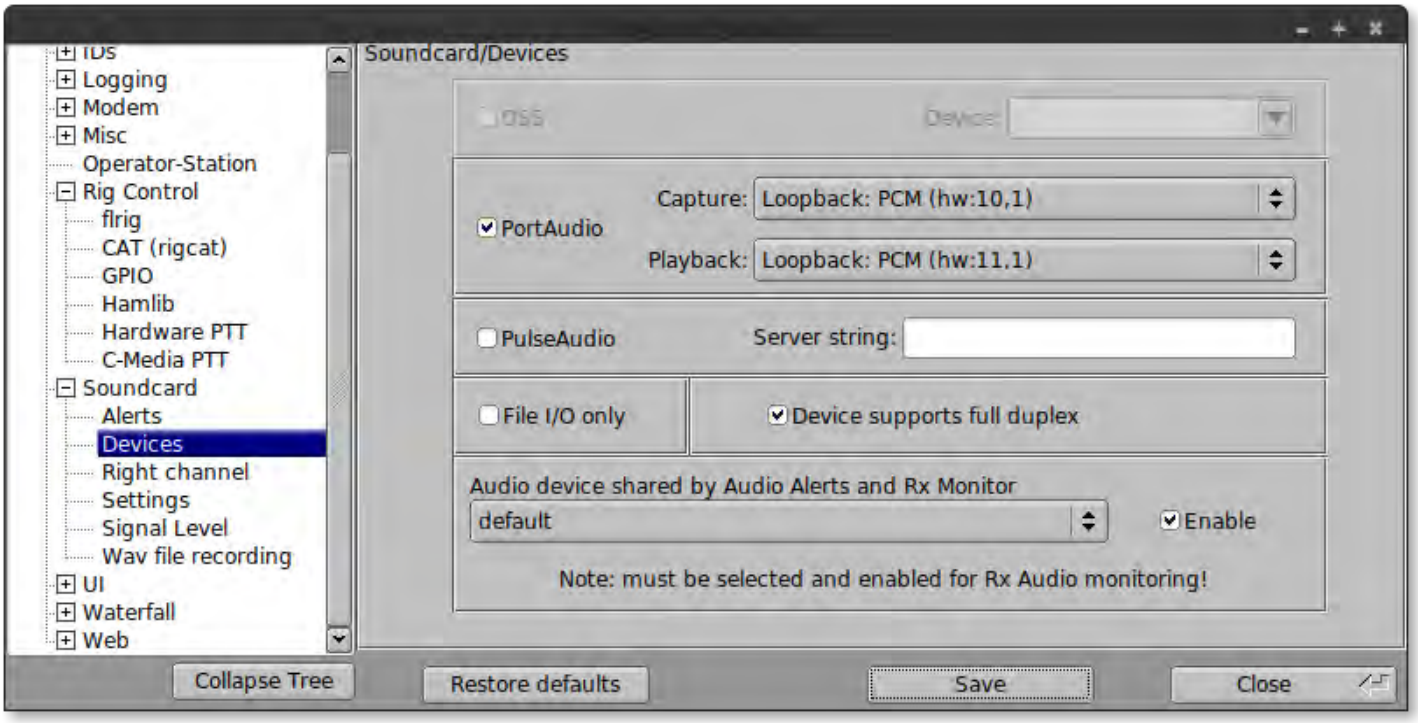
```
hw:CARD=Loopback_1,DEV=1
```

The WSJT-X Audio tab should look similar to this:

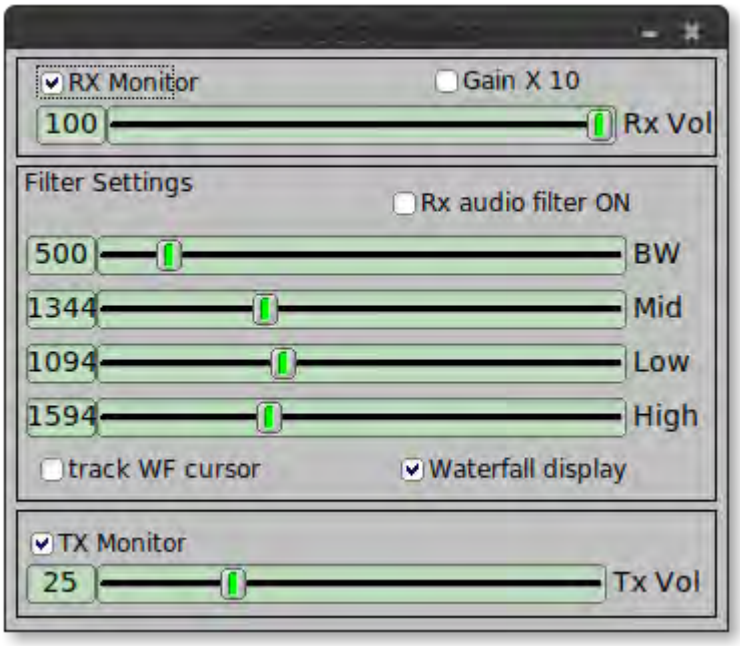


fldigi Audio configuration

After following the above guide for creating a loopback device, configure fldigi's audio as shown (there will be some differences depending upon the platform). If you enable the "RX Monitor" feature, then you can select your speakers and hear the receive audio from the radio *and* keep the audio going into fldigi for decoding! Be sure to select the correct instance of each loopback device. The ",1" part means you are on the second side of each loopback device (DEV=1 basically). Note that the capture device is connected to the wfview Audio Output device, and the Playback device is connected to the wfview Audio Input device.



If you'd like to hear the audio in the loopback, use the RX Audio Monitor inside fldigi. It can be accessed under the View menu and is called "Rx Audio Dialog". From there, you may enable the "RX Monitor" and "TX Monitor" (TX Monitor is available in versions 4.12.24 and greater). The TX Monitor is a great way to hear your CW sidetone if you do CW in fldigi.



Remote Operation (server)

wfview has a built-in radio server. This guide will help you configure the built-in server, which you can use to remotely access radios which lack a built-in server.

At this time, the built-in server functionality of wfview only supports Icom radios. We do plan to support other brands in the future.

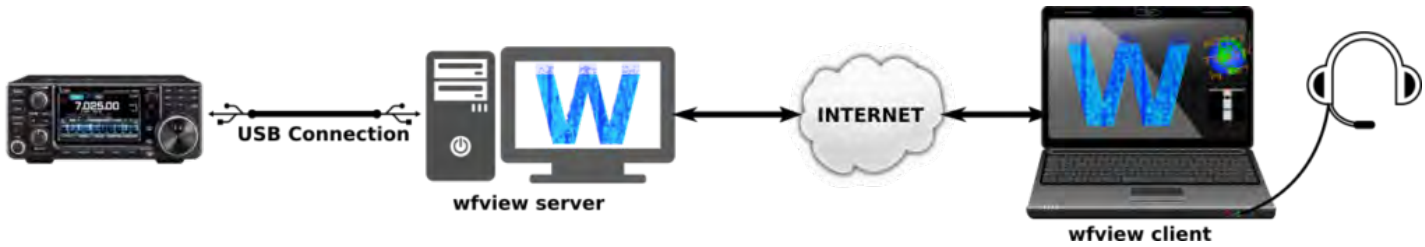
Scope

This article **only** applies to USB and Serial radios. For radios with built-in ethernet or wifi, simply connect wfview to the radio's ip address. See the Getting Started chapter. So, to be clear, this article and the entire "Radio Server" page are not needed if your radio already has a network connection.

The most commonly used radio with this setup is the IC-7300, but many older rigs will work great with the wfview built-in server, which handles getting full-duplex audio and control over the network.

wfview's built-in server handles the radio's controls as well as full-duplex audio. Simply run one copy of wfview on a server computer, which can be anything from a humble Raspberry Pi to a full desktop PC, and run a second copy of wfview on your remote computer or laptop. The controls are very responsive and audio latency is generally under 75ms (LAN) or under 200ms (WAN). The normal graphical wfview program you're familiar with has the server capability already. With this method, there is no "screen sharing" or "remote desktop" or "VNC". One copy of wfview connects over IP to the other copy of wfview.

Concept



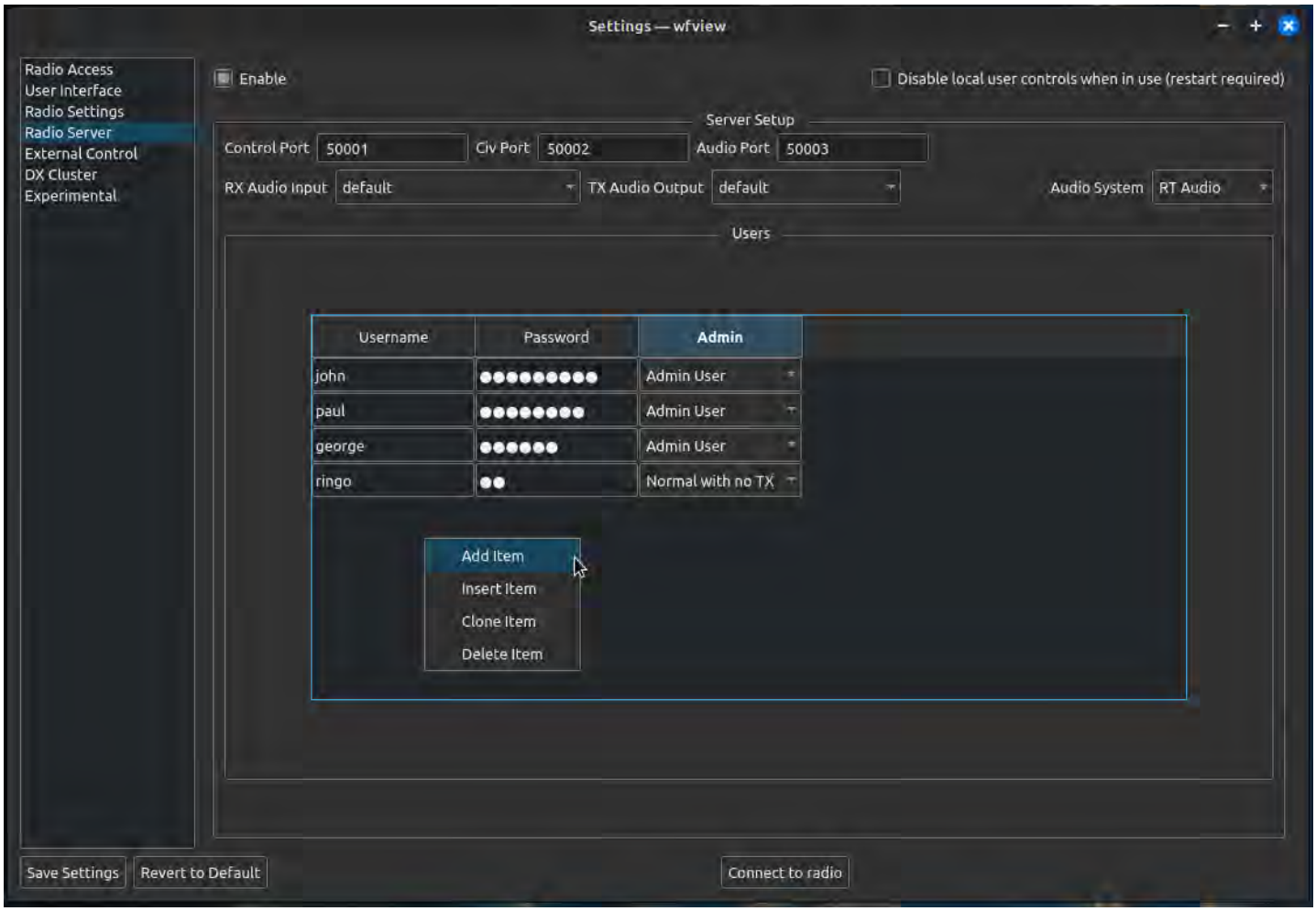
To remotely operate a radio, you will need to run one copy of wfview on a computer connected to the radio with a USB cable (or other appropriate cables). It can be a small, single-board computer like a Raspberry Pi, or a normal desktop computer. Currently, a desktop environment is required, but a command-line server version will eventually be made available. Once connected and configured, client computers running wfview can connect over the network (internet or LAN) to the computer hosting the radio.

Server Setup

For the server computer, begin by installing wfview. Connect the radio, and then launch wfview. Don't forget to disable sleep — you want the server to always be available for remote connection.

Follow the Getting Started guide for serial port radios. Make sure, on the radio, that CI-V Transceive is turned on.

Once wfview is running and properly controlling the rig, head over to the Settings tab of wfview, and click on the “Radio Server” page.



The standard ports, as shown in the screenshot, are 50001, 50002, and 50003. You should use these ports unless you have a reason to use something else.

Select the radio’s receive audio device under “RX Audio Input”, and select the radio’s transmit audio device under “TX Audio Output”. If these drop-down menus are grayed out, it means that wfview has not connected to the radio yet; head over to Radio Access and make sure the Serial port has been set up correctly and then press “Connect to Radio”.

Under Linux, these two names (or similar names) should work for the server audio on the IC-7300:

- Audio Input: `alsa_input.usb-Burr-Brown_from_TI_USB_Audio_CODEC-00.analog-stereo`
- Audio Output: `alsa_output.usb-Burr-Brown_from_TI_USB_Audio_CODEC-00.analog-stereo`

Generally speaking, on Windows, the audio device will say “USB Audio CODEC” in the name and often also “Burr-Brown”. Remember, you are looking for a USB device. If you see many USB Audio CODEC devices, try unplugging other USB audio devices and any other radios, and then re-launch wfview and see if you see fewer, it may help.

The Audio System should generally be set to “Qt Audio” unless there are problems with this audio system. Note that you cannot change the audio system once a client is connected; therefore, it is necessary to make sure all clients are disconnected prior to changing the audio system.

Next, define a username and password for a “Full User” account. As of wfview 1.5, user privileges are not enforced, but it is something we may do in the near future. Since the passwords are not well-encrypted, do not use a sensitive password.

If you are satisfied with the settings, press “Save Settings” and then re-check the “Enable” checkbox to turn on the

server. You should see “udp.server: starting udp server” in the Log window. You should also see a message about the Audio Input and Audio Output devices; make sure they match your settings.

Check the “Disable local controls” checkbox to lower unnecessary radio polling from the server and create a more responsive experience.

At this point, server setup is complete.

Client Setup

setting up a second computer to connect to your wfview server is simple. Launch wfview on the second computer, and follow the Getting Started guide. For the radio connection type, specify “Network”. Supply the ip address of your wfview server computer for the Hostname, along with the same control port (50001 by default), username, and password.

For the TX and RX Codec, “Opus 1ch” is the recommended choice, as it uses about 4 to 10 times less bandwidth than the 16-bit PCM formats. Note that Opus will only work if you are running a wfview server; it is a special benefit of this software. For VPN users, you will likely need to use an 8-bit format such as “8-Bit uLaw” due to typical VPN packet mangling. The uLaw codec uses logarithmic bit spacing and provides surprisingly good quality.

The audio devices should be chosen for your client computer, typically a headset or speakers. Use the Qt Audio system unless you have problems with it. If you need to loop audio into a second program such as wsjt-x or fldigi, read our audio configuration page.

Most of the details for these controls and options are covered in the Settings manual page.

Radio Settings

Your radio needs to be told to source audio from the USB or ACC connector, as applicable to your setup, so head over to the Radio Settings page of the Settings Tab, and select the audio interface that the radio should use for transmit audio. IC-7300 users should select “USB”. Older radios, lacking a USB port, will likely be connected to the computer using an ACC adapter. Some older radios do not have a command for changing the input, so the input must be changed using buttons on the radio.

See this page for details.

Router Setup

To access wfview from outside your home network, you will need to port forward UDP traffic on the three ports mentioned above. Important: DO NOT map the ports to any other port numbers. They must be the same port on the router and the computer running wfview server. This is because the port numbers are a part of the communication protocol itself.

Troubleshooting

Running a server is an inherently tricky task, and it is even more difficult when streaming audio is involved! You will need to be familiar with concepts such as TCP/IP and be comfortable with topics like finding your ip address.

Audio

There are three places to look: Radio audio settings (input selection, AF gain, Mic gain, etc), Server (TX and RX audio source/sink selection), and Client (mic and speaker connection). The best strategy is to begin with a simple USB headset on the client. You can enable TxAudio as a type of secondary meter and you should be able to see the raw headset audio on the meter even when receiving. If you don’t see it, adjust the gain for the headset using

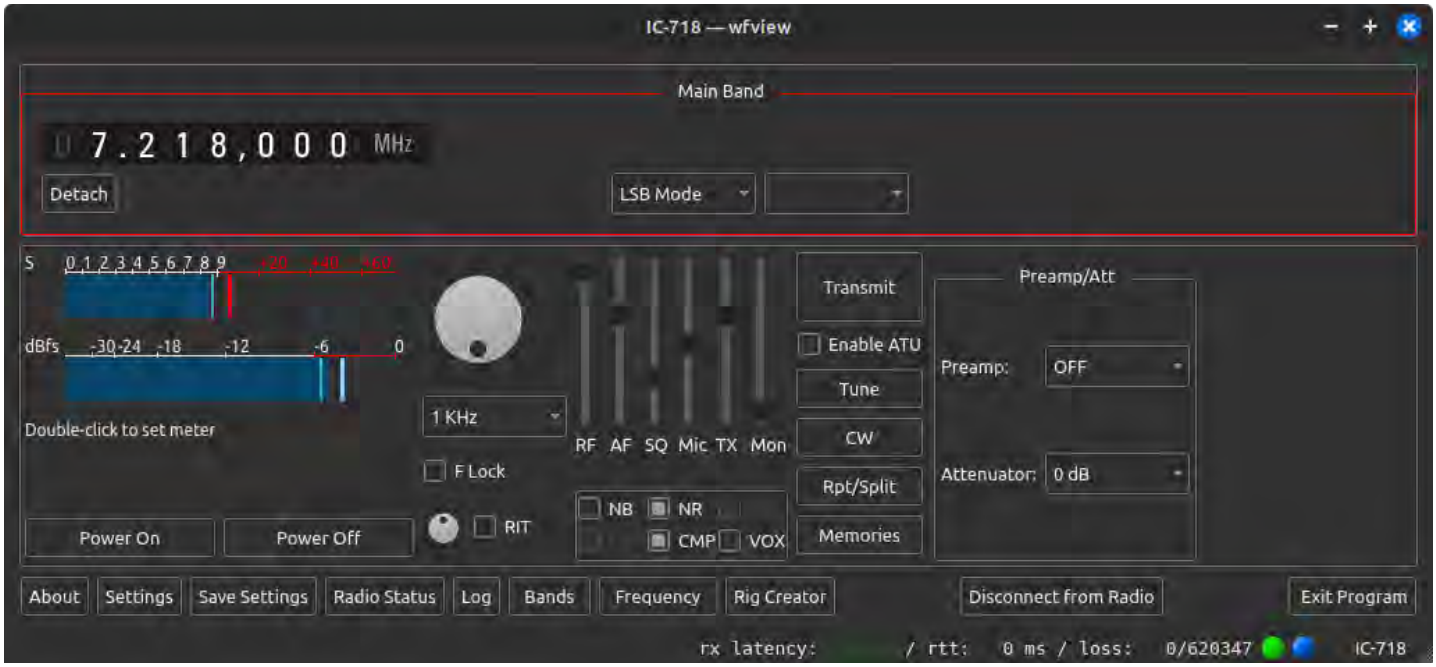
your operating system's audio settings control panel. On the server, make sure the right audio devices are not only selected in the GUI but also being ultimately used by the server. The server audio modules will load upon the first client connection, and you should see this in the log (press "Log"). Check what device is being used. Radio devices are generally easy to spot. Linux users may try this script to list serial and audio devices.

Connecting

First, make the connection between two computers on the same LAN. This will eliminate router configuration and port forwarding from your initial attempts.

Make sure you can ping from one computer to the other. Verify the IP address of the server. Check the log on the server (press the Log button). See if you can see the remote login activity.

Here is a screenshot of an IC-718 accessed via the internet:



Inexpensive Remote Server

This page tells how to set up a simple remote server so that radios without network access can be used remotely. A popular radio for this purpose is the IC-7300, but other radios from Icom will work just as well.

For this how-to, I will be using the $35 Inovato Quadra. A Raspberry Pi 3, Orange Pi, or any modern SBC ("single-board computer") should work fine as well. The Quadra comes with built-in storage and a case, so you do not need to buy those things. The included power supply is not appropriate for ham radio as it makes hash all throughout the HF bands (ask me how I know…). However, in testing, I observed that the maximum current on the 5V supply was about 812mA, briefly during booting, and that at idle the current is only 235mA (with the desktop open and wfview running). There are many low-noise supplies which can support this SBC at these low currents. A 7805 regulator may even be sufficient (with heatsink and capacitors of course).

You should start by reading our server setup page to get the idea of what we're about to do. The process is very similar, but we will be using a small and dedicated computer instead of a normal desktop. You will need to be comfortable editing files and running commands in the terminal. Familiarity with network setup is a must.

You will need cables to connect your radio using the USB port. Many radios, like the IC-7300, have a built-in USB port, and for that radio you just need a USB cable. For older radios, you will need something like a RigBlaster or XGG Comms adapter. Adapters with transformer isolation are recommended.

Operating System Configuration

The Quadra comes with armbian, which is a Linux operating system based on Debian Bullseye. The Raspberry Pi also runs a debian-based distribution. Any Debian-like system should be fine.

Start with the usual. Change the default password (change it for *both* the root user and the quadra user — make sure to do both!) and make sure the network access is solid. I used ethernet for my setup as it is definitely ideal for streaming service such as this. Also, on some platforms the WiFi over USB causes issues with how we stream over USB. Therefore, use Ethernet if at all possible. Optionally run a system update and reboot. ssh is enabled out of the box on port 22.

With armbian, I had to remove `brltty` to stop it from stealing the serial port of the radio. If you don't do this, you won't see the serial CI-V device! Some systems may also require removal of the ModemManager.

```
sudo apt-get remove brltty
sudo apt-get remove modemmanager
```

Reboot once that is complete. I also added x11vnc and sshd so that I could easily remotely view the display. You can do this entire tutorial using ssh and ssh with x11vnc if you wish, or you can sit at the SBC and do it that way. Either way though, this utility is quite handy.

Install sshd:

```
sudo apt-get install openssh-server
```

And install x11vnc:

```
sudo apt-get install x11vnc
```

You can ssh to the box and run x11vnc, and then connect over VNC to see the local display. If the machine is stuck at the login manager, then you need to run it as sudo with the "`-auth guess`" argument:

```
sudo x11vnc -auth guess
```

Automatic Graphical Login

I configured the machine to automatically log in the primary user. If you switch later to the non-graphical wfserver version, you won't need to do this, but for now it's quite simple and effective.

I followed this guide to configure the machine for automatic login. Here's the change to make (replace "quadra" with whatever username you are using, perhaps "pi" for raspberry pi users):

Add the following two lines to `/etc/lightdm/lightdm.conf.d/11-armbian.conf`:

```
autologin-user=quadra
autologin-user-timeout=0
```

Note, on the Raspberry Pi, this file will have a different name. If your linux distro does not use lightdm, consider either switching to lightdm or googling how to enable automatic login for your graphical login manager.

Installing wfview

Installing wfview is simple. You can use our installer script to automate the entire process, including installing dependencies, compiling, and installing the binary. Follow the directions carefully, and watch the video if you have questions. One thing you will need to change (before you run the script) in the script is to cause the build process to use only one core. This prevents smaller computers like the Raspberry Pi and Quadra from overheating or over-currenting. You can actually skip this step, however, the computer will reach 64C, so maybe it's worth doing. Edit the script using your favorite text editor, and change this line (approximately line 159):

```
time make -j2
```

to this:

```
time make
```

Then run the script. The build process took about 15 minutes on my Quadra. Once installed, you can find wfview under the Applications menu in the “Internet” category.

Configure wfview

This part is simple. Connect the radio. Open wfview. Configure the access to the radio under Radio Access. Be sure to select the serial port (it should be /dev/ttyUSB0). Enable “Use RTS for PTT” if your radio needs it (many older ones do. The 7300 does not.) Configure the baud rate. Turn on CI-V Transceive on the radio, and make a note of the radio’s CI-V address. Newer radios, like the IC-7100, IC-718, etc, respond to our CI-V bus query and identify themselves. Older radios will need the address manually specified. Press “Connect” once you have the settings in. Be sure to press “Save Settings” once you get it configured successfully.

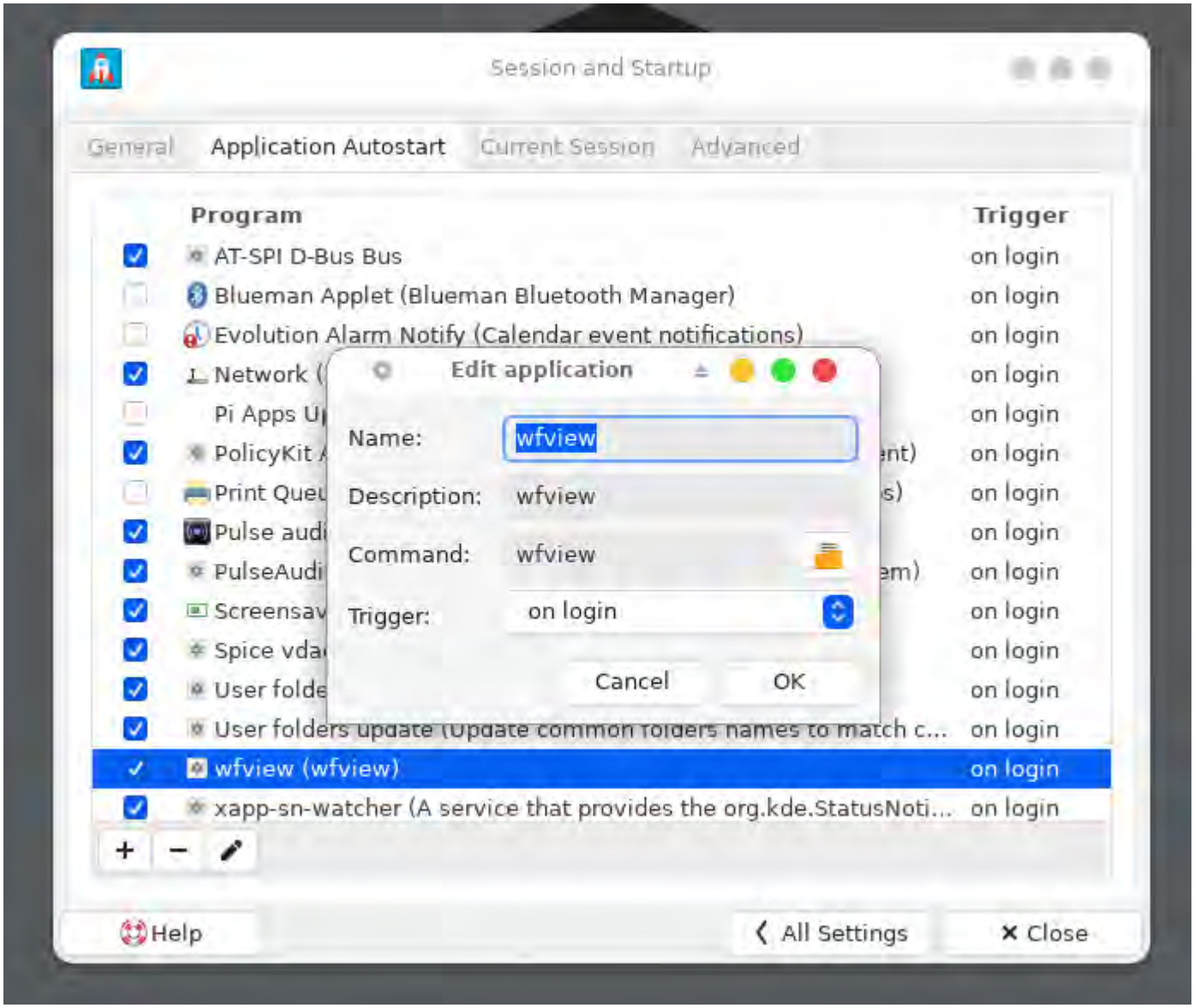
Under “Radio Server”, check the box to turn on the radio server, and then press “Disconnect” to disconnect from the radio. Define a user account. Select “Qt Audio” as the audio system. Select your RX and TX audio sources. Make a note of the three ports listed above (by default, 50001, 50002, and 50003).

Under “User Interface”, set the polling to manual and 250ms interval. This will keep the server side from polling the radio to death. Since the server and client do not sync their polling, sometimes you can have issues with older radios that have slower internal CPUs. The client side polling can remain automatic (20-75ms).

Press “Save Settings”, and then press “Connect to radio”. Check the Log to make sure the audio devices were used properly.

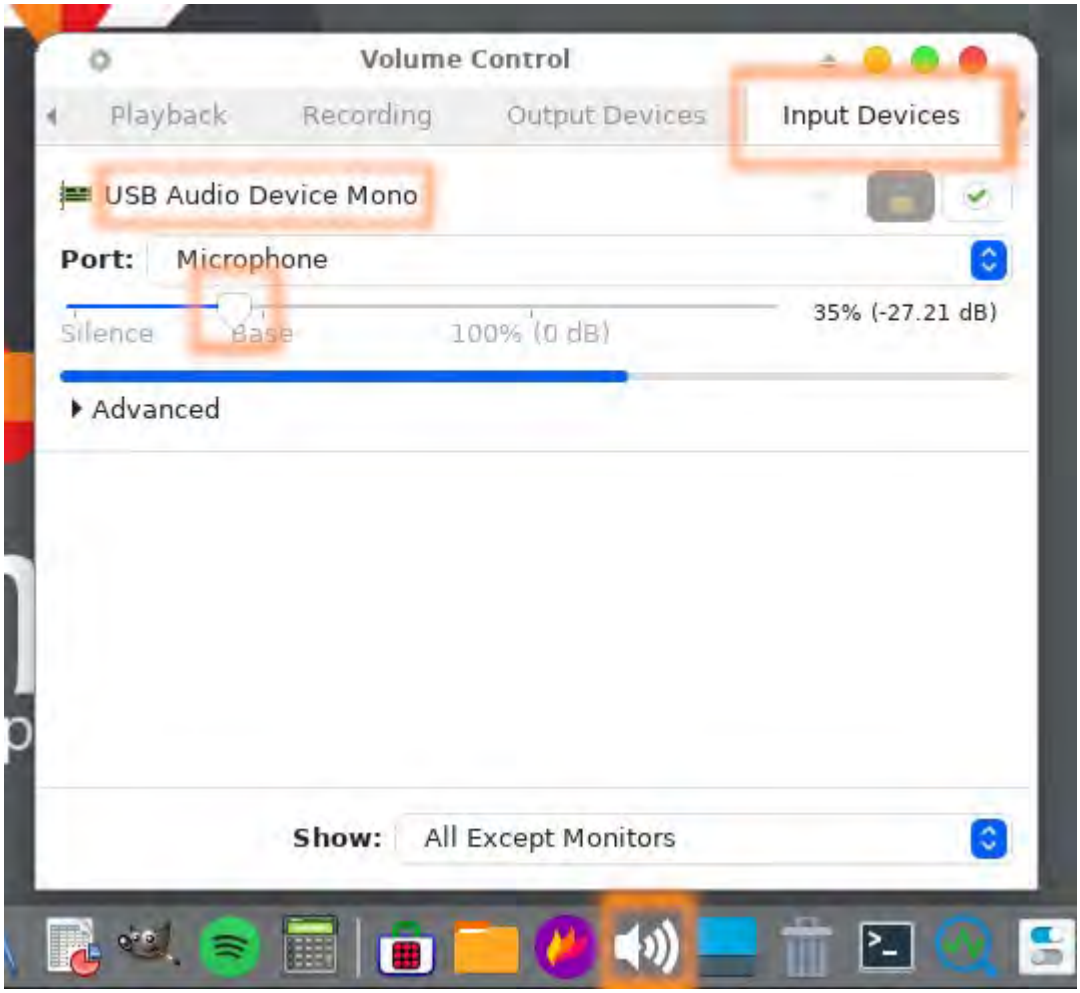
Additional System Configuration

Open the “Settings Manager” of xfce. It looks like a set of switches in the dock widget at the bottom of the screen. Scroll down to “Session and Startup” (look for the rocket ship icon). Then go under “Application Autostart” and add wfview as a startup application. You can also uncheck Bluetooth and Evolution Alarm Notifier if you don’t need those things. For the Raspberry Pi, follow this guide for starting a graphical program on login.



Tune the radio to an active station. Connect to your wfview server using a client-side wfview instance. On the client, set the secondary meter to “TxRxAudio” (Settings, User Interface, Secondary Meter).

Set the audio level from the radio’s receiver into the computer by using the Volume widget in the dock on the server. Click the speaker icon and then select “Audio mixer…”. Now click the tab “Input Devices”. If you do not click this tab, then you will be scaling the audio and not adjusting the analog level at the converter. So don’t forget! Under input devices, you should see something like “USB Audio Device”. Use the slider to adjust the level until there isn’t any distortion. I like to see levels peaking at -6dB fs. I don’t know how to configure this to save the gain setting, sometimes it seems to on its own and other times it doesn’t.



Final Thoughts

On my system, wfview uses less than 1% when there isn't an active connection. This is with my IC-718, mind you. If you are doing this with a 7300, it will be higher due to the server drawing the spectrum. It's worth a shot to configure wfserver once you have a working configuration file. wfserver does not have any tool to help you configure anything, it's purely command-line and without any user interaction. But, it understands the same configuration file.

When a user connects, I see that the total CPU usage jumps to 11% (and I'm using the opus codec, which should be higher than the PCM and uLaw options). The CPU temperature is generally around 46C, and I have not added a fan or anything to help with the thermal environment. I am seeing that the CPU idles at 480 MHz, and it will jump to 1704 MHz and then down to 720 MHz when a client connects to wfview. You can check using `sudo armbianmonitor -m`.

On storage, the quadra arrived with 8.8 GB available and it's down to 8.7 GB available with wfview installed. This is nice, the default image included a lot of the usual tools and libraries and still has plenty of space. htop shows memory usage is "490M/1.94G", and wfview us using 4.6% of that.

Overall what this means is that the computer isn't overly taxed. It can handle this task easily.

I remotely accessed my IC-718 using the Opus codec over my cellular hot spot from the park recently. Audio was smooth and control was consistently responsive. I did need to run the latency up a bit higher, around 250ms seemed to be good.

View
Band
Frequency
Settings
14.240000 MHz
Mode:
USB
Data
Receive Filter
1
500 Hz
F Lock
RIT
Show More
Power On
Power Off
RF
AF
SQ
TX
Mic
Transmit
Repeater
Preamp/Att
Preamp:
Disabled
Attenuator:
0dB
Antenna:
About
Save Settings
Radio Status
Log
Disconnect from Radio
Exit Program
rx latency: 240 ms / rtt: 1 ms / loss: 0/700
IC-718

wfview

Open Source interface for Icom and Kenwood transceivers

How to send a logfile

This page will tell you how to send your logfile to the wfview development team.

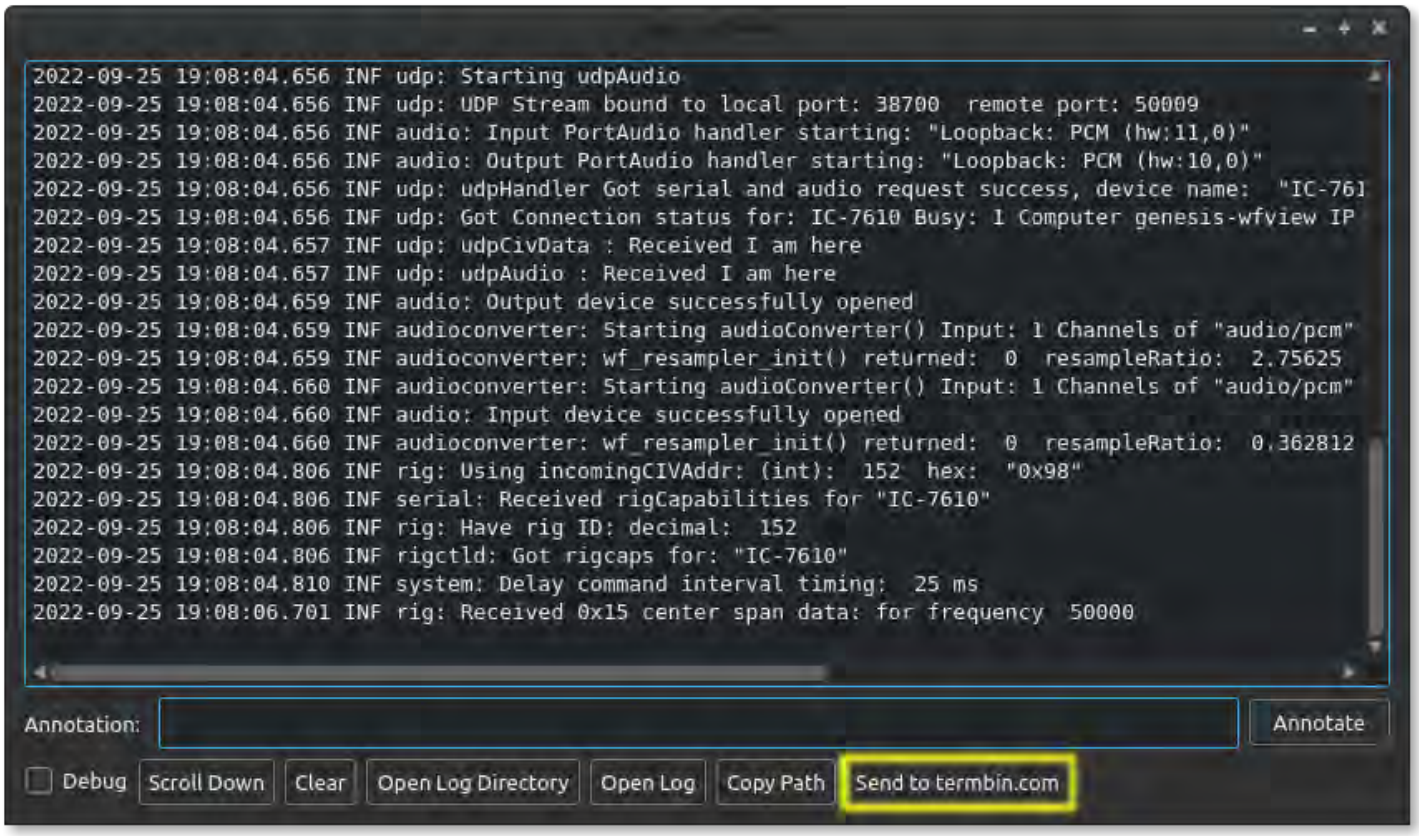
The steps are simple, but they must be followed carefully.

Begin by closing wfview and then re-opening it and replicating the issue as promptly as possible. This important so that you do not have a log file that is overly long or contains too much irrelevant information.

Once the issue has occurred, press the “Log” button in wfview.

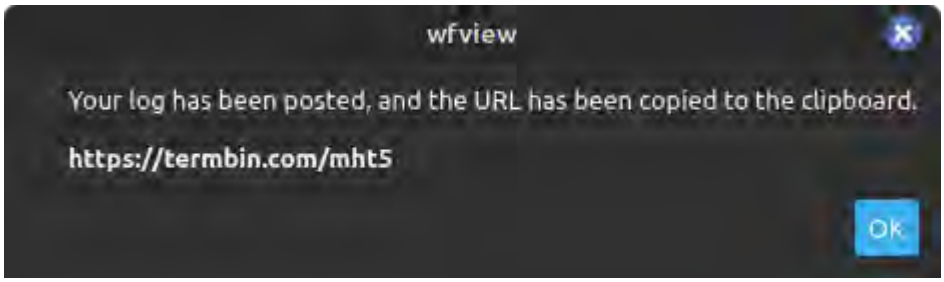
This will cause the log window to open up. The log contains lots of useful information for the developers, but you may want to have a look through it to see if you see any issues you can address yourself.

Here is an example log window:



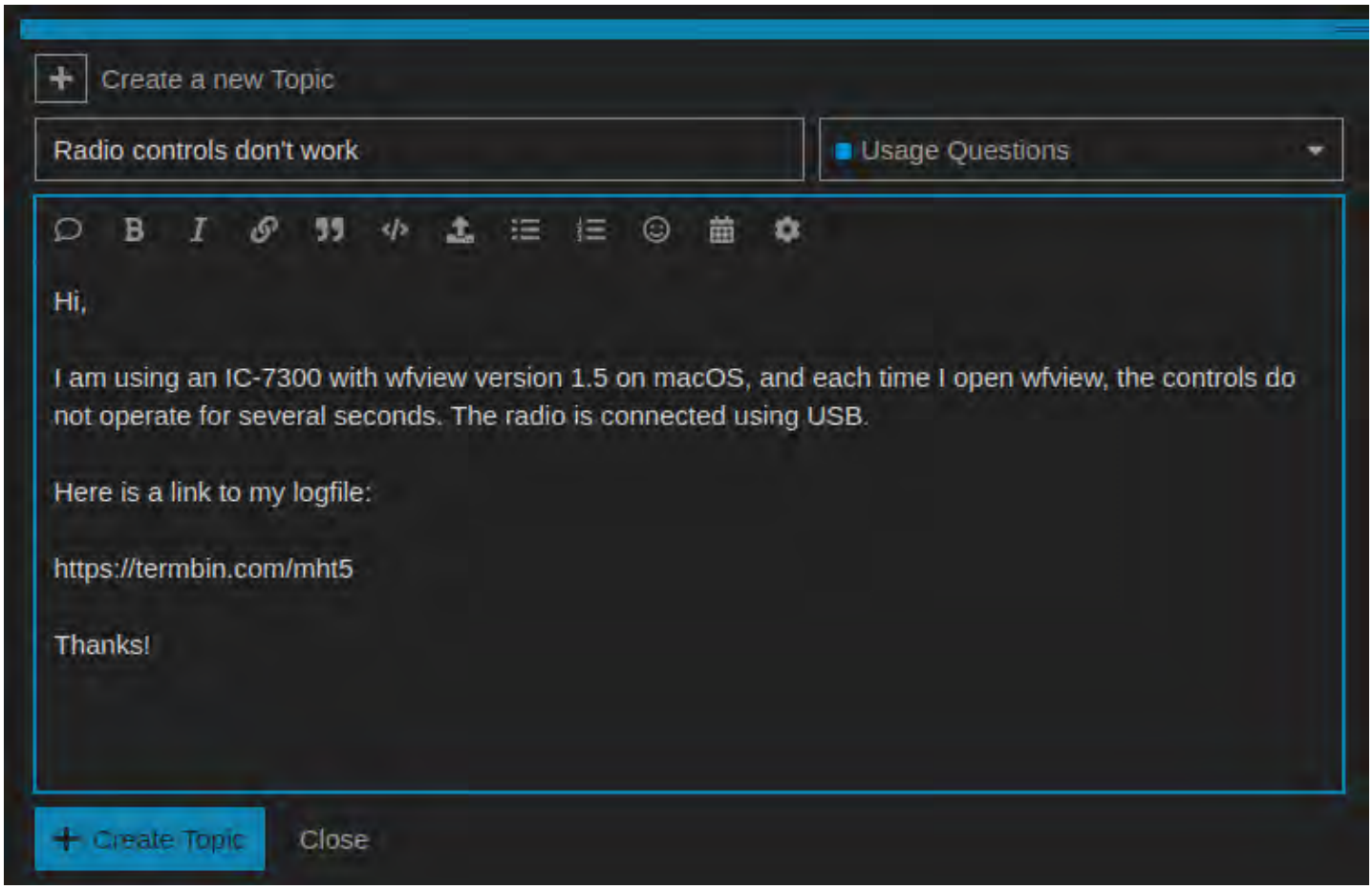
Review the information and make sure you are comfortable sharing it. There isn't much information that is personally-identifiable, but you should check just to be sure. If you want to annotate the log with a message, you may, simply by typing a message into the log window and then pressing "Annotate". For example, you might type in "This is wfview running on my PC with my IC-7300." This is very helpful if you are going to send us several log files, for example, from a server and client set or for several radios.

Once ready, **press** "Send to termbin.com". This will cause the logfile to be sent to termbin. Note that *we do not operate termbin.com*. Termbin is a service that many other programs use. Thus, please understand, we cannot see your log until you share the URL from the pop-up with us. The pop-up will appear about a second or two after you send the log file.



The resulting URL is already copied to your clipboard. What you need to do is then **send us the URL by pasting it** into a message on our support forum. Once you send us that URL, we can all look at it together, which is very convenient.

For example, you might post it like this:



APRS decode from IC-9700 over Network

Objective

The objective is to be able to decode standard 1200 baud AFSK APRS packets using the network access from the IC-9700 and wfview's audio. This how-to focuses on Linux, however similar methods will work on other operating systems.

Setup

First, follow the Audio Setup guide for linux, creating two loopback devices, each with two entry points. Follow the document carefully. After you reboot, launch wfview and select the audio loopback devices. Make sure to carefully select the audio ports *per the linked page*. If you select the wrong ports, you won't have any audio. Select the ports and then Save Settings and close wfview. Optionally, use a program like Audacity to verify the audio.

Check that you can see the loopback devices via this command:

```
eliggett@genesis:~$ arecord -l
**** List of CAPTURE Hardware Devices ****
card 1: Generic_1 [HD-Audio Generic], device 0: ALC257 Analog [ALC257 Analog]
Subdevices: 1/1
Subdevice #0: subdevice #0
card 10: Loopback [Loopback], device 0: Loopback PCM [Loopback PCM]
Subdevices: 8/8
Subdevice #0: subdevice #0
Subdevice #1: subdevice #1
Subdevice #2: subdevice #2
Subdevice #3: subdevice #3
Subdevice #4: subdevice #4
Subdevice #5: subdevice #5
Subdevice #6: subdevice #6
Subdevice #7: subdevice #7
card 10: Loopback [Loopback], device 1: Loopback PCM [Loopback PCM]
Subdevices: 7/8
Subdevice #0: subdevice #0
Subdevice #1: subdevice #1
Subdevice #2: subdevice #2
Subdevice #3: subdevice #3
Subdevice #4: subdevice #4
Subdevice #5: subdevice #5
Subdevice #6: subdevice #6
Subdevice #7: subdevice #7
card 11: Loopback_1 [Loopback], device 0: Loopback PCM [Loopback PCM]
Subdevices: 7/8
Subdevice #0: subdevice #0
```

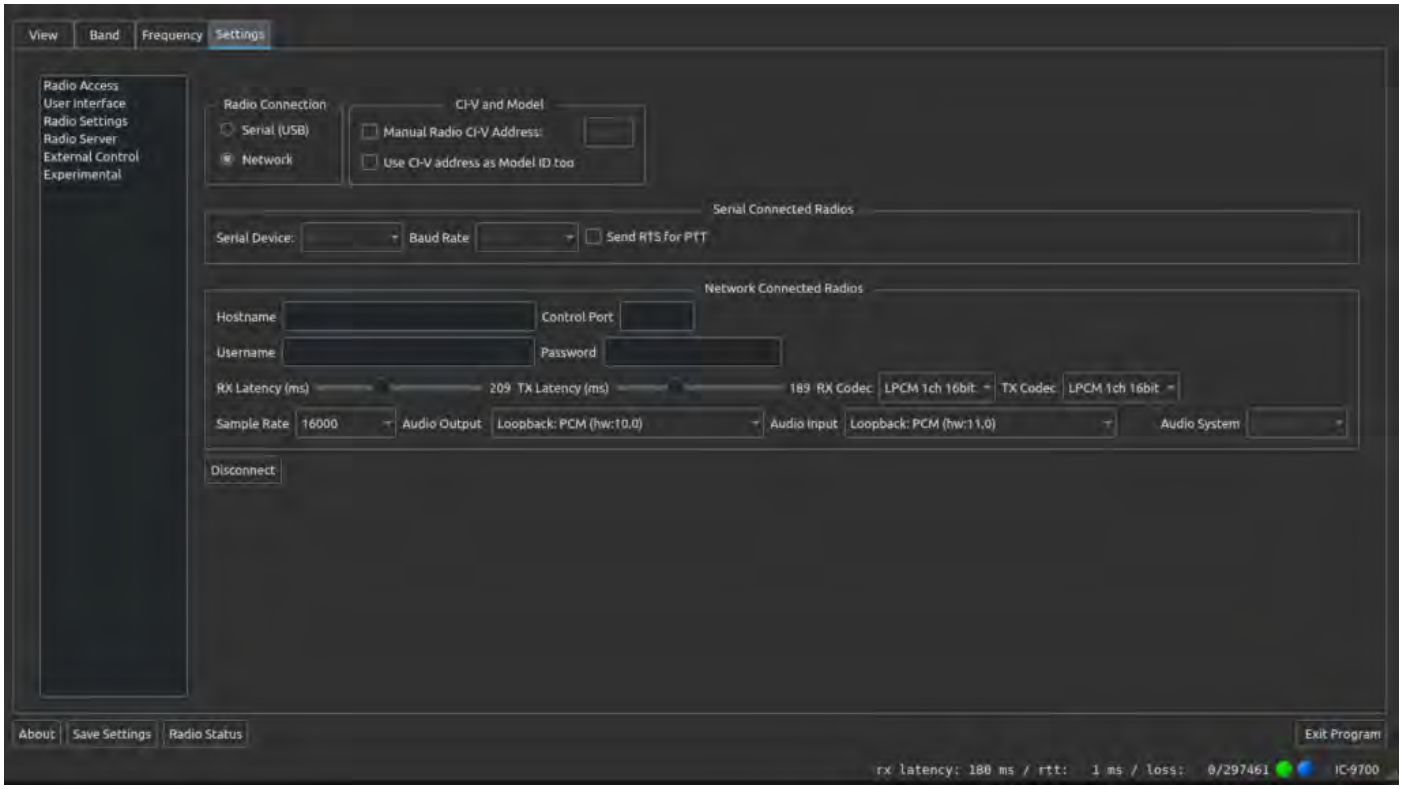
```
Subdevice #1: subdevice #1
Subdevice #2: subdevice #2
Subdevice #3: subdevice #3
Subdevice #4: subdevice #4
Subdevice #5: subdevice #5
Subdevice #6: subdevice #6
Subdevice #7: subdevice #7
card 11: Loopback_1 [Loopback], device 1: Loopback PCM [Loopback PCM]
Subdevices: 8/8
Subdevice #0: subdevice #0
Subdevice #1: subdevice #1
Subdevice #2: subdevice #2
Subdevice #3: subdevice #3
Subdevice #4: subdevice #4
Subdevice #5: subdevice #5
Subdevice #6: subdevice #6
Subdevice #7: subdevice #7
eliggett@genesis:~$
```

You will need to install either direwolf or multimon-ng, or both, as well as sox:

```
$sudo apt-get install direwolf multimon-ng sox
```

Note that direwolf will provide a cleaner output and also allow you to contribute what you receive to the igate server, which is pretty cool. Don't be intimidated by the direwolf setup; for our purpose the setup is merely 7 lines, detailed below.

Minus the obvious hostname, username, and password, here is how my wfview Radio Access setup looks. Be sure to set the CODEC to LPCM 16bit. The sample rate should be at least 16k (you may have more success setting this to 48k with direwolf and 22050 for multimon-ng, but I did not experiment. The sox commands below will set the loopback to 48k in both cases). To be clear, the Audio Output should be "Loopback: PCM (hw:10,0)" and the Audio Input should be "Loopback: PCM (hw:11,0)". This is per the Audio Configuration linked above.



wfview audio setup for APRS

Setup the APRS program:

Note: It is far easier to start the APRS chain *prior* to opening wfview. This is because the APRS programs (and sox) are less-flexible about sample rates and audio formats in general, so we start sox and the APRS programs first, so as to define the audio format which wfview will then follow.

For direwolf, you need to follow the most basic setup first. I made the following direwolf config file, which I called “direwolf.conf”:

```
ADEVICE null null
CHANNEL 0
MYCALL W6EL
IGSERVER noam.aprs2.net
IGLOGIN W6EL PASSWORDHERE
MODEM 1200

PBEACON sendto=IG delay=0:40 every=20 symbol=R& alt=475 lat=34.23386N long=118.25866W comment="iG
ATE AAPRS rx-only"
```

You should use this link to get your APRS password. I don’t really understand how this works but in any case, it worked for me.

For multimon-ng, there isn’t any setup.

Starting the APRS programs:

For direwolf, use this command (copy-paste carefully! Be sure to enter the path for the direwolf config file correctly.):

```
sox -q -t alsa hw:10,1,0 -e float -b 32 -r 48000 -t raw -b 16 -c 1 -e signed-integer -r 48000 - |
 direwolf -c ./direwolf.conf -r 48000 -D 1 -t 0 -
```

For multimon-ng:

```
sox -q -t alsa hw:10,1,0 -e float -b 32 -r 48000 -t raw -b 16 -c 1 -e signed-integer -r 22050 - |
 multimon-ng -t raw -a AFSK1200 -A /dev/stdin
```

Note, if you wish, you can start sox without the “-q” flag and without the pipe to the APRS program. In this mode, you can see the audio level in a small bar-graph. There won’t be any decoding, but it is fine for testing the audio pipe.

I will note here that it took me about two hours to come up with those two commands. The sample rate for multimon-ng must be 22050, and multimon-ng doesn’t seem to provide any way to select another audio device. Direwolf is flexible with sample rates. sox must be setup with the loopback audio format matching any format that has already been applied to the loopback device, and it won’t tell you what that is! This is why I start sox first, before I open wfview, so that sox may *define* the audio format, and wfview will *follow* whatever format is applied.

Monitoring APRS packets

With one of the above commands (sox + direwolf or sox + multimon-ng), launch wfview and confirm the loopback devices are selected. If necessary, press Disconnect, select the loopback devices per the audio configuration how-to, and then press Save Settings and hit Connect. Now tune wfview in to your local APRS frequency. 144.39 FM is used in SoCal, for example. Turn up the AF Gain slider all the way (if the signal is too loud or distorted, turning this

down will just make it quiet-yet-distorted, so adjust the LAN RX level at the radio if needed). I recommend setting the squelch so that it opens with the slightest signal, or all the way down if desired. Use the “RxAudio” meter to monitor the audio coming from the radio.

To visualize the station locations, you might wish to try xastir or YAAC. I have not tried either yet.

```
APRS: WC6Q>APWA01,N6EX-4*:@183156h3412.06N/11834.80Wk090/010 /A=000650 Fore 103% (30/25) WC6Q.com
APRS: KK6OZ-15>APT311,N6EX-4*:>TinyTrak3 v1.1
APRS: WC6Q>APWA01,N6EX-4*:@183216h3412.06N/11834.80Wk/A=000649 STOP 102%(18) (24/20) WC6Q.com
APRS: WA6IAJ-1>APMI06,N6EX-4*:T#130,187,077,012,098,159,00000000
APRS: KC6UXQ-5>APDR16,N6EX-4*:=3346.56N\11749.96Wk071/048/A=000131 APRSDroid
APRS: KF6NMZ-1>APTT4,N6EX-4*:T#310,127,154,255,041,013,00101011
APRS: KC6UXQ-5>APDR16,N6EX-4*:=3346.73N\11749.84Wk356/052/A=000121 APRSDroid
APRS: KE6VRK-9>SSUQTV,N6EX-4*:`.1cl"z>/`"46}_1
APRS: KE6VRK-9>SSUQTV,W6SCE-10*:`.1cl"z>/`"46}_1
APRS: KC6OYN>APOTW1,N6EX-4*:!3429.67N/11942.76W_132/004g005t092P000h30b10207OTW1
APRS: W6MAF>APTW01,OAKHLS*,WIDE1*,KELLER*,N6EX-4*: 09061132c195s005g005t098r000p000P000h10b10158tU2k
APRS: W6MAF>APTW01,OAKHLS*,WIDE1*,KELLER*,N6EX-4*:!3422.10N/11724.50W_PHG6150/A=003980/Oak Hills WXTrak
APRS: KM6FDI-5>APMI0A,KELLER*,WIDE1*,N6EX-4*:!3718.37N/12153.11W>APRS Voyager U=14.6V, T=??.?C.
APRS: WB0AL-9>ST1QUX,W6SCE-10*,N6EX-4*:`.S+lTnu/`"6,}Over the road truck driver at Panther Premium_(
APRS: HEAPS>APN383,N6EX-4*:!3414.09N/11708.44W_PHG4390/Heaps Peak/A=006420/k6tvi@cal-net.org
APRS: WA6MHA-11>APOTW1,W6SCE-10*,N6EX-4*:!3410.50N/11828.90W_092/003g009t101P000h26b10125V138OTW1
APRS: KF6NMZ-1>APTT4,N6EX-4*:/183404h/=9r70VEN>"!_/TinyTrak4 Alpha
APRS: KM6FDI-5>APMI0A,KELLER*,WIDE1*,N6EX-4*:T#041,199,000,000,000,000,00000000
APRS: W6PWM>S4RYQS,W6SCE-10*,N6EX-4*:`.B.l K\]"92}I Monitor PAPA1&DARN10=
APRS: NQ6X-10>S4QRRX,W6SCE-10*,N6EX-4*:`.O)!|6>/]"6l}=
APRS: KM6AHX-12>APOTU0,N6EX-4*:> CHG STN ON
APRS: KM6AHX-12>APOTU0,N6EX-4*:!3411.20N/11813.02W_ 14.2V
APRS: KM6AHX-12>APOTU0,N6EX-4*:!3411.20N/11813.02W_057/001g...t094p000h43T2WX
APRS: WB0AL-9>ST1QVQ,W6SCE-10*,N6EX-4*:`.S=mr8u/`"6.}Over the road truck driver at Panther Premium_(
APRS: KM6FDI-5>3T1QUU,KELLER*,WIDE1*,N6EX-4*:`-1E3y\"9T}In conference KM6FDI-9
APRS: KM6FDI-5>3T1QUU,KELLER*,WIDE1*,W6SCE-10*:`-1E3y\"9T}In conference KM6FDI-9
APRS: KK6OZ-15>APT311,N6EX-4*:/061835z3355.03N/11759.58Wv269/032/A=000291
APRS: WC6Q>APWA01,W6SCE-10*,N6EX-4*:@183605h3413.02N/11834.79Wk000/031 /A=000701 Fore 103% (30/25) WC6Q.com
APRS: KF6NMZ-4>APBPQ1,N6EX-4*:}KA6ECT>LAECTC,TCPIP,KF6NMZ-4*:;C06113316*061836z3413.88N\11824.18W!1125-Traffic Haz
ard   I5 S / Laurel Canyon Blvd
APRS: KM6FDI-5>3T1QUU,KELLER*,WIDE1*,N6EX-4*:`-1E3y\"9T}In conference KM6FDI-9
APRS: KM6FDI-5>3T1QUU,KELLER*,WIDE1*,W6SCE-10*:`-1E3y\"9T}In conference KM6FDI-9
APRS: WC6Q>APWA01,N6EX-4*:@183630h3413.20N/11834.79Wk/A=000708 Fore 100% (31/25) WC6Q.com
APRS: XE2SI-10>APMI06,WA6ZSN*,N6EX-4*:@061836z3228.44N/11653.81W-PHG53304-WX3IN1
APRS: KE6VRK-9>SSUQTV,N6EX-4*:`.1cl_W>/`"46}_1
APRS: NQ6X-10>S4QPVQ,CAMRIO*,WIDE1*,N6EX-4*:`.Of!";>/]"6\}=
APRS: NQ6X-10>S4QPVQ,CAMRIO*,WIDE1*,W6SCE-10*:`.Of!";>/]"6\}=
APRS: KF6NMZ-1>APTT4,N6EX-4*:T#312,127,154,255,040,012,10111011
APRS: KM6AHX-12>APOTU0,N6EX-4*:> CHG STN ON
APRS: KM6AHX-12>APOTU0,N6EX-4*:!3411.20N/11813.02W_ 14.2V
```

View | Band | Frequency | Settings

Spectrum

100 80 60 40 20

144 144.2 144.4 144.6

0 100 200 300 400 500

Spectrum Mode: Center | Span: ±500k | Edge 1 | ToFixed | Clear Peaks | Enable WF | Theme: Ion

144.390000 MHz

S 0 1 2 3 4 5 6 7 8 9 +20 +40 +60 dB

dBfs -30 -24 -18 -12 -6 0

Mode: FM | Data | Receive Filter 1 | Show More

100 Hz | F Lock | RIT

Power On | Power Off

RF AF SQ TX LAN Ref Len Top Bot

Transmit | Repeater

Preamp/Att | Preamp: Pre #1 | Attenuator: 0dB | Antenna:

About | Save Settings | Radio Status

rx latency: 180 ms

Serial Port Management

This page will help you determine which serial port is your radio, and in some instances, how to give the port a more obvious name. For Linux, there are detailed directions on the use of udev.

Jump to: macOS, Windows, Linux

macOS Configuration

macOS users wishing for more consistent device names may wish to check out Discover 7610 and CI-V Redirect, both by Kok Chen, W7AY. Note that while these tools are designed for the 7610, the method outlined on the pages shows how to use the System Profiler to see which device is the radio, and that method should be equally valid for other radio models. You can access the System Profiler (also known as “System Information”) by choosing “About this Mac” from the apple menu and then pressing “System Profiler”. You can also use the system_profiler and ioreg commands, as outlined here.

Windows Configuration

Windows users can use the Device Manager to determine which COM device is the radio. To connect to your rig via USB, the (USB) serial port will appear in the list of available devices in the Serial Device: combobox of wfview. Where your rig has two serial ports (A/B) you must select port A (this is usually the lower numbered of the two.

Linux Configuration

With linux, the user logged in may or may not actually have write-access to the serial port, which is needed in order to run software like wfview. There are also times where it is useful to know which serial devices are connected to radios (or perhaps to which radios).

Check for radio ports

This shell script will list serial ports and attempt to work out the vendor and serial number of each:

https://gitlab.com/eliggett/scripts/-/blob/master/radioscan.sh

Run it like this:

```
chmod 755 radioscan.sh
./radioscan.sh
```

Check out how the script works, these commands will be handy for what follows.

Basic Serial Port Ownership and Group Membership:

A quick command may be run to temporarly enable full access, however, this is not a good long-term solution as the command must be run every time:

```
sudo chown $USER /dev/ttyUSB*
```

A better solution is to modify the current user's account, adding membership into the group that the system defaults to for serial ports.

First determine the name of the group:

```
ls -l /dev/ttyUSB*
```

The group name will be the second name, after the owner's name. Here is an example output:

```
eliggett@zep:~$ ls -l /dev/ttyUSB*
crw-rw---- 1 root dialout 188, 0 May 2 01:48 /dev/ttyUSB0
crw-rw---- 1 root dialout 188, 1 Apr 3 14:38 /dev/ttyUSB1
crw-rw---- 1 root dialout 188, 2 Apr 3 14:38 /dev/ttyUSB2
```

Here, the owner is root and the group owner is "dialout". On some systems, this might be "modem".

To append group membership, issue the following command:

```
sudo usermod -aG dialout $USER
```

You may need to log out and log back in for this to take effect.

On some systems, you really will need to be the owner of the port, and this depends upon how the system security is configured. In this next section, you'll have the opportunity to become the default serial port owner.

Using UDEV for nice port names and default ownership:

Linux provides a subsystem called "udev" which, among other things, helps name entries in /dev when devices are connected to the computer. Using a custom udev entry, we can create symlinks to nice device names. For

example, on my system, /dev/IC9700 is my IC-9700 serial port, and /dev/IC7300 is my IC-7300 serial port. These names are much easier to remember and to use than their default ttyUSB names, which will have different numbers when you reboot or re-connect a radio.

To do this, we'll first need to determine the USB device serial number for each radio.

Issue the following command to show the device ids:

```
ls -l /dev/serial/by-id
```

Note, some linux kernels do not provide /dev/serial/by-id. For those systems, use the following command for each serial device (/dev/ttyUSB*) and look at the "ID_SERIAL" attribute:

```
udevadm info /dev/ttyUSB0
```

You will see output similar to what follows:

```
lrwxrwxrwx 1 root root 13 Feb 3 19:25 usb-Silicon_Labs_CP2102N_USB_to_UART_Bridge_Controller_IC-9
700_13001202_A-if00-port0 -> ../../ttyUSB3
lrwxrwxrwx 1 root root 13 Feb 3 19:25 usb-Silicon_Labs_CP2102N_USB_to_UART_Bridge_Controller_IC-9
700_13001202_B-if00-port0 -> ../../ttyUSB2
lrwxrwxrwx 1 root root 13 Feb 3 19:25 usb-Silicon_Labs_CP2102_USB_to_UART_Bridge_Controller_IC-73
00_03001507-if00-port0 -> ../../ttyUSB4
lrwxrwxrwx 1 root root 13 Feb 3 19:25 usb-Silicon_Labs_CP2102_USB_to_UART_Bridge_Controller_IC-78
51_03001140_A-if00-port0 -> ../../ttyUSB0
lrwxrwxrwx 1 root root 13 Feb 3 19:25 usb-Silicon_Labs_CP2102_USB_to_UART_Bridge_Controller_IC-78
51_03001140_B-if00-port0 -> ../../ttyUSB1
```

The part we're concerned with here isn't how many nice radios this person has, but rather, the name and serial numbers, for example "IC-7851_03001140_A". Here, the model is IC-7851, and the serial number is 03001140. Some Icom radios present more than one serial port, and in most cases, we are interested in the "A" port for rig control. The "B" port is for other uses.

Now we need to make a new udev rule. Open up a text document and prepare the text that we will put in. Start with this:

```
SUBSYSTEM=="tty", ATTRS{idVendor}=="10c4", ATTRS{idProduct}=="ea60", ATTRS{serial}=="IC-7851 0300
1140 A", SYMLINK+="IC7851A", OWNER="eliggett"
SUBSYSTEM=="tty", ATTRS{idVendor}=="10c4", ATTRS{idProduct}=="ea60", ATTRS{serial}=="IC-7851 0300
1140 B", SYMLINK+="IC7851B", OWNER="eliggett"
SUBSYSTEM=="tty", ATTRS{idVendor}=="10c4", ATTRS{idProduct}=="ea60", ATTRS{serial}=="IC-7300 0300
1507", SYMLINK+="IC7300", OWNER="eliggett"
SUBSYSTEM=="tty", ATTRS{idVendor}=="10c4", ATTRS{idProduct}=="ea60", ATTRS{serial}=="IC-9700 1300
1202 A", SYMLINK+="IC9700A", OWNER="eliggett"
SUBSYSTEM=="tty", ATTRS{idVendor}=="10c4", ATTRS{idProduct}=="ea60", ATTRS{serial}=="IC-9700 1300
1202 B", SYMLINK+="IC9700B", OWNER="eliggett"
```

Edit the file so that your serial numbers, model numbers, and username are reflected. Remove any lines for radios you don't own (yet).

Save the file to /tmp/rules.txt (or any other location you can easily get to)

Now from the terminal, copy the file in to the udev rule location for your system (commonly as shown):

```
sudo cp /tmp/rules.txt /etc/udev/rules.d/90-persistent-usb.rules
```

To test it out, disconnect the USB cables from your rigs, and then reconnect. You should now see nice entries in

/dev for your radios:

```
eliggett@zep45:~$ ls -l /dev/IC*
lrwxrwxrwx 1 root root 7 Apr 3 14:38 /dev/IC7300 -> ttyUSB0
lrwxrwxrwx 1 root root 7 Apr 3 14:38 /dev/IC9700 -> ttyUSB1
lrwxrwxrwx 1 root root 7 Apr 3 14:38 /dev/IC9700-data -> ttyUSB2
```

(Note that I named my “B” port “-data”. You can select whatever you like for yours.)

You can make individual shortcuts to wfview which use settings files that specify different serial ports. See the Preferences File manual page as well as the Command-Line Arguments page for more information.

Command-Line Arguments

This page will describe various ways to start wfview from a command-line with command-line arguments. These arguments are most useful to start wfview from a specific settings file or to enable debug-level logging.

How to supply arguments

To supply arguments, one must generally open a terminal and then enter in the name of the program followed by the arguments. For example, from linux, assuming wfview is within your current $PATH:

`wfview --argument`

From macOS, there are two ways. One way is to change directory to the inside of the application bundle, and then to issue commands like so:

`/Applications/wfview.app/Contents/MacOS/wfview --argument`

You can drag the wfview.app icon into the terminal and edit the text.

However, it may be easier within macOS do it like this:

`open /Applications/wfview.app --args --argument`

With Microsoft Windows, you will need to open a `cmd.exe` window. From there, you can drag the wfview.exe icon into the cmd window. This should enter in the full path of the executable. Once entered, you may add arguments to the text.

Click here to jump down to a list of all arguments.

Most commonly used arguments

The most commonly used arguments are as follows:

Settings file

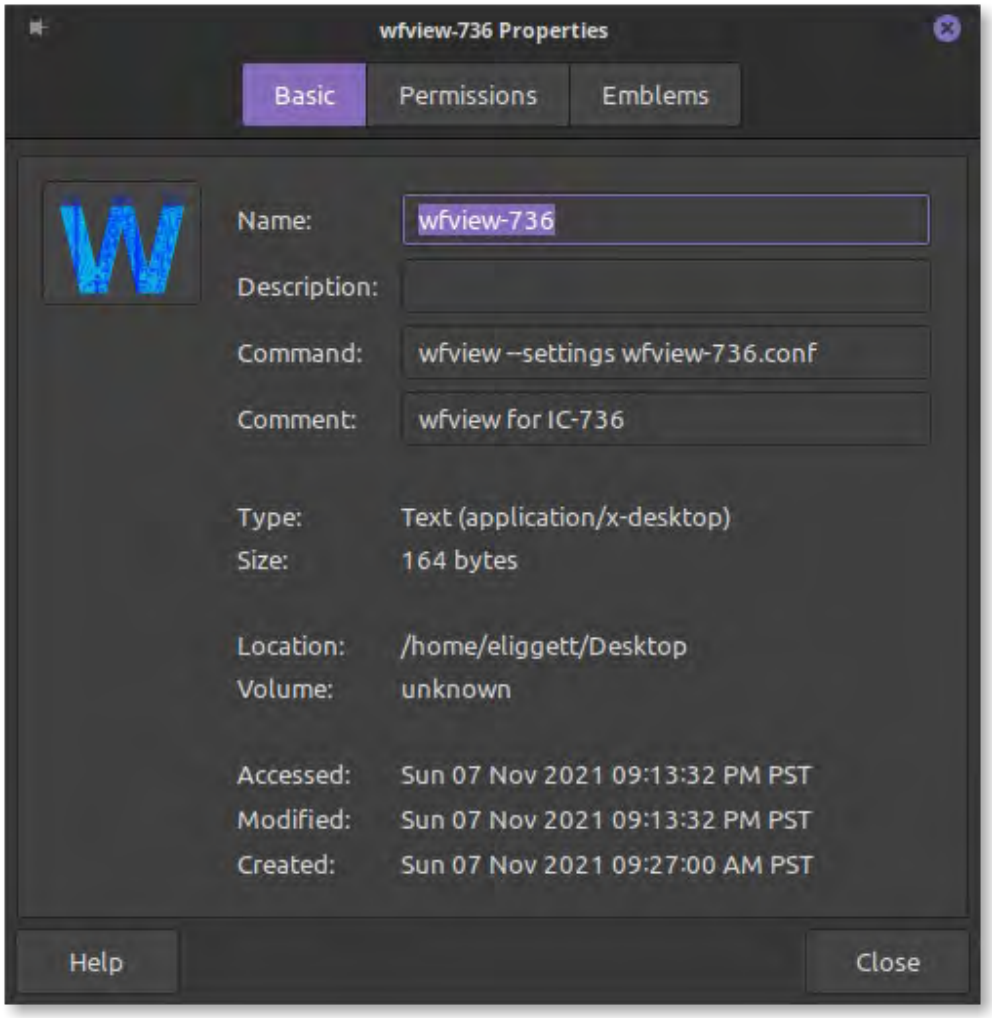
To specify an alternate settings file, the `--settings` argument may be supplied followed by either an absolute path (generally starting with a slash), a relative path (starting from the current working directory), or simply a filename.

Specifying **only a filename** (without any path) is the preferred and simplest method; wfview will automatically generate a full filename to the operating system's user-writable application settings folder. On linux, this automatically-added path would be `~/.local/wfview/wfview`. **The file doesn't need to initially exist**, wfview will create the specified file when you press "Save Settings".

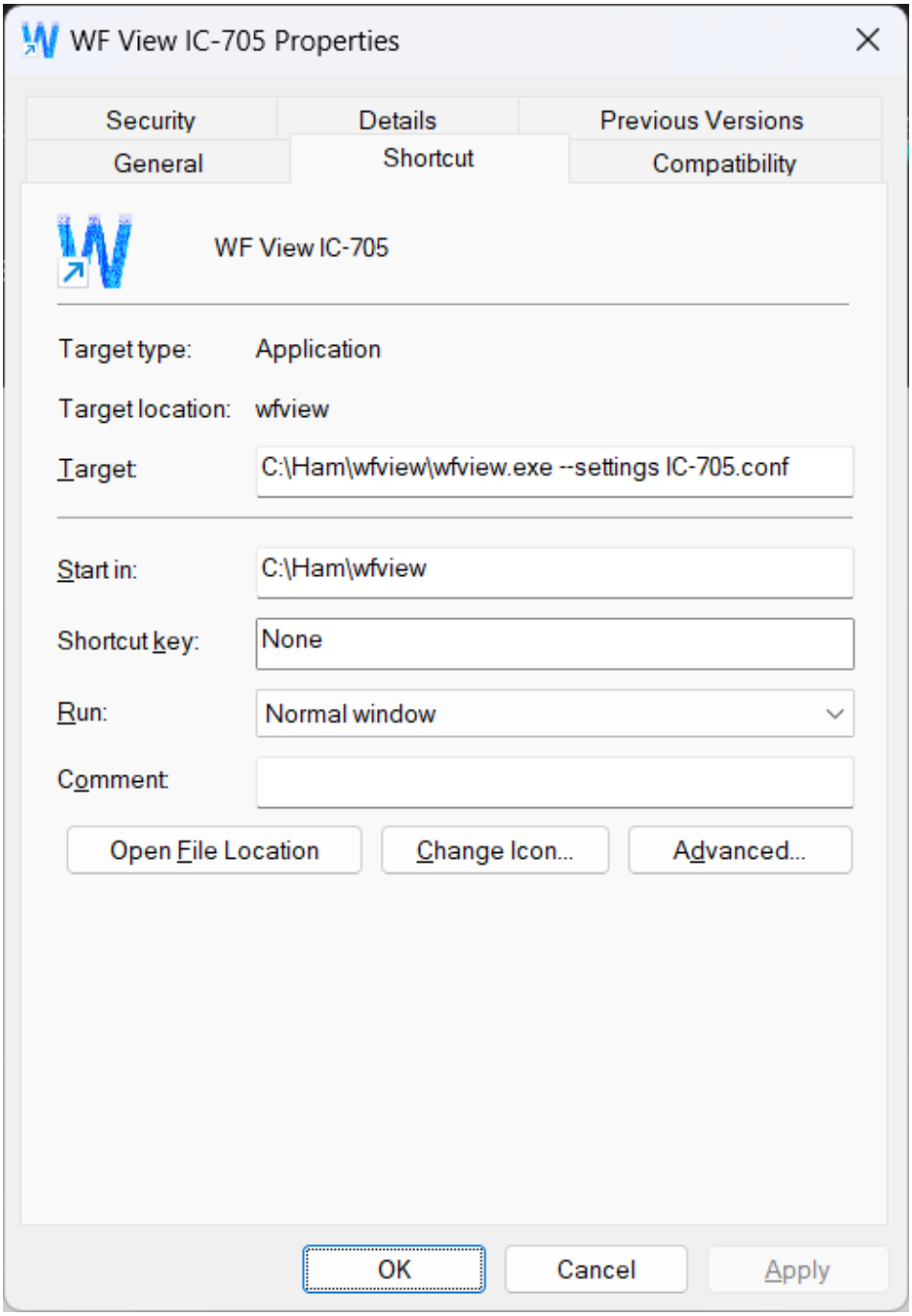
You can read more about the settings file format here. A general strategy is to simply supply `--settings` followed by your radio's name:

`wfview --settings ic-718.conf`

Once wfview is set up (under the Settings tab), do not forget to press “Save Settings”. wfview will then write the settings file to disk, and you will from then on be able to immediately open wfview to your various radios simply by making desktop shortcuts that include the needed settings file within the arguments. Here is how one of mine looks for linux. I simply made it by right-clicking on the wfview icon within the applications menu, and choosing “add to desktop”. From there, I customized the launcher to include the needed command-line arguments:



Desktop launcher for Linux with specific settings file specified.



This is an example wfview shortcut under windows which specifies a `--settings` argument.

Sometimes you may end up with an unusual configuration which you need to remove in order to launch wfview. You can do so using the `--clearconfig CONFIRM` argument to wfview, which will cause all your settings to revert to their defaults on start.

Debug Mode

To enter debug mode, simply pass `--debug` to wfview. This will cause wfview to log much more information to the log file, including the contents of most commands and most radio replies (except for periodic polling commands and spectrum data).

```
wfview --debug
```

Full List of Supported Command-Line Arguments

The following is a list of all supported command-line arguments. Many are intended for development use only and are not advisable to be used. The commands are listed in their short and long formats. The parsing of the commands takes place in main.cpp.

Some arguments have a “short” and “long” version. Do not supply both. Just supply either the short or the long form.

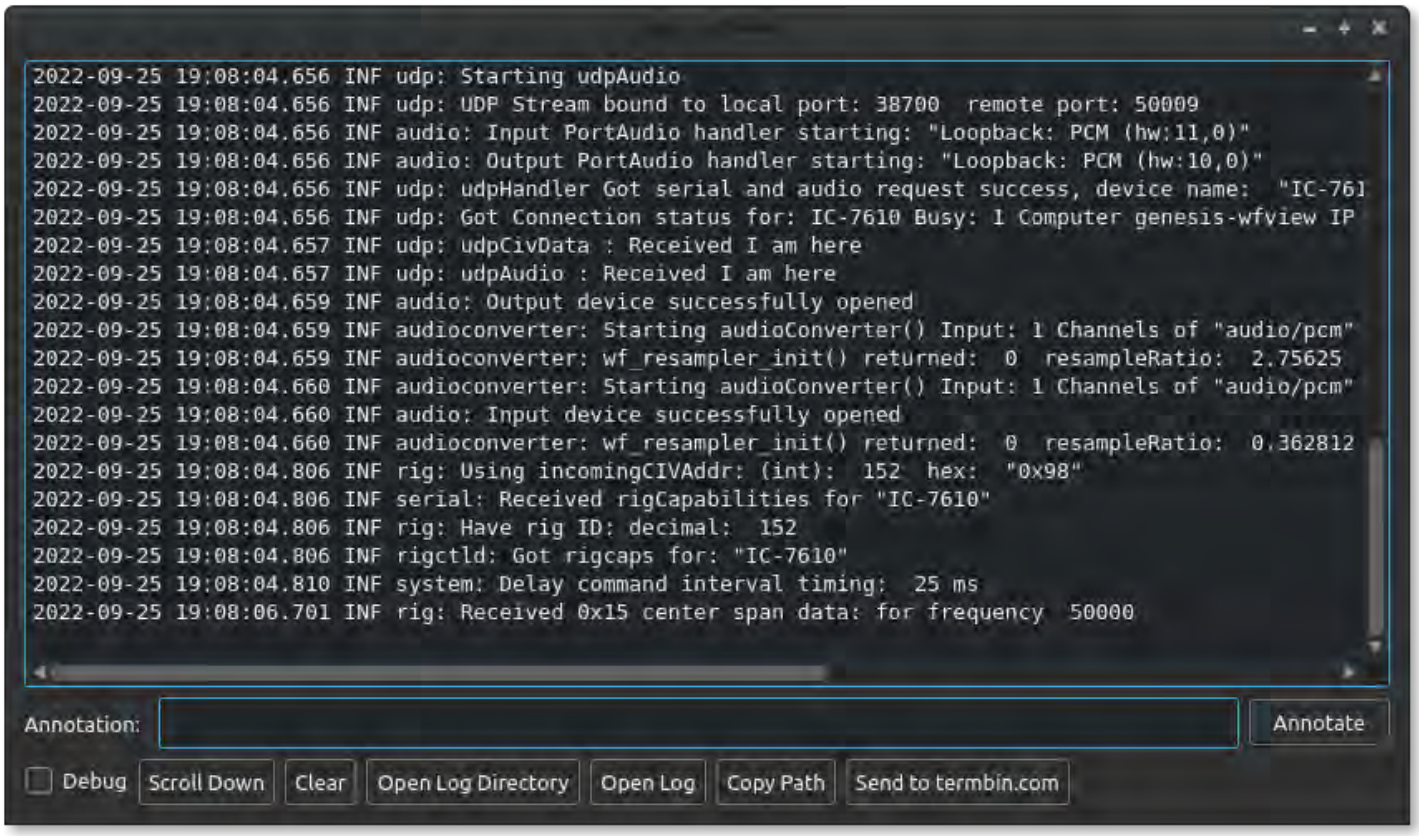
- `-d --debug`
  - Enter debug mode. Causes more verbose logging and additional features to become visible.
- `-l --logfile /tmp/my_preferred_log_filename.txt`
  - Specify alternate location for the log file. This isn’t usually necessary but it can be helpful for organizing logs, especially while debugging. A log file is *always* created.
- `-s --settings ic718.conf`
  - Specify settings file. Path may be absolute, starting with a slash, may be relative from the current working directory (not recommended), or may be a simple filename (recommended), to which wfview will determine the correct and complete path for user-accessible application data for your operating system.
  - We recommend **not** specifying a path, merely specify a *filename*. A file that doesn’t exist yet will be *automatically created* when you Save Settings. wfview knows where to place settings files, and it’s best to just let it handle the location.
  - **DO NOT** specify path names using “`~`“, as this will create a directory within the application data folder named “~”. This is a Qt bug.
  - Good example: `wfview -s ic718.conf` This will create and/or load settings from the file “ic718.conf” which is (assumed) to be located within the user-accessible settings directory indicated by your operating system. Again, the file does not need to exist yet, it’ll get created once you press Save Settings.
- `--clearconfig CONFIRM`
  - This resets your settings to their defaults on startup. You must include “`CONFIRM`“.
- `-h --help`
  - Show help text and exit (will not work on Windows)
- `-platform offscreen`
  - Technically a Qt option, this will make wfview run without displaying a window.
- `-platform vnc`
  - Also a Qt option, this creates a minimalist VNC server to run the app inside. There is no window manager with this option, but sometimes it’s useful. The default port (5900) can be changed by exporting a variable like this: `export QT_VNC_PORT=5901` prior to running wfview.

Log file

wfview writes a log file, which can be useful to look at when things don't work. When you witness unexpected program behavior, click the Log button. This is like your gauges in a car — look here first when there is trouble. When you post to our support forum, it'll save everyone a step if you check the log first. If you see anything interesting, use the "send to termbin" function and paste a link to the log into your support forum post. It helps a lot!

Since wfview version 1.5, the program has a button, "Log" in the bottom row.

Press this button to open the log window, which will display the log and allows for several useful actions:



Here is what each button does:

- Annotation: Type any text that you would like to enter into the program's log file. This is useful to take notes. You might want to take notes about malfunctions you experience, or note what you are trying to do just

before an action takes place.
- Debug: Checking here enables more verbose logging
- Scroll Down: Scrolls the contents of the window down, useful if you have scrolled up and just want to jump to the bottom
- Clear: Clears the display. Does not clear the contents of the log file.
- Open Log Directory: Opens the directory where the log is stored.
- Open Log: Opens the log file in your operating system's text editor.
- Copy Path: Copies the path of the log file to the clip board
- Send to termbin.com: Sends the displayed log entries to termbin.com, and provides you with a URL from which you can then access and share the log file.

About Send to termbin.com: When you press this button, the log (as displayed) will be sent to termbin.com. This is the same website used by many linux commands to help people share debugging information. Keep in mind that the log file may contain information such as your user name and ip address. When you are looking for help with wfview, if you press this button, a URL will be copied to the clipboard automatically, which you can then paste into a post on our support forum. This is by far the easiest way to share a log file.

Default Log File Location

Remember, you can copy the log file path using the "Copy Path" button in wfview.

- Linux: /tmp
- Windows: %TEMP%/wfview
- macOS: /private/$TMPDIR/org.wfview.wfview (use the "Copy Path" button in wfview's log GUI to obtain the path quickly)

The log file name is, by default, in for form of "`wfview-yyyymmddhhmmss.log`". The time stamp refers to the moment the program was started. You can override this behavior by specifying the `--logfile` argument when you start wfview, for example:

```
wfview --logfile /tmp/IC-7300.log
```

would cause the log file to be named "/tmp/IC-7300.log". Paths are relative to the current working directory if an absolute path isn't given.

Debug Logging

To aid in debugging, wfview may be started with a debugging flag, which causes much more data to be logged to the log file. To start wfview with debug logging, run wfview with either a `-d` flag or `--debug`. For example:

```
wfview --debug
```

Once done, the log file will contain a much greater amount of information. For macOS, run the following terminal commands:

```
cd /Applications
open wfview.app --args --debug
```

Note that you can turn on debug logging from within wfview, by checking the "Debug" checkbox in the Log window. However, starting wfview with debug logging will generate some more messages concerning the startup sequence.

Logging Features

When wfview starts, it will begin with “INF system: Starting wfview”. Each line has the current date, time, logging category, and subsystem label. Logging categories include:

- INF (informative)
- DBG (debug)
- WRN (warning)

Subsystems (logical units of code basically) include:

- system (wfmain.cpp)
- audio (audio handler)
- audioconverter (sample rate conversion, channel count, bit depth, etc)
- udpaudio (network audio)
- udp (communication to and from a radio over network)
- rig (rigCommander)
- serial (pseudo-term and serial-connected rigs)
- gui (gui-specific parts of wfmain.cpp)
- user (user annotations)

Useful Log Messages:

Briefly, here are some useful things to look for:

Rig-ID: This is how the radio is identified:

```
2021-05-15 16:38:41.230 DBG rig: ---Rig FOUND from broadcast query:
2021-05-15 16:38:41.230 INF rig: Using incomingCIVAddr: (int): 148 hex: 94
2021-05-15 16:38:41.230 INF rig: Have rig ID: decimal: 148
2021-05-15 16:38:41.230 DBG system: Rig name: "IC-7300"
2021-05-15 16:38:41.230 DBG system: Has LAN capabilities: false
2021-05-15 16:38:41.230 DBG system: Rig ID received into wfmain: spectLenMax: 475
2021-05-15 16:38:41.230 DBG system: Rig ID received into wfmain: spectAmpMax: 160
2021-05-15 16:38:41.230 DBG system: Rig ID received into wfmain: spectSeqMax: 11
2021-05-15 16:38:41.230 DBG system: Rig ID received into wfmain: hasSpectrum: true
```

In the above case, the rig was identified by means of a broadcast request (to CI-V address 0x00).

With debug logging enabled, the following additional data are available:

Traffic to the radio: wfview logs data to the radio, except for regularly polled queries (such as the s-meter). The format is as follows:

```
2021-05-15 16:38:41.267 DBG rig: Final payload in rig commander to be sent to rig:
2021-05-15 16:38:41.267 DBG rig: ---- Begin hex dump -----:
2021-05-15 16:38:41.267 DBG rig: "INDEX: 00 01 02 03 04 05 "
2021-05-15 16:38:41.267 DBG rig: "DATA:  fe fe 94 e1 03 fd "
2021-05-15 16:38:41.267 DBG rig: ----- End hex dump -----
```

Here, we are requesting the current frequency (command 0x03) from the radio at address 0x94.

Traffic from radio: wfview logs any data received from the radio, except for spectrum and metering data, since these requests happen so frequently that logging them takes a lot of resources. The format is as follows:

```
2021-05-15 16:38:41.270 DBG rig: ---- Begin hex dump -----:
2021-05-15 16:38:41.271 DBG rig: "INDEX: 00 01 02 03 04 05 06 "
2021-05-15 16:38:41.271 DBG rig: "DATA:  03 00 20 28 07 00 fd "
2021-05-15 16:38:41.271 DBG rig: ----- End hex dump -----
```

You can see that the CI-V preamble and to/from address portion is already stripped out. In this case, the rig is responding to a frequency query request (command 0x03). The indicated frequency is 7.282000 MHz.

Preferences File

The preferences file contains data that are read upon opening wfview. The contents are saved upon pressing the Save Settings button under Settings.

These data include settings for the program's appearance, radio connection, window position, and more. Editing this file is not recommended; do so at your own risk, and only if you need to do so.

With each release of wfview, we do retire and add settings, so you may see some attributes from prior versions, which are ignored.

Don't forget that you can reset your preferences easily by pressing "Revert to Default" in the Settings window.

If you are unable to open wfview and wish to reset the settings, you may pass the following command-line flag to wfview: `--clearconfig CONFIRM`

Accessing:

The preference file may be accessed as follows:

- macOS: Use the defaults command, specifying the org as org.wfview.org
- Linux: Read the plain text file from ~/.config/wfview/wfview.conf
- Windows: Configuration is stored in the Windows Registry under **"HKEY_CURRENT_USER\SOFTWARE\wfview\wfview"**. Please be careful when editing the Windows Registry as incorrect usage of the regedit.exe utility can seriously damage your Windows installation. DO NOT change anything outside the registry key above unless you know what you are doing!

Note: You may specify an alternate settings file using the `--settings` command-line argument. See here for a list of arguments.

Preference File Sections and Attributes:

Each section is as follows (note that the qt library does not allow wfview to determine the exact order these will appear in the file, and thus, they may be re-ordered by the user for clarity):

[Controls]

These options are from options under the Settings tab:

- EnablePTT: Set to 'true' or 'false' to enable or disable the PTT controls of wfview. Does not change the radio's own PTT lock controls.
- NiceTS: Set to 'true' or 'false' to enable or disable the frequency rounding that can occur when tuning the frequency or double-clicking in the spectrum.

[Interface]

Most of these options are from checkboxes under the Settings tab:

- ConfirmPowerOff: Boolean value to disable the “Are you sure you want to power off” dialog box.
- ConfirmExit: Boolean value to disable the “Are you sure you want to quit?” dialog box
- StylesheetPath: Set to the path of a custom stylesheet, relative to /usr/share/wfview/stylesheets
- UseFullScreen: Set to ‘true’ or ‘false’ to enable full screen on startup.
- UseSystemTheme: Set to ‘true’ or ‘false’ to enable or disable using the system’s theme (versus a stylesheet — included or user-added)
- splitter: The position of the draggable plot-to-waterfall vertical space splitter. Probably best not to edit this by hand.
- windowState: The state of the window (position and maximized status)
- windowGeometry: The location and size of the window. Remove this variable and the one above if you have difficulty seeing the wfview window, this will restore the default.
- WFAntiAlias: boolean for waterfall anti-alias
- WFInterpolate: boolean for interpolation between the waterfall data points and the pixels on the screen
- WFLength: length of waterfall display. Values should be from 160 to 1023.
- Meter2Type: Selected type of meter for the secondary meter. Integers represent the meter number per the enumerated types in the source code.
- plotCeiling: value of the “top” slider
- plotFloor: value of the “bot” slider
- WFTheme: Selected color scheme theme for the waterfall
- underlayMode: The selected mode of plot underlay:
    - 0: none
    - 1: Peak Hold
    - 2: Buffered Peaks
    - 3: Average
- underlayBufferSize: The size of the buffer. Valid numbers are 2 to 128.

[Radio]

These options are specific to serial port access:

- RigCIVuInt: Use this parameter to manually set the CI-V address of the radio, in unsigned integer format. If set, wfview will not broadcast a request for all radios to identify, instead, it will send a rig ID request only to the specified CI-V address. This is useful if you have a radio that does not respond to broadcast queries on the CI-V bus, or, if you have more than one radio on the same physical CI-V bus (very rare in the post-USB-adapter decades). If unset or set to 0, wfview will automatically find any connected radio (regardless of if that radio is a known model) and attempt to control it. The default value is 0, and is recommended for most users.
- CIVisRadioModel: Boolean value. When set to “true” causes wfview to assume the radio model is the same as the manual set CI-V above. For older radios that do not answer Rig ID queries, supply the default CI-V address above and then set this parameter to true.
- SerialPortBaud: Use this attribute to specify a custom baud rate, often needed on older radios. The default is 115200, required by the newer radios that provide spectrum data over serial.
- SerialPortRadio: Use this attribute to specify a custom path to the serial port. This can be used, for example, to specify a pseudo-terminal if you are using socat to forward serial traffic from a remote host. If unset, or set to “auto”, wfview will automatically find directly-connected supported Icom radios on the linux platform. On other platforms, it will attempt to use the first available serial port.
- localAFgain: Unsigned char (0-255) used to store the user’s local AF gain (volume). Does not change the volume at the radio, just how loud it sounds at the computer.
- VirtualSerialPort: String path to the virtual pseudo-terminal created for other programs to use as a serial port.

[Server]

These options are for the built-in OEM-compatible server, useful for placing non-ethernet rigs onto a network.

- ServerEnabled: true/false whether to enable wfview's internal server which emulates an Icom rig and allows multiple clients to connect. This is still quite experimental
- ServerControlPort: The UDP port that clients will make an initial connection to.
- ServerCivPort: The UDP port that the server announces to clients, 50002 is the default.
- ServerAudioPort: The UDP port that the server announces to clients, 50003 is the default.
- ServerNumUsers: The number of users currently configured.
- ServerUsername_x, ServerPassword_x, serverUserType_x: contain the user information.

[LAN]

These options are for configuring the network access to radios that either have built-in network, are connected to a PC running Icom's software, or are connected to a host running wfview with the above [Server] enabled.

- EnableLAN: true or false to enable LAN connection to the radio (rather than USB)
- AudioInput: Name of audio input device, this is best set via the Settings tab as it must be spelled exactly
- AudioOutput: Name of audio output device, this is best set via the Settings tab as it must be spelled exactly
- AudioRXCodec: This is the receive codec that is sent to the radio, 0x01 = single channel uLaw, 0x02 = single channel 8bit PCM, 0x04 = single channel 16bit PCM, 0x08 is two channel 8bit PCM, 0x10 is two channel 16bit PCM and 0x20 is two channel uLaw. It is anticipated that other modes, potentially including compression will be added at some point for wfview-wfview communications.
- AudioTXCodec: This is the receive codec that is sent to the radio, 0x01 = single channel uLaw, 0x02 = single channel 8bit PCM, 0x04 = single channel 16bit PCM. It is anticipated that other modes, potentially including compression will be added at some point for wfview-wfview communications.
- AudioRXLatency: Use this to specify the number of milliseconds of latency to be allowed in RX.
- AudioTXLatency: Use this to specify the number of milliseconds of latency to be allowed in TX.
- AudioRXSampleRate: This is the sample rate to tell the radio to use for RX audio. 48000 is the default but 8000, 16000 and 24000 are also available. Any other rate will likely be rejected by the radio. Smaller sample rates will reduce the required network bandwidth but may reduce quality.
- AudioTXSampleRate: This is the sample rate to tell the radio to use for TX audio. 48000 is the default but 8000, 16000 and 24000 are also available. Any other rate will likely be rejected by the radio. Smaller sample rates will reduce the required network bandwidth but may reduce quality.
- ClientName: This will usually be the "name" of your computer but can be replaced with any other name. This cannot be changed from within wfview, so must be changed by manually editing the file/registry entry.
- ControlLANPort: This is the "control" port that wfview will use to communicate with the radio and defaults to 50001. Any other required ports are configured by the radio.
- EnableRigCtlD: true or false to enable an emulation of rigctld within wfview. This is highly experimental and currently only supports a small subset of rigctld commands. This is disabled by default and canot be enabled from within the wfview GUI.
- RigCtlPort: Port that the internal rigctld emulation should listen on. Default is 4533 so as to not conflict with any other rigctld running on the machine.
- IPAddress: IP Address of the radio
- Username: Username configured within Network menu of the radio
- Password: Password configured for the above user in the radio.
- ResampleQuality: This is a number from 0-10 which determines the quality of wfview's internal audio resampler, with 0 being the lowest quality. This is only used if a sample rate of anything other than 48000 is used and the default of 4 has been chosen to be middle-of-the-road in terms of quality and speed.
- WaterfallFormat: The format of the waterfall data to be provided.
    - 0: do nothing
    - 1: serial-port style (multiple segments)

    - 2: LAN style (single segment)
- tcpServerPort: Port for raw CI-V access over TCP/IP, such as is used by N1MM+. A value of zero disables this feature.

[Memory]

This section contains the 99 memories managed by wfview. Don't bother making a parser for this format, we will change to CSV in future versions.

[Color Presets]

(in progress)

These lines define the color presets a user has entered. The data are read in as strings, and the expected format is #AARRGGBB, where AA is the alpha channel (0xff = opaque, 0x00 = transparent), and RRGGBB is your standard red green blue HTML color. Colors that are fully transparent (AA = 00) will basically hide an element of the interface. PresetName is used for the preset combo box and must be 10 characters or less. Unicode characters should work although this hasn't been tested. The presetNum attribute is automatically assigned and should not be changed.

Any color that is not read in properly is reverted to the default wfview colors.

You can see more about these colors in this section of the manual.

Example for the first color preset:

```
ColorPreset\1\axisColor=#ffffffff
ColorPreset\1\gridColor=#ff000000
ColorPreset\1\meterAverage=#ff3fb7cd
ColorPreset\1\meterLevel=#ff0000ff
ColorPreset\1\meterLowText=#ffeff0f1
ColorPreset\1\meterLowerLine=#ffeff0f1
ColorPreset\1\meterPeakLevel=#ffaaffff
ColorPreset\1\meterPeakScale=#ffff0000
ColorPreset\1\passband=#80006fa7
ColorPreset\1\plotBackground=#ff000000
ColorPreset\1\presetName=Dark
ColorPreset\1\presetNum=0
ColorPreset\1\spectrumFill=#00000000
ColorPreset\1\spectrumLine=#ffffff00
ColorPreset\1\textColor=#ffffffff
ColorPreset\1\tuningLine=#ff55ffff
ColorPreset\1\underlayFill=#96465e96
ColorPreset\1\underlayLine=#9633ff55
ColorPreset\1\wfAxis=#ffffffff
ColorPreset\1\wfBackground=#ff000000
ColorPreset\1\wfGrid=#00000000
ColorPreset\1\wfText=#ffffffff
```

Using Older Radios

Scope

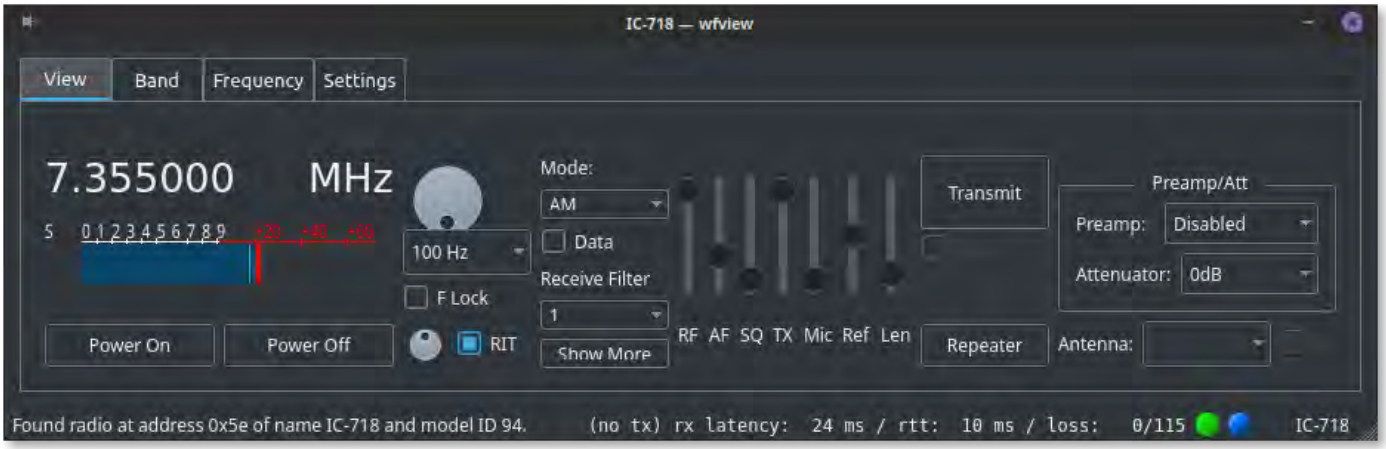
This guide applies to radios that are not in the core supported radios.

Considerations

Older radios may have several limitations:

- Lack waterfall data
  - wfview gets waterfall data from the rig, so if your rig doesn't have it, or doesn't provide it over CI-V (IC-7700, IC-7600), then this part of the interface will be blank. Sadly many radios which present waterfall data on the screen of the radio do not export this data to the connected computer.
- Difficult physical connection constraints
  - May need a DIY level converter circuit
  - May need a custom audio interface
  - Some rigs have integrated USB, and this is generally preferred.
- May require manual baud rate assignment in wfview
- May require manual CI-V address assignment in wfview
- May not answer our rig ID broadcast (discovery) request (Ic-706 and older)
  - You can manually assign the CI-V address in the Settings tab, and then select "Use as model too". Only works if the radio is using the default CI-V address.
- May have unusual metering scale, or no metering at all
- May lack PTT command (IC-780, IC-736, IC-718, IC-706)
  - Note, we have added the capability to use **RTS** (serial port signal "Ready To Send") as a way to PTT
  - For these radios, the usual approach is to use a USB to CI-V converter that includes the ability to toggle the PTT pin on the ACC port by means of the internal serial adapter's RTS pin.
    - These two devices include both audio to USB as well as CI-V and PTT:
      - XGG Comms Digimode-4-Icom
      - West Mountain Radio's Rig Blaster-series boxes
- May use non-standard frequency format (IC-713, currently not supported)

Despite these limitations, if you are willing to do some work, you can operate your rig remotely using wfview for the CI-V data and the audio.



Look how beautiful wfview is with a radio such as the Icom IC-718! I am accessing this radio *remotely* too!

Setup

First, it will be wise to launch wfview and set some basic settings:

- Set the serial port to the correct port for the radio.
- Set the baud rate to the highest baud rate the radio reliably works under
- Manually enter the CI-V address of the radio. For example, for the IC-7100, the default CI-V address is “88”. Some radios don’t require this step, in particular newer radios from the past two decades. But it doesn’t hurt if you know the CI-V address to just put it in.
- If the radio does not have a “Read Transceiver ID” command (0x19 0x00), then check the “**Use as Model too**” box. This checkbox causes wfview to identify the radio by the CI-V address you provided, trusting that the radio is set to the default CI-V address, which other radios use as their default CI-V and reply to Rig ID queries. Basically, every radio made prior to 1998 lacks this command. So if you have an IC-736, check this box.

Now press “Save Settings” and then press “Connect”.

On the radio, the recommended settings, when available, are:

- CI-V Echo: off
- CI-V baud rate: maximum supported
- CI-V Transceive: ON
- CI-V address: Use the default value (and note it)

Once these parameters are set, launch wfview. Verify that, for supported radios, the bottom-right corner lists the model number (or Rig ID) and not “NONE”.

If you wish to serve the radio’s control and audio over the network, please see the Remote Operation guide.

Adding a Radio

With the release of wfview 2.0 comes a new tool called RigCreator, which allows the user to create a rig file. This is a graphical tool which presents to the user a list of all the features and commands for a radio.

The details are covered on this page.

Briefly:

1. Start up wfview and enable RigCreator in the User Interface settings
2. Press RigCreator at the bottom of wfview’s main window
3. Select “Icom” or “Kenwood” as the manufacturer

4. Press “Default Rigs” and select an existing rig model that is close to your radio in terms of when it came out.
5. Edit the rig file. You can right-click in the Commands table to add a new entry, and then select from the drop-down menu the command you wish to add.
6. Make sure to edit the three fields at the top of the window:
    1. Model
    2. CI-V address (or CAT identifier for Kenwood)
    3. RigCtlD Model (model number from Hamlib)
7. Review the changes carefully, and then press “Save File”. This will place the file in the correct user-writable location for your platform. You can read about that more here.
8. Restart wfview
9. If successful, please consider sending your rig file to the wfview team, either on our support forum or with a Pull Request to our gitlab project.

Supported Radios and Features

For a list of currently-supported radios, please see this area of our source repository. There are no older radios which provide spectrum (waterfall) data over CI-V or CAT.

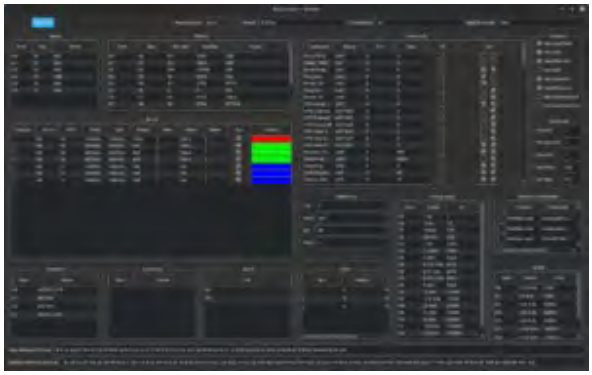
As of 2025-01-27:

Manufacturer	Model	CIV Address	can transmit	full-duplex comms	PTT Command
Icom	DEFAULT	0x0	True	False	YES
Icom	IC-705	0xA4	True	True	YES
Icom	IC-706 (MkIIG)	0x58	True	False	YES
Icom	IC-706 (Mk1/Mk2)	0x48	True	False	YES
Icom	IC-7100	0x88	True	False	YES
Icom	IC-718	0x5E	True	False	YES
Icom	IC-7200	0x76	True	False	YES
Icom	IC-7300	0x94	True	True	YES
Icom	IC-746 PRO	0x66	True	False	YES
Icom	IC-746	0x56	True	False	RTS/DTR
Icom	IC-756 PRO III	0x6E	True	False	YES
Icom	IC-756 PRO II	0x64	True	False	YES
Icom	IC-756 PRO	0x5C	True	False	YES
Icom	IC-756	0x50	True	False	RTS/DTR
Icom	IC-7610	0x98	True	True	YES
Icom	IC-785x	0x8E	True	True	YES
Icom	IC-905	0xAC	True	True	YES
Icom	IC-9100	0x7C	True	True	YES
Icom	IC-9700	0xA2	True	True	YES
Icom	IC-R7100	0x34	False	False	NO

Icom	IC-R8500	0x4A	False	False	NO
Icom	IC-R8600	0x96	False	True	NO
Icom	ID-31A/E	0x84	True	False	YES

Rig Files and Rig Creator

wfview version 2 introduces a new feature called “rig files”, along with a graphical editing tool called Rig Creator.



wfview’s Rig Creator interface, click to expand

Rig files are plain text files which contain a description of the radio’s features and commands. The utility of the rig file is that radios can be added and edited easily without waiting for a new version of wfview to be distributed.

Here’s what you need to know:

1. New rig files can be downloaded here from our official repository. Check now and then to see if a newer rig definition file is available for your radio.
2. Rig files are on your computer in the following location(s):
    1. Linux:
        1. System: `/usr/local/share/wfview/rigs/` (possibly without “local” depending on the package maintainer)
        2. User: `~/.local/share/wfview/wfview/rigs/`
    2. Windows:
        1. System (64-bit): `C:\Program Files\wfview\rigs`
        2. System (32-bit): `C:\Program Files (x86)\wfview\rigs`
        3. User: `C:\Users\<USER>\AppData\Roaming\wfview\wfview\rigs`
    3. macOS:
        1. System: `/Applications/wfview.app/Contents/MacOS/rigs` (Right-click on wfview.app and select “Show Contents” or use the terminal.)
        2. User: `~/Library/Application Support/`
3. Rig files can be edited with a graphical tool within wfview, or with a text editor. Editing a rig file is only for advanced users. See below.

Strategy for Updates

wfview will always load your user-writable rig files over those of the system-level rig files, when available. This means you can store changes and per-rig manual updates in your user-writable rig folder if you wish. However,

when you update wfview, we will send you new rig files which are always installed at the system-level.

Thus, when updating wfview, please **remove** your personal user-writable rig files so that wfview will instead use fresh files from the wfview team.

If you have created or updated any rig files, please consider sharing them with the community. You can use the support forum or a gitlab pull/merge request to submit files.

Note: Do not keep duplicate “backup” copies of rig files in these standard rig file locations on your computer, wfview will load both versions and there will be somewhat undefined behavior as to which one matches the radio. Instead, make backup copies outside the standard file locations.

Editing Rig Files within wfview (advanced users only)

Warning: This is an advanced feature intended only for advanced users and developers.

wfview includes a graphical editor for editing rig files called Rig Creator. This editor makes it easy to create new rig definition files and also to edit existing rig definition files.

To enable Rig Creator:

Open wfview and press “Settings”. Next, under the “User Interface” category, check the box “Enable Rig Creator Feature (use with care)”. Once this setting is enabled, you will find a new button at the bottom of the wfview main window called “Rig Creator”. This button is in between “Frequency” and “Connect to Radio”.

Keep in mind that your user needs permission to write to the location where the rig file is stored, so it is often advisable to use the user-writable location rather than the program’s main rig file location. Also, you will need to re-launch wfview after you have altered a rig file.

Getting Started:

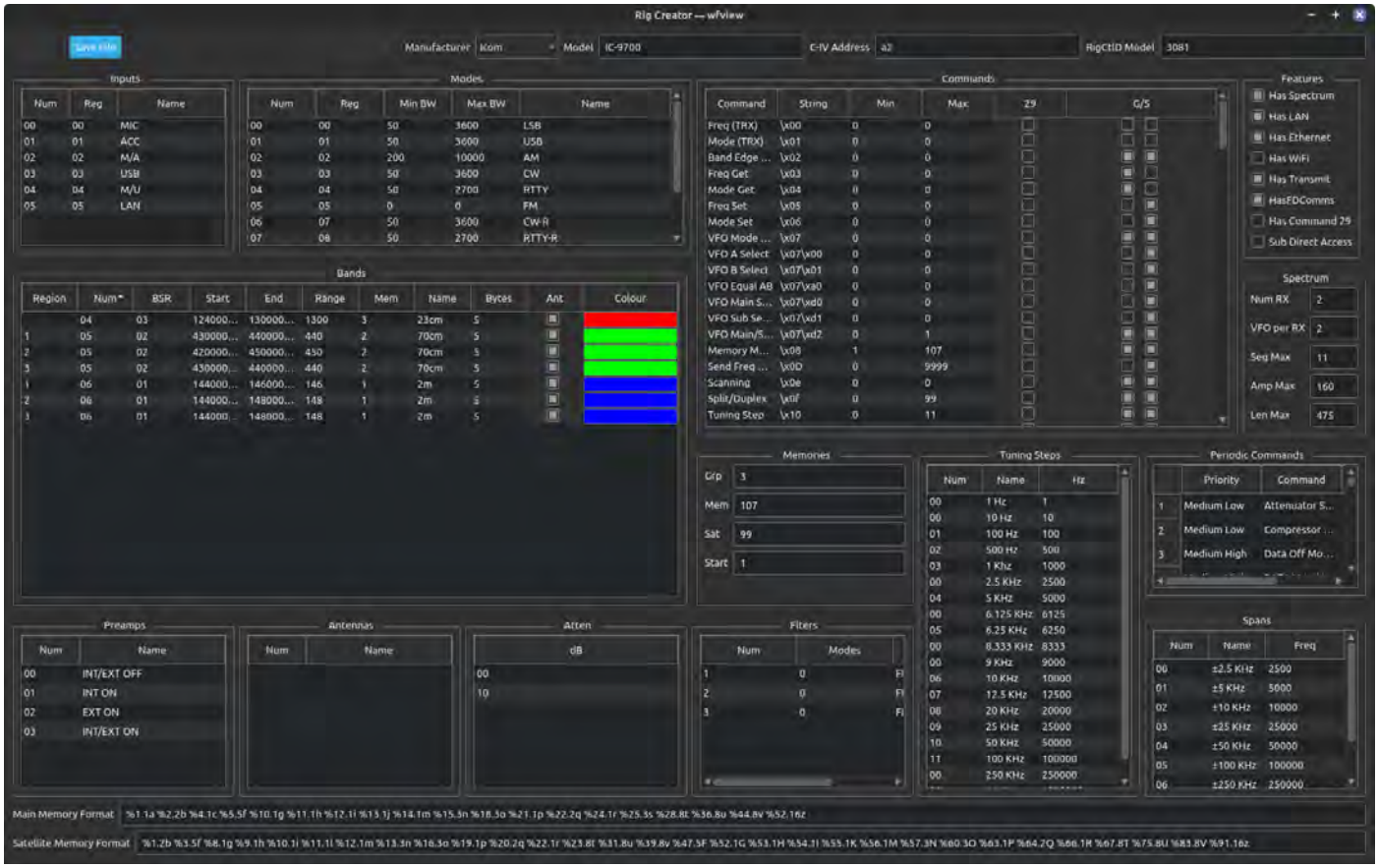
It’s usually simplest to begin with an existing rig. Press the button “Default Rig” (if it is grayed out, close Rig Creator and open it again), which will bring up a list of existing rig files. So, as a starting point, select a radio with similar features to the one you are adding.

Be sure to change the CI-V Address (Icom radios, hex format) or Radio Identifier (Kenwood), as well as the RigCtlD Model number (you can look these up here).

To add a new command, right-click in the command area and select “Add Item”

Make all required changes and press “Save File” when you are ready; wfview will prompt for a filename and default to the user-writable location for your operating system. Save the file in the default user-writable location, which the save dialog box will default to.

When finished, close Rig Creator and close wfview.



wfview Rig Creator interface

Supported Radios and Features

wfview is designed with compatibility in mind. Where possible, commands are used with a rich history dating back sometimes as far as 20 years. And while the more advanced features, such as the waterfall, are only available on newer radios, older radios can still enjoy rapid radio control and remote operation using the built-in server.

The following radios are considered the “core” supported radios for wfview, and thus have the most supported features:

- Icom:
  - IC-705
  - IC-905
  - IC-7300
  - IC-7610
  - IC-7850 / IC-7851
  - IC-9700
  - IC-R8600
- Kenwood:
  - TS-590S (USB)
  - TS-890S (USB and Network)
- Xiegu:
  - X6100 (limited support possible)
  - X6200 (limited support possible)

If your radio is not on the above list, please see the chapter for Using Older Radios.

Support for these older radios is coded into wfview. Some radios have more features implemented than others:

- ~~IC-703 (PTT using RTS)~~
- IC-706 (PTT using RTS)
- IC-706 MKGII (PTT using RTS)
- IC-718 (PTT using RTS)
- IC-746
- IC-746 Pro
- IC-756
- IC-756 Pro I/II/III
- ~~IC-736 (PTT using RTS)~~
- ~~IC-737 (PTT using RTS)~~
- ~~IC-738 (PTT using RTS)~~
- ~~IC-910H (needs more testing)~~
- IC-7100
- IC-7200
- ~~IC-7410~~
- ~~IC-7000 (needs more testing)~~
- ~~IC-7600 (needs more testing)~~

- ~~IC-7700 (needs more testing)~~
- ~~IC-7800 (needs more testing)~~
- IC-9100
- ID-31
- IC-R7100
- IC-R8500

NB: Crossed out radios have not been ported yet to wfview version 2. These radios have some support in version 1.64.

For the most up-to-date list, you can look in the code and see the currently supported radios.

Remember, even though you may not have as many features as the newer radios, wfview can get these radios to work remotely, for audio and control, with ease.

If you'd like to help add a new radio model to wfview, we have a new tool for just this purpose called RigCreator. You can read more about it here.

Supported Radio Features

Provided the radio supports these commands, you can expect the following radio features controllable from wfview:

- Spectrum and Waterfall (core radios only)
- Tuning control
- RIT
- Mode Selection
- Receive Filter selection
- Twin Pass Band Filter adjustment
- Receiver Filter width adjustment
- RF Gain (Receiver)
- AF volume (currently however, network-access instances assign this slider to the local computer audio volume)
- Squelch
- Transmit Power Level
- Mic, USB, ACC, or LAN modulation level (depending upon the source selected)
- Waterfall reference level
- Transmit (via command or RTS/DTR on some radios)
- Enable/Disable Automatic Antenna Tuner ("ATU")
- Begin ATU tuning cycle
- Repeater duplex/simplex selection
- Repeater offset programming
- Repeater tone, both transmit and receive
- Repeater tone selection, both CTCSS and DCS (aka "DTCS") for supported radios
- DCS inversion
- Split mode programming and activation
- Quick Split
- Main/Sub and A/B VFO functions
- Preamp selection
- Attenuator selection
- Antenna selection
- Bandscope Mode (center, fixed, scroll-c, scroll-f)
- Bandscope Span
- Bandscope Edge selection
- "To Fixed" and "Custom Edge" bandscope edge programming

- Bandscope enable/disable
- Band Stacking Register (“BSR”) read access to jump to previous band positions
- Modulation source selection (Mic, ACC, LAN, USB, etc)
- Metering:
    - S-meter
    - Power Output
    - SWR
    - ALC
    - Compression
    - Vd (scale may be incorrect on some radios)
    - Id (scale may be incorrect on some radios)
    - Center Tune (IC-R8600 only)
- Frequency Calibration, fine and course (IC-9700 only) (under Radio Settings).
- IF-Shift and/or Tunable Pass-Band Filter adjustment (press “Show More” on the View tab)
- Send CW using the radio’s built-in keyer

wfview

Open Source interface for Icom and Kenwood transceivers

Headless Server

This page describes how to setup a command-line wfview server. This topic is for **advanced users only**. For most users, the normal graphical server setup is sufficient.

The following documentation is very “beta” and is not ready yet… but feel free to have a look.

Building the Server

The normal wfview repository should first be cloned. (You can clone from other branches if you wish though — check our branch list.)

```
git clone --depth 1 --recurse-submodules https://gitlab.com/eliggett/wfview.git
```

Next, create a build directory and enter the build directory:

```
mkdir build
cd build
```

Now run qmake. Be careful to specify the correct pro file:

```
qmake ../wfview/wfserver.pro
```

Now run make. You can run make -j4 for a faster build using four cores (or more) if your hardware is up to the task:

```
make -j4
```

Configuring

wfserver uses the same type of configuration file as wfview, with many of the same options. The default location for the settings file is `~/.config/wfview/wfserver.ini`. The first time wfserver is run, it will create a basic configuration file if one does not exist yet. This is the simplest way to get started.

```
[General]
AudioSystem=0

[Radios]
1\AudioInput=default
1\AudioOutput=default
1\ForceRTSasPTT=false
1\GUID={ee2725b2-ca44-41c7-93f5-d9efa0bc4445}
1\RigCIVuInt=0
1\RigName=<NONE>
```

```
1\SerialPortBaud=115200
1\SerialPortRadio=auto
1\WaterfallFormat=0
size=1

[Server]
ServerAudioPort=50003
ServerCivPort=50002
ServerControlPort=50001
ServerEnabled=true
Users\1\Password=ks/\xyzzyw
Users\1\UserType=0
Users\1\Username=user
Users\size=1
```

[General]:

- AudioSystem sets the audio system to:
  - 0: QtAudio, default, most compatible and best usual choice
  - 1: Port Audio
  - 2: RT Audio

[Radios] (these start with “1\” since this is for the first radio):

- AudioInput: This is the input to wfview, from the radio’s receiver. “default” may work on some installs but it is best to set this variable manually. If possible, use wfview’s usual GUI client to see what the audio device would be called for your radio. You can also run wfserver with the debug flag to see a list of audio devices found:
  - `wfserver --debug`
- AudioOutput: This is the output of wfview, used for the radio’s transmit audio.
- ForceRTSasPTT: Set this to true if your radio interface needs the RTS pin used for PTT. For radios that have a PTT CI-V command (such as the 7300), set this to false.
- GUID: Internally-generated ID, do not alter
- RigCIVuInt: The CI-V address, in unsigned integer, for the radio. Note: If your radio has CI-V Transceive enabled, then you should not need to set this (set it to zero). wfserver will automatically find the radio. For older radios, set the CI-V address here to match the radio.
- RigName: String, radio name used in some debug messages. Recommended format is the same as the model — “IC-7300” for example. If set to “<NONE>”, wfview will fill this in with the model number of the radio discovered.
- SerialPortBaud: Set this to the same baud rate as the radio. Higher baud rates are recommended but not always better. For the IC-7300, you must use 115200 for the waterfall to work.
- SerialPortRadio: Set this to the device path for the serial port. “auto” will simply grab `/dev/ttyUSB0`.
- WaterfallFormat: Using this variable, wfview can force the waterfall into the “standard” ethernet or serial-style format. This is useful because some radio control programs expect the waterfall one or the other format.
  - 1: “combined” format, which Icom radios send over network
  - 0: “serial” format (default), which Icom radios send over serial. In this format, a single line of spectrum data is split up into about 11 chunks.
- size: Indicates how many radios are defined. Yes, it may be possible one day to handle several radios from one server. For now, this value should be set to 1.

[Server]:

- Server X Port: These are the three UDP ports that wfserver will use. Remember, do not remap the ports with a router, always keep the ports the same number on both sides of the router.

- ServerEnabled: Set to true to enable the server
- Users and passwords: Set the users and passwords here. Passwords can either be plaintext or “pseudo-encrypted” format. Use wfview to generate them on a GUI client.
- UserType:
    - 0 = Full User, has transmit capability
    - 1 = Full User with No Transmit
    - 2 = Monitor only, no controls

Audio Levels

You can adjust the input (capture) levels in the terminal using alsamixer and amixer. Be careful, sometimes there are adjustments that merely scale the level and then there are adjustments that actually set the ADC conversion gain. We want the conversion gain.

To bring up the capture levels for card #1, use this command (try card #0 and #2 as well until you see the controls you expect):

```
alsamixer -c 1 --view=capture
```

Once adjusted, make a note of the adjustment level. You can then script amixer to make this adjustment for you just prior to starting wfserver using a startup script:

~~amixer –set-volume=33 -c 1 –device=”capture”~~ (note this is wrong, I need to figure this one out)

Running Automatically

You can create a systemd script to automatically run wfserver on startup. **DO NOT EVER run wfserver as root.** This is entirely unnecessary and not supported at all. If you have permission errors, try and dig to the root of the issues (ie, serial port ownership or group membership, typically).

To run wfserver in a terminal with persistence, for now, it is recommended to use the “screen” session manager program. Simply hit control-a, d, to detach the screen session and keep wfserver running. We will add a background fork mechanism later.

wfview

Open Source interface for Icom and Kenwood transceivers

Developer’s Corner

Welcome Developers!

wfview is, and will remain, Open Source Software, developed by many people and licensed GNU GPL v3. By keeping the source open and available, wfview honors the spirit of amateur radio, providing material for the end-user to understand, modify, and contribute to. If you wish to use the wfview code in another project, you must keep the project also licensed by the GNU GPL v3. If you have questions, please ask.

The source code is available here: https://gitlab.com/eliggett/wfview/

Issue Tracker: https://gitlab.com/eliggett/wfview/-/issues

Compiling:

- Windows Compile Guide
- Windows Buildscript (requires setup)
- Linux Compile Guide
- Linux Buildscript
- macOS Build Directions Thanks, G0UQT.
- macOS Universal Qt6 automated build workflow
- Linux x86 Qt5 AppImage automated build workflow

Debug Build

To build with debugging symbols enabled, run qmake like this:

```
qmake ../wfview/wfview.pro CONFIG+=debug
```

More information is on the Backtrace page.

Developer Tutorials

With all the work on V2, our old tutorials were quite irrelevant. Stay tuned for more information as we get a chance to provide it.

For Radio Developers and OEMs

We’ve added a new page called “Built for wfview“, which shows what commands to implement in order to support wfview and/or wfserver on new radio designs.

Contributing

We welcome contributors, from both first-time and seasoned developers. To contribute:

1. Make an account at gitlab.com
2. Fork the wfview code into your own repository
3. Test your idea.
4. Cleanly and surgically make your final change and commit back to your own repository
5. Submit a “Pull Request” and optionally, drop us an email or a note on the forum to explain what you’re changing

Coding Style

wfview does not have a specific coding style. When contributing, try and make your changes easy to read. Add comments where needed. Try and stick with the sort of style you see within the source code. If you recognize our current style as something you’ve seen before, please let us know and we’ll change this paragraph to indicate it.

Older User Manual Pages

Note: This page contains outdated information for older version of wfview.

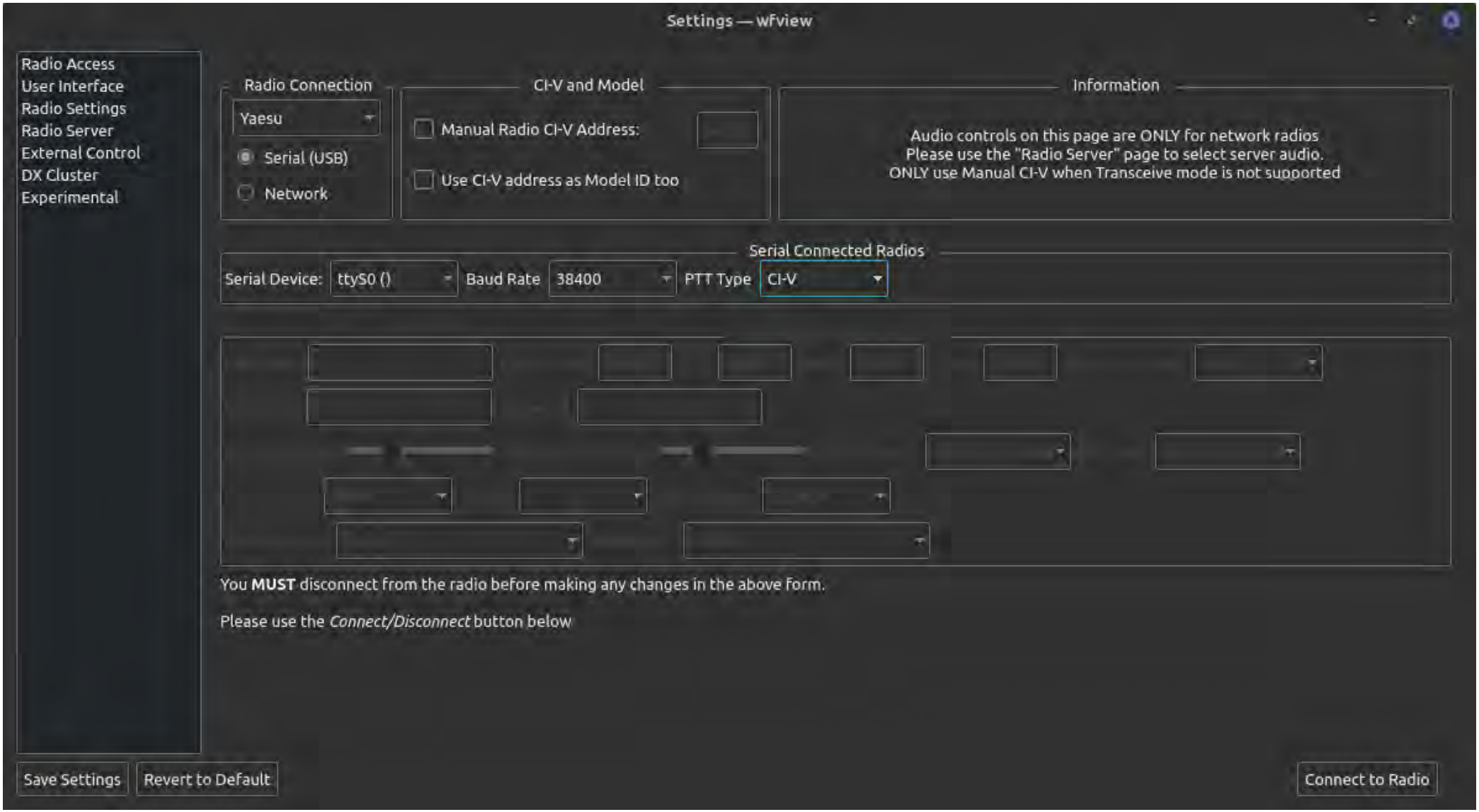
Current User Manual pages may be found here.

Versions 1.1-1.2:

- Getting Started
- View Tab
- SettingsTab
- Appearance
- Preferences File
- Remote Operation

Version 1.5-1.65:

- Getting Started
- View (main) Tab
- Settings Tab


Settings — wfview
Radio Access
User Interface
Radio Settings
Radio Server
External Control
DX Cluster
Experimental
Radio Connection
Yaesu
Serial (USB)
Network
CI-V and Model
Manual Radio CI-V Address:
Use CI-V address as Model ID too
Information
Audio controls on this page are ONLY for network radios
Please use the "Radio Server" page to select server audio.
ONLY use Manual CI-V when Transceive mode is not supported
Serial Connected Radios
Serial Device:
ttyS0 ()
Baud Rate
38400
PTT Type
CI-V
You MUST disconnect from the radio before making any changes in the above form.
Please use the Connect/Disconnect button below
Save Settings
Revert to Default
Connect to Radio

wfview

Open Source interface for Icom and Kenwood transceivers

Videos

The following is a list of useful videos we found on the web. Please be sure and show your support to these excellent presenters here.

In this video, Kevin Loughin (KB9RLW) demonstrates using wfview on linux, including controlling an IC-705 and a remote IC-9700 with a sideband QSO.

In this video, VK3FS shows how to operate the IC-705 with wfview for remote operation with WSJT-X.

In this video, DL6JN shows how to set up a Raspberry Pi as a server for remote operation of an IC-7300 with a windows client.

In this video, VK3FS shows how to forward the ports from icom radios through home routers to allow for remote control.

In this video, F8DSN gives an excellent overview of the program, including many options for customizing the appearance.

In this video, DL2YMR demonstrates wfview using a Mac connected to an IC-7300, IC-9700, and an IC-705.

In this video, W6EL shows how to install wfview on linux using the official build script. This is a great way to get the latest version of wfview quickly on debian-based platforms, including Mint, Ubuntu, and Raspberry Pi OS.

In this video, W6EL shows how to customize the appearance of wfview version 1.5.

In this podcast interview, members of the wfview team take a deep dive into the world of wfview software development. Here is a link to the podcast landing page. wfview is covered in episode 412.

wfview

Open Source interface for Icom and Kenwood transceivers

Download

wfview is free and open source software, licensed under the GNU/GPL V3.

Our source is controlled in a gitlab repository at the following URL: https://gitlab.com/eliggett/wfview

Gitlab source-referenced releases can be viewed here.

Linux and Raspberry Pi users: please see here.

Errata is available for the V2 release.

IMPORTANT: We have discovered that the Windows release installer does not work if your desktop is in “Dark mode”. You can read more about it in the Errata. Please temporarily change your desktop to the normal “bright” theme and then the installer will work correctly.

Latest Official Release (Windows and macOS)

Windows (x86) release v2.21 39.5 MB 155 downloads

Released 2nd Feb 2026 - Only install if your PC is Windows 7 and/or 32bit sha256:...

Download

Windows (x64) release v2.21 71.9 MB 808 downloads

Released 2nd Feb 2026 - Most users with modern Windows (10 or above) will want to...

Download

MacOS (universal) v2.20 53.6 MB 134 downloads

Universal Binary for MacOS 12 and above Released 1st Feb 2026 sha256sum: afa585ec30f0cf3a8f40b800b690ac7a83a6345c243a6fac5692a126df7f670d ...

Download

MacOS (universal) v1.64 (legacy) 64.50 MB 5123 downloads

(legacy version) Universal Binary for MacOS 15 and above if issues with v2 ...

Download

Windows (x86) release v1.64 (legacy) 38.1 MB 7773 downloads

(legacy version) Released 16th July 2023 - Windows 7 and lower or if v2 causes problems ...

Download

We have discovered that one feature of Icom rigs that wfview relies on, CI-V transceive, is disabled on a number of users radios. This MUST be enabled for wfview to function correctly and can be configured via MENU/Set/Connectors/CI-V/CI-V Transceive = ON. Please see the Getting Started chapter of the User Manual for more details.

macOS users may also choose to install using macports (although it may be somewhat outdated).

Weekly Automated Builds (Windows and Linux currently)

For weekly automated builds, please make sure you have installed one of the builds from this page first. Verify that the official release is working with your computer and radio. If so, you may download a weekly automated build from this link:

https://www.wfview.org/public_builds/

Please see our page on how to run the automated builds for more information.

Our Patreon community members receive nightly builds. Sign up, membership starts at only $2 per month, and helps support the software we are all enjoying.

Other files that may be useful

These files may be useful for some users, you may be pointed to download one of them by a core team member

Printable Manual 2023-12-30 3.52 MB 6174 downloads

OBSOLETE This printable manual is perfect for situations where you might not have...

Download

Portaudio lib and include files for building wfview in Windows 8.40 MB 3034 downloads

...

Download

Older versions may be found here.

Notes:

The Windows build requires the latest Microsoft Visual C++ Runtime (x86) which is included in the download but is also available from:**https://aka.ms/vs/16/release/vc_redist.x86.exe**

The MacOS build (since v1.2c) is signed and notarised by Apple so should not require approval to run. If you have previously downloaded this build and were unable to get rig audio working, please re-download as we have (hopefully) fixed this. Currently, we recommend trying higher latency on the receive audio if there are issues, and/or using an external audio interface (may or may not help).

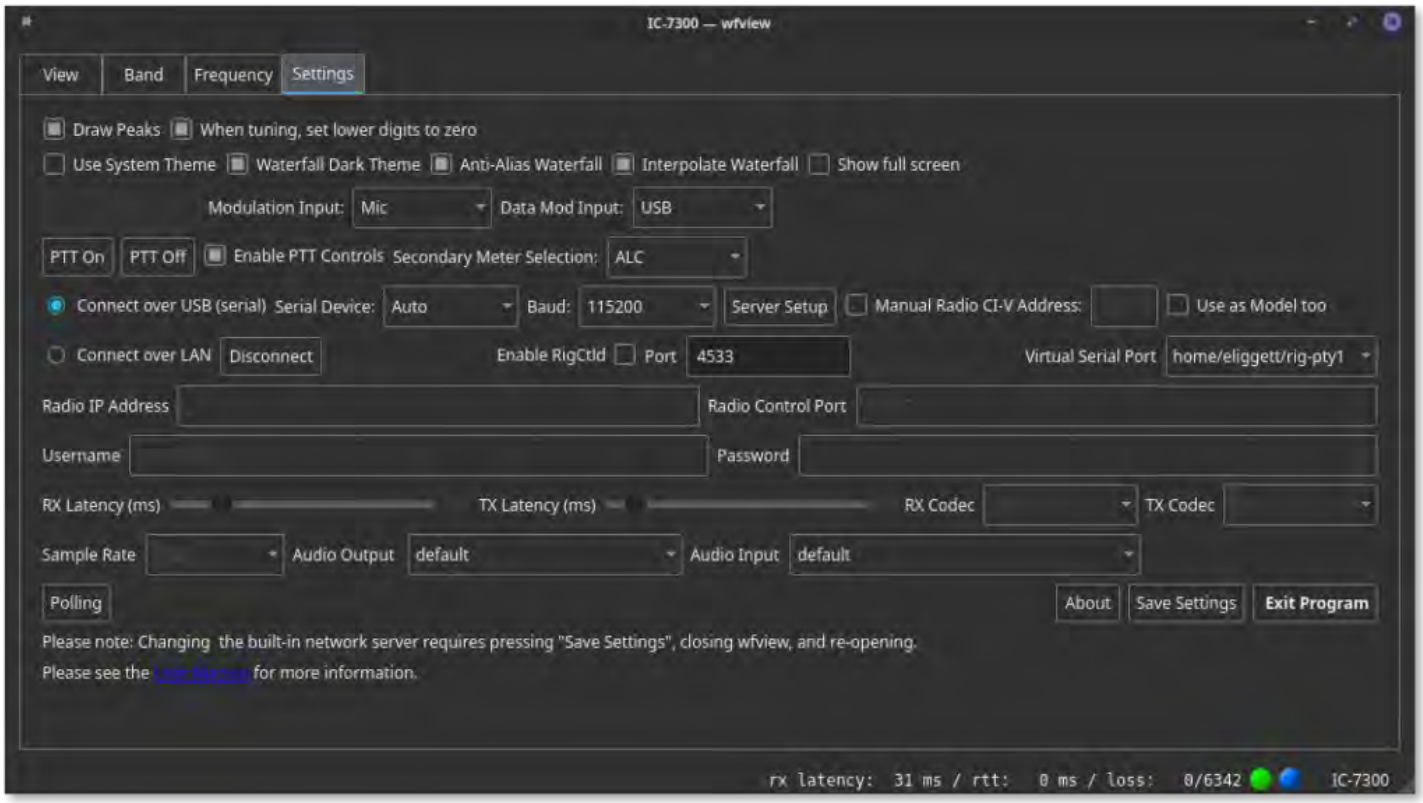
If you experience any issues with these downloads please sign up on the wfview support and discussion forum at https://forum.wfview.org/. Once you sign up, you can use either email or the web interface to post questions and discuss wfview.

Building From Source:

Please see our Developer's Corner

Settings Tab (V1.1)

This is the Settings Tab:



From the settings tab, numerous program and radio settings may be adjusted.

Radio Connection Settings:

These controls are for the user to select and connect to the radio. When changing these controls, make sure to press “Save Settings” to save the settings for next time. Also, if a different rig is being selected after connecting to a rig, save the settings and re-launch wfview.

Serial Connections:

If an Icom radio is connected with a USB cable, wfview will generally find and use this connection automatically, without any intervention from the user. If you are having difficulty getting a USB connection to work, you may need to change port permissions. We have a section of the users guide for this, but to temporary try changing permissions, use the following command:

sudo chown $USER /dev/ttyUSB*

The serial device may also be specified using the Serial Device combo box. “Auto” is for automatic detection. wfview automatically populates this list with normal serial devices. If you are using a pseudo-terminal (such as from socat), you may select “Manual…” and enter in the serial port desired. Be sure and press “Save Settings” if you use

this feature. And note that you may edit the serial port manually in the preference file, as detailed in the preferences section of this manual.

- Baud: For radios with waterfall view, such as the 7300, you must select 115200 at both the radio and within wfview, otherwise the radio will not generate a waterfall view. For all radios, make sure you match the radio's baud rate setting.
- Manual CI-V address: Use this box only if you cannot enable CI-V Transceive on the radio. It is strongly recommended to keep CI-V Transceive ON, and to not check this box. Some older radios do not respond to our Rig-ID requests, and for those radios, please supply the CI-V address here. The address is supplied in hex, which is the same way the CI-V address is given in the radio's menu.
- Use as Model too: Check this box for older radios that do not have a Rig Identification command. wfview will assume that your supplied CI-V address is the manufacturer's default address and use this address as the radio's model number, which is what modern radios do. Don't check this box unless you need to.
- Server Setup: This button enables a built-in CI-V and audio server. Use this feature to take a radio without an OEM network connection and "serve" it to the network. This button is intended for use with the IC-7300, but may be used on any CI-V radio. Once the server is running, you can launch a separate instance of wfview on another computer and connect to the radio. The other instance of wfview will have access to the radio's audio and control.

Network Connections:

To connect to a radio over WiFi or Ethernet, click on the "Enable LAN" radio button. This button also enables the text boxes so that radio information may be entered. You will need to supply the radio's IP address (or hostname), Radio Control Port (the default is 50001), Username, and Password. These settings are defined at the physical radio. Also be sure to enable network control. Once filled in, press "Connect". If successful, don't forget to press "Save Settings".

- Connect/Disconnect: This button causes the connection to be made to the rig, or, for the existing connection to drop.
- RX Latency (ms): Adjusts the buffer size for the receive audio. If you encounter hiccups in the receive audio, you may want to increase this value. For lower latency, such as an at-home ethernet connection, lower values may be used. The default is 150ms.
- TX Latency (ms): This is an adjustment for the transmit audio buffer. Longer values are needed for lower-quality connections. The default is 150ms.
- RX Codec: Select the encoding method for the receive audio. LDPCM is uncompressed 16-bit audio, like a CD. uLaw is a lower-bitrate and slightly lower-quality option involving logarithmic spaced bits. If, however, you are connecting to a wfview server, you may also select "Opus" which is a high quality lossy compression option with significantly lower bandwidth requirements. Opus is recommended when available, but remember, there are no radios with this option built-in; it will only work with radios using a local wfview server (such as an IC-7300).
- TX Codec: Same as above.
- Sample Rate: This is the sample rate applied to both transmit and receive audio streams. 48k is the default. Lower sample rates will lower the required network bandwidth, and generally sound fine for most usage.
- Audio Output: Use this control to select the audio output device for your system. Linux users may see several devices here. wfview attempts to select the "default" system device on startup, and that should generally be used if at all possible. Using the system's control panels to select and test the default device (for example, a USB headset), prior to the launching of wfview, is strongly advised.
- Audio Input: Same as above but for the transmit audio source.

PTT:

Use these controls to manually toggle the PTT (transmit) on and off. The "Enable PTT Controls" checkbox must be checked in order to send transmit commands. Note that this control does not lock out the transmit button on the radio (although that can be done from the physical radio's settings). wfview also has an automatic five minute

transmit timeout. For radios with a PTT command (most), wfview will send the PTT command over CI-V. For radios without this command, wfview will toggle the RTS pin of the serial converter.

Appearance and Behavior Settings:

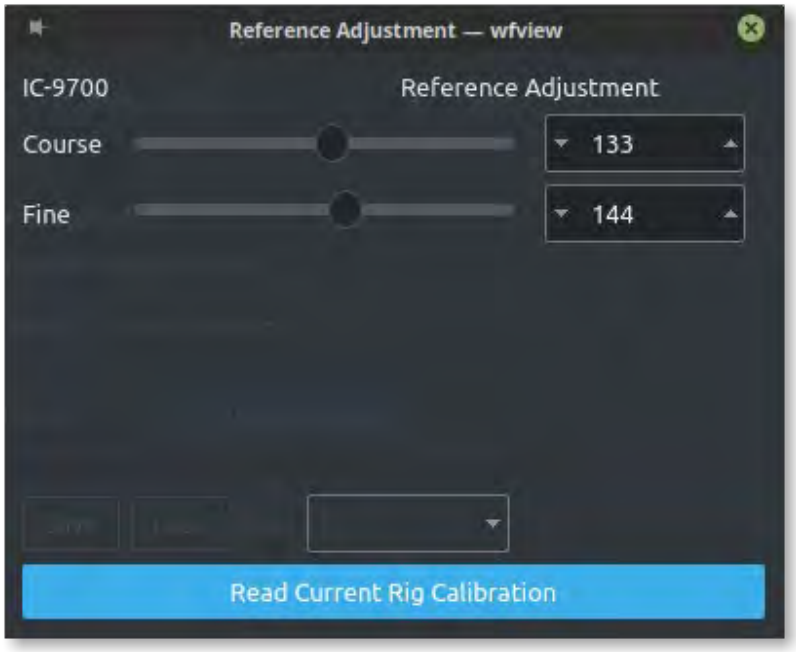
- “Draw Peaks” enables a backdrop on the spectrum plot, which holds the maximum value for each frequency. The peak data is cleared upon changing the scope bounds or pressing the “Clear Peaks” button.
- “Waterfall Dark Theme” toggles between a bright theme and a dark theme for the waterfall and spectrum plot. The colors can be adjusted by editing the preference file. This is detailed more in the preferences section of the manual.
- “Use System Theme” toggles between a built-in theme (qdarkstyle) and the operating system’s supplied theme. For linux users, the operating system theme can usually be adjusted using your system preferences. The built-in theme may be customized by editing the installed stylesheet (linux only) or specifying a custom stylesheet in the user preference file.
- “Show full screen” toggles between windowed and full screen mode. Pressing F11 also toggles this state.
- “When tuning, set lower digits to zero”: This checkbox causes the frequency to round to the selected tuning step. It also applies for double-clicking in the waterfall.
- Anti-Alias Waterfall: Toggles the anti-alias smoothing algorithm for the waterfall view. Recommended.
- Interpolate Waterfall: Toggles the interpolation of in-between pixels on the waterfall view. Useful with large screens.

Modulation (mic):

These two combo menus allow the user to select the modulation (transmit audio) source, for both normal and data modes. For remote operation using the OEM network connection, select “LAN” for both. For rigs using a USB connection where the audio is being served using the wfview server, select USB. The modulation level for the selected source can be adjusted on the View tab, and will be labeled with the same name as the source.

Calibration (Reference):

Adjust Reference (IC-9700 only): This button brings up the Adjust Reference dialog box. From this box, the radio’s internal reference may be adjusted. Unlike the radio’s user interface, this allows you to view the waterfall display while making adjustments to the reference. Calibrating the radio to a known-good source on the air, such as a beacon, is a snap.



IC-9700 Reference Adjustment

To adjust the reference, tune into a known-good station or connect appropriate test signal generation equipment. Now press “Read Current Rig Calibration”. wfview will read the radio’s current settings and update the UI controls. From here, it is recommended to switch to the View tab of wfview. You can now drag the Course and Fine adjustments and see the spectrum change. For best results, begin with the “Center” spectrum mode. Bring wfview in to the 5 KHz span. Be mindful of the mode selected. In sideband mode, the IC-9700 will center the display at the passband center. In wfview, the spectrum’s axis always indicate real frequency, so use this as a guide. Once zoomed in, press “ToFixed” to fix the spectrum bounds. Now adjust the Course reference until the test signal’s waterfall mark lands on the prescribed frequency.

Other

Polling: Click here to set the polling rate. The rate is automatically computed from the baud rate upon rig connection, however, there can be circumstances where slower polling is useful, such as when sharing the connection to the radio with other programs.

Secondary Meter Selection: Pick the type of secondary meter desired (if any). The secondary meter is shown under the primary meter on the main tab of the program.

Save Settings: Press this button to save the current wfview configuration to the settings file. Attributes saved include:

1. Radio connection type and interface
2. Appearance settings
3. Window size
4. Full screen status
5. waterfall and spectrum plot divider position
6. Frequency memories

The preferences are stored plaintext, and contain many other useful things, as is explained in the preferences page.

About: Shows the about dialog box, including program build date and version

Exit Program: Closes the program

Debug (debug builds only): Runs debug functions. This button is disabled on release builds.

Satellite Ops: Incomplete feature to facilitate satellite operation on the IC-9700. This button is disabled on release builds.

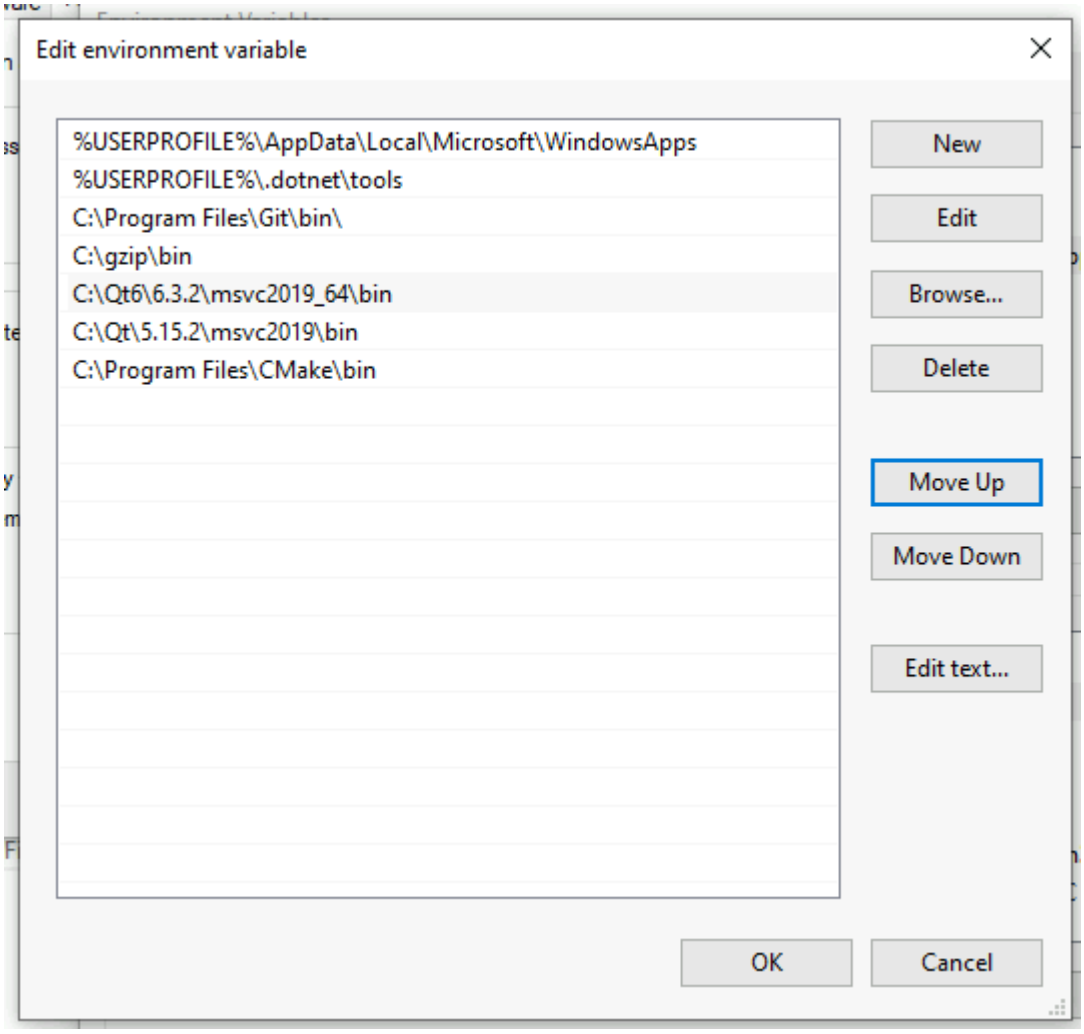
How to compile with Windows

Prerequisites

You should install the following software first:

- git (add to %PATH%)
- gzip binary (add to %PATH%), needed to decompress the QCustomPlot source *from the command line*
- Microsoft Visual Studio with nmake, C++ compiler, and debug tools. 2019 recommended (other versions may work too).
- For 32-bit: qt5, version 5.15.2 (scroll down to “Downloads for open source users”, there is no need to pay)
    - Make sure to install these optional components too:
        - qt creator
        - qmake (add to %PATH%)
- For 64-bit: qt6, version 6.3.2. Place the preferred qt binary folder within the %PATH% variable ahead of any other qt versions. The folder is named like this: `C:\Qt6\6.3.2\msvc2019_64\bin`
- QCustomPlot (source files), version 2.0.1 (other versions may work too)
- Opus (use our build)
- RT Audio (clone from https://github.com/thestk/rtaudio, checkout ‘**tags/5.2.0**‘. No need to build, just clone it)
- PortAudio (use our build)
- Eigen (see Setup below)

Make sure the version of qt and MSVC are compatible. My environmental PATH variable looks like this:



Setup

The following directory structure is recommended and how the wfview.pro file is designed to work. Please verify that the following files and folders exist. Assume the git command has been run from `Documents/projects/wfview-master`:

wfview source code: `wfview-master/wfview`

qcp header: `wfview-master/qcustomplot/qcustomplot.h`

qcp source: `wfview-master/qcustomplot/qcustomplot.cpp`

opus include: `wfview-master/opus/include`

opus library: `wfview-master/opus/win32`

RT Audio: `wfview-master/rtaudio`

Port Audio: `wfview-master/portaudio/msvc/Win32/Release/portaudio_x86.dll`

Eigen: `wfview-master/eigen`

(from the wfview-master directory, clone the repo: `git clone https://gitlab.com/libeigen/eigen.git`)

build directory: `wfview-master/build`

Clone the repository

```
cd wfview-master
```

```
git clone https://gitlab.com/eliggett/wfview.git
```

Command-line Build

To build from the command line, you can use build.bat, or follow these steps:

`call "C:\Program Files (x86)\Microsoft Visual Studio\2019\Community\VC\Auxiliary\Build\vcvars32.bat"`

This sets many of the necessary environment variables. (You can also just start by launching the “x86 Native Tools Command Prompt for VS 2019”)

Now make the directory for building, enter the directory, run qmake, and then nmake (nmake is apparently microsoft’s make):

```
mkdir build
cd build
qmake ..\wfview\wfview.pro
nmake
```

The resulting file is release\wfview.exe

qtcreator Build

Launch qtcreator

Open wfview.pro

Configure the project using the standard options.

Press Build. qtcreator will create a build directory for either Debug or Release, with the objects and executable.

Debug and Stack Tracing

This page will show how to generate a debug build and how to handle core dumps.

Concept:

Build wfview with the debugging symbols intact. This will guide gdb.

If this build of wfview crashes, load up gdb and tell it to generate a stack trace from the core dump file.

Build:

Follow the usual build directions, except, change the qmake step to this:

```
qmake ../wfview/wfview.pro CONFIG+=debug
```

You may need to obtain a debug-enabled version of qcustomplot. If this is difficult (it is), just change the wfview.pro file so that instead of “libqcustomplotd” it shows “libqcustomplot”

Install this version of wfview, or keep it uninstalled and run it from this directory. Make sure you have installed the rig files recently though.

Run wfview:

Now run wfview and recreate the crash. Ideally, pass the `--debug` flag to wfview so that additional logging is created. When wfview crashes, you will see a message like this:

```
Segmentation fault (core dumped)
```

View the core:

The core file will likely be in `/var/lib/systemd/coredump` or in your home directory, depending on how your system is setup.

If the file ends in “zst” you will need to decompress the file first. For example:

```
zstd -core core.wfview.1000.54c8d1d763e54b76938af40630022cdf.12386.1754757191000000.zst -o /tmp/c
ore.dump
```

Now launch gdb as shown, with the path to your debug build wfview and core file:

`gdb /home/w6el/Downloads/wfview--20250803--1202222/debugbuild/wfview /tmp/core.dump`

You will see:

`Reading symbols from /home/eliggett/Downloads/wfview--20250809--08-37-25/debug3/wfview...`

You may need to hit enter to complete the initial text output.

Now issue the “bt” command to show the backtrace:

```
--Type <RET> for more, q to quit, c to continue without paging--
[Current thread is 1 (Thread 0x7ff2f100d640 (LWP 12412))]
(gdb) bt
#0 0x00007ff33e9e2f5e in snd_pcm_avail_update () at /lib/x86_64-linux-gnu/libasound.so.2
#1 0x00007ff33804c6ac in () at /usr/lib/x86_64-linux-gnu/qt5/plugins/audio/libqtaudio_alsa.so
#2 0x00005625c0bfd1b6 in audioHandler::convertedOutput(audioPacket) (this=0x7ff2c8005c90, packet=
...) at ../wfview/audiohandler.cpp:312
#3 0x00005625c0d5a22e in audioHandler::qt_static_metacall(QObject*, QMetaObject::Call, int, void*
*)
(_o=0x7ff2c8005c90, _c=QMetaObject::InvokeMetaMethod, _id=11, _a=0x7ff2c001f188) at moc_audiohand
ler.cpp:165
#4 0x00007ff33f2d341e in QObject::event(QEvent*) () at /lib/x86_64-linux-gnu/libQt5Core.so.5
#5 0x00007ff33fff3713 in QApplicationPrivate::notify_helper(QObject*, QEvent*) () at /lib/x86_64-
linux-gnu/libQt5Widgets.so.5
#6 0x00007ff33f2a5e3a in QCoreApplication::notifyInternal2(QObject*, QEvent*) () at /lib/x86_64-l
inux-gnu/libQt5Core.so.5
#7 0x00007ff33f2a8f27 in QCoreApplicationPrivate::sendPostedEvents(QObject*, int, QThreadData*) (
) at /lib/x86_64-linux-gnu/libQt5Core.so.5
#8 0x00007ff33f2ffa67 in () at /lib/x86_64-linux-gnu/libQt5Core.so.5
#9 0x00007ff33dedbd3b in g_main_context_dispatch () at /lib/x86_64-linux-gnu/libglib-2.0.so.0
#10 0x00007ff33df312b8 in () at /lib/x86_64-linux-gnu/libglib-2.0.so.0
#11 0x00007ff33ded93e3 in g_main_context_iteration () at /lib/x86_64-linux-gnu/libglib-2.0.so.0
#12 0x00007ff33f2ff112 in QEventDispatcherGlib::processEvents(QFlags<QEventLoop::ProcessEventsFla
g>) ()
at /lib/x86_64-linux-gnu/libQt5Core.so.5
#13 0x00007ff33804dda3 in () at /usr/lib/x86_64-linux-gnu/qt5/plugins/audio/libqtaudio_alsa.so
#14 0x00007ff33804de1d in () at /usr/lib/x86_64-linux-gnu/qt5/plugins/audio/libqtaudio_alsa.so
#15 0x00007ff3405ad605 in QAudioOutput::~QAudioOutput() () at /lib/x86_64-linux-gnu/libQt5Multime
dia.so.5
#16 0x00007ff3405ad61d in QAudioOutput::~QAudioOutput() () at /lib/x86_64-linux-gnu/libQt5Multime
dia.so.5
#17 0x00005625c0bfb274 in audioHandler::~audioHandler() (this=0x7ff2c8005c90, __in_chrg=<optimize
d out>) at ../wfview/audiohandler.cpp:35
#18 0x00005625c0bfb316 in audioHandler::~audioHandler() (this=0x7ff2c8005c90, __in_chrg=<optimize
d out>) at ../wfview/audiohandler.cpp:39
#19 0x00007ff33f2d31c3 in QObject::event(QEvent*) () at /lib/x86_64-linux-gnu/libQt5Core.so.5
#20 0x00007ff33fff3713 in QApplicationPrivate::notify_helper(QObject*, QEvent*) () at /lib/x86_64
-linux-gnu/libQt5Widgets.so.5
#21 0x00007ff33f2a5e3a in QCoreApplication::notifyInternal2(QObject*, QEvent*) () at /lib/x86_64-
linux-gnu/libQt5Core.so.5
#22 0x00007ff33f2a8f27 in QCoreApplicationPrivate::sendPostedEvents(QObject*, int, QThreadData*)
() at /lib/x86_64-linux-gnu/libQt5Core.so.5
#23 0x00007ff33f0b8389 in () at /lib/x86_64-linux-gnu/libQt5Core.so.5
#24 0x00007ff33f0b8ca9 in () at /lib/x86_64-linux-gnu/libQt5Core.so.5
#25 0x00007ff33eb22ac3 in start_thread (arg=<optimized out>) at ./nptl/pthread_create.c:442
#26 0x00007ff33ebb4850 in clone3 () at ../sysdeps/unix/sysv/linux/x86_64/clone3.S:81
(gdb)
```

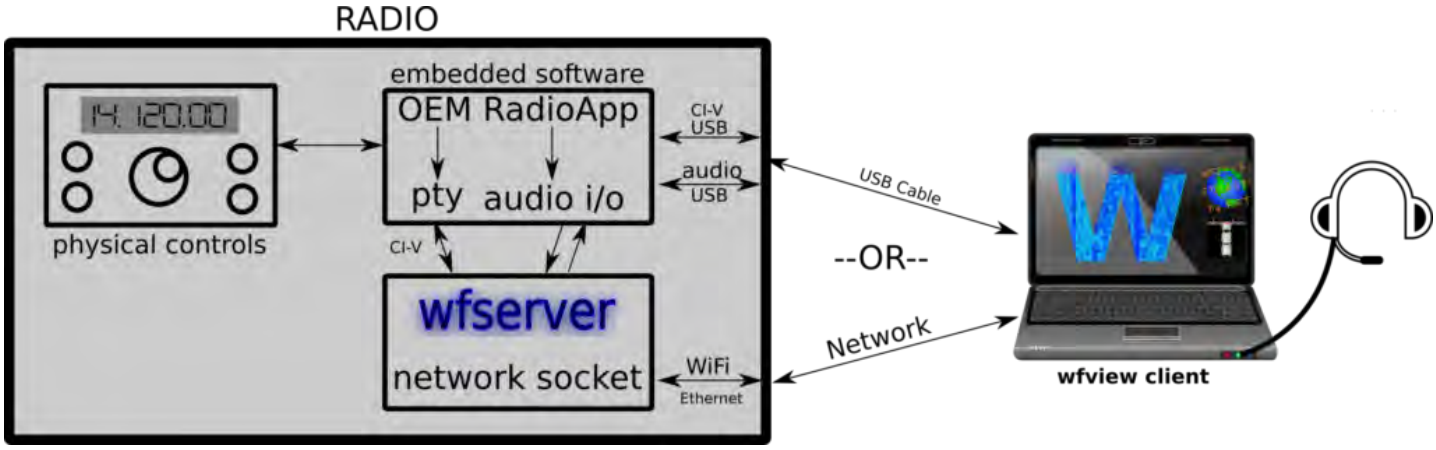
wfview

Open Source interface for Icom and Kenwood transceivers

Built for wfview

This page describes how to design a radio to work best with wfview. wfview is, in part, designed around the CI-V protocol, which is used by over a hundred existing radios. The CI-V protocol is simple, human-readable, and ideal for modern radios. A CI-V bus allows for multiple radios and computers on the same CI-V physical (or software) interface, and is thus ideal for the modern world of software controlled radios and networking. You can read more about the CI-V protocol here.

Update: we have added the Kenwood protocol (as used on the TS-890S for remote access) to wfview. This protocol is likely simpler to implement as it simply uses ASCII TCP/IP for commands and RTP for the audio (Signed LE16-bit LPCM). You might consider the Kenwood protocol instead of Icom's CI-V method of networking. The Kenwood protocol is documented in their TS-890S computer connectivity guide. This page was written before wfview supported the kenwood protocol, but the ideas and concepts remain the same.



Connectivity

There are two primary methods wfview can connect to a radio.

1. Using a USB port, which emulates a basic serial port
2. Using a network connection (WiFi or Ethernet typically)

For network connections, wfserver can be run directly on the radio to serve the CI-V control and full-duplex audio over the network. wfserver needs a CI-V bus to connect to, typically a pseudo-port ("pty") hosted on the device, and an audio device to use (or a loopback to an audio device). With these two things made available, running wfserver on the radio will permit network control from wfview and some other popular software programs.

Features to implement

wfview is built off a set of somewhat "classic" CI-V commands dating back to the 1980s. These commands were selected for the highest level of compatibility with the most radios on the market and compatibility with most software programs. Using this set of commands assures your radio can be pretty easily controlled, not just by wfview, but by many other programs.

One of the most popular and fun features to implement is of course spectrum output (waterfall). wfview can contend with any reasonable width of spectrum. Care must be taken not to exceed the bandwidth of the serial port. The exact details of the data are described below.

Please be sure to give the user a way to set the audio input to use the LAN audio versus the MIC audio. Without this selection the user does not know which audio source will be transmitted. Follow the example from other radios which provide this functionality.

Other features include frequency, mode, PTT, metering (S-meter, power, ALC, SWR, etc), Rig Identification, attenuators, preamp, mic gain, RF Gain, TX Power, Antenna switch, input mod source, etc.

The CI-V bus should fully support “CI-V Transceive” which means two principle things:

1. The radio will respond to data sent to address “0x00” (“Broadcast”)
    1. This is how wfview and many other software programs “find” connected radios without making the user fill out CI-V address and model number data.
2. The radio will, without a query command, send out changes to frequency and mode, when these changes occur from the radio’s physical controls (ie, a knob twist).
    1. This makes the radio and software connection much more responsive and up-to-date. wfview does not hammer the radio for information since the radio freely sends applicable changes over CI-V.

CI-V Transceive should be able to be disabled for legacy applications.

Command Set

The following tables show the expected command sets that are supported by wfview.

Common to all radios:

Command Name	Core Syntax	Notes
Frequency	0x00	
Mode	0x06	
PTT	0xC1 0x00	
S-Meter	0x15 0x02	
Power Meter	0x15 0x11	
Rig ID	0x19	Important: Pick unique reply
Attenuator	0x11	
Preamp	0x16 0x02	
Mic Gain	0x14 0x0B	Or: Copy IC-9700 level commands for MIC, USB, LAN
RF Gain (receive)	0x14 0x02	
TX Power	0x14 0x0A	
Squelch	0x14 0x03	

Unique to each radio:

For these features, wfview has unique commands set up for popular radios. Please pick a popular and supported radio to “clone” these commands from rather than making your own set. The IC-9700, 7300, and 705 are great examples for these commands.

- Antenna Switch
- Input (MOD) source select (MIC, ACC, USB, LAN, etc)
- Modulation Level. Either just use the above mic gain for all, or, implement the full set from the IC-9700 for each of MIC, ACC, USB, and LAN, as applicable.
- Spectrum Data (see below)

Spectrum (waterfall) Data

Following in the footsteps of giants such as the IC-7300, radios should support sending spectrum data (“waterfall”) out at regular and automatic intervals. The IC-7300 spectrum commands are recommended as a baseline for any implementations.

Commands:

Command Name	Command	Notes
Scope Data Prefix	0x27 0x00	actual scope data
Scope Data Output enable/disable	0x27 0x11	
Scope Mode (center, span, scroll) set/get	0x27 0x15	
Scope Span get/set	0x27 0x15	
Scope Edge get/set	0x27 0x16	
Scope Reference Level get/set	0x27 0x19	

How the spectrum data is sent:

Spectrum data is sent one “line” at a time, which, when combined, form a waterfall display. Icom cuts these lines into smaller “chunks” for slower serial port links (11 chunks per line, to be exact), however this is not chiefly necessary for modern hardware. The spectrum data lines include the frequency range, either given as a +- “span” or as the start and end frequencies, depending on the mode. Spectrum amplitude is sent as numbers ranging from 0-200 (lower ranges are fine as well), with one bytes per pixel sent.

Typical data rates over network are around 30 lines per second. For USB connections, typical rates are 5-10 Hz.The messages are like this:

Start:
Main/Sub[00=main 01=sub] SequenceNumber[01-11] TotalSeqneuceSize[01=LAN, 11=USB]

For LAN radios, the next part is like this:
Center/Fixed [00=cent, 01=fixed]

Now it will send slightly different information depending on if we are in “center” or “fixed” mode:

Center:

CenterFreq[encoded like standard icom frequencies] Span [encoded similar to standard frequency encoding, see IC-9700 CI-V manual, page 24]

Fixed:
StartFrequency[encoded like standard icom frequencies] EndFrequency[encoded like standard icom frequencies]

Range information, to tell the user if they are out of the scope’s range somehow:
00 = In range, 01 = Out of range

Now waveform data. This is the data taken from the magnitude of the Fourier Transform, sometimes called the “FFT” data. The data are 475 points long (each “point” is basically a bin composed of a segment of the scope span). Each point may have values from 0 to 160 (0x00 to 0xA0). We believe these are logarithmic spaced, although I don’t know of anyone that has checked it.

For USB connections, it is very similar. The first “chunk” is a message with only the “information” about the scope settings (center/fixed, start/stop freq, span). The next 10 chunks are pieces of the 475 waveform bits. These 10 chunks include the Main/Sub, Sequence number (2 through 11), and of course the wave data.

New radio designs could send wider spectrum and also potentially use numbers a bit more than 200 for more dynamic range. wfview can easily adapt to changes such as these, provided the radio identifies itself in a way we can understand.

Example Spectrum Data

Here is an example of spectrum data from the IC-9700. The data are not split into chunks in this example:

```
27 00 00 01 01 00 30 99 19 44 01 00 00 25 00 00 00 0d 0c 00 06 04 08 04 00 00 08 10 0c 00 07 0a 05
07 06 00 00 00 00 09 07 0a 01 0c 00 0e 10 00 00 00 02 02 00 00 06 08 00 0f 14 08 02 00 03 00 00 00
1c 23 06 09 00 00 00 00 0c 09 05 00 00 00 04 04 08 02 07 03 0c 28 2c 13 09 00 01 02 00 00 03 05 07
05 04 03 06 10 0e 00 00 00 09 08 11 0f 03 03 00 00 00 00 00 0b 0e 09 0e 0c 0a 00 00 09 0a 00 00 02
00 09 0c 09 0e 0a 03 06 04 04 0b 03 07 01 00 00 02 00 00 08 00 05 07 07 02 0c 0c 10 10 08 00 04 00
12 0f 05 02 00 00 05 0b 06 00 00 01 0d 0b 04 03 00 05 04 0d 09 00 00 08 00 00 01 03 00 0b 0c 08 09
00 00 06 00 00 00 05 04 00 02 00 00 06 11 13 00 00 0f 0c 00 02 08 00 06 00 02 04 00 02 0d 11 00 00
00 00 00 00 05 0b 00 00 00 07 05 0e 0b 00 00 05 00 03 05 00 0a 08 00 00 00 00 00 06 04 00 12 19 0f
07 0a 06 0f 15 13 02 01 01 0e 00 0c 0d 07 04 00 09 09 06 00 02 02 00 00 07 0b 00 00 0d 05 02 00 00
00 00 04 08 00 06 05 00 01 07 0c 02 00 00 00 00 00 00 00 00 04 00 00 06 00 09 00 00 03 04 04 03 03
03 07 07 00 06 00 00 00 03 01 03 00 06 00 00 06 0a 00 00 01 07 08 00 0a 10 05 00 00 00 01 00 01 00
00 00 00 04 00 08 0d 07 02 04 05 06 0f 0f 08 00 00 00 07 03 00 04 00 11 0d 00 00 00 00 00 0b 0a 00
01 00 00 00 05 02 00 0e 0f 07 0a 00 02 00 00 00 00 03 10 0f 00 00 04 00 06 06 00 08 00 0b 06 00 02
00 00 02 0f 25 24 15 1c 0d 09 03 00 06 04 10 0f 0d 05 01 00 00 00 08 0b 06 08 00 00 00 00 00 00 06
07 07 00 00 04 07 04 06 09 08 05 00 00 00 00 04 00 00 08 00 0c 00 08 07 01 06 00 00 09 0b fd
```

Here is an example of spectrum data from the IC-7300, which is split into chunks:

Chunk #1 includes center/fixed mode at [05]. No pixels.

```
INDEX: 00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17
```

```
DATA:  27 00 00 01 11 01 00 00 00 14 00 00 00 35 14 00 00 fd
```

Chunks #2 to #10, which represent 50 pixels of spectrum data each:

```
INDEX: 00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55
```

```
DATA:  27 00 00 07 11 27 13 15 01 00 22 21 09 08 06 19 0e 20 23 25 2c 2d 17 27 29 16 14 1b 1b 21 27
1a 18 17 1e 21 1b 24 21 22 23 13 19 23 2f 2d 25 25 0a 0e 1e 20 1f 1a 0c fd
```

Note: Chunk number is identified at index 03, with the total number of chunks at index 04. In this example, chunk 7 out of 11 is indicated.

Chunk #11, final chunk, 25 pixels:

```
INDEX: 00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
```

```
DATA:  27 00 00 11 11 0b 13 21 23 1a 1b 22 1e 1a 1d 13 21 1d 26 28 1f 19 1a 18 09 2c 2c 2c 1a 1b fd
```

As you can see, the “single piece” format used on the IC-9700 for network spectrum is much easier.

Licensing

wfserver is distributed under the GNU/GPL Version 3. This means that you may include it inside your radio provided that it is not modified, or, provided that any modifications to wfserver are made available to the end-user as source code. Most likely you will not need to modify wfserver any, and may simply use it as-is. If you find modifications are needed, please contact our team and we may actually make the modifications part of the standard wfserver (and thus relieve you of needing to distribute the source code), depending upon what has been changed.

Getting Started v1.1

NOTE: This page is for an **older** version of wfview (v1.1). For the current version, please see this page.

This page is here for archival purposes.

Before launching wfview, it is important to make sure the radio is set up properly.

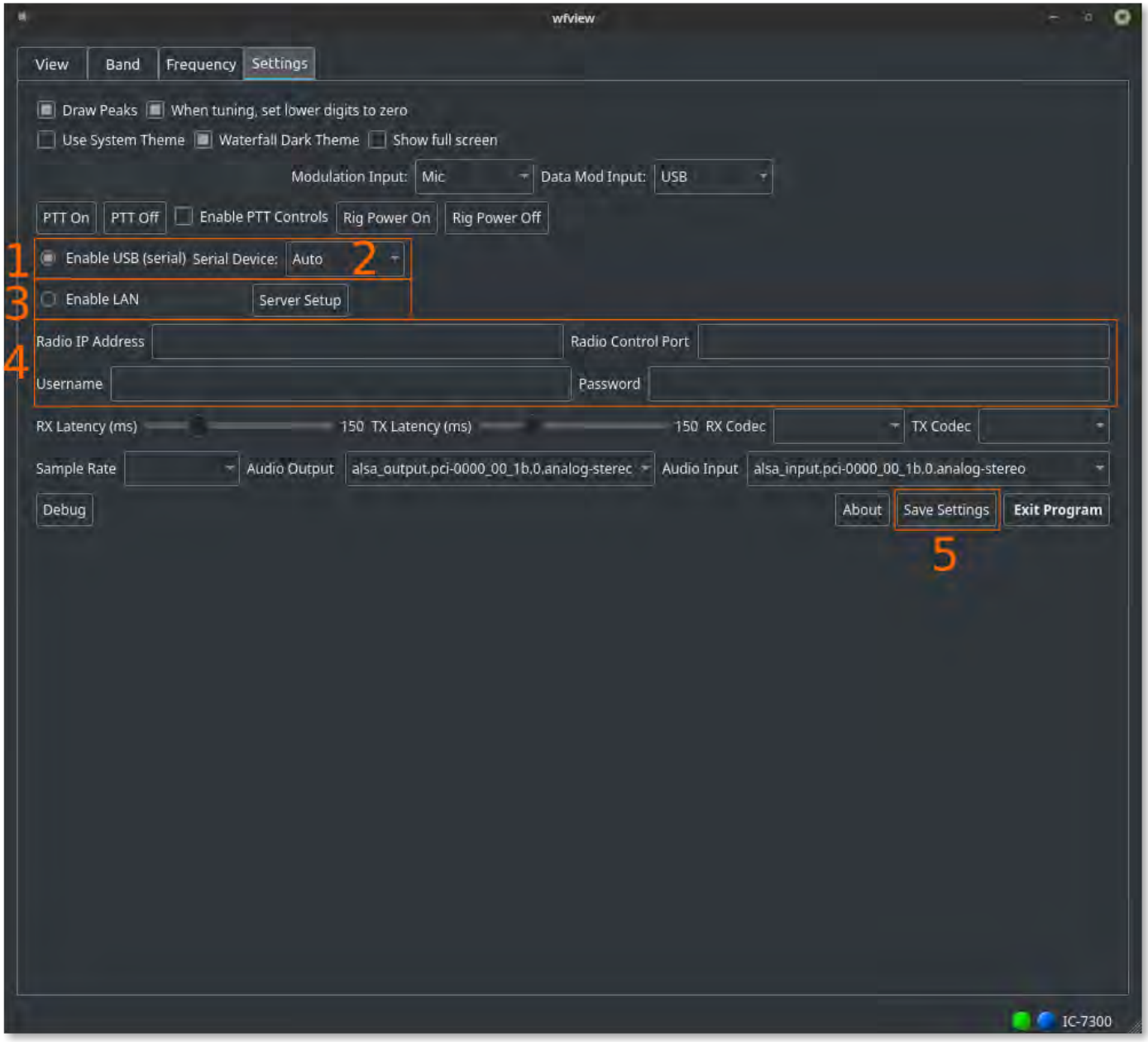
Radio Settings:

For **network access radios** (IC-9700, IC-7850, IC-7610, IC-705), make sure that the network is set up (SET > Network), and that **Network Control is On**. You'll also need a "User1" ID and a User1 password. Take note of the Control Port (UDP) value, which is by default 50001. Also note the IP address of the radio. Lastly, you must enable **CI-V Transceive** (SET > Connectors > CI-V). Many other programs will send commands to disable CI-V transceive, so keep an eye on it.

For **serial port access radios** (IC-7300 and basically any radio connected with USB or with a serial port), you will need to set the CI-V "**Unlink from [REMOTE]**" to (SET > Connectors > CI-V) **true**. Next, set the "**CI-V USB Baud Rate"** to 115200 (required for spectrum access). Finally, **enable** the "**CI-V Transceive**" option, which provides wfview with a telemetry stream from the radio.

wfview Settings:

wfview is designed to be simple and intuitive. Common controls are on the "View" tab, which is what the program shows when it opens. When you first open wfview, it will attempt to find and use a rig connected via USB (serial). If you don't see your rig, or your rig uses a network connection, you'll need to enter it in on the Settings tab. Note that the user doesn't need to specify the rig model number or CI-V address — wfview will auto-discover this information.



Annotated settings

On the settings tab, you can select from [1]"Enable USB (serial)" or [3]"Enable LAN".

Use the "**Enable USB (serial)**" option if the rig is connected directly to your computer via a USB cable. You can then select the USB serial device fro the combo menu to the left [2], or specify one manually if needed. For most cases, wfview will find the port connected to an icom radio automatically. Do note, that your account must have permission to access the port. If the connection is successful, make sure to press [5]"Save Settings" for future use.

If you're accessing the radio over **network connection**, select [3]"**Enable LAN**". Note that you must first enable Network Control on the radio. Next, enter [4] in the Radio's IP address (or hostname), Radio control port (the default is 50001), Username, and Password (as set on the rig's screen). The remaining controls for latency and sources should be left at their default values. Press "Connect", and if the connection is successful, make sure to press [5]"Save Settings" for future use.

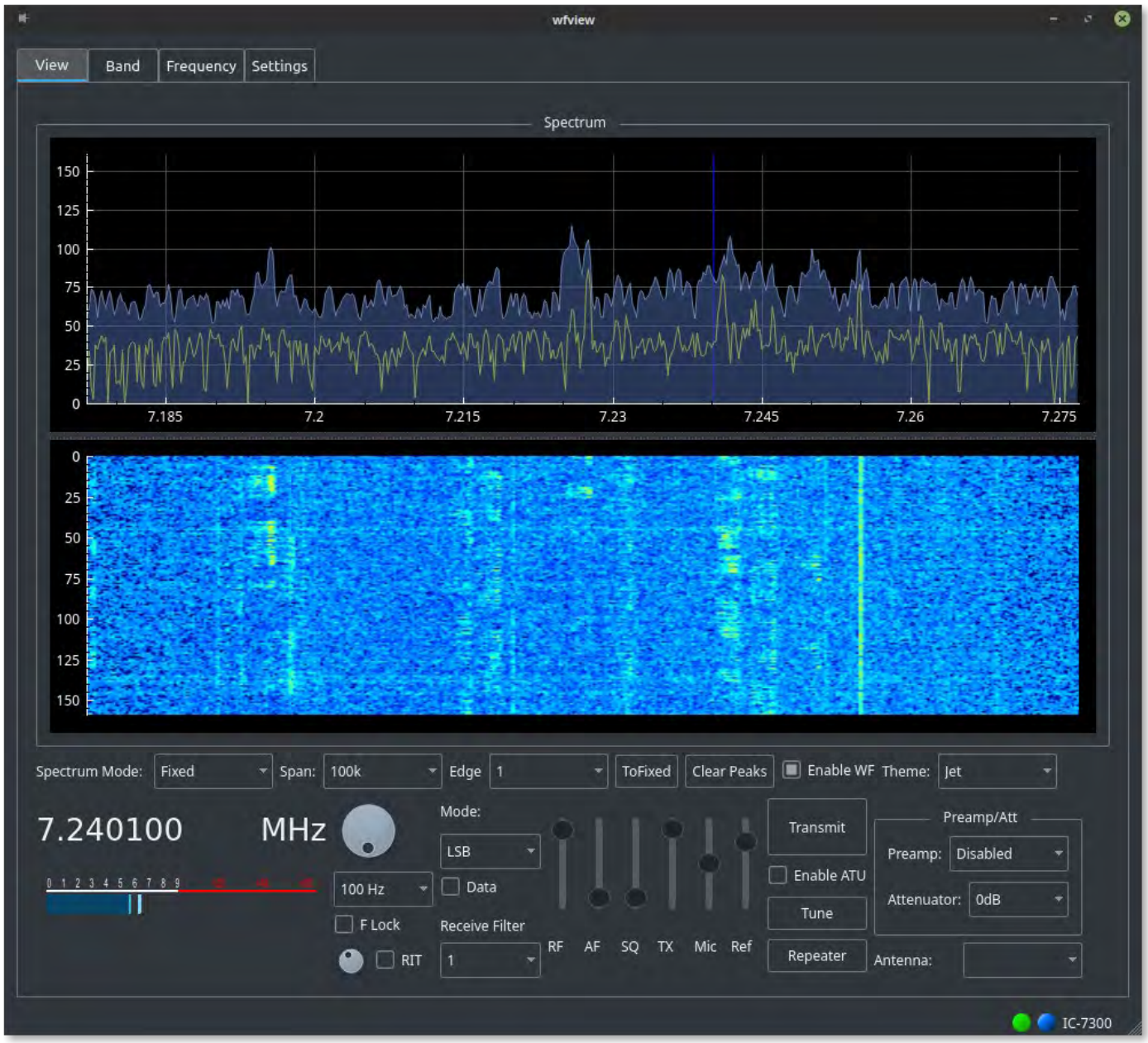
Once connected, you'll probably want to go back to the View tab. From here, the radio may be adjusted as per the many controls found on the page.

wfview

Open Source interface for Icom and Kenwood transceivers

View Tab (V1.1)

This is the “View” tab of wfview:



View tab of wfview

This tab of wfview is the main tab, where the waterfall display can be viewed and the rig’s most common controls are located. During a QSO or while browsing the airwaves, this is the tab that will be usually used. You can quickly

jump to this tab any time by pressing F1.

The Spectrum and Waterfall:

The top visual section is the spectrum plot. This plot shows the intensity of signals at each frequency. There is also a “peak” plot behind the main plot, which holds the highest signal level received at a given frequency. The blue vertical line shows the tuned frequency.

The lower section is a waterfall display, showing the “history” of signals received. In the middle, between the two plots, is a divider that may be dragged up and down.

Using the scroll wheel in either plot tunes the radio per the selected tuning step. Holding shift divides the tuning step by ten, and holding control multiplies the tuning step by ten.

Clicking within either plot reports the clicked frequency to the status bar at the bottom of the window. Double-clicking causes the radio to tune to the selected frequency.

Spectrum controls:

- Spectrum Mode: Select the desired mode, either Fixed or Center. For rigs with updated firmware, Scroll-C and Scroll-F are also available.
- Span: This selects the span, to the left and right, of the center frequency.
- Edge: Selects the edge. Rigs with updated firmware support four edge presets.
- ToFixed: When you press this button, the currently shown range of spectrum is placed into the currently selected Edge, thus creating a fixed range from the current centered range. This is useful when you want to change frequencies and don’t want the spectrum display to slide. Keep in mind this will override whatever edge is currently programmed in the selected edge slot. Once you are comfortable with this feature, it will become second nature.
- Clear Peaks: This button clears the held peaks. The peaks also automatically clear whenever the spectrum range changes.
- Enable WF: This control toggles the spectrum plot and waterfall data on and off at the rig. Use this if you need “quieter” CI-V traffic or need lower bandwidth on the graphics. This control is also useful to “hold” the current display if you wish to take a screenshot of an interesting signal.
- Theme: this combo box is a menu of preset color themes for the waterfall display. The themes come from the plotting library, QCustomPlot, and are not currently editable (though that may change).

The S-Meter:

This meter shows the instantaneous signal level of the received signal, as well as a peak value (bright line) and an average (thin blue line). The polling rate is 10ms. During transmit, the meter shows transmit power (watts), however, at this time, the scale does not change to reflect the transmit units (this will be done with future version of wfview).

The Frequency controls:

The frequency controls include a tuning knob, tuning step selection combo box, frequency lock, and RIT.

- Tuning Knob: This knob may be either “turned” with the mouse button, or, operated using the scroll wheel of the mouse when the mouse is over the knob. Tuning steps are as selected in the tuning step combo box, and can be modified with control or shift to select ten times wider steps or ten times smaller steps, respectively.
- Tuning Steps: The tuning steps combo box provides many common tuning steps to select between. Keep in mind that there is an option under Settings to automatically round the frequency to the nearest tuning step.
- F Lock: Locks the frequency tuning controls of wfview. This does not lock the controls at the rig, and in fact, the rig’s frequency lock control doesn’t lock the controls of wfview. Thus, they are independently lockable.

RIT: Receive Incremental Tuning: Use this control to specify a receive-only offset, often useful for tuning in stations on sideband that may be slightly off-frequency. The RIT checkbox must be checked to use the knob. The RIT range is programmed to allow for +/- 500 Hz.

Mode and Receive Filter:

Controls in this section allow for the selection of various modes and receive filters.

- Mode: This combo box allows the user to select from the available modes on the radio.
- Data: This checkbox toggles “data” mode on the radio, which is generally used to toggle between different modulation inputs and receive filters.
- Receive Filter: Use this combo box to pick between the rig’s three receive filter presets. (Future version of wfview will allow the user to adjust these presets.)

Slider Adjustments:

These controls adjust the radio’s various level parameters. The names are short so as to allow the controls to be close together, however, there is generally useful mouse-over help text.

- RF: This control adjusts the receiver’s RF Gain.
- AF: For USB-connected radios, this control adjusts the rig’s speaker volume. For LAN-connected radios, this control adjusts the local computer’s receive volume.
- SQ: Receiver Squelch.
- TX: Transmit power. Keep in mind that some radios have different power levels for each band, so you may need to bump this control as you change bands. Future version of wfview will be more aware of this level and track it as the bands change.
- Mic: This slider adjusts the transmit (modulation) audio level. The name of the slider changes depending upon what input source is selected under the Settings tab.
- Ref: This slider adjusts the reference level for the spectrum display.

Transmit and Receive (PTT):

The “Transmit” button keys the transmitter. Once the transmitter sends wfview confirmation that it is keyed, the button changes to a receive button with “Receive” as the text. It is similar to a latching PTT, just as the radio has on the front panel. The keyboard commands Control-R (receive) and Copntrol-T (transmit) are always available. Additionally, in QSO, if you have last clicked the Transmit button, then pressing the space bar will generally press the button again (the button retains focus). Keep in mind that the PTT must be enabled on the Settings tab (it is disabled by default). There are also dedicated PTT ON and PTT OFF buttons under settings, in case there becomes a need to manually send either command (as opposed to the toggle action of this main button). It’s also worth noting that wfview has a 5 minute timer on transmit time. If you hit the 5 minute mark, wfview will return the rig to receive mode automatically. This is just to help you in case your cat sits on the space bar of your computer while you’re out getting coffee.

Antenna Tuner Unit (ATU):

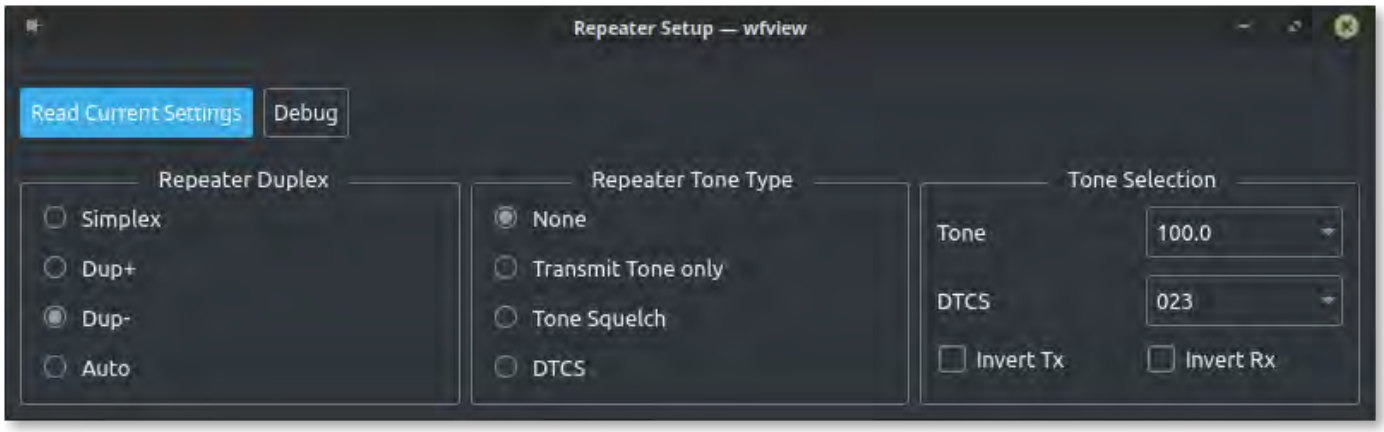
For rigs equiped with either an internal ATU or an attached compatible ATU, the following controls apply:

- Enable ATU: Enables the ATU.
- Tune: Runs the ATU’s tuning cycle. wfview will probe the rig for tuner status, so watch the status bar in the program to see if the tuning is successful. If not successful, the ATU enabled checkbox is unchecked.

Repeater Button:

The “Repeater” button brings up a dialog box with controls needed to operate repeaters, including PL tone, Tone

Squelch, DCS/DTCS, and Offset. Of course, some of these controls can be useful for dedicated receivers, and they all can be used for simplex and split operation.



Repeater Setup controls for Tone and Offset

Repeater Duplex:

These controls toggle the duplex of the radio. The “Auto” option uses the automatic mode (generally only correct in the US). Keep in mind that with the Auto selection, the rig won’t compute an automatic offset until the frequency is changed. Note, Icom uses “DTCS” as their DCS name. This is probably done to lower confusion with the DD/DV mode digital squelch.

Repeater Tone Type:

This is the selection for the type of tone to be transmitted and/or received:

- None: No tone transmitted, and no tone required for receive
- Transmit Tone only: Transmit the selected PL tone. No tonesquelch.
- Tone Squelch: Transmit the selected PL tone, and also, require this tone for the tonesquelch. There is no provision currently for mismatched tones.
- DTCS: Digital Continuous Squelch (“DCS”). This enables transmitting a DCS code as well as a DCS squelch. There is no provision currently for mismatched tones.

Tone Selection:

- Tone: Use this to select the PL tone. Once clicked, arrow keys can be used to rapidly try different tones as a sort of “tone scanner”. Tones can also be typed in once the menu is clicked.
- DTCS: Use this control to select the DCS tone.
- Invert Tx and Invert Rx: These controls invert their respective DCS tones. Most systems do not require these to be checked.

Preamp and Attenuator:

For rigs with built-in attenuators and preamps, these selection boxes allow for selection. Keep in mind, some rigs do not allow preamp and attenuator at the same time, and so, you may see one selection changing in response to another. For these controls, wfview simply sends the command and then checks the status of both parameters, following the rig’s status closely.

Antenna Selection:

For radios with multiple antenna options, this combo box allows the current antenna to change. This combo box is not sufficient for the IC-7850/IC-7851, as this radio allows for more complicated arrangements, but may be

sufficient for other radios. Future version of wfview will include more options for this selection.

Status Bar:

The status bar shows two “LEDs”, the rig connected, and any useful status messages. The LEDs indicate PTT status (green for receive, red for transmit — this is the LED on the left), and rig connectivity (blue means connected).

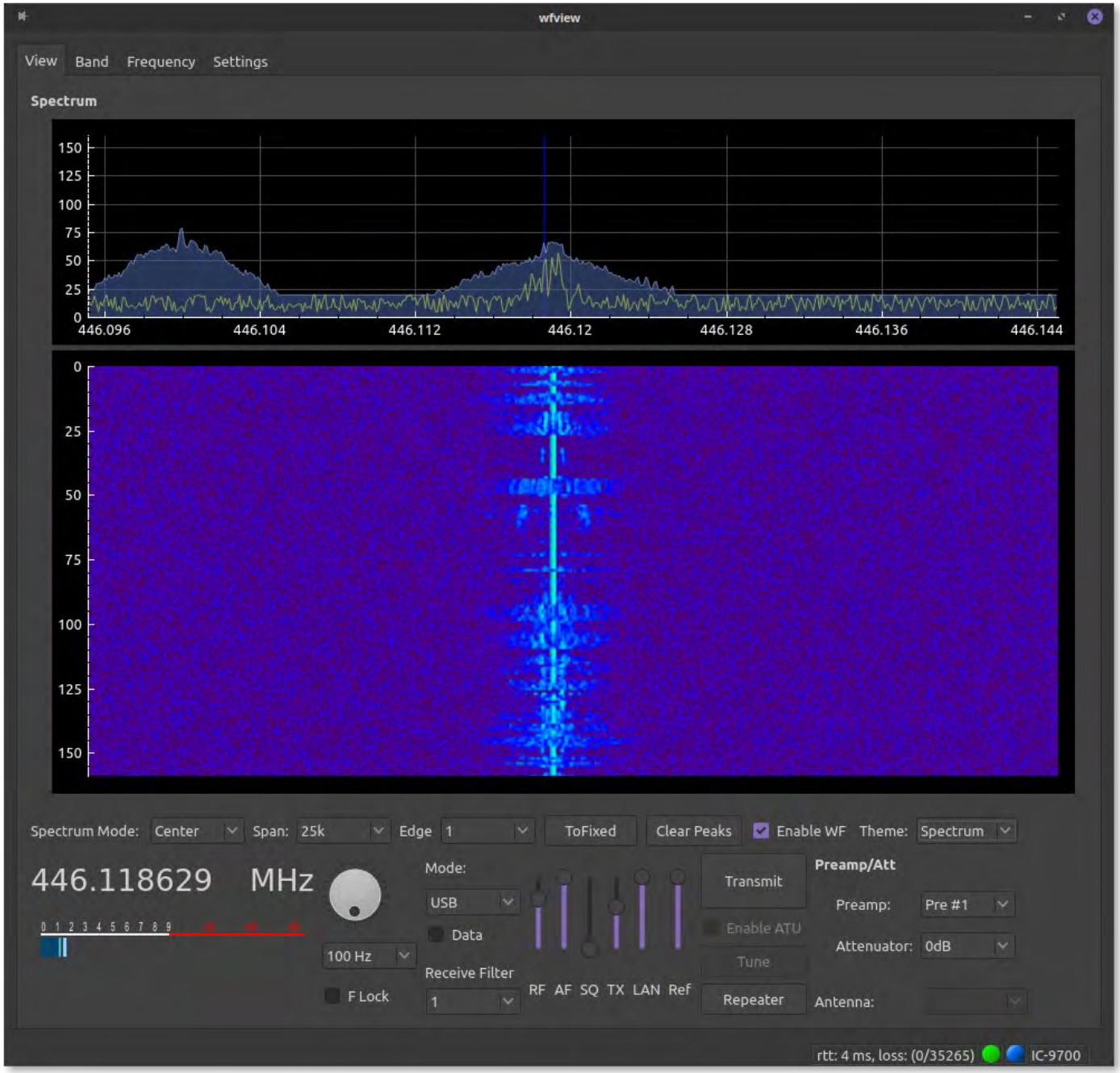
Appearance Customization Examples (V1.1)

Methods of Customization:

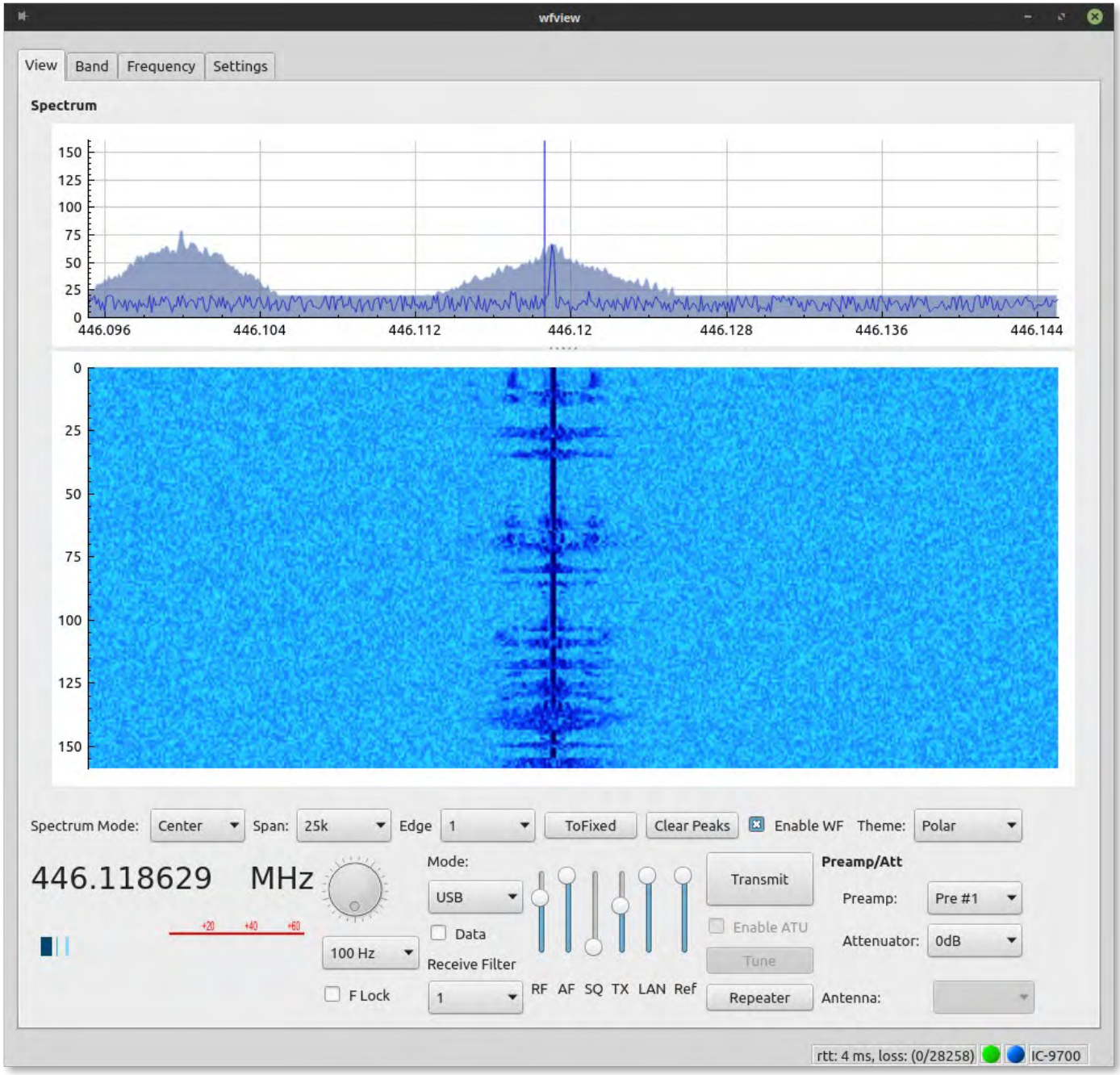
wfview's appearance can be customized through several means:

1. Select the "Use system theme" checkbox, and then change the theme using your system's theme selector. Depending upon how your system is setup, the theme selector should generate GTK and qt themes (we use qt). These themes get applied to the window background and widgets (buttons, menus, etc).
2. Check or un-check the "Waterfall Dark Theme" box, which switches between the dark and light preset themes (which are also read from the preference file)
3. Select a theme for the waterfall view using the Theme combo box on the View tab (bottom-right of the waterfall display).
4. Customize the colors for the plot in the preference file, and then re-launch.
5. Customize the stylesheet. We're using QDarkStyleSheet, and it is installed in /usr/share/wfview/stylesheets on linux. For mac and windows builds, at this time, the file is bundled into the application and can't be edited after the program is compiled.
6. Specify an alternate stylesheet in the preferences file (the attribute is "StylesheetPath"), and then re-launch. Note, images referenced in a custom stylesheet *must* be given a full path, ie, "/usr/share/wfview/stylesheets/oldschoolknobs/images/knob.png". This is because qt will assume the images are based out of the current working directory, not the location of the stylesheet.

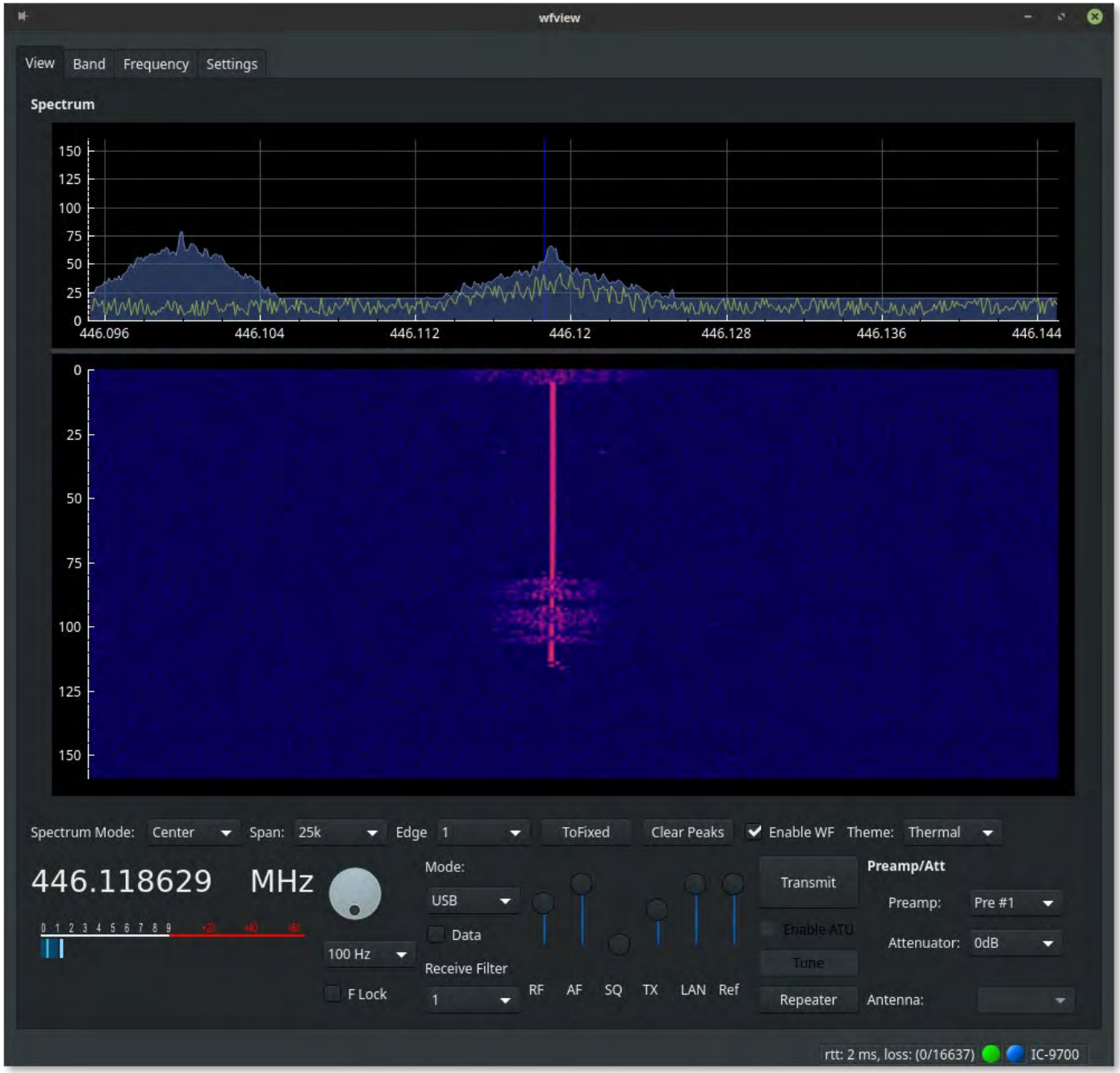
Examples:



Mint's purple theme with the "Spectrum" color scheme selected for the waterfall, and dark plot theme.



Mint's Aqua theme with the "Polar" theme selected for the waterfall, and light plot scheme.



Mint's Dark theme with the "Thermal" waterfall color scheme, and dark waterfall color selected.

Preferences File (V1.1)

The preferences file contains data that are read upon opening wfview. The contents are saved upon pressing the Save Settings button under Settings.

These data include settings for the program's appearance, radio connection, window position, and more. Editing this file is not recommended; do so at your own risk, and only if you need to do so.

Accessing:

The preference file may be accessed as follows:

macOS: Use the defaults command, specifying the org as org.wfview.org

Linux: Read the plain text file from ~/.config/wfview/wfview.conf

Windows: Configuration is stored in the Windows Registry under **"HKEY_CURRENT_USER\SOFTWARE\wfview\wfview"**. Please be careful when editing the Windows Registry as incorrect usage of the regedit.exe utility can seriously damage your Windows installation. DO NOT change anything outside the registry key above unless you know what you are doing!

Preference File Sections and Attributes:

Each section is as follows (note that the qt library does not allow these to be ordered and they may appear in any order):

[Controls]

These options are from options under the Settings tab:

- EnablePTT: Set to 'true' or 'false' to enable or disable the PTT controls of wfview. Does not change the radio's own PTT lock controls.
- NiceTS: Set to 'true' or 'false' to enable or disable the frequency rounding that can occur when tuning the frequency or double-clicking in the spectrum.

[Interface]

Most of these options are from checkboxes under the Settings tab:

- DrawPeaks: Set to 'true' or 'false' to enable or disable the peak plot behind the spectrum plot
- ConfirmPowerOff: Boolean value to disable the "Are you sure you want to power off" dialog box.
- ConfirmExit: Boolean value to disable the "Are you sure you want to quit?" dialog box
- StylesheetPath: Set to the path of a custom stylesheet, relative to /usr/share/wfview/stylesheets
- UseDarkMode: Set to 'true' or 'false' to set the dark mode option for the spectrum and waterfall plot areas
- UseFullScreen: Set to 'true' or 'false' to enable full screen on startup.

- UseSystemTheme: Set to ‘true’ or ‘false’ to enable or disable using the system’s theme (versus a stylesheet — included or user-added)
- splitter: The position of the draggable plot-to-waterfall splitter. Probably best not to edit this by hand.
- windowState: The state of the window (position and maximized status)
- windowGeometry: The location and size of the window. Remove this variable and the one above if you have difficulty seeing the wfview window, this will restore the default.
- WFAntiAlias: boolean for waterfall anti-alias
- WFInterpolate: boolean for interpolation between the waterfall data points and the pixels on the screen
- WFLength: length of waterfall display. Values should be from 160 to 1023.
- Meter2Type: Selected type of meter for the secondary meter. Integers represent the meter number per the enumerated types in the source code.

[Radio]

These options are specific to serial port access:

- RigCIVuInt: Use this parameter to manually set the CI-V address of the radio, in unsigned integer format. If set, wfview will not broadcast a request for all radios to identify, instead, it will send a rig ID request only to the specified CI-V address. This is useful if you have a radio that does not respond to broadcast queries on the CI-V bus, or, if you have more than one radio on the same physical CI-V bus (very rare in the post-USB-adapter decades). If unset or set to 0, wfview will automatically find any connected radio (regardless of if that radio is a known model) and attempt to control it. The default value is 0, and is recommended for most users.
- CIVisRadioModel: Boolean value. When set to “true” causes wfview to assume the radio model is the same as the manual set CI-V above. For older radios that do not answer Rig ID queries, supply the default CI-V address above and then set this parameter to true.
- SerialPortBaud: Use this attribute to specify a custom baud rate, often needed on older radios. The default is 115200, required by the newer radios that provide spectrum data over serial.
- SerialPortRadio: Use this attribute to specify a custom path to the serial port. This can be used, for example, to specify a pseudo-terminal if you are using socat to forward serial traffic from a remote host. If unset, or set to “auto”, wfview will automatically find directly-connected supported Icom radios on the linux platform. On other platforms, it will attempt to use the first available serial port.
- localAFgain: Unsigned char (0-255) used to store the user’s local AF gain (volume). Does not change the volume at the radio, just how loud it sounds at the computer.
- VirtualSerialPort: String path to the virtual pseudo-terminal created for other programs to use as a serial port.

[Server]

These options are for the built-in OEM-compatible server, useful for placing non-ethernet rigs onto a network.

- ServerEnabled: true/false whether to enable wfview’s internal server which emulates an Icom rig and allows multiple clients to connect. This is still quite experimental
- ServerControlPort: The UDP port that clients will make an initial connection to.
- ServerCivPort: The UDP port that the server announces to clients, 50002 is the default.
- ServerAudioPort: The UDP port that the server announces to clients, 50003 is the default.
- ServerNumUsers: The number of users currently configured.
- ServerUsername_x, ServerPassword_x, serverUserType_x: contain the user information.

[LAN]

These options are for configuring the network access to radios that either have built-in network, are connected to a PC running Icom’s software, or are connected to a host running wfview with the above [Server] enabled.

- EnableLAN: true or false to enable LAN connection to the radio (rather than USB)
- AudioInput: Name of audio input device, this is best set via the Settings tab as it must be spelled exactly

- AudioOutput: Name of audio output device, this is best set via the Settings tab as it must be spelled exactly
- AudioRXCodec: This is the receive codec that is sent to the radio, 0x01 = single channel uLaw, 0x02 = single channel 8bit PCM, 0x04 = single channel 16bit PCM, 0x08 is two channel 8bit PCM, 0x10 is two channel 16bit PCM and 0x20 is two channel uLaw. It is anticipated that other modes, potentially including compression will be added at some point for wfview-wfview communications.
- AudioTXCodec: This is the receive codec that is sent to the radio, 0x01 = single channel uLaw, 0x02 = single channel 8bit PCM, 0x04 = single channel 16bit PCM. It is anticipated that other modes, potentially including compression will be added at some point for wfview-wfview communications.
- AudioRXLatency: Use this to specify the number of milliseconds of latency to be allowed in RX.
- AudioTXLatency: Use this to specify the number of milliseconds of latency to be allowed in TX.
- AudioRXSampleRate: This is the sample rate to tell the radio to use for RX audio. 48000 is the default but 8000, 16000 and 24000 are also available. Any other rate will likely be rejected by the radio. Smaller sample rates will reduce the required network bandwidth but may reduce quality.
- AudioTXSampleRate: This is the sample rate to tell the radio to use for TX audio. 48000 is the default but 8000, 16000 and 24000 are also available. Any other rate will likely be rejected by the radio. Smaller sample rates will reduce the required network bandwidth but may reduce quality.
- ClientName: This will usually be the “name” of your computer but can be replaced with any other name. This cannot be changed from within wfview, so must be changed by manually editing the file/registry entry.
- ControlLANPort: This is the “control” port that wfview will use to communicate with the radio and defaults to 50001. Any other required ports are configured by the radio.
- EnableRigCtlD: true or false to enable an emulation of rigctld within wfview. This is highly experimental and currently only supports a small subset of rigctld commands. This is disabled by default and canot be enabled from within the wfview GUI.
- RigCtlPort: Port that the internal rigctld emulation should listen on. Default is 4533 so as to not conflict with any other rigctld running on the machine.
- IPAddress: IP Address of the radio
- Username: Username configured within Network menu of the radio
- Password: Password configured for the above user in the radio.
- ResampleQuality: This is a number from 0-10 which determines the quality of wfview’s internal audio resampler, with 0 being the lowest quality. This is only used if a sample rate of anything other than 48000 is used and the default of 4 has been chosen to be middle-of-the-road in terms of quality and speed.

[Memory]

This section contains the 99 memories managed by wfview. Don’t bother making a parser for this format, we will change to CSV in future versions.

[LightColors]

More to come here. These colors are unsigned integers made from numbers in the format 0xAARRGGBB, where AA = opaqueness, RR = red, GG = green, and BB = Blue

Like HTML colors, but with the AA for the opaqueness. FF = fully opaque, 00= transparent. The hex numbers must be converted to unsigned int, which you can do using python or many websites with number conversion tools.

```
Light_PeakPlotLine=4278190335
Light_PlotAxisPen=4291348680
Light_PlotBackground=4294967295
Light_PlotBasePen=4278190080
Light_PlotLegendBorderPen=4278190080
Light_PlotLegendBrush=3372220415
Light_PlotLegendTextColor=4278190080
Light_PlotTickLabel=4278190080
```

```
Light_PlotTickPen=4278190080
Light_TuningLine=4278190335
```

[DarkColors]

```
Dark_PeakPlotLine=4294967040
Dark_PlotAxisPen=4283124555
Dark_PlotBackground=4278190080
Dark_PlotBasePen=4294967295
Dark_PlotLegendBorderPen=4294967295
Dark_PlotLegendBrush=3355443200
Dark_PlotLegendTextColor=4294967295
Dark_PlotTickLabel=4294967295
Dark_PlotTickPen=4294967295
Dark_TuningLine=4278190335
```

[StandardColors]

These colors are not read by wfview. They are provided as an example to demonstrate how the colors are formatted in this file. The colors in the file are stored as unsigned integer, 32-bit, and contain the following bytes: AARRGGBB, where A is the alpha (opaque-ness) channel, R is Red, G is Green, and B is Blue. The values may be converted easily using the python hex() function, for example: hex(2164195328) gives: ‘0x80ff0000’, which is 50% translucent red.

```
black=4278190080
blue_opaque=4278190335
blue_translucent=2147483903
cyan=4278255615
green_opaque=4278255360
green_translucent=2147548928
magenta=4294902015
red_opaque=4294901760
red_translucent=2164195328
white=4294967295
yellow=4294967040
```

wfview

Open Source interface for Icom and Kenwood transceivers

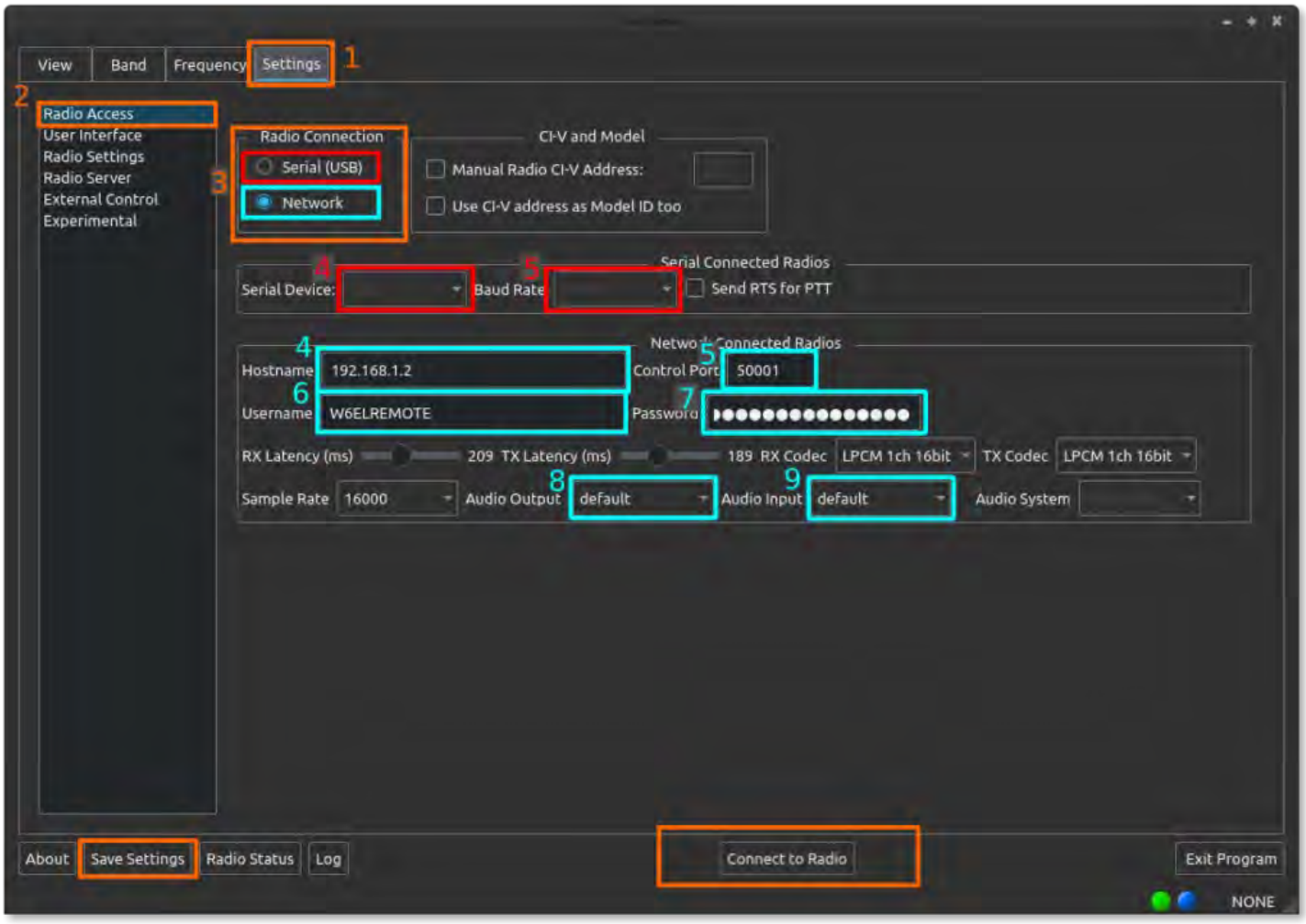
Getting Started (V 1.50-1.65)

Once wfview has been downloaded, you will need to set up the radio and the program.

Radio Settings (via the Radio's Menus):

- Set the radio to the default CI-V address.
- Enable CI-V Transceive.
- For radios with ethernet or wifi: Set a known username and password, and **enable** Network Control. Make a note of the radio's ip address, which is visible in the Network Settings screen on the radio.
- For radios with Serial or USB: Set the baud rate to the highest supported rate, and make a note of the rate. Here is an in-depth guide to IC-7300 radio settings.

wfview settings:



First:

Press [1] “Settings” and then press [2] “Radio Access”. Select [3] either Serial (USB) or Network.

For Serial (USB) radios (7300):

- Select the [4] Serial Device
- Select the [5] Baud Rate
- **Do not** check “Manual Radio CI-V Address”
- **Do not** check “Use CI-V address as Model ID too”

For Network (ethernet or wifi) radios (705, 7610, 9700):

- Type in the [4] IP address of the radio or radio server. If you know the hostname, you may supply the hostname instead.
- Enter the [5] Control Port. The default is 50001 on most radios.
- Enter the [6] username as set on the radio.
- Enter the [7] password
- Select the [8] audio output device you wish to use. Typically this is your headset or speakers. Linux users may simply select “default” for their desktop environment’s default audio choice.
- Select the [9] audio input device you wish to use. Typically this is your microphone. Linux users may simply select “default” for their desktop environment’s default audio choice.
- **Do not** check “Manual Radio CI-V Address”
- **Do not** check “Use CI-V address as Model ID too”

Next, for both types:

Press “Connect to Radio” (bottom of the window). If successful, the radio model number should appear in the window title bar and the lower-right corner of the window. Once working, **press “Save Settings”** (bottom left of the window).

You can read the entire User Manual here.

Settings Tab (v1.64)

NOTE: This is for wfview version 1.64

The Settings tab has several “pages” of settings:

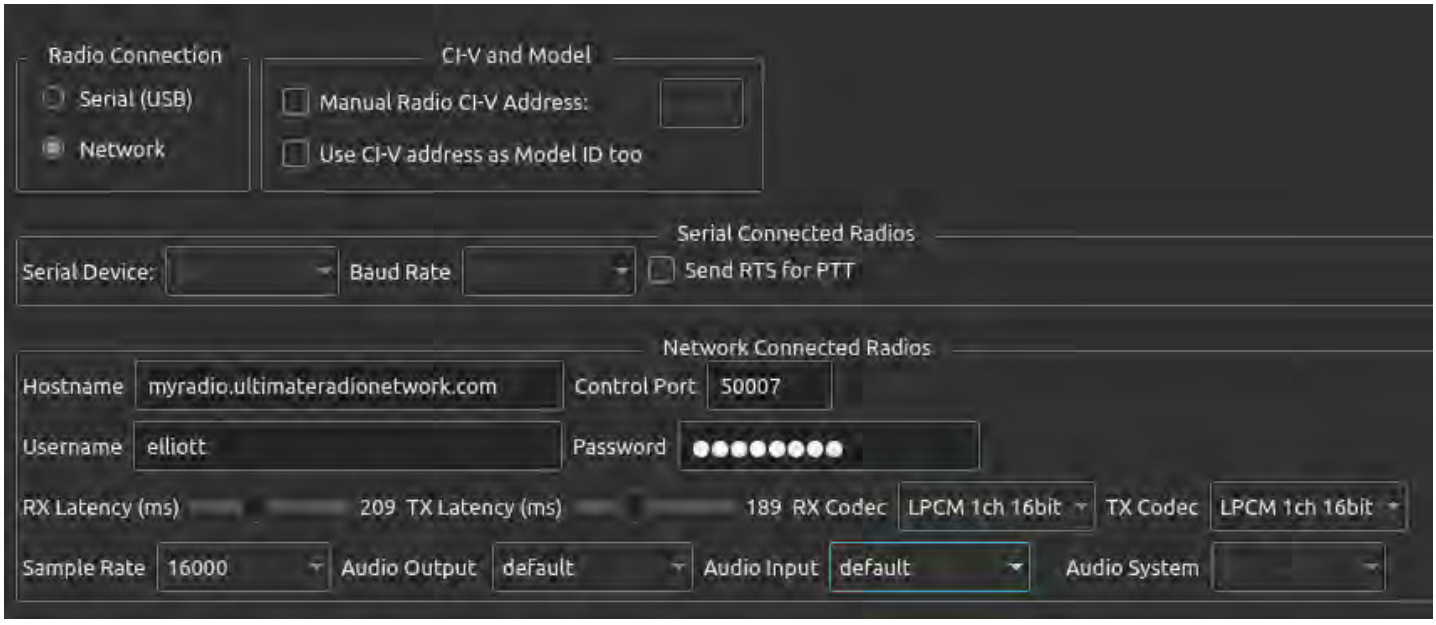
Each page contains categories of settings. The settings are saved in a format detailed here.

Quickly jump to each page:

Radio Access, **User Interface**, **Radio Settings**, **Radio Server**, **External Control**, **Experimental**

Radio Access

Under Radio Access are controls relating to how wfview access your radio:



Radio Connection:

- Serial (USB): Select this option if your radio connects to this computer using a Serial port, USB port, or Serial CI-V USB adapter.
- Network: Select this option if you are accessing your radio over a network

CI-V and Model:

These controls should not be used except for very old radios that do not support CI-V Radio ID queries or for unusual circumstances. wfview automatically figures out the CI-V address and model number for most radios. Don't forget to turn on CI-V Transceive, as we use that feature to automatically find your radio.

- Manual Radio CI-V Address: Check this box to manually set the radio's CI-V address. The input format is hex, which is the same as the radio's menu and documentation. For example, if you have an Icom IC-756 Pro, enter "5C" here for the default CI-V address on this model. **We recommend not checking this box** unless absolutely necessary. Enable CI-V Transceive on your radio instead.
- Use CI-V address as Model ID too: Check this box to cause wfview to infer the radio's capabilities by assuming the provided CI-V address is the same as the model number. This is needed on older radios that do not respond to Radio ID queries (command 0x19 0x00). Set the radio to the default CI-V address and enter it in as a manual address above. This is not needed on most radios made after the IC-718 (late 90s). Just enable CI-V Transceive on the radio and leave this unchecked.
- Usage Tip: You can also use these two boxes to cause a radio to masquerade as a different model, which is useful if wfview doesn't have support for your radio but you know that the capabilities are similar enough to another model. Simply set the radio's CI-V address on the radio to the default address of a supported model, and then enter this address in manually in wfview, and check the "Use CI-V address as Model ID too" box.

Serial Connected Radios:

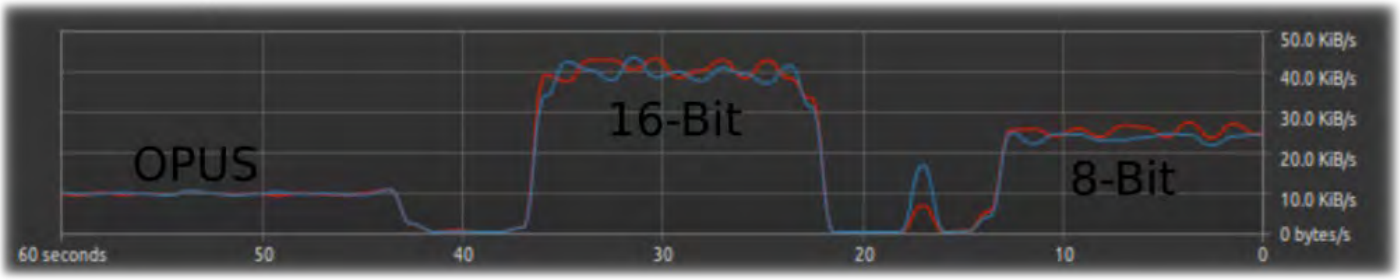
- Serial Device: Select the serial device entry here. On Linux, "Auto" will automatically select OEM Icom radios with built-in USB ports. Use "Manual" if you need to manually enter a port that isn't on the list (for example, if you are using a mysterious serial port)
- Baud Rate: Select the radio's baud rate. It's recommended to set the radio to the highest-supported baud rate unless there are problems. For the IC-7300, you must set the baud rate to 115200 in order to get spectrum data.
- Send RTS for PTT: check this box if the radio does not have a PTT command. This will cause wfview to toggle the RTS serial line to indicate PTT, which is supported on many rig interface boxes. wfview will intercept any PTT command from external programs and toggle the RTS as well.
- Tip: Linux users can automatically set useful names for USB serial devices with reasonable default permissions. See this page on how to do it.

Network Connected Radios:

- Hostname: Type in the radio's (or radio server's) IP address or hostname here.
- Control Port: Type in the radio's control port here. The default value is 50001. The other ports used will be ascertained via the control channel automatically.
- Username: Type in the user name as is set on the remote radio here.
- Password: Type in the password as is set on the remote radio here.
- RX Latency (ms): This is the size of the receive audio buffer, in milliseconds. Larger values are needed for less-stable connections.
- TX Latency (ms): This is the size of the transmit audio buffer, in milliseconds. Larger values are needed for less-stable connections. Tip: Set the RX latency control to the lowest value and advance until there are no dropouts. Advance about 20ms further, and then set the TX Latency to the same value.
- RX Codec and TX Codec: These select the codec for the audio going to and from the radio:
    - LPCM 1/2 ch 16bit: PCM Lossless encoding, one (or two) channel, 16-bits per sample
    - LPCM 1/2 ch 8 bit: PCM Lossless encoding, one (or two) channel, 8-bits per sample
    - uLaw 1/2 ch 8 bit: Logarithmic bit space 8-bit per sample. This is a higher-quality format than LPCM 8-

bit, but still has the advantage of only needing 8 bits per sample.
    - Opus: This is a high-quality perceptually-lossless (but technically lossy) format, which provides a 4:1 reduction in audio bandwidth. Opus is only supported when the radio server is another computer running a wfview server, or the Xiego X6100. Opus is the recommended choice when the other side is wfview. If selected on an unsupported direct-to-radio connection, wfview will default back to LPCM 16 bit.
    - Tip: Some VPNs and ISPs re-order and mangle UDP packets, causing 16-bit packets to get split. Using uLaw 8 bit or Opus will solve this problem.

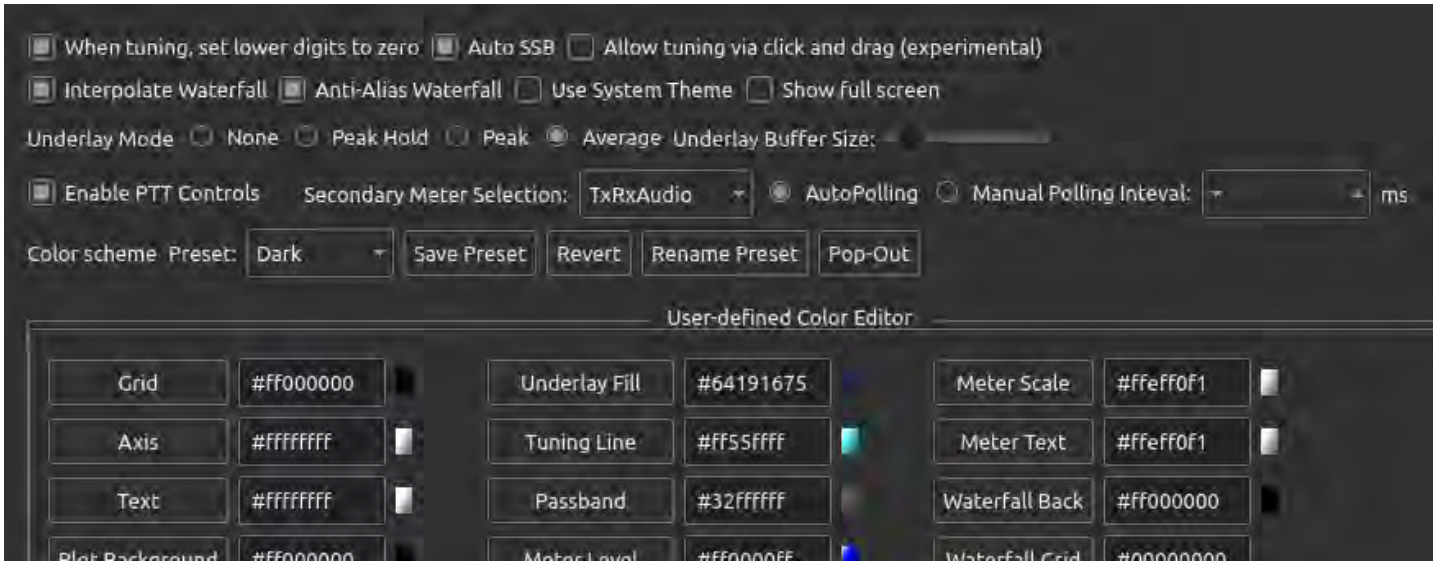
Here is a graph showing the network bandwidth in kilobytes per second required for the Opus codec, 16-bit LPCM, and 8-Bit LPCM (same as 8-bit uLaw). Each was taken with a 1-channel format with a sample rate of 16 KHz:



- **Please note**, the audio configuration cannot be changed while connected to a radio; please press “Disconnect from Radio” prior to making changes.
- Sample Rate: Select the desired sample rate here. Remember, the audio will never exceed half the sample rate. For example, using 16 KHz as a sample rate will limit the audio to just under 8 KHz, but will also require only 1/3rd the bandwidth of a 48 KHz sample rate connection. Lowering the sample rate is an excellent way to reduce network bandwidth. Values below 16 KHz may sound a bit lower in quality, but at 16 KHz and above is generally favorable. wfview takes care of converting the audio for your computer’s hardware sample rates.
- Audio Output: Select your computer’s speakers or headset speakers here. Some platforms will provide a “default” device, which is generally acceptable.
- Audio Input: Select your computer’s microphone or headset microphone here. Again, some platforms provide a “default” device.
- Audio System: This lets you select from the three available audio systems. The differences between the different systems can include latency, device sharing, and reliability. To change audio systems, first disconnect from the radio, select an audio system, and then, very important, re-visit the Audio Input and Audio Output combo boxes. Do not forget to click on these boxes and make sure to re-select the device. The devices tend to get re-ordered and renamed, so it is worth clicking and reading all the available choices each time. Once selected, you can then re-connect to the radio.
    - Qt Audio: This is the default choice and should work on all platforms.
    - Port Audio: This system may provide a more consistent stream of audio for some users.
    - RT Audio: For some people, this is a lower-latency choice
    - Tip: If your current audio system works, just leave this alone and don’t worry about it :-).

User Interface

These settings determine many aspects of the interface and user experience in wfview.



- When tuning, set lower digits to zero: This control causes the tuning dial to automatically set the below-tuning-step digits to zero. This is generally desired. Uncheck this control if you want to dial in and keep an offset applied to all your tuning, for example if you want frequencies always ending in 5 KHz for some reason.
- Auto SSB: This setting will automatically select the “correct” sideband when switching frequencies. This is only done if you are already in LSB or USB mode, and only if you change frequencies using the Frequency tab (or hit the star/asterisk key).
- Allow tuning via click and drag: This checkbox made it into the master branch accidentally. If enabled, you can try clicking and dragging on the spectrum to tune. It is an experimental option and still under development.
- Interpolate Waterfall: If you have a big screen, check this box and the pixels of the waterfall will not look quite as blocky and pixelated. Checking this box can add significant CPU use, so do not do this on a humble platform like a Raspberry Pi.
- Anti-Alias Waterfall: Check this box for anti-aliasing on the waterfall display. This seems to have almost no impact on the performance or appearance! But it’s an option if you want it.
- Use System Theme: Check this box to switch from wfview’s built-in theme (QDarkStyleSheet) to your desktop environment’s default appearance. Read more about this here.
- Show full screen: Check this box for full-screen mode. You can also use F11 to toggle full screen. Tip: For a dedicated wfview display, use a raspberry pi set to launch wfview on startup under the default user. Check the full screen box and save the settings.
- Underlay Mode: These four options select an “underlay” for the spectrum plot. The underlay is an additional plot shown under the radio’s spectrum data. Two of the underlay options obtain data from a variable-size buffer of received spectra, in which one value is deleted each time a new value is received. The size can be set using the buffer size slider. The available underlay options are:
    - None: no underlay drawn
    - Peak Hold: An underlay is drawn showing the maximum value received. The peaks do not expire except when tuning.
    - Peak: A buffered peak mode. The peak value in the buffer (as set with the Buffer Size slider) is plotted for each frequency.
    - Average: The average value for each frequency is plotted. This provides a “smoothed” spectrum mode which often helps beat down broad-band noise such as RFI.
- Underlay Buffer Size: Sets the size of the underlay buffer. Lower values will be more responsive, higher values will be more “sticky”. Tip: Press “Pop-Out” to see the effects of adjusting this slider in real-time as you make the adjustments.
- Enable PTT Controls: Enables wfview’s transmit button. This checkbox has no effect on the radio’s physical PTT buttons, nor does it block any other radio control programs from sending PTT commands. It is designed to prevent inadvertent transmission, and could be useful during open radio demonstrations. wfview has a

built-in 3 minute timeout on transmit.

- Secondary Meter Selection: Select a desired secondary meter here. Some radios do not support some meter types.
    - Available meters: None, SWR, ALC, Compression, Voltage, Current, Center (IC-R8600 only), TxRxAudio (switches Tx/Rx automatically), RxAudio (RX audio all the time), TxAudio (TX audio all the time)
    - See more about the meter under the View Tab manual page.
- Polling: wfview automatically determines the optimal polling rate for the radio. However, faster polling will make the meters more responsive and is generally fine so long as a secondary program isn't polling as well. Slower polling rates may increase reliability, especially with older radios using single-wire CI-V. If you are running wfview as a server as well, it can make sense to set the polling period to a high value on the server so that the server isn't contributing much to the overall polling activity.

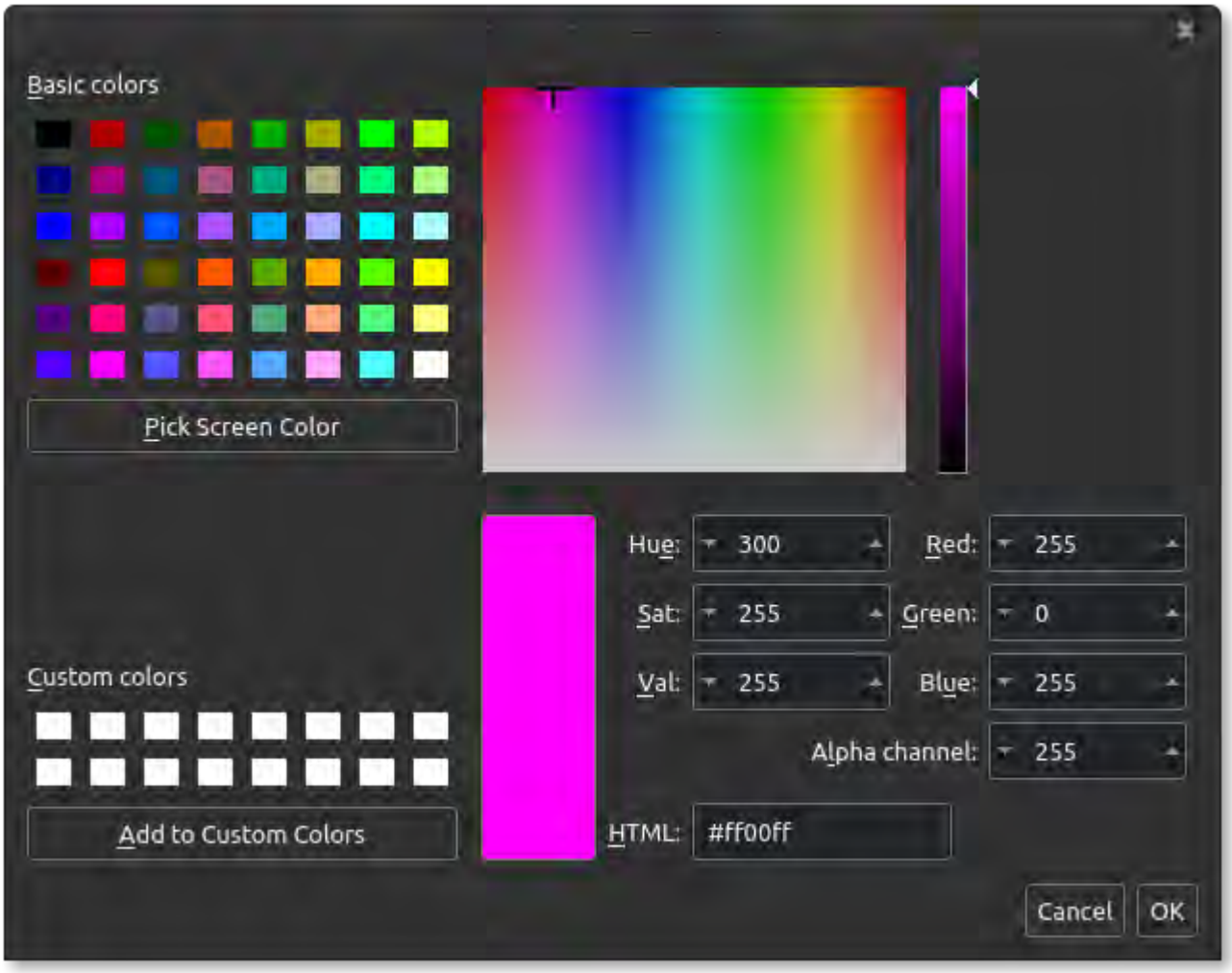
Color Scheme:

Use these controls to adjust the color scheme. Pressing "Pop-Out" will let you see the color scheme changes as you are working on them. More information on customizing the appearance is available here.

- Preset: This control is for recalling the five available color scheme presets. Presets are saved with the settings and may be renamed and reverted.
- Save Preset: This button will save the selected preset's colors and name to the settings file.
- Revert: Press this button to revert the selected preset to the default values.
- Rename Preset: Press here to rename the selected preset. Names may be up to ten characters long and may contain spaces and probably even unicode characters.
- Pop-Out: Read carefully, Jumangi players. Pressing this button pops the Settings tab out into a separate window. This will let you see the changes you are making on the View tab. When you are done making changes, you must press the button a second time to re-dock the Settings window back into a tab. The button will say "re-attach" in this state, and the pop'd out Settings window *cannot be closed until you press this button*. The first player to reach the golden city and yell Jumangi wins.

User-Defined Color Editor

Use these buttons to change the colors of many of the user-interface elements of wfview. The color editor allows a custom color to be selected. Simply press the button for a given attribute (such as "Grid", "Axis", "Text", etc) and the color picker will be shown:



The color can then be previewed. Important: It is easy to accidentally set the Alpha channel to a low (transparent) value and not realize it, since the picker does not demonstrate the transparency. So be mindful. On the other hand, you can set an element to fully transparent (value 0) to completely hide it. wfview will warn the user if a color is set to fully transparent, but it is fine to do so.

You can also simply type in an HTML-style color in the text boxes. The format is #AARRGGBB, where AA is the transparency (00 for fully transparent, ff for fully opaque), RR is red, GG is green, and BB is blue. The “#” is optional and will be added automatically. Don’t forget to save the preset you just worked on, otherwise it will be gone next time you open wfview.

Radio Settings

These settings are convenience functions to help with radio operation.

- Modulation Input: Select from the available transmit audio choices:
  - LAN: Network audio for radios with built-in network support (this is what most network users will want)
  - USB: USB port for radios with a USB port (this is what IC-7300 users will need for remote IC-7300 access using wfview’s built-in server)
  - MIC: Microphone jack on the radio

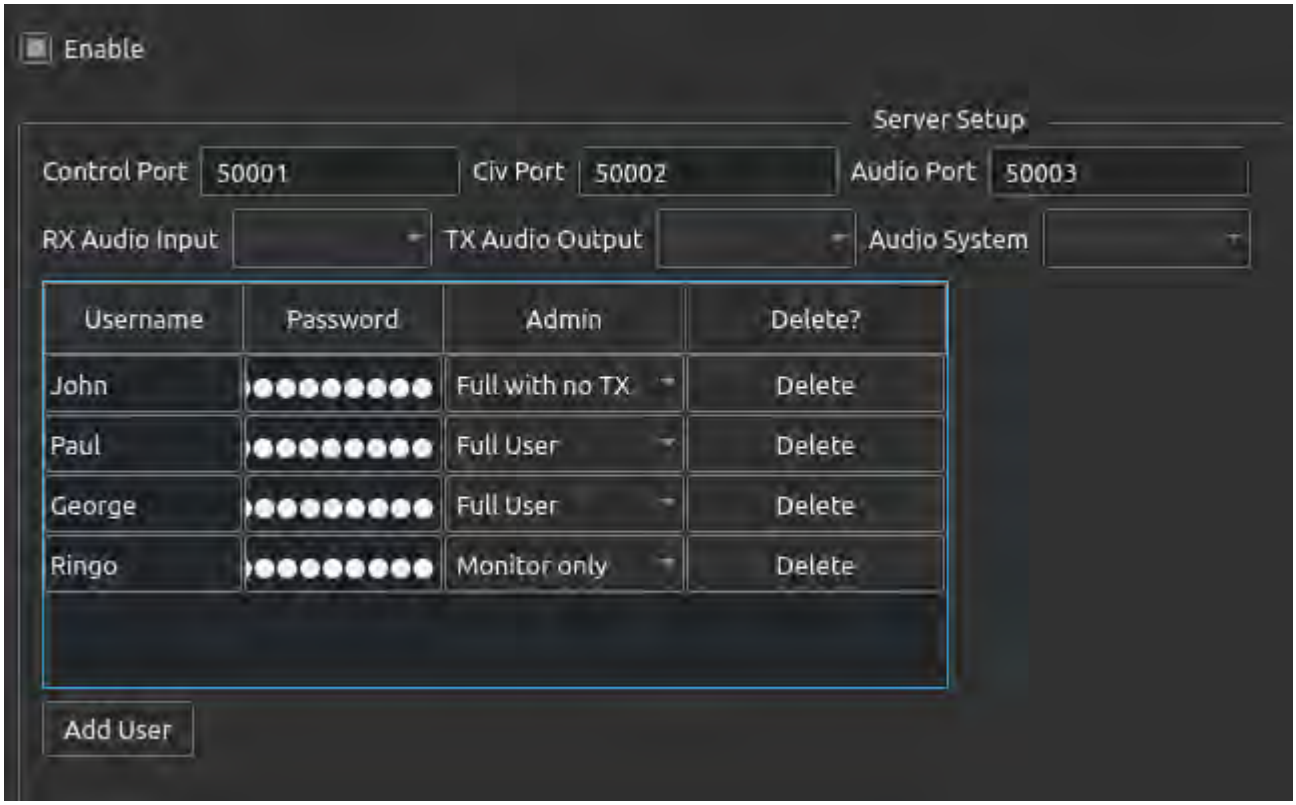
    - ACC: Accessory Port
- Data Mod Input: This is the transmit audio source for data modes. Generally not set to MIC. Usually you want this set to LAN or USB.
- Set Clock: Sets the radio's clock to the same as the computer's clock. wfview will send the time command at the next change-of-minute. Watch the status bar carefully and you may catch it.
- Use UTC: Sets the radio's clock to the current UTC time, again using the computer's clock as a reference.
- Adjust Reference (IC-9700 only): Brings up the reference adjustment window. It is very useful to switch to the view tab and zoom in on a reliable signal while adjusting these controls.
- Manual PTT Toggle: Use these buttons to manually toggle the PTT on or off. Useful if the View tab's Transmit/Receive button becomes out-of-sync with the world.

Radio Server

This page contains settings for the built-in radio server. This server is designed to take a serial or USB port radio, with audio, and allow it to be accessed over a network by another copy of wfview running on another computer. The radio server handles full-duplex streaming audio and radio control. The server is largely compatible with the Icom OEM protocol, but has been extended to allow for some additional functionality.

The wfview team has tested this feature extensively, with radios as old as the IC-736 and as modern as the IC-7300. Even across the ocean, audio and control can work quite well.

See this page of the manual for more information about remote control.

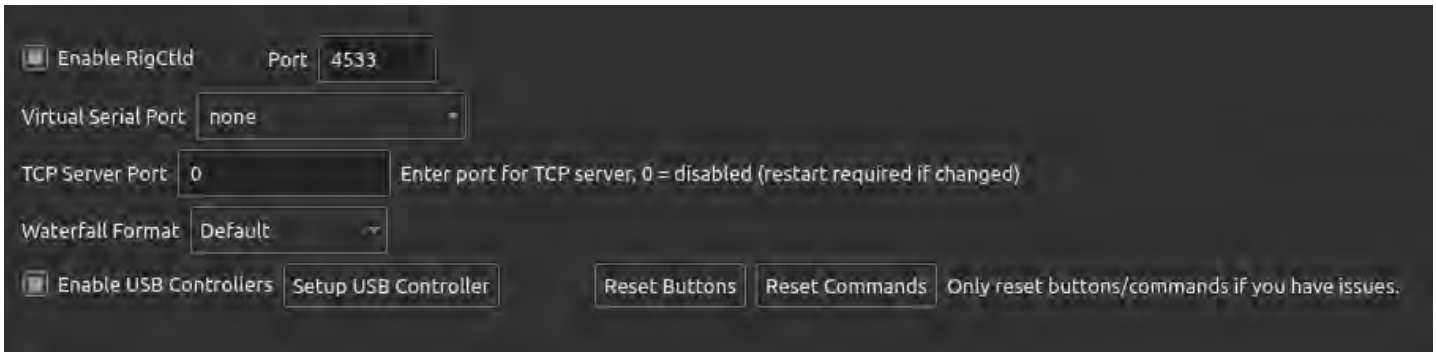


Brief description of controls:

- Enable: Check this box to enable the built-in server
- Control Port: This is the primary port for the server, used for server-related control tasks. The default is 50001.
- CI-V Port: This is the port through which CI-V traffic is passed. The default is 50002
- Audio Port: This is the port through which audio is passed. The default is 50003.
- Note: All traffic is UDP

External Control

These settings help other programs control the radio while wfview is open. This is useful if you like to use digital modes or logging programs. The best and most simple-to-use option is wfview's excellent Hamlib Rigctld server, which emulates the hamlib rigctld program for TCP/IP control. Almost any program written to support Hamlib can utilize this mode of control. When TCP/IP isn't an option, the built-inVirtual Serial Port (pseudo-terminal) support will create a fake serial port which other programs can connect to as though connecting to a genuine radio's serial port directly. The Virtual Serial Port can only accept one program at a time, so if you need more than one, try and use the rigctld option, as it can allow for multiple clients at once.



Tip: The Hamlib model number for network control is #2, called "Hamlib NET rigctl (stable)". The device string is "127.0.0.1:4532" (substitute whatever port is in wfview). fldigi can connect to wfview using this method if it is compiled with hamlib support. The port most programs expect is 4532.

- Enable Rigctld: Enables the built-in rigctld server.
- Port: This is the port that wfview's built-in rigctld server will listen on. The port most programs expect is 4532.
- Virtual Serial Port: Select a virtual serial port here through which other programs designed for a serial port can connect. See here for more details.
- TCP Server Port: This is a raw TCP/IP port for CI-V traffic and is compatible with N1MM+ and may work with RUMLogNG. To disable this feature, set it to zero.
- Waterfall Format: wfview can arrange the waterfall spectrum data so that the data are arranged as multiple separate chunks (serial connection style) or as single long messages (network radio style). This is useful if you are using a program on the Virtual Serial Port that expects serial connection style data, for example.
- Enable USB Controllers: This checkbox enables the use of external hardware USB controllers, such as the Icom RC-28, Contour Shuttle Express, Contour ShuttlePRO V2, and XBox. Please see this page for more information on how to setup and use external USB controllers.
- Setup USB Controller: Press here to open the USB Controller Setup page
- Reset Buttons: Press here to clear all button assignments
- Reset Commands: Press here to clear all button assignments and to reset your controller preferences. Use this option if the assigned button commands are malfunctioning. Once clicked, Save Settings and exit wfview.

See this page for more information on USB Controllers.

See this page of the manual for more information about sharing radio control. Also see this page on sharing audio with other programs.

Experimental

This page contains experimental and undocumented options. Currently, there aren't any!